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TITLE: NASA Awards Global Modeling, Assimilation Support Contract

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TITLE: High-Tech Computing

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TITLE: SCaN.

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Jet Propulsion LaboratoryFeb 11, 2025ArticleContentsDawn of Twilight CloudsCloud MysteryMastcam's Partial ViewMore About CuriosityNews Media ContactsTo view this video please enable JavaScript, and consider upgrading to a web browser thatsupports HTML5 videoNASA's Curiosity Mars rover captured these drifting noctilucent, or twilight, clouds in a 16-minute recording on Jan. 17. (This looping clip has been speeded up about 480 times.) The white plumes falling out of the clouds are carbon dioxide ice that would evaporate closer to the Martian surface.NASA/JPL-Caltech/MSSS/SSIWhile the Martian clouds may look like the kind seen in Earth's skies, they include frozen carbon dioxide, or dry ice.Red-and-green-tinted clouds drift through the Martian sky in a new set of images captured by NASA's Curiosity rover using its Mastcam — its main set of "eyes." Taken over 16 minutes on Jan. 17 (the 4,426th Martian day, or sol, of Curiosity's mission), the images show the latest observations of what are called noctilucent (Latin for "night shining"), or twilight clouds, tinged with color by scattering light from the setting Sun.Sometimes these clouds even create arainbow of colors, producing iridescent, or "mother-of-pearl" clouds. Too faint to be seen in daylight, they're only visible when the clouds are especially high and evening has fallen.Martian clouds are made of either water ice or, at higher altitudes and lower temperatures, carbon dioxide ice. (Mars' atmosphere is more than 95% carbon dioxide.) The latter are the only kind of clouds observed at Mars producing iridescence, and they can be seen near the top of the new images at an altitude of around 37 to 50 miles (60 to 80 kilometers). They're also visible as white plumes falling through the atmosphere, traveling as low as 31 miles (50 kilometers) above the surface before evaporating because of rising temperatures. Appearing briefly at the bottom of the images are water-ice clouds traveling in the opposite direction roughly 31 miles (50 kilometers) above the rover.Dawn of Twilight CloudsTwilight clouds were first seen on Mars byNASA's Pathfinder missionin 1997;Curiositydidn't spot them until 2019, when it acquired itsfirst-ever imagesof iridescence in the clouds. This is the fourth Mars year the rover has observed the phenomenon, which occurs during early fall in the southern hemisphere.Mark Lemmon, an atmospheric scientist with the Space Science Institute in Boulder, Colorado, led apapersummarizing Curiosity's first two seasons of twilight cloud observations, which published late last year in Geophysical Research Letters. "I'll always remember the first time I saw those iridescent clouds and was sure at first it was some color artifact," he said. "Now it's become so predictable that we can plan our shots in advance; the clouds show up at exactly the same time of year."Each sighting is an opportunity to learn more about the particle size and growth rate in Martian clouds. That, in turn, provides more information about the planet's atmosphere.Cloud MysteryOne big mystery is why twilight clouds made of carbon dioxide ice haven't been spotted in other locations on Mars. Curiosity, which landed in 2012, is on Mount Sharp in Gale Crater, just south of the Martian equator. Pathfinder landed in Ares Vallis, north of the equator.NASA's Perseverancerover, located in the northern hemisphere’s Jezero Crater, hasn’t seen any carbon dioxide ice twilight clouds since its 2021 landing. 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For more about Curiosity, visit:

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TITLE: La NASA identifica causa de pérdida de material del escudo térmico de Orion de Artemis I

11 min readLa NASA identifica causa de pérdida de material del escudo térmico de Orion de Artemis I

Erika PetersDec 06, 2024ArticleContentsLos hallazgosEl proceso de investigación de la NASAAvances en el escudo térmicoEl 28 de junio de 2024, la nave espacial Orion de Artemis II es retirada de la Celda de Ensamblaje Final y Pruebas del Sistema (FAST, por sus siglas en inglés) y colocada en la cámara de altitud oeste dentro del Edificio de Operaciones y Revisión del Centro Espacial Kennedy de la NASA en Florida. Dentro de la cámara de altitud, la nave espacial se sometió a una serie de pruebas que simulaban las condiciones de vacío del espacio profundo.Crédito de la foto: NASA / Rad SinyakRead this story in Englishhere.Tras extensos análisis y pruebas, la NASA ha identificado la causa técnica de la pérdida imprevista de material carbonizado en el escudo térmico de la nave espacial Orion de Artemis I.Los ingenieros determinaron que, cuando Orion regresaba de su misión sin tripulación alrededor de la Luna, los gases generados dentro del material ablativo exterior del escudo térmico, denominado Avcoat, no pudieron ventilarse y disiparse como estaba previsto. Esto permitió que se acumulara presión y se produjeran grietas, lo que causó que parte del material carbonizado se desprendiera en varios lugares.“Nuestros primeros vuelos de Artemis son una campaña de prueba, y el vuelo de prueba de Artemis I nos dio la oportunidad de comprobar nuestros sistemas en el entorno del espacio profundo antes de incorporar a la tripulación en futuras misiones”, dijo Amit Kshatriya, administrador asociado adjunto de la Oficina del programa De la Luna a Marte, en la sede de la NASA en Washington. “La investigación sobre el escudo térmico ayudó a garantizar que comprendiéramos completamente la causa y la naturaleza del problema, así como el riesgo que les pedimos a nuestras tripulaciones que asuman cuando emprendan su viaje a la Luna”.Los hallazgosLos equipos técnicos adoptaron un enfoque metódico para comprender e identificar el origen del problema de pérdida de material carbonizado, incluyendo el muestreo detallado del escudo térmico de Artemis I, la revisión de las imágenes y los datos de los sensores de la nave espacial, y pruebas y análisis exhaustivos en tierra.Durante Artemis I, los ingenieros utilizaron la técnica de guiamiento de reentrada atmosférica doble para el regreso de Orion a la Tierra. Esta técnica ofrece más flexibilidad ya que amplía el alcance del vuelo de Orion después del punto de reentrada para llevarlo hasta un lugar de amerizaje en el océano Pacífico. Con esta maniobra, Orion se sumergió en la parte superior de la atmósfera de la Tierra y utilizó la resistencia atmosférica para reducir su velocidad. A continuación, Orion utilizó la sustentación aerodinámica de la cápsula para rebotar y salir de nuevo de la atmósfera, para luego volver a entrar en el descenso final con paracaídas para su amerizaje.Utilizando los datos de la respuesta del material Avcoat de Artemis I, el equipo de investigación pudo simular el entorno de la trayectoria de entrada de Artemis I —una parte clave para comprender la causa del problema— dentro de la instalación de chorro en arco del Centro de Investigación Ames de la NASA en California. Observaron que, durante el período entre las inmersiones en la atmósfera, las tasas de calentamiento disminuyeron y la energía térmica se acumuló dentro del material Avcoat del escudo térmico. Esto condujo a la acumulación de gases que forman parte del proceso de ablación (desgaste) previsto. Debido a que el Avcoat no tenía “permeabilidad”, la presión interna se acumuló y produjo el agrietamiento y el desprendimiento desigual de la capa exterior.Después de que la nave espacial Orion de la NASA fuera recuperada al término del vuelo de prueba Artemis I y transportada al Centro Espacial Kennedy de la NASA en Florida, su escudo térmico fue separado del módulo de la tripulación en el interior del Edificio de Operaciones y Revisión y rotado para su inspección.Crédito: NASALos equipos técnicos realizaron extensas pruebas en tierra para simular el fenómeno de rebote en la reentrada antes de la misión Artemis I. Sin embargo, hicieron pruebas a velocidades de calentamiento mucho más altas que las que la nave espacial experimentó durante su vuelo. Las altas velocidades de calentamiento puestas a prueba en tierra permitieron que el material carbonizado permeable se formara y se desgastara como estaba previsto, liberando la presión del gas. El calentamiento menos severo observado durante la reentrada real de Artemis I desaceleró el proceso de formación de material carbonizado, al tiempo que siguió creando gases en esta capa de material. La presión del gas se acumuló hasta el punto de agrietar el Avcoat y liberar partes de la capa carbonizada. Las mejoras recientes en la instalación de chorro en arco han permitido una reproducción más precisa de los entornos de vuelo registrados por Artemis I, de modo que este comportamiento de agrietamiento pudo demostrarse en pruebas en tierra.Si bien Artemis I no estaba tripulado, los datos del vuelo mostraron que, si la tripulación hubiera estado a bordo, habría estado a salvo. Los datos de la temperatura de los sistemas del módulo de tripulación dentro de la cabina también estaban dentro de los límites y se mantenían estables, con temperaturas alrededor de los 24 grados centígrados (75 grados Fahrenheit). El desempeño del escudo térmico superó las expectativas.Los ingenieros comprenden tanto el fenómeno material como el entorno con el que interactúan los materiales durante la entrada a la atmósfera. Al cambiar el material o el entorno, pueden predecir cómo responderá la nave espacial. Los equipos de la NASA acordaron por unanimidad que la agencia puede desarrollar un análisis de vuelo aceptable que mantenga a la tripulación segura utilizando el actual escudo térmico de Artemis II con cambios operativos para su entrada en la atmósfera.El proceso de investigación de la NASAPoco después de que los ingenieros de la NASA descubrieran las condiciones del escudo térmico de Artemis I, la agencia comenzó un extenso proceso de investigación, el cual contó con un equipo multidisciplinario de expertos en sistemas de protección térmica, aerotermodinámica, pruebas y análisis térmicos, análisis de estrés (fatiga de materiales), pruebas y análisis de materiales, y muchos otros campos técnicos relacionados. El Centro de Ingeniería y Seguridad de la NASA también participó para aportar su experiencia técnica, incluyendo evaluación no destructiva, análisis térmico y estructural, análisis de árbol de fallas y otros métodos de respaldo de las pruebas.“Nos tomamos muy en serio nuestro proceso de investigación del escudo térmico, con la seguridad de la tripulación como la fuerza impulsora que mueve esta investigación", dijo Howard Hu, gerente del Programa Orion del Centro Espacial Johnson de la NASA en Houston. "El proceso fue extenso. Le dimos al equipo el tiempo necesario para investigar todas las causas posibles, y trabajaron incansablemente para asegurarse de que entendiéramos el fenómeno y los pasos necesarios para mitigar este problema en futuras misiones".El escudo térmico de Artemis I estaba muy cargado de instrumentos para este vuelo, e incluía sensores de presión, extensómetros y termopares a diferentes profundidades del material ablativo. Los datos de estos instrumentos acrecentaron el análisis de muestras físicas, lo que permitió al equipo validar modelos informáticos, crear reconstrucciones de entornos, proporcionar perfiles de temperatura interna y dar información sobre el momento de la pérdida de material carbonizado.Alrededor de 200 muestras de Avcoat fueron extraídas del escudo térmico de Artemis I en el Centro de Vuelo Espacial Marshall de la NASA en Alabama para su análisis e inspección. El equipo llevó a cabo una evaluación no destructiva para "ver" dentro del escudo térmico.Uno de los hallazgos más importantes que arrojó el examen de estas muestras fue que algunas superficies en la zona del Avcoat permeable, las cuales habían sido identificadas antes del vuelo, no sufrieron agrietamiento ni pérdida de material carbonizado. Dado que estas superficies eran permeables al comienzo de la entrada en la atmósfera, los gases producidos por la ablación pudieron ventilarse adecuadamente, eliminando la acumulación de la presión, el agrietamiento y la pérdida de material carbonizado.Un bloque de pruebas de Avcoat se somete a una prueba de pulso de calor dentro de una cámara de pruebas de chorro de plasma en arco en el Centro de Investigación Ames de la NASA en California. El elemento de prueba, configurado con secciones Avcoat permeables (superior) y no permeables (inferior) para su comparación, ayudó a confirmar la comprensión de la causa raíz de la pérdida de material Avcoat carbonizado que los ingenieros vieron en la nave espacial Orion tras el vuelo de prueba Artemis I más allá de la Luna.Crédito: NASALos ingenieros hicieron ocho campañas separadas de pruebas térmicas posteriores al vuelo para respaldar el análisis del origen de estas condiciones, y completaron 121 pruebas individuales. Estas pruebas fueron llevadas a cabo en instalaciones en diferentes lugares de Estados Unidos que cuentan con capacidades únicas, entre ellas: la Instalación de Calentamiento Aerodinámico en el Complejo de Chorro en Arco del centro Ames, para poner a prueba perfiles de calentamiento convectivo con diversos gases de prueba; el Laboratorio de Evaluación de Materiales Endurecidos por Láser en la Base de la Fuerza Aérea Patterson-Wright en Ohio, con el fin de poner a prueba perfiles de calentamiento radiativo y proporcionar radiografías en tiempo real; y la Instalación de Calentamiento por Interacción del centro Ames, para poner a prueba perfiles combinados de calentamiento convectivo y radiativo en el aire en bloques completos, esto es, aplicando todas las pruebas en cada bloque de material.Los expertos en aerotermia también completaron dos campañas de pruebas en el túnel de viento hipersónico del Centro de Investigación Langley de la NASA en Virginia y en las instalaciones de pruebas aerodinámicas del CUBRC en Buffalo, Nueva York, para realizar pruebas con una diversidad de configuraciones de pérdida de material carbonizado, y mejorar y validar los modelos analíticos. También se realizaron pruebas de permeabilidad en el centro Kratos en Alabama, en la Universidad de Kentucky y en el centro Ames para caracterizar aún mejor el volumen elemental y la porosidad del Avcoat. La instalación de pruebas del centro de investigaciones Advanced Light Source, una instalación para usuarios científicos del Departamento de Energía de Estados Unidos en el Laboratorio Nacional Lawrence Berkeley, también fue utilizada por los ingenieros para examinar el comportamiento del calentamiento del Avcoat a nivel microestructural.En la primavera de 2024, la NASA creó un equipo de revisión independiente que realizó una revisión exhaustiva del proceso de investigación, los hallazgos y los resultados de la agencia. La revisión independiente fue dirigida por Paul Hill, un exdirectivo de la NASA que se desempeñó como director principal de vuelo del transbordador espacial para el programa Return to Flight (Regreso a los vuelos) después del accidente del Columbia, quien también dirigió la Dirección de Operaciones de Misiones de la NASA y es miembro actual del Panel Asesor de Seguridad Aeroespacial de la agencia. La revisión se llevó a cabo durante un período de tres meses a fin de evaluar las condiciones del escudo térmico posteriores al vuelo, los datos del entorno para la entrada a la atmósfera, la respuesta térmica del material ablativo y el avance de las investigaciones de la NASA. El equipo de revisión estuvo de acuerdo con los hallazgos de la NASA sobre la causa técnica del comportamiento físico del escudo térmico.Avances en el escudo térmicoAl saber que la permeabilidad de Avcoat es un parámetro clave para evitar o minimizar la pérdida de material carbonizado, la NASA tiene la información correcta para garantizar la seguridad de la tripulación y mejorar el desempeño de los futuros escudos térmicos del programa Artemis. A lo largo de su historia, la NASA ha aprendido de cada uno de sus vuelos e incorporado mejoras en el hardware y las operaciones. Los datos recopilados a lo largo del vuelo de prueba de Artemis I han proporcionado a los ingenieros información valiosísima para guiar futuros diseños y refinamientos. Los datos de desempeño del vuelo de retorno lunar y un sólido programa de calificación de pruebas en tierra, mejorado después de la experiencia del vuelo de Artemis I, están respaldando las mejoras en la producción del escudo térmico de Orion. Los futuros escudos térmicos para el regreso de Orion en las misiones de alunizaje de Artemis están en producción para lograr una uniformidad y permeabilidad consistente. El programa de calificación se está completando actualmente, junto con la producción de bloques de Avcoat más permeables, en la Instalación de Ensamblaje Michoud de la NASA en Nueva Orleans.Para obtener más información sobre las campañas Artemis de la NASA, visita el sitio web (en inglés):https://www.nasa.gov/artemis-fin-Meira Bernstein / Rachel Kraft / María José ViñasSede, Washington202-358-1600meira.b.bernstein@nasa.gov / rachel.h.kraft@nasa.gov / maria-jose.vinasgarcia@nasa.gov

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El 28 de junio de 2024, la nave espacial Orion de Artemis II es retirada de la Celda de Ensamblaje Final y Pruebas del Sistema (FAST, por sus siglas en inglés) y colocada en la cámara de altitud oeste dentro del Edificio de Operaciones y Revisión del Centro Espacial Kennedy de la NASA en Florida. Dentro de la cámara de altitud, la nave espacial se sometió a una serie de pruebas que simulaban las condiciones de vacío del espacio profundo.Crédito de la foto: NASA / Rad SinyakRead this story in Englishhere.Tras extensos análisis y pruebas, la NASA ha identificado la causa técnica de la pérdida imprevista de material carbonizado en el escudo térmico de la nave espacial Orion de Artemis I.Los ingenieros determinaron que, cuando Orion regresaba de su misión sin tripulación alrededor de la Luna, los gases generados dentro del material ablativo exterior del escudo térmico, denominado Avcoat, no pudieron ventilarse y disiparse como estaba previsto. Esto permitió que se acumulara presión y se produjeran grietas, lo que causó que parte del material carbonizado se desprendiera en varios lugares.“Nuestros primeros vuelos de Artemis son una campaña de prueba, y el vuelo de prueba de Artemis I nos dio la oportunidad de comprobar nuestros sistemas en el entorno del espacio profundo antes de incorporar a la tripulación en futuras misiones”, dijo Amit Kshatriya, administrador asociado adjunto de la Oficina del programa De la Luna a Marte, en la sede de la NASA en Washington. “La investigación sobre el escudo térmico ayudó a garantizar que comprendiéramos completamente la causa y la naturaleza del problema, así como el riesgo que les pedimos a nuestras tripulaciones que asuman cuando emprendan su viaje a la Luna”.Los hallazgosLos equipos técnicos adoptaron un enfoque metódico para comprender e identificar el origen del problema de pérdida de material carbonizado, incluyendo el muestreo detallado del escudo térmico de Artemis I, la revisión de las imágenes y los datos de los sensores de la nave espacial, y pruebas y análisis exhaustivos en tierra.Durante Artemis I, los ingenieros utilizaron la técnica de guiamiento de reentrada atmosférica doble para el regreso de Orion a la Tierra. Esta técnica ofrece más flexibilidad ya que amplía el alcance del vuelo de Orion después del punto de reentrada para llevarlo hasta un lugar de amerizaje en el océano Pacífico. Con esta maniobra, Orion se sumergió en la parte superior de la atmósfera de la Tierra y utilizó la resistencia atmosférica para reducir su velocidad. A continuación, Orion utilizó la sustentación aerodinámica de la cápsula para rebotar y salir de nuevo de la atmósfera, para luego volver a entrar en el descenso final con paracaídas para su amerizaje.Utilizando los datos de la respuesta del material Avcoat de Artemis I, el equipo de investigación pudo simular el entorno de la trayectoria de entrada de Artemis I —una parte clave para comprender la causa del problema— dentro de la instalación de chorro en arco del Centro de Investigación Ames de la NASA en California. Observaron que, durante el período entre las inmersiones en la atmósfera, las tasas de calentamiento disminuyeron y la energía térmica se acumuló dentro del material Avcoat del escudo térmico. Esto condujo a la acumulación de gases que forman parte del proceso de ablación (desgaste) previsto. Debido a que el Avcoat no tenía “permeabilidad”, la presión interna se acumuló y produjo el agrietamiento y el desprendimiento desigual de la capa exterior.Después de que la nave espacial Orion de la NASA fuera recuperada al término del vuelo de prueba Artemis I y transportada al Centro Espacial Kennedy de la NASA en Florida, su escudo térmico fue separado del módulo de la tripulación en el interior del Edificio de Operaciones y Revisión y rotado para su inspección.Crédito: NASALos equipos técnicos realizaron extensas pruebas en tierra para simular el fenómeno de rebote en la reentrada antes de la misión Artemis I. Sin embargo, hicieron pruebas a velocidades de calentamiento mucho más altas que las que la nave espacial experimentó durante su vuelo. Las altas velocidades de calentamiento puestas a prueba en tierra permitieron que el material carbonizado permeable se formara y se desgastara como estaba previsto, liberando la presión del gas. El calentamiento menos severo observado durante la reentrada real de Artemis I desaceleró el proceso de formación de material carbonizado, al tiempo que siguió creando gases en esta capa de material. La presión del gas se acumuló hasta el punto de agrietar el Avcoat y liberar partes de la capa carbonizada. Las mejoras recientes en la instalación de chorro en arco han permitido una reproducción más precisa de los entornos de vuelo registrados por Artemis I, de modo que este comportamiento de agrietamiento pudo demostrarse en pruebas en tierra.Si bien Artemis I no estaba tripulado, los datos del vuelo mostraron que, si la tripulación hubiera estado a bordo, habría estado a salvo. Los datos de la temperatura de los sistemas del módulo de tripulación dentro de la cabina también estaban dentro de los límites y se mantenían estables, con temperaturas alrededor de los 24 grados centígrados (75 grados Fahrenheit). El desempeño del escudo térmico superó las expectativas.Los ingenieros comprenden tanto el fenómeno material como el entorno con el que interactúan los materiales durante la entrada a la atmósfera. Al cambiar el material o el entorno, pueden predecir cómo responderá la nave espacial. Los equipos de la NASA acordaron por unanimidad que la agencia puede desarrollar un análisis de vuelo aceptable que mantenga a la tripulación segura utilizando el actual escudo térmico de Artemis II con cambios operativos para su entrada en la atmósfera.El proceso de investigación de la NASAPoco después de que los ingenieros de la NASA descubrieran las condiciones del escudo térmico de Artemis I, la agencia comenzó un extenso proceso de investigación, el cual contó con un equipo multidisciplinario de expertos en sistemas de protección térmica, aerotermodinámica, pruebas y análisis térmicos, análisis de estrés (fatiga de materiales), pruebas y análisis de materiales, y muchos otros campos técnicos relacionados. El Centro de Ingeniería y Seguridad de la NASA también participó para aportar su experiencia técnica, incluyendo evaluación no destructiva, análisis térmico y estructural, análisis de árbol de fallas y otros métodos de respaldo de las pruebas.“Nos tomamos muy en serio nuestro proceso de investigación del escudo térmico, con la seguridad de la tripulación como la fuerza impulsora que mueve esta investigación”, dijo Howard Hu, gerente del Programa Orion del Centro Espacial Johnson de la NASA en Houston. “El proceso fue extenso. Le dimos al equipo el tiempo necesario para investigar todas las causas posibles, y trabajaron incansablemente para asegurarse de que entendiéramos el fenómeno y los pasos necesarios para mitigar este problema en futuras misiones”.El escudo térmico de Artemis I estaba muy cargado de instrumentos para este vuelo, e incluía sensores de presión, extensómetros y termopares a diferentes profundidades del material ablativo. Los datos de estos instrumentos acrecentaron el análisis de muestras físicas, lo que permitió al equipo validar modelos informáticos, crear reconstrucciones de entornos, proporcionar perfiles de temperatura interna y dar información sobre el momento de la pérdida de material carbonizado.Alrededor de 200 muestras de Avcoat fueron extraídas del escudo térmico de Artemis I en el Centro de Vuelo Espacial Marshall de la NASA en Alabama para su análisis e inspección. El equipo llevó a cabo una evaluación no destructiva para “ver” dentro del escudo térmico.Uno de los hallazgos más importantes que arrojó el examen de estas muestras fue que algunas superficies en la zona del Avcoat permeable, las cuales habían sido identificadas antes del vuelo, no sufrieron agrietamiento ni pérdida de material carbonizado. Dado que estas superficies eran permeables al comienzo de la entrada en la atmósfera, los gases producidos por la ablación pudieron ventilarse adecuadamente, eliminando la acumulación de la presión, el agrietamiento y la pérdida de material carbonizado.Un bloque de pruebas de Avcoat se somete a una prueba de pulso de calor dentro de una cámara de pruebas de chorro de plasma en arco en el Centro de Investigación Ames de la NASA en California. El elemento de prueba, configurado con secciones Avcoat permeables (superior) y no permeables (inferior) para su comparación, ayudó a confirmar la comprensión de la causa raíz de la pérdida de material Avcoat carbonizado que los ingenieros vieron en la nave espacial Orion tras el vuelo de prueba Artemis I más allá de la Luna.Crédito: NASALos ingenieros hicieron ocho campañas separadas de pruebas térmicas posteriores al vuelo para respaldar el análisis del origen de estas condiciones, y completaron 121 pruebas individuales. Estas pruebas fueron llevadas a cabo en instalaciones en diferentes lugares de Estados Unidos que cuentan con capacidades únicas, entre ellas: la Instalación de Calentamiento Aerodinámico en el Complejo de Chorro en Arco del centro Ames, para poner a prueba perfiles de calentamiento convectivo con diversos gases de prueba; el Laboratorio de Evaluación de Materiales Endurecidos por Láser en la Base de la Fuerza Aérea Patterson-Wright en Ohio, con el fin de poner a prueba perfiles de calentamiento radiativo y proporcionar radiografías en tiempo real; y la Instalación de Calentamiento por Interacción del centro Ames, para poner a prueba perfiles combinados de calentamiento convectivo y radiativo en el aire en bloques completos, esto es, aplicando todas las pruebas en cada bloque de material.Los expertos en aerotermia también completaron dos campañas de pruebas en el túnel de viento hipersónico del Centro de Investigación Langley de la NASA en Virginia y en las instalaciones de pruebas aerodinámicas del CUBRC en Buffalo, Nueva York, para realizar pruebas con una diversidad de configuraciones de pérdida de material carbonizado, y mejorar y validar los modelos analíticos. También se realizaron pruebas de permeabilidad en el centro Kratos en Alabama, en la Universidad de Kentucky y en el centro Ames para caracterizar aún mejor el volumen elemental y la porosidad del Avcoat. La instalación de pruebas del centro de investigaciones Advanced Light Source, una instalación para usuarios científicos del Departamento de Energía de Estados Unidos en el Laboratorio Nacional Lawrence Berkeley, también fue utilizada por los ingenieros para examinar el comportamiento del calentamiento del Avcoat a nivel microestructural.En la primavera de 2024, la NASA creó un equipo de revisión independiente que realizó una revisión exhaustiva del proceso de investigación, los hallazgos y los resultados de la agencia. La revisión independiente fue dirigida por Paul Hill, un exdirectivo de la NASA que se desempeñó como director principal de vuelo del transbordador espacial para el programa Return to Flight (Regreso a los vuelos) después del accidente del Columbia, quien también dirigió la Dirección de Operaciones de Misiones de la NASA y es miembro actual del Panel Asesor de Seguridad Aeroespacial de la agencia. La revisión se llevó a cabo durante un período de tres meses a fin de evaluar las condiciones del escudo térmico posteriores al vuelo, los datos del entorno para la entrada a la atmósfera, la respuesta térmica del material ablativo y el avance de las investigaciones de la NASA. El equipo de revisión estuvo de acuerdo con los hallazgos de la NASA sobre la causa técnica del comportamiento físico del escudo térmico.Avances en el escudo térmicoAl saber que la permeabilidad de Avcoat es un parámetro clave para evitar o minimizar la pérdida de material carbonizado, la NASA tiene la información correcta para garantizar la seguridad de la tripulación y mejorar el desempeño de los futuros escudos térmicos del programa Artemis. A lo largo de su historia, la NASA ha aprendido de cada uno de sus vuelos e incorporado mejoras en el hardware y las operaciones. Los datos recopilados a lo largo del vuelo de prueba de Artemis I han proporcionado a los ingenieros información valiosísima para guiar futuros diseños y refinamientos. Los datos de desempeño del vuelo de retorno lunar y un sólido programa de calificación de pruebas en tierra, mejorado después de la experiencia del vuelo de Artemis I, están respaldando las mejoras en la producción del escudo térmico de Orion. Los futuros escudos térmicos para el regreso de Orion en las misiones de alunizaje de Artemis están en producción para lograr una uniformidad y permeabilidad consistente. El programa de calificación se está completando actualmente, junto con la producción de bloques de Avcoat más permeables, en la Instalación de Ensamblaje Michoud de la NASA en Nueva Orleans.Para obtener más información sobre las campañas Artemis de la NASA, visita el sitio web (en inglés):https://www.nasa.gov/artemis-fin-Meira Bernstein / Rachel Kraft / María José ViñasSede, Washington202-358-1600meira.b.bernstein@nasa.gov / rachel.h.kraft@nasa.gov / maria-jose.vinasgarcia@nasa.gov

El 28 de junio de 2024, la nave espacial Orion de Artemis II es retirada de la Celda de Ensamblaje Final y Pruebas del Sistema (FAST, por sus siglas en inglés) y colocada en la cámara de altitud oeste dentro del Edificio de Operaciones y Revisión del Centro Espacial Kennedy de la NASA en Florida. Dentro de la cámara de altitud, la nave espacial se sometió a una serie de pruebas que simulaban las condiciones de vacío del espacio profundo.Crédito de la foto: NASA / Rad Sinyak

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Crédito de la foto: NASA / Rad Sinyak

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Tras extensos análisis y pruebas, la NASA ha identificado la causa técnica de la pérdida imprevista de material carbonizado en el escudo térmico de la nave espacial Orion de Artemis I.

Los ingenieros determinaron que, cuando Orion regresaba de su misión sin tripulación alrededor de la Luna, los gases generados dentro del material ablativo exterior del escudo térmico, denominado Avcoat, no pudieron ventilarse y disiparse como estaba previsto. Esto permitió que se acumulara presión y se produjeran grietas, lo que causó que parte del material carbonizado se desprendiera en varios lugares.

“Nuestros primeros vuelos de Artemis son una campaña de prueba, y el vuelo de prueba de Artemis I nos dio la oportunidad de comprobar nuestros sistemas en el entorno del espacio profundo antes de incorporar a la tripulación en futuras misiones”, dijo Amit Kshatriya, administrador asociado adjunto de la Oficina del programa De la Luna a Marte, en la sede de la NASA en Washington. “La investigación sobre el escudo térmico ayudó a garantizar que comprendiéramos completamente la causa y la naturaleza del problema, así como el riesgo que les pedimos a nuestras tripulaciones que asuman cuando emprendan su viaje a la Luna”.

Los equipos técnicos adoptaron un enfoque metódico para comprender e identificar el origen del problema de pérdida de material carbonizado, incluyendo el muestreo detallado del escudo térmico de Artemis I, la revisión de las imágenes y los datos de los sensores de la nave espacial, y pruebas y análisis exhaustivos en tierra.

Durante Artemis I, los ingenieros utilizaron la técnica de guiamiento de reentrada atmosférica doble para el regreso de Orion a la Tierra. Esta técnica ofrece más flexibilidad ya que amplía el alcance del vuelo de Orion después del punto de reentrada para llevarlo hasta un lugar de amerizaje en el océano Pacífico. Con esta maniobra, Orion se sumergió en la parte superior de la atmósfera de la Tierra y utilizó la resistencia atmosférica para reducir su velocidad. A continuación, Orion utilizó la sustentación aerodinámica de la cápsula para rebotar y salir de nuevo de la atmósfera, para luego volver a entrar en el descenso final con paracaídas para su amerizaje.

Utilizando los datos de la respuesta del material Avcoat de Artemis I, el equipo de investigación pudo simular el entorno de la trayectoria de entrada de Artemis I —una parte clave para comprender la causa del problema— dentro de la instalación de chorro en arco del Centro de Investigación Ames de la NASA en California. Observaron que, durante el período entre las inmersiones en la atmósfera, las tasas de calentamiento disminuyeron y la energía térmica se acumuló dentro del material Avcoat del escudo térmico. Esto condujo a la acumulación de gases que forman parte del proceso de ablación (desgaste) previsto. Debido a que el Avcoat no tenía “permeabilidad”, la presión interna se acumuló y produjo el agrietamiento y el desprendimiento desigual de la capa exterior.

Después de que la nave espacial Orion de la NASA fuera recuperada al término del vuelo de prueba Artemis I y transportada al Centro Espacial Kennedy de la NASA en Florida, su escudo térmico fue separado del módulo de la tripulación en el interior del Edificio de Operaciones y Revisión y rotado para su inspección.Crédito: NASA

Después de que la nave espacial Orion de la NASA fuera recuperada al término del vuelo de prueba Artemis I y transportada al Centro Espacial Kennedy de la NASA en Florida, su escudo térmico fue separado del módulo de la tripulación en el interior del Edificio de Operaciones y Revisión y rotado para su inspección.

Crédito: NASA

Los equipos técnicos realizaron extensas pruebas en tierra para simular el fenómeno de rebote en la reentrada antes de la misión Artemis I. Sin embargo, hicieron pruebas a velocidades de calentamiento mucho más altas que las que la nave espacial experimentó durante su vuelo. Las altas velocidades de calentamiento puestas a prueba en tierra permitieron que el material carbonizado permeable se formara y se desgastara como estaba previsto, liberando la presión del gas. El calentamiento menos severo observado durante la reentrada real de Artemis I desaceleró el proceso de formación de material carbonizado, al tiempo que siguió creando gases en esta capa de material. La presión del gas se acumuló hasta el punto de agrietar el Avcoat y liberar partes de la capa carbonizada. Las mejoras recientes en la instalación de chorro en arco han permitido una reproducción más precisa de los entornos de vuelo registrados por Artemis I, de modo que este comportamiento de agrietamiento pudo demostrarse en pruebas en tierra.

Si bien Artemis I no estaba tripulado, los datos del vuelo mostraron que, si la tripulación hubiera estado a bordo, habría estado a salvo. Los datos de la temperatura de los sistemas del módulo de tripulación dentro de la cabina también estaban dentro de los límites y se mantenían estables, con temperaturas alrededor de los 24 grados centígrados (75 grados Fahrenheit). El desempeño del escudo térmico superó las expectativas.

Los ingenieros comprenden tanto el fenómeno material como el entorno con el que interactúan los materiales durante la entrada a la atmósfera. Al cambiar el material o el entorno, pueden predecir cómo responderá la nave espacial. Los equipos de la NASA acordaron por unanimidad que la agencia puede desarrollar un análisis de vuelo aceptable que mantenga a la tripulación segura utilizando el actual escudo térmico de Artemis II con cambios operativos para su entrada en la atmósfera.

Poco después de que los ingenieros de la NASA descubrieran las condiciones del escudo térmico de Artemis I, la agencia comenzó un extenso proceso de investigación, el cual contó con un equipo multidisciplinario de expertos en sistemas de protección térmica, aerotermodinámica, pruebas y análisis térmicos, análisis de estrés (fatiga de materiales), pruebas y análisis de materiales, y muchos otros campos técnicos relacionados. El Centro de Ingeniería y Seguridad de la NASA también participó para aportar su experiencia técnica, incluyendo evaluación no destructiva, análisis térmico y estructural, análisis de árbol de fallas y otros métodos de respaldo de las pruebas.

"Nos tomamos muy en serio nuestro proceso de investigación del escudo térmico, con la seguridad de la tripulación como la fuerza impulsora que mueve esta investigación", dijo Howard Hu, gerente del Programa Orion del Centro Espacial Johnson de la NASA en Houston. "El proceso fue extenso. Le dimos al equipo el tiempo necesario para investigar todas las causas posibles, y trabajaron incansablemente para asegurarse de que entendiéramos el fenómeno y los pasos necesarios para mitigar este problema en futuras misiones".

El escudo térmico de Artemis I estaba muy cargado de instrumentos para este vuelo, e incluía sensores de presión, extensómetros y termopares a diferentes profundidades del material ablativo. Los datos de estos instrumentos acrecentaron el análisis de muestras físicas, lo que permitió al equipo validar modelos informáticos, crear reconstrucciones de entornos, proporcionar perfiles de temperatura interna y dar información sobre el momento de la pérdida de material carbonizado.

Alrededor de 200 muestras de Avcoat fueron extraídas del escudo térmico de Artemis I en el Centro de Vuelo Espacial Marshall de la NASA en Alabama para su análisis e inspección. El equipo llevó a cabo una evaluación no destructiva para "ver" dentro del escudo térmico.

Uno de los hallazgos más importantes que arrojó el examen de estas muestras fue que algunas superficies en la zona del Avcoat permeable, las cuales habían sido identificadas antes del vuelo, no sufrieron agrietamiento ni pérdida de material carbonizado. Dado que estas superficies eran permeables al comienzo de la entrada en la atmósfera, los gases producidos por la ablación pudieron ventilarse adecuadamente, eliminando la acumulación de la presión, el agrietamiento y la pérdida de material carbonizado.

Un bloque de pruebas de Avcoat se somete a una prueba de pulso de calor dentro de una cámara de pruebas de chorro de plasma en arco en el Centro de Investigación Ames de la NASA en California. El elemento de prueba, configurado con secciones Avcoat permeables (superior) y no permeables (inferior) para su comparación, ayudó a confirmar la comprensión de la causa raíz de la pérdida de material Avcoat carbonizado que los ingenieros vieron en la nave espacial Orion tras el vuelo de prueba Artemis I más allá de la Luna.Crédito: NASA

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Los ingenieros hicieron ocho campañas separadas de pruebas térmicas posteriores al vuelo para respaldar el análisis del origen de estas condiciones, y completaron 121 pruebas individuales. Estas pruebas fueron llevadas a cabo en instalaciones en diferentes lugares de Estados Unidos que cuentan con capacidades únicas, entre ellas: la Instalación de Calentamiento Aerodinámico en el Complejo de Chorro en Arco del centro Ames, para poner a prueba perfiles de calentamiento convectivo con diversos gases de prueba; el Laboratorio de Evaluación de Materiales Endurecidos por Láser en la Base de la Fuerza Aérea Patterson-Wright en Ohio, con el fin de poner a prueba perfiles de calentamiento radiativo y proporcionar radiografías en tiempo real; y la Instalación de Calentamiento por Interacción del centro Ames, para poner a prueba perfiles combinados de calentamiento convectivo y radiativo en el aire en bloques completos, esto es, aplicando todas las pruebas en cada bloque de material.

Los expertos en aerotermia también completaron dos campañas de pruebas en el túnel de viento hipersónico del Centro de Investigación Langley de la NASA en Virginia y en las instalaciones de pruebas aerodinámicas del CUBRC en Buffalo, Nueva York, para realizar pruebas con una diversidad de configuraciones de pérdida de material carbonizado, y mejorar y validar los modelos analíticos. También se realizaron pruebas de permeabilidad en el centro Kratos en Alabama, en la Universidad de Kentucky y en el centro Ames para caracterizar aún mejor el volumen elemental y la porosidad del Avcoat. La instalación de pruebas del centro de investigaciones Advanced Light Source, una instalación para usuarios científicos del Departamento de Energía de Estados Unidos en el Laboratorio Nacional Lawrence Berkeley, también fue utilizada por los ingenieros para examinar el comportamiento del calentamiento del Avcoat a nivel microestructural.

En la primavera de 2024, la NASA creó un equipo de revisión independiente que realizó una revisión exhaustiva del proceso de investigación, los hallazgos y los resultados de la agencia. La revisión independiente fue dirigida por Paul Hill, un exdirectivo de la NASA que se desempeñó como director principal de vuelo del transbordador espacial para el programa Return to Flight (Regreso a los vuelos) después del accidente del Columbia, quien también dirigió la Dirección de Operaciones de Misiones de la NASA y es miembro actual del Panel Asesor de Seguridad Aeroespacial de la agencia. La revisión se llevó a cabo durante un período de tres meses a fin de evaluar las condiciones del escudo térmico posteriores al vuelo, los datos del entorno para la entrada a la atmósfera, la respuesta térmica del material ablativo y el avance de las investigaciones de la NASA. El equipo de revisión estuvo de acuerdo con los hallazgos de la NASA sobre la causa técnica del comportamiento físico del escudo térmico.

Al saber que la permeabilidad de Avcoat es un parámetro clave para evitar o minimizar la pérdida de material carbonizado, la NASA tiene la información correcta para garantizar la seguridad de la tripulación y mejorar el desempeño de los futuros escudos térmicos del programa Artemis. A lo largo de su historia, la NASA ha aprendido de cada uno de sus vuelos e incorporado mejoras en el hardware y las operaciones. Los datos recopilados a lo largo del vuelo de prueba de Artemis I han proporcionado a los ingenieros información valiosísima para guiar futuros diseños y refinamientos. Los datos de desempeño del vuelo de retorno lunar y un sólido programa de calificación de pruebas en tierra, mejorado después de la experiencia del vuelo de Artemis I, están respaldando las mejoras en la producción del escudo térmico de Orion. Los futuros escudos térmicos para el regreso de Orion en las misiones de alunizaje de Artemis están en producción para lograr una uniformidad y permeabilidad consistente. El programa de calificación se está completando actualmente, junto con la producción de bloques de Avcoat más permeables, en la Instalación de Ensamblaje Michoud de la NASA en Nueva Orleans.

Para obtener más información sobre las campañas Artemis de la NASA, visita el sitio web (en inglés):

https://www.nasa.gov/artemis

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"If so, it will be the first planet ever found orbiting a hypervelocity star."A paper describing the results, led by Terry, was published in The Astronomical Journal on February 10.A Star on the MoveThe pair of objects wasfirst spotted indirectly in 2011thanks to a chance alignment. A team of scientists combed through archived data from MOA (Microlensing Observations in Astrophysics) – a collaborative project focused on a microlensing survey conducted using the University of Canterbury Mount John Observatory in New Zealand — in search of light signals that betray the presence of exoplanets, or planets outside our solar system.Microlensing occurs because the presence of mass warps the fabric of space-time. Any time an intervening object appears to drift near a background star, light from the star curves as it travels through the warped space-time around the nearer object. If the alignment is especially close, the warping around the object can act like a natural lens, amplifying the background star's light.This artist's concept visualizes stars near the center of our Milky Way galaxy. Each has a colorful trail indicating its speed — the longer and redder the trail, the faster the star is moving. NASA scientists recently discovered a candidate for a particularly speedy star, visualized near the center of this image, with an orbiting planet. If confirmed, the pair sets a record for fastest known exoplanet system.NASA/JPL-Caltech/R. Hurt (Caltech-IPAC)In this case, microlensing signals revealed a pair of celestial bodies. Scientists determined their relative masses (one is about 2,300 times heavier than the other), but their exact masses depend on how far away they are from Earth. It's sort of like how the magnification changes if you hold a magnifying glass over a page and move it up and down."Determining the mass ratio is easy," said David Bennett, a senior research scientist at the University of Maryland, College Park and NASA Goddard, who co-authored the new paper and led the original study in 2011. "It's much more difficult to calculate their actual masses."The 2011 discovery team suspected the microlensed objects were either a star about 20 percent as massive as our Sun and a planet roughly 29 times heavier than Earth, or a nearer "rogue" planet about four times Jupiter's mass with a moon smaller than Earth.To figure out which explanation is more likely, astronomers searched through data from theKeck Observatoryin Hawaii and ESA's (European Space Agency's)Gaia satellite. If the pair were a rogue planet and moon, they'd be effectively invisible – dark objects lost in the inky void of space. But scientists might be able to identify the star if the alternative explanation were correct (though the orbiting planet would be much too faint to see).They found a strong suspect located about 24,000 light-years away, putting it within the Milky Way's galactic bulge — the central hub where stars are more densely packed. By comparing the star's location in 2011 and 2021, the team calculated its high speed.This Hubble Space Telescope image shows a bow shock around a very young star called LL Ori. Named for the crescent-shaped wave made by a ship as it moves through water, a bow shock can be created in space when two streams of gas collide. Scientists think a similar feature may be present around a newfound star that could be traveling at least 1.2 million miles per hour, or 540 kilometers per second. 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Apollo 13 backup Lunar Module Pilot Charlie Duke, left, scoops up soil at the Kennedy Space Center while backup Commander John Young looks on.

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TITLE: Dark Energy

Explore the UniverseDark EnergyContentsA Brief HistoryIt All Started With CepheidsDiscovering an Expanding UniverseExpansion is Speeding Up, Supernovae ShowWhat Exactly is Dark Energy?Vacuum EnergyQuintessenceSpace WrinklesA Flaw in General RelativityThe FutureSome 13.8 billion years ago, the universe began with a rapid expansion we call the big bang. After this initial expansion, which lasted a fraction of a second, gravity started to slow the universe down. But the cosmos wouldn't stay this way. Nine billion years after the universe began, its expansion started to speed up, driven by an unknown force that scientists have nameddark energy.But what exactlyisdark energy?The short answer is: We don't know. But we do know that it exists, it's making the universe expand at an accelerating rate, and approximately 68.3 to 70% of the universe is dark energy.The history of the universe is outlined in this infographic.NASAA Brief HistoryIt All Started With CepheidsDark energy wasn't discovered until the late 1990s. But its origin in scientific study stretches all the way back to 1912 when American astronomer Henrietta Swan Leavitt made an important discovery using Cepheid variables, a class of stars whose brightness fluctuates with a regularity that depends on the star's brightness.All Cepheid stars with a certain period (a Cepheid's period is the time it takes to go from bright, to dim, and bright again) have the same absolute magnitude, or luminosity – the amount of light they put out. Leavitt measured these stars and proved that there is a relationship between their regular period of brightness and luminosity. Leavitt's findings made it possible for astronomers to use a star's period and luminosity to measure the distances between us and Cepheid stars in far-off galaxies (and our own Milky Way).Around this same time in history, astronomer Vesto Slipher observed spiral galaxies using his telescope's spectrograph, a device that splits light into the colors that make it up, much like the way a prism splits light into a rainbow. He used the spectrograph, a relatively recent invention at the time, to see the different wavelengths of light coming from the galaxies in different spectral lines. With his observations, Silpher was the first astronomer to observe how quickly the galaxy was moving away from us, called redshift, in distant galaxies. These observations would prove to be critical for many future scientific breakthroughs, including the discovery of dark energy.Redshift is a term used when astronomical objects are moving away from us and the light coming from those objects stretches out. Light behaves like a wave, and red light has the longest wavelength. So, the light coming from objects moving away from us has a longer wavelength, stretching to the "red end" of the electromagnetic.Discovering an Expanding UniverseThe discovery of galactic redshift, the period-luminosity relation of Cepheid variables, and a newfound ability to gauge a star or galaxy's distance eventually played a role in astronomers observing that galaxies were getting farther away from us over time, which showed how the universe was expanding. In the years that followed, different scientists around the world started to put the pieces of an expanding universe together.In 1922, Russian scientist and mathematician Alexander Friedmann published a paper detailing multiple possibilities for the history of the universe. The paper, which was based on Albert Einstein's theory of general relativity published in 1917, included the possibility that the universe is expanding.In 1927, Belgian astronomer Georges Lemaître, who is said to have been unaware of Friedmann's work, published a paper also factoring in Einstein's theory of general relativity. And, while Einstein stated in his theory that the universe was static, Lemaître showed how the equations in Einstein's theory actually support the idea that the universe is not static but, in fact, is actually expanding.Astronomer Edwin Hubble confirmed that the universe was expanding in 1929 using observations made by his associate, astronomer Milton Humason. Humason measured the redshift of spiral galaxies. Hubble and Humason then studied Cepheid stars in those galaxies, using the stars to determine the distance of their galaxies (or nebulae, as they called them). They compared the distances of these galaxies to their redshift and tracked how the farther away an object is, the bigger its redshift and the faster it is moving away from us. The pair found that objects like galaxies are moving away from Earth faster the farther away they are, at upwards of hundreds of thousands of miles per second – an observation now known as Hubble's Law, or the Hubble-Lemaître law. The universe, they confirmed, is really expanding.This composite image features one of the most complicated and dramatic collisions between galaxy clusters ever seen. Known officially as Abell 2744, this system has been dubbed Pandora's Cluster because of the wide variety of different structures found. Data from NASA's Chandra X-ray Observatory (red) show gas with temperatures of millions of degrees. In blue is a map showing the total mass concentration (mostly dark matter) based on data from NASA's Hubble Space Telescope, the VLT (Very Large Telescope), and the Subaru telescope. Optical data from Hubble and VLT also show the constituent galaxies of the clusters. Astronomers think at least four galaxy clusters coming from a variety of directions are involved with this collision.NASA, ESA, J. Merten (Institute for Theoretical Astrophysics, Heidelberg/Astronomical Observatory of Bologna), and D. Coe (STScI)Expansion is Speeding Up, Supernovae ShowScientists previously thought that the universe's expansion would likely be slowed down by gravity over time, an expectation backed by Einstein's theory of general relativity. But in 1998, everything changed when two different teams of astronomers observing far-off supernovae noticed that (at a certain redshift) the stellar explosions were dimmer than expected. These groups were led by astronomers Adam Riess, Saul Perlmutter, and Brian Schmidt. This trio won the 2011 Nobel Prize in Physics for this work.While dim supernovae might not seem like a major find, these astronomers were looking atType 1a supernovae, which are known to have a certain level of luminosity. So they knew that there must be another factor making these objects appear dimmer. Scientists can determine distance (and speed) using an objects' brightness, and dimmer objects are typically farther away (though surrounding dust and other factors can cause an object to dim).This led the scientists to conclude that these supernovae were just much farther away than they expected by looking at their redshifts.Using the objects' brightness, the researchers determined the distance of these supernovae. And using the spectrum, they were able to figure out the objects' redshift and, therefore, how fast they were moving away from us. They found that the supernovae were not as close as expected, meaning they had traveled farther away from us faster than anticipated. These observations led scientists to ultimately conclude that the universe itself must be expanding faster over time.While other possible explanations for these observations have been explored, astronomers studying even more distant supernovae or other cosmic phenomena in more recent years continued to gather evidence and build support for the idea that the universe is expanding faster over time, a phenomenon now called cosmic acceleration.But, as scientists built up a case for cosmic acceleration, they also asked: Why? What could be driving the universe to stretch out faster over time?Enter dark energy.What ExactlyisDark Energy?Right now, dark energy is just the name that astronomers gave to the mysterious "something" that is causing the universe to expand at an accelerated rate.Dark energy has been described by some as having the effect of a negative pressure that is pushing space outward. However, we don't know if dark energy has the effect of any type of force at all. There are many ideas floating around about what dark energy could possibly be. Here are four leading explanations for dark energy. Keep in mind that it's possible it's something else entirely.Vacuum EnergySome scientists think that dark energy is a fundamental, ever-present background energy in space known as vacuum energy, which could be equal to the cosmological constant, a mathematical term in the equations of Einstein's theory of general relativity. Originally, the constant existed to counterbalance gravity, resulting in a static universe. But when Hubble confirmed that the universe was actually expanding, Einstein removed the constant, calling it "my biggest blunder," according to physicist George Gamow.But when it was later discovered that the universe's expansion was actually accelerating, some scientists suggested that there might actually be a non-zero value to the previously-discredited cosmological constant. They suggested that this additional force would be necessary to accelerate the expansion of the universe. This theorized that this mystery component could be attributed to something called "vacuum energy," which is a theoretical background energy permeating all of space.Space is never exactly empty. According to quantum field theory, there are virtual particles, or pairs of particles and antiparticles. It's thought that these virtual particles cancel each other out almost as soon as they crop up in the universe, and that this act of popping in and out of existence could be made possible by "vacuum energy" that fills the cosmos and pushes space outward.While this theory has been a popular topic of discussion, scientists investigating this option have calculated how much vacuum energy there should theoretically be in space. They showed that there should either be so much vacuum energy that, at the very beginning, the universe would have expanded outwards so quickly and with so much force that no stars or galaxies could have formed, or... there should be absolutely none. This means that the amount of vacuum energy in the cosmos must be much smaller than it is in these predictions. However, this discrepancy has yet to be solved and has even earned the moniker "the cosmological constant problem."QuintessenceSome scientists think that dark energy could be a type of energy fluid or field that fills space, behaves in an opposite way to normal matter, and can vary in its amount and distribution throughout both time and space. This hypothesized version of dark energy has been nicknamed quintessence after the theoretical fifth element discussed by ancient Greek philosophers.It's even been suggested by some scientists that quintessence could be some combination of dark energy and dark matter, though the two are currently considered completely separate from one another. While the two are both major mysteries to scientists, dark matter is thought to make up about 85% of all matter in the universe.Space WrinklesSome scientists think that dark energy could be a sort of defect in the fabric of the universe itself; defects like cosmic strings, which are hypothetical one-dimensional "wrinkles" thought to have formed in the early universe.A Flaw in General RelativitySome scientists think that dark energy isn't something physical that we can discover. Rather, they think there could be an issue with general relativity and Einstein's theory of gravity and how it works on the scale of the observable universe. Within this explanation, scientists think that it's possible to modify our understanding of gravity in a way that explains observations of the universe made without the need for dark energy. Einstein actually proposed such an idea in 1919 called unimodular gravity, a modified version of general relativity that scientists today think wouldn't require dark energy to make sense of the universe.The FutureDark energy is one of the great mysteries of the universe. For decades, scientists have theorized about our expanding universe. Now, for the first time ever, we have tools powerful enough to put these theories to the test and really investigate the big question: “what is dark energy?”NASA plays a critical role in the ESA (European Space Agency) missionEuclid(launched in 2023), which will make a 3D map of the universe to see how matter has been pulled apart by dark energy over time. 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Explore the Universe

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Astronomer Edwin Hubble confirmed that the universe was expanding in 1929 using observations made by his associate, astronomer Milton Humason. Humason measured the redshift of spiral galaxies. Hubble and Humason then studied Cepheid stars in those galaxies, using the stars to determine the distance of their galaxies (or nebulae, as they called them). They compared the distances of these galaxies to their redshift and tracked how the farther away an object is, the bigger its redshift and the faster it is moving away from us. The pair found that objects like galaxies are moving away from Earth faster the farther away they are, at upwards of hundreds of thousands of miles per second – an observation now known as Hubble's Law, or the Hubble-Lemaître law. The universe, they confirmed, is really expanding.

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Scientists previously thought that the universe's expansion would likely be slowed down by gravity over time, an expectation backed by Einstein's theory of general relativity. But in 1998, everything changed when two different teams of astronomers observing far-off supernovae noticed that (at a certain redshift) the stellar explosions were dimmer than expected. These groups were led by astronomers Adam Riess, Saul Perlmutter, and Brian Schmidt. This trio won the 2011 Nobel Prize in Physics for this work.

While dim supernovae might not seem like a major find, these astronomers were looking atType 1a supernovae, which are known to have a certain level of luminosity. So they knew that there must be another factor making these objects appear dimmer. Scientists can determine distance (and speed) using an objects' brightness, and dimmer objects are typically farther away (though surrounding dust and other factors can cause an object to dim).

This led the scientists to conclude that these supernovae were just much farther away than they expected by looking at their redshifts.

Using the objects' brightness, the researchers determined the distance of these supernovae. And using the spectrum, they were able to figure out the objects' redshift and, therefore, how fast they were moving away from us. They found that the supernovae were not as close as expected, meaning they had traveled farther away from us faster than anticipated. These observations led scientists to ultimately conclude that the universe itself must be expanding faster over time.

While other possible explanations for these observations have been explored, astronomers studying even more distant supernovae or other cosmic phenomena in more recent years continued to gather evidence and build support for the idea that the universe is expanding faster over time, a phenomenon now called cosmic acceleration.

But, as scientists built up a case for cosmic acceleration, they also asked: Why? What could be driving the universe to stretch out faster over time?

Enter dark energy.

What ExactlyisDark Energy?

Right now, dark energy is just the name that astronomers gave to the mysterious "something" that is causing the universe to expand at an accelerated rate.

Dark energy has been described by some as having the effect of a negative pressure that is pushing space outward. However, we don't know if dark energy has the effect of any type of force at all. There are many ideas floating around about what dark energy could possibly be. Here are four leading explanations for dark energy. Keep in mind that it's possible it's something else entirely.

Some scientists think that dark energy is a fundamental, ever-present background energy in space known as vacuum energy, which could be equal to the cosmological constant, a mathematical term in the equations of Einstein's theory of general relativity. Originally, the constant existed to counterbalance gravity, resulting in a static universe. But when Hubble confirmed that the universe was actually expanding, Einstein removed the constant, calling it “my biggest blunder,” according to physicist George Gamow.

But when it was later discovered that the universe’s expansion was actually accelerating, some scientists suggested that there might actually be a non-zero value to the previously-discredited cosmological constant. They suggested that this additional force would be necessary to accelerate the expansion of the universe. This theorized that this mystery component could be attributed to something called “vacuum energy,” which is a theoretical background energy permeating all of space.

Space is never exactly empty. According to quantum field theory, there are virtual particles, or pairs of particles and antiparticles. It's thought that these virtual particles cancel each other out almost as soon as they crop up in the universe, and that this act of popping in and out of existence could be made possible by “vacuum energy” that fills the cosmos and pushes space outward.

While this theory has been a popular topic of discussion, scientists investigating this option have calculated how much vacuum energy there should theoretically be in space. They showed that there should either be so much vacuum energy that, at the very beginning, the universe would have expanded outwards so quickly and with so much force that no stars or galaxies could have formed, or... there should be absolutely none. This means that the amount of vacuum energy in the cosmos must be much smaller than it is in these predictions. However, this discrepancy has yet to be solved and has even earned the moniker "the cosmological constant problem."

Some scientists think that dark energy could be a type of energy fluid or field that fills space, behaves in an opposite way to normal matter, and can vary in its amount and distribution throughout both time and space. This hypothesized version of dark energy has been nicknamed quintessence after the theoretical fifth element discussed by ancient Greek philosophers.

It's even been suggested by some scientists that quintessence could be some combination of dark energy and dark matter, though the two are currently considered completely separate from one another. While the two are both major mysteries to scientists, dark matter is thought to make up about 85% of all matter in the universe.

Some scientists think that dark energy could be a sort of defect in the fabric of the universe itself; defects like cosmic strings, which are hypothetical one-dimensional "wrinkles" thought to have formed in the early universe.

Some scientists think that dark energy isn't something physical that we can discover. Rather, they think there could be an issue with general relativity and Einstein's theory of gravity and how it works on the scale of the observable universe. Within this explanation, scientists think that it's possible to modify our understanding of gravity in a way that explains observations of the universe made without the need for dark energy. Einstein actually proposed such an idea in 1919 called unimodular gravity, a modified version of general relativity that scientists today think wouldn't require dark energy to make sense of the universe.

Dark energy is one of the great mysteries of the universe. For decades, scientists have theorized about our expanding universe. Now, for the first time ever, we have tools powerful enough to put these theories to the test and really investigate the big question: “what is dark energy?”

NASA plays a critical role in the ESA (European Space Agency) missionEuclid(launched in 2023), which will make a 3D map of the universe to see how matter has been pulled apart by dark energy over time. This map will include observations of billions of galaxies found up to 10 billion light-years from Earth.

NASA'sNancy Grace Roman Space Telescope, set to launch by May 2027, is designed to investigate dark energy, among many other science topics, and will also create a 3D dark matter map. Roman's resolution will be as sharp as NASA’s Hubble Space Telescope's, but with a field of view 100 times larger, allowing it to capture more expansive images of the universe. This will allow scientists to map how matter is structured and spread across the universe and explore how dark energy behaves and has changed over time. Roman will also conduct an additional survey to detect Type Ia supernovae.

In addition to NASA's missions and efforts, the Vera C. Rubin Observatory, supported by a large collaboration that includes the U.S. National Science Foundation, which is currently under construction in Chile, is also poised to support our growing understanding of dark energy. The ground-based observatory is expected to be operational in 2025.

The combined efforts of Euclid, Roman, and Rubin will usher in a new "golden age" of cosmology, in which scientists will collect more detailed information than ever about the great mysteries of dark energy.Additionally, NASA'sJames Webb Space Telescope(launched in 2021), the world's most powerful and largest space telescope, aims to make contributions to several areas of research, and will contribute to studies of dark energy.

NASA'sSPHEREx(the Spectro-Photometer for the History of the Universe, Epoch of Reionization, and Ices Explorer) mission, scheduled to launch no later than April 2025, aims to investigate the origins of the universe. Scientists expect that the data collected with SPHEREx, which will survey the entire sky in near-infrared light, including over 450 million galaxies, could help to further our understanding of dark energy.

NASA also supports a citizen science project calledDark Energy Explorers, which enables anyone in the world, even those who have no scientific training, to help in the search for dark energy answers.

A brief note

Lastly, to clarify, dark energy is not the same as dark matter. Their main similarity is that we don't yet know what they are!

By Chelsea GohdNASA's Jet Propulsion Laboratory

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In an era where data security is critical to aviation safety, NASA is exploring bold new solutions.

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Katrina LeeMolly KearnsDec 17, 2025ArticleContentsAmazonSES Space & DefenseSpaceXTelesatViasatEditor's Note:This story was updated on Dec. 18, 2025, to change SES to SES Space and Defense.NASA's commercial partners are actively demonstrating next-generation satellite relay capabilities for spaceflight missions, marking a significant step toward retiring the agency's Tracking and Data Relay Satellite (TDRS) system and adopting commercial services. The demonstrations – ranging from real-time spacecraft tracking during launch to transmitting mission commands and scientific data – are part of NASA'sCommunications Services Project, which is modernizing how the agency communicates with its science missions in near-Earth orbit.Managed by the agency'sSCaN(Space Communications and Navigation) Program, the project awarded funded Space Act Agreements in 2022 tosix U.S. companiesthat are developing and testing commercial satellite communications services. The initiative supports NASA's broader strategy to retire the TDRS constellation and adopt a commercial-first model for near-Earth communications."In collaboration with our commercial partners, SCaN is ushering in a new era of space exploration that will deliver powerful, forward-thinking solutions that reduce cost, increase adaptability, and increase mission success," said Kevin Coggins, deputy associate administrator for SCaN at NASA Headquarters in Washington. "This work advances our commitment to expanding the low Earth orbit economy, and our commercial space partners are leading the charge through these groundbreaking demonstrations, proving for the first time that commercial satellite relay services can work for NASA missions."This work advances our commitment to expanding the low Earth orbit economy, and our commercial space partners are leading the charge through these groundbreaking demonstrations.Kevin CogginsDeputy Associate Administrator for SCaNBy leveraging private-sector innovation, NASA aims to establish a more flexible, cost-effective, and scalable communications infrastructure for future science missions.AmazonAmazon Leo for Government, a subsidiary of Amazon, is demonstrating high-rate data exchanges over optical links using its satellite network in low Earth orbitAmazon has developed the hardware and software components necessary to support optical communication links within its Amazon Leo satellite relay network. Optical communications, also known as laser communications, use infrared light to transmit data at a higher rate compared to standard radio frequency systems. The Amazon Leo demonstrations, scheduled to begin in early 2026, will test the pointing, acquisition, and tracking capabilities of their optical communications systems to ensure the technology can accurately locate, lock onto, and stay connected with a mission as it travels through space.An image of the view from an Amazon Leo satellite overlooking the Earth.Credit: AmazonSES Space & DefenseSES Space & Defense is demonstrating high-rate data exchanges as well as tracking, telemetry, and command services using its O3b mPOWER satellite network in medium Earth orbit and its satellites in geosynchronous Earth orbit.Over the last two months, in collaboration with

Planet Labs, SES Space & Defense conducted multiple flight tests of its near-Earth space relay services. These demonstrations showcased uninterrupted, high-capacity connectivity between a Planet Labs satellite in low Earth orbit and SES communications satellites in geosynchronous Earth orbit and medium Earth orbit, demonstrating the ability to deliver real-time data relay across multiple orbits. SES Space & Defense has validated two relay services, one for low-rate tracking, telemetry, and command applications via its geostationary C-band satellites, and one for high-rate data applications over its O3b mPOWER Ka-band constellation. Additional flight demonstrations are planned for early 2026.An artist's concept of SES Space and Defense's satellite relay demonstration.Credit: SES Space and DefenseSpaceXSpaceX is demonstrating high-rate data exchanges over optical links using its Starlink network in low Earth orbit.Since 2024, SpaceX has completed multiple demonstrations of on-orbit optical communications services. During two human spaceflight missions, Polaris Dawn and Fram2, SpaceX leveraged the Starlink satellite constellation and an optical communications terminal installed on the Dragon spacecraft to demonstrate high-rate data relay services. Optical communications technology is not currently available through TDRS. By demonstrating optical relay services with multiple commercial partners, the agency is unlocking new capabilities for emerging missions.An artist's concept of SpaceX's commercial satellite relay demonstration using the Dragon spacecraft and Starlink network.Credit: SpaceXTelesatTelesat U.S. Services LLC, doing business as Telesat Government Solutions, is demonstrating high-rate data exchanges over optical links using its anticipated Telesat Lightspeed network in low Earth orbit.Development of the Telesat Lightspeed satellite network is currently underway, with satellite launches planned for late 2026. These satellites will use innovative technologies, like optical inter-satellite links and advanced onboard processing, to establish a global, mesh network in space. Software-defined networks aim to enable robust and reliable routing of traffic from a space-based or terrestrial terminal to its final destination autonomously. In 2027, Telesat plans to complete multiple demonstrations of space-to-space connectivity, including an optical data exchange from a Planet Labs spacecraft in low Earth orbit to the Telesat Lightspeed constellation. The data will then be routed over optical links before getting downlinked to a Telesat landing station on Earth, representing a full end-to-end capability.An artist illustration of Telesat's planned commercial relay demonstration using its Lightspeed satellite network.Credit: TelesatViasatViasat Inc. is demonstrating launch, tracking, telemetry, command, and high-data rate exchanges for launch vehicles and low Earth orbit operations. In May 2023, Viasat completed the acquisition of Inmarsat, the sixth satellite communications company to win a contract award from NASA, combining the resources of both companies to form a unified global communications provider.Viasat's space demonstrations will use its established satellite networks in geostationary orbit to validate three primary capabilities: launch telemetry over the L-band radio frequency to track and monitor spacecraft during ascent; command and control over L-band to maintain continuous spacecraft custody and enable real-time operations; and high-speed Ka-band data relay to transfer large volumes of mission data through next-generation spacecraft terminals. Flights test began in November, when Viasat used its satellite network to successfully track the telemetry of Blue Origin's New Glenn rocket as it launched into low Earth orbit. Follow-on demonstrations are planned for 2026, including additional L-band launch services as well as high-capacity services over Ka-band frequencies.An artist's concept outlining Viasat's satellite relay capabilities.Credit: ViasatCommercializing communications services for future near-Earth science missions enables NASA to focus resources on deep space missions to the Moon as part of the Artemis campaign, in preparation for future human missions to Mars. The agency will continue to work with these commercial partners to demonstrate next-generation services through 2027. By 2031, NASA plans to purchase satellite relay services for science missions from one or more U.S. satellite communications providers.To learn more about the decision to use commercial satellite relay services in low Earth orbit, visit:Embracing Commercial Relay Services – NASAThe Communications Services Project is managed by NASA's Glenn Research Center in Cleveland, under the direction of the Space Communications and Navigation Program within NASA's Space Operations Mission Directorate.

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By leveraging private-sector innovation, NASA aims to establish a more flexible, cost-effective, and scalable communications infrastructure for future science missions.

Amazon Leo for Government, a subsidiary of Amazon, is demonstrating high-rate data exchanges over optical links using its satellite network in low Earth orbit

Amazon has developed the hardware and software components necessary to support optical communication links within its Amazon Leo satellite relay network. Optical communications, also known as laser communications, use infrared light to transmit data at a higher rate compared to standard radio frequency systems. The Amazon Leo demonstrations, scheduled to begin in early 2026, will test the pointing, acquisition, and tracking capabilities of their optical communications systems to ensure the technology can accurately locate, lock onto, and stay connected with a mission as it travels through space.

An image of the view from an Amazon Leo satellite overlooking the Earth.Credit: Amazon

An image of the view from an Amazon Leo satellite overlooking the Earth.

Credit: Amazon

SES Space & Defense is demonstrating high-rate data exchanges as well as tracking, telemetry, and command services using its O3b mPOWER satellite network in medium Earth orbit and its satellites in geosynchronous Earth orbit.

Over the last two months, in collaboration with Planet Labs, SES Space & Defense conducted multiple flight tests of its near-Earth space relay services. These demonstrations showcased uninterrupted, high-capacity connectivity between a Planet Labs satellite in low Earth orbit and SES communications satellites in geosynchronous Earth orbit and medium Earth orbit, demonstrating the ability to deliver real-time data relay across multiple orbits. SES Space & Defense has validated two relay services, one for low-rate tracking, telemetry, and command applications via its geostationary C-band satellites, and one for high-rate data applications over its O3b mPOWER Ka-band constellation. Additional flight demonstrations are planned for early 2026.

An artist's concept of SES Space and Defense's satellite relay demonstration.Credit: SES Space and Defense

SpaceX is demonstrating high-rate data exchanges over optical links using its Starlink network in low Earth orbit.

Since 2024, SpaceX has completed multiple demonstrations of on-orbit optical communications services. During two human spaceflight missions, Polaris Dawn and Fram2, SpaceX leveraged the Starlink satellite constellation and an optical communications terminal installed on the Dragon spacecraft to demonstrate high-rate data relay services. Optical communications technology is not currently available through TDRS. By demonstrating optical relay services with multiple commercial partners, the agency is unlocking new capabilities for emerging missions.

An artist's concept of SpaceX's commercial satellite relay demonstration using the Dragon spacecraft and Starlink network.Credit: SpaceX

Telesat U.S. Services LLC, doing business as Telesat Government Solutions, is demonstrating high-rate data exchanges over optical links using its anticipated Telesat Lightspeed network in low Earth orbit.

Development of the Telesat Lightspeed satellite network is currently underway, with satellite launches planned for late 2026. These satellites will use innovative technologies, like optical inter-satellite links and advanced onboard processing, to establish a global, mesh network in space. Software-defined networks aim to enable robust and reliable routing of traffic from a space-based or terrestrial terminal to its final destination autonomously. In 2027, Telesat plans to complete multiple demonstrations of space-to-space connectivity, including an optical data exchange from a Planet Labs spacecraft in low Earth orbit to the Telesat Lightspeed constellation. The data will then be routed over optical links before getting downlinked to a Telesat landing station on Earth, representing a full end-to-end capability.

An artist illustration of Telesat's planned commercial relay demonstration using its Lightspeed satellite network.Credit: Telesat

An artist illustration of Telesat's planned commercial relay demonstration using its Lightspeed satellite network.

Credit: Telesat

Viasat Inc. is demonstrating launch, tracking, telemetry, command, and high-data rate exchanges for launch vehicles and low Earth orbit operations. In May 2023, Viasat completed the acquisition of Inmarsat, the sixth satellite communications company to win a contract award from NASA, combining the resources of both companies to form a unified global communications provider.

Viasat's space demonstrations will use its established satellite networks in geostationary orbit to validate three primary capabilities: launch telemetry over the L-band radio frequency to track and monitor spacecraft during ascent; command and control over L-band to maintain continuous spacecraft custody and enable real-time operations; and high-speed Ka-band data relay to transfer large volumes of mission data through next-generation spacecraft terminals. Flights test began in November, when Viasat used its satellite network to successfully track the telemetry of Blue Origin's New Glenn rocket as it launched into low Earth orbit. Follow-on demonstrations are planned for 2026, including additional L-band launch services as well as high-capacity services over Ka-band frequencies.

An artist's concept outlining Viasat's satellite relay capabilities.Credit: Viasat

Commercializing communications services for future near-Earth science missions enables NASA to focus resources on deep space missions to the Moon as part of the Artemis campaign, in preparation for future human missions to Mars. The agency will continue to work with these commercial partners to demonstrate next-generation services through 2027. By 2031, NASA plans to purchase satellite relay services for science missions from one or more U.S. satellite communications providers.

To learn more about the decision to use commercial satellite relay services in low Earth orbit, visit:

Embracing Commercial Relay Services – NASA

The Communications Services Project is managed by NASA's Glenn Research Center in Cleveland, under the direction of the Space Communications and Navigation Program within NASA's Space Operations Mission Directorate.

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TITLE: NASA, SpaceX Continue Crew-2 Space Station Undocking

Brandi DeanNovember 6, 2021CategoriesNASA, SpaceX Continue Crew-2 Space Station UndockingAstronauts (from left) Thomas Pesquet, Megan McArthur, Shane Kimbrough and Akihiko Hoshide talk to journalists on Earth before their return to Earth aboard the SpaceX Crew Dragon Endeavour.NASA’s SpaceX Crew-2 mission continues to target a return to Earth no earlier than 7:14 a.m. EST Monday, Nov. 8, with a splashdown off the coast of Florida. The Crew Dragon spacecraft, named Endeavour, is scheduled to undock from theInternational Space Stationat 12:04 p.m. Sunday, Nov. 7, to begin the journey home. Mission teams completed an undocking weather review on Saturday and are ‘go’ to proceed to the next weather review about six hours prior to the scheduled undocking time. Winds remain the main watch item for the return of the mission.NASA will provide coverage of the mission on NASA Television, theNASA app, and the agency’swebsite.NASA astronautsShane KimbroughandMegan McArthur, JAXA (Japan Aerospace Exploration Agency) astronautAki Hoshide, and ESA (European Space Agency) astronautThomas Pesquetwill complete 199 days in space at the conclusion of their mission. The spacecraft also will return to Earth with about 530 pounds of hardware and scientific investigations.Endeavour now will forego a fly around maneuver to photograph the exterior of the International Space Station to allow for additional alternate splashdown locations off the coast of Florida.NASA and SpaceX also have a backup undocking and splashdown opportunity available Monday, Nov. 8, if weather conditions are not favorable for the primary opportunity.The NASA and SpaceX teams will determine a primary and alternate splashdown location from the seven possible landing locations prior to return, factoring in weather, crew rescue, and recovery operations. Additional decision milestones take place prior to undocking, during free flight, and before Crew Dragon performs the deorbit burn.NASA and SpaceX closely coordinate with the U.S. Coast Guard to establish a safety zone around the expected splashdown location to ensure safety for the public and for those involved in the recovery operations, as well as the crew aboard the returning spacecraft.With Crew-2 splashdown Monday, Nov. 8, NASA’s SpaceX Crew-3 mission is targeting launch no earlier than 9:03 p.m. Wednesday, Nov. 10, on a SpaceX Falcon 9 rocket from Launch Complex 39A at NASA’s Kennedy Space Center in Florida. For this launch opportunity, the Crew Dragon Endurance is scheduled to dock to the space station around 7:10 p.m. Thursday, Nov. 11.NASA’s SpaceX Crew-2 return coverage is as follows:Sunday, Nov. 79:45 a.m. EST– Coverage begins for 10:15 a.m. hatch closure11:45 a.m. EST– Coverage begins for 12:04 p.m. undocking (NASA will provide continuous coverage from undocking to splashdown)Monday, Nov. 87:14 a.m. EST– SplashdownCrew-2 is the second of six NASA and SpaceX crewed missions to fly as part of the agency’s Commercial Crew Program, which is working with the U.S. aerospace industry to launch astronauts on American rockets and spacecraft from American soil.NASA’s Commercial Crew Program has delivered on its goal of safe, reliable, and cost-effective transportation to and from the International Space Station from the U.S. through a partnership with American private industry. This partnership is changing the arc of human spaceflight history by opening access to low-Earth orbit and the International Space Station to more people, more science, and more commercial opportunities. The space station remains the springboard to NASA’s next great leap in space exploration, including future missions to the Moon and, eventually, to Mars.Brandi DeanNovember 6, 2021Categories

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NASA astronautsShane KimbroughandMegan McArthur, JAXA (Japan Aerospace Exploration Agency) astronautAki Hoshide, and ESA (European Space Agency) astronautThomas Pesquetwill complete 199 days in space at the conclusion of their mission. The spacecraft also will return to Earth with about 530 pounds of hardware and scientific investigations.

Endeavour now will forego a fly around maneuver to photograph the exterior of the International Space Station to allow for additional alternate splashdown locations off the coast of Florida.

NASA and SpaceX also have a backup undocking and splashdown opportunity available Monday, Nov. 8, if weather conditions are not favorable for the primary opportunity.

The NASA and SpaceX teams will determine a primary and alternate splashdown location from the seven possible landing locations prior to return, factoring in weather, crew rescue, and recovery operations. Additional decision milestones take place prior to undocking, during free flight, and before Crew Dragon performs the deorbit burn.

NASA and SpaceX closely coordinate with the U.S. Coast Guard to establish a safety zone around the expected splashdown location to ensure safety for the public and for those involved in the recovery operations, as well as the crew aboard the returning spacecraft.

With Crew-2 splashdown Monday, Nov. 8, NASA's SpaceX Crew-3 mission is targeting launch no earlier than 9:03 p.m. Wednesday, Nov. 10, on a SpaceX Falcon 9 rocket from Launch Complex 39A at NASA's Kennedy Space Center in Florida. For this launch opportunity, the Crew Dragon Endurance is scheduled to dock to the space station around 7:10 p.m. Thursday, Nov. 11.

NASA's SpaceX Crew-2 return coverage is as follows:

Sunday, Nov. 79:45 a.m. EST– Coverage begins for 10:15 a.m. hatch closure

11:45 a.m. EST– Coverage begins for 12:04 p.m. undocking (NASA will provide continuous coverage from undocking to splashdown)

Monday, Nov. 87:14 a.m. EST– Splashdown

Crew-2 is the second of six NASA and SpaceX crewed missions to fly as part of the agency's Commercial Crew Program, which is working with the U.S. aerospace industry to launch astronauts on American rockets and spacecraft from American soil.

NASA's Commercial Crew Program has delivered on its goal of safe, reliable, and cost-effective transportation to and from the International Space Station from the U.S. through a partnership with American private industry. This partnership is changing the arc of human spaceflight history by opening access to low-Earth orbit and the International Space Station to more people, more science, and more commercial opportunities. The space station remains the springboard to NASA's next great leap in space exploration, including future missions to the Moon and, eventually, to Mars.

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TITLE: What Is Climate Change?

NASA Kids ScienceWhat Is Climate Change?Climate change describes a change in the average conditions over a long period of time. These include changes to things like temperature or rainfall. Scientists have been observing Earth for a long time. They use satellites and other instruments to collect information about Earth. This information shows that the planet is warming. And that warming is impacting Earth's climate.Weather vs. ClimateNASA/JPL-CaltechWeatherdescribes the conditions outside right now in a specific place. For example, if you see that it's raining outside right now, that's a way to describe today's weather. Rain, snow, wind, hurricanes, tornadoes — these are all weather events.Climate, on the other hand, is more than just one or two rainy days. Climate describes the weather conditions that are expected in a region.Is it usually rainy or usually dry? Is it typically hot or typically cold? A region's climate is determined by observing its weather over a period of many years. Generally 30 years or more is used.So, for example, a week of rainyweatherwouldn't change the fact that Phoenix typically has a dry, desertclimate. Even though it's rainy right now, we still expect Phoenix to be dry because that's what is usually the case.Want to know more about the difference between weather and climate? Take a look at this video.What Is Climate Change?Climate change describes a change in the average conditions over a long period of time. These include changes to things like temperature or rainfall. For example, 20,000 years ago during the last ice age, much of the United States was covered in glaciers. That would be a much colder climate than what we are used to today.When changes happen that impact the entire planet, it is called global climate change. These changes include warming global temperatures and shifts in precipitation. Plus, when the planet warms, it affects other parts of the Earth, such as:Rising sea levelsShrinking mountain glaciersIce melting at a faster rate than usual in Greenland, Antarctica, and the ArcticChanges in flower and plant blooming times.Earth's climate has constantly been changing — even long before humans came into the picture. However, scientists have observed a shift in the speed at which Earth's climate is changing. Previously, it has taken thousands of years for Earth's climate to switch between more ice and less ice. But now, observations show that Earth's temperature is increasing much more quickly.How Do Scientists Know Earth’s Climate is Changing?Scientists use instruments in space, on airplanes, and the ground to track Earth. These can be things like weather stations or even space lasers that measure the height of land and ice. When collected over time, these measurements help scientists understand what changes are happening.Watch this video to learn how scientists can use new and old pictures to keep track of Earth’s changes over time!To better understand how changes today compare to those in the past, scientists have to dig a little deeper. Earth keeps its own climate record for us through things that have been around a really long time. These are things like the bottom of the ocean or the Greenland and Antarctic ice sheets.Over time, these places build layers that capture information from when they formed. Scientists will drill down and take cylinder-shaped samples. These are known assediment coresandice cores. Sediment cores come from the bottoms of lakes or the ocean floor. Ice cores are from deep — sometimes miles deep — below the surface of the ice in places like Antarctica.The layers in an ice core are frozen solid. These layers of ice give clues about every year of Earth's history back to the time the deepest layer was formed. The ice contains bubbles of the air from each year. Scientists analyze the bubbles in each layer to find out how much of the air contained certain gases, like carbon dioxide. They can also learn about temperature and if it was a wet or dry year.NASA/JPL-CaltechHow Is Earth’s Climate Changing Right Now?Some parts of Earth are warming faster than others. But on average, global air temperatures have gone up more than 2 degrees Fahrenheit in the past 100 years. In fact, the past five years have been the warmest five years in centuries.Alaska's Muir glacier in August 1941 and August 2004. The glacier has retreated significantly in the 63 years between these two photos.USGSAs Earth continues to warm, it impacts Earth’s weather and climate. Scientists have already observed changes in several ways. The warmer temperatures have led to more intense and more common heat waves. Melting ice sheets and glaciers and ocean warming have led to sea level rise along the coast. And changes in rainfall have made droughts and floods more intense.What Causes Climate Change?There are lots of factors that can contribute to changes in Earth’s climate. However, scientists agree that Earth has been getting warmer over the past 100 years. They found that this warming is due to an increase in greenhouse gases in the atmosphere.Greenhouse gases in Earth’s atmosphere trap some of the heat from sunlight. This is called thegreenhouse effect. These gases keep Earth warm like the glass in a greenhouse keeps plants warm.Things we do, such as burning fuel to power factories, cars and buses, are changing the natural greenhouse. These changes cause the atmosphere to trap more heat than it used to, leading to a warmer Earth.How Do Scientists Use Satellites to Understand Climate Change?NASA’s Earth observing-satellites collect information about our planet. 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Weather vs. Climate

NASA/JPL-Caltech

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How Do Scientists Use Satellites to Understand Climate Change?

NASA's Earth observing-satellites collect information about our planet. These include theatmosphere,waterand land. By looking at this information, scientists can see how Earth is changing over time. This helps us understand how Earth is connected through its global climate.

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Explore This SectionWhat Is Climate Change?Climate change is a long-term change in the average weather patterns that have come to define Earth’s local, regional and global climates. These changes have a broad range of observed effects that are synonymous with the term.Changes observed in Earth’s climate since the mid-20th century are driven by human activities, particularly fossil fuel burning, which increases heat-trapping greenhouse gas levels in Earth’s atmosphere, raising Earth’s average surface temperature. Natural processes, which have been overwhelmed by human activities, can also contribute to climate change, including internal variability (e.g., cyclical ocean patterns like El Niño, La Niña and the Pacific Decadal Oscillation) and external forcings (e.g., volcanic activity,changes in the Sun’s energy output,variations in Earth’s orbit).Scientists use observations from the ground, air, and space, along withcomputermodels, to monitor and study past, present, and future climate change. Climate data records provide evidence of climate change key indicators, such as global land and ocean temperature increases; rising sea levels; ice loss at Earth’s poles and in mountain glaciers; frequency and severity changes in extreme weather such as hurricanes, heatwaves, wildfires, droughts, floods, and precipitation; and cloud and vegetation cover changes.“Climate change” and “global warming” are often used interchangeably but have distinct meanings. Similarly, the terms "weather" and "climate" are sometimes confused, though they refer to events with broadly different spatial- and timescales.What Is Global Warming?This graph illustrates the change in global surface temperature relative to 1951-1980 average temperatures, with the year 2020 statistically tying with 2016 for hottest on record (Source:NASA's Goddard Institute for Space Studies). Learn more about global surface temperaturehere.NASA/JPL-CaltechGlobal warming is the long-term heating of Earth’s surface observed since the pre-industrial period (between 1850 and 1900) due to human activities, primarily fossil fuel burning, which increases heat-trapping greenhouse gas levels in Earth’s atmosphere. This term is not interchangeable with the term "climate change."Since the pre-industrial period, human activities are estimated to have increased Earth’s global average temperature by about 1 degree Celsius (1.8 degrees Fahrenheit), a number that is currently increasing by more than 0.2 degrees Celsius (0.36 degrees Fahrenheit) per decade. The current warming trend is unequivocally the result of human activity since the 1950s and is proceeding at an unprecedented rate over millennia.Weather vs. Climate“If you don’t like the weather in New England, just wait a few minutes.”
- Mark TwainWeather refers to atmospheric conditions that occur locally over short periods of time—from minutes to hours or days. Familiar examples include rain, snow, clouds, winds, floods, or thunderstorms.Climate, on the other hand, refers to the long-term (usually at least 30 years) regional or even global average of temperature, humidity, and rainfall patterns over seasons, years, or decades.Find Out More: A Guide to NASA’s Global Climate Change WebsiteThis website provides a high-level overview of some of the known causes, effects and indications of global climate change:Evidence.Brief descriptions of some of the key scientific observations that our planet is undergoing abrupt climate change.Causes.A concise discussion of the primary climate change causes on our planet.Effects.A look at some of the likely future effects of climate change, including U.S. regional effects.Vital Signs.Graphs and animated time series showing real-time climate change data, including atmospheric carbon dioxide, global temperature, sea ice extent, and ice sheet volume.Earth Minute.This fun video series explains various Earth science topics, including some climate change topics.Other NASA ResourcesGoddard Scientific Visualization Studio.An extensive collection of animated climate change and Earth science visualizations.Sea Level Change Portal.NASA's portal for an in-depth look at the science behind sea level change.NASA’s Earth Observatory.Satellite imagery, feature articles and scientific information about our home planet, with a focus on Earth’s climate and environmental change.Header image is of Apusiaajik Glacier, and was taken near Kulusuk, Greenland, on Aug. 26, 2018, during NASA's Oceans Melting Greenland (OMG) field operations. Learn morehere. Credit: NASA/JPL-CaltechKeep ExploringDiscover More Topics From NASAExplore Earth ScienceEarth Science in ActionNASA's unique vantage point helps us inform solutions to enhance decision-making, improve livelihoods, and protect our planet.Earth Science DataFacts About Earth

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What Is Global Warming?

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Climate, on the other hand, refers to the long-term (usually at least 30 years) regional or even global average of temperature, humidity, and rainfall patterns over seasons, years, or decades.

Find Out More: A Guide to NASA’s Global Climate Change Website

This website provides a high-level overview of some of the known causes, effects and indications of global climate change:

Evidence.Brief descriptions of some of the key scientific observations that our planet is undergoing abrupt climate change.

Causes.A concise discussion of the primary climate change causes on our planet.

Effects.A look at some of the likely future effects of climate change, including U.S. regional effects.

Vital Signs.Graphs and animated time series showing real-time climate change data, including atmospheric carbon dioxide, global temperature, sea ice extent, and ice sheet volume.

Earth Minute.This fun video series explains various Earth science topics, including some climate change topics.

Other NASA Resources

Goddard Scientific Visualization Studio.An extensive collection of animated climate change and Earth science visualizations.

Sea Level Change Portal.NASA's portal for an in-depth look at the science behind sea level change.

NASA’s Earth Observatory.Satellite imagery, feature articles and scientific information about our home planet, with a focus on Earth’s climate and environmental change.

Header image is of Apusiaajik Glacier, and was taken near Kulusuk, Greenland, on Aug. 26, 2018, during NASA's Oceans Melting Greenland (OMG) field operations. Learn morehere. Credit: NASA/JPL-Caltech

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While listening to and collaborating with industry leaders, international partners, academic institutions, and other users of our data, we deepen knowledge of our planet, drive innovations, and deliver science to help inform decisions that benefit the nation and the world.Learn more about how NASA Earth science helps Americans respond to challenges.Wildfires and Climate ChangeEarth's warming climate is amplifying wildland fire activity, particularly in northern and temperate forests. When fires ignite the landscape, NASA’s satellites and instruments can detect and track them. 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Learn more about how NASA Earth science helps Americans respond to challenges.

Wildfires and Climate ChangeEarth's warming climate is amplifying wildland fire activity, particularly in northern and temperate forests. When fires ignite the landscape, NASA’s satellites and instruments can detect and track them. This information helps communities and land managers around the worldprepare for and respondto fires, and also providesa rich data sourceto help scientists better understand this growing risk.Learn Moreabout Wildfires and Climate Change

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Climate Data Resources

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TITLE: Ultra-Efficient Aviation

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TITLE: What Is the Kuiper Belt?

kuiper-beltWhat Is the Kuiper Belt?The Short Answer:The Kuiper Belt is a ring of icy bodies just outside of Neptune's orbit. Pluto is the most famous Kuiper Belt Object.The Sun is at the center of our solar system. It is orbited by eight planets: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. But what's past Neptune?Just outside of Neptune's orbit is a ring of icy bodies. We call it theKuiper Belt. It's pronouncedky-purr.This is where you'll find dwarf planet Pluto. It's the most famous of the objects floating in the Kuiper Belt, which are also calledKuiper Belt Objects, or KBOs.Why is it named Kuiper?The Kuiper Belt is named after a scientist namedGerard Kuiper. In 1951 he had the idea that a belt of icy bodies might have existed beyond Neptune when the solar system formed. He was trying to explain where comets with small orbits came from. No one had seen anything out there yet because it's hard to see small comets past Neptune even with the best telescopes. But even without being able to see it with his own eyes, Kuiper made a prediction. And it turned out to be right.What's out there?There are bits of rock and ice, comets, and dwarf planets. Besides Pluto, two other interesting Kuiper Belt Objects are Eris and Haumea.ErisEris is a Kuiper Belt Object a little smaller than Pluto. It is so far away it takes 557 years to orbit the Sun. Eris has a small moon named Dysnomia.Learn more about ErisHaumeaAnother interesting Kuiper Belt Object is Haumea. It's shaped like a squashed American football about 1,200 miles (1,931 km) long. It spins end over end every few hours. The strange shape and rotation were caused by a collision with an object about half its size. When Haumea and this other object smashed into each other, the impact blasted away big pieces of ice and sent Haumea spinning.Haumea also has two moons named Hi'iaka and Namaka.Learn more about HaumeaArrokothMove the slider to the left to see a real image of Arrokoth from the New Horizons spacecraft. Credit: NASA/Johns Hopkins University Applied Physics Laboratory/Southwest Research InstituteA billion miles past Pluto there is a relatively small snowman-shaped object named Arrokoth. Originally called MU69, this Kuiper Belt Object was first spotted in 2014 by the Hubble Space Telescope. Then in 2019, the New Horizons spacecraft actually flew right by it and captured some pictures!Arrokoth is the farthest world ever studied up close by a spacecraft. Its name is a Native American term meaning “sky” in the Powhatan/Algonquian language. Scientists think Arrokoth might hold clues about the formation of our planet and the origin of life on Earth.Learn more about ArrokothMore to LearnThe Kuiper Belt is still a very mysterious place, and we have a lot to learn about it. New Horizons will keep exploring the Kuiper Belt and sending us more information about it.Learn more about the Kuiper Beltarticle last updated June 3, 2020More about the solar systemWhat's past the Kuiper Belt?Why are planets round?Explore each planet!If you liked this, you may like:All About PlutoThirsty? Have a comet!What Is a Meteor Shower?GamesCraftsActivitiesVideosGlossaryMystery

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Eris

Eris is a Kuiper Belt Object a little smaller than Pluto. It is so far away it takes 557 years to orbit the Sun. Eris has a small moon named Dysnomia.

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Haumea

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Learn more about Haumea

Arrokoth

Move the slider to the left to see a real image of Arrokoth from the New Horizons spacecraft. Credit: NASA/Johns Hopkins University Applied Physics Laboratory/Southwest Research Institute

A billion miles past Pluto there is a relatively small snowman-shaped object named Arrokoth. Originally called MU69, this Kuiper Belt Object was first spotted in 2014 by the Hubble Space Telescope. Then in 2019, the New Horizons spacecraft actually flew right by it and captured some pictures!

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Physical Sciences Program, which solicits and conducts research using the space environment as a tool to provide transformational insights in physics and engineering science, and to understand how physical systems respond to spaceflight environments, particularly weightlessness and the partial gravity of planetary bodies.

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NASA Science Editorial TeamJan 23, 2025ArticleNASA's Mars rover Curiosity acquired this image using its Left Navigation Camera on sol 4429 — or Martian day 4,429 of the Mars Science Laboratory mission — on Jan. 20, 2025, at 23:05:33 UTC.NASA/JPL-CaltechEarth planning date: Tuesday, Jan. 21, 2025Over the weekend Curiosity completed a 38-meter drive (about 125 feet), and the rover is now perched on the rim of a crater named “Rustic Canyon.” Today we planned contact science to search for any unique textures or compositions associated with the crater, and a lot of imaging to assess the morphology of the crater to learn about degradation rates and processes. Today is a three-sol plan on a Tuesday, to account for the U.S. holiday yesterday, Martin Luther King Jr. Day.The plan starts with remote sensing observations of the rocks in our workspace. ChemCam and Mastcam will investigate two dark blocks named “Slide Peak” and “Rocky Peak” to assess their composition and texture. Then Curiosity will use the DRT to clear the dust at “Bee Rock,” a block in our workspace that looks smoother than the surrounding areas, followed by MAHLI and APXS on the same block. Curiosity will also use MAHLI and APXS to assess a dark block at “Pleasant View Ridge” to see how these ejecta blocks compare to the surrounding rocks of the sulfate unit. Mastcam will be busy in this plan, acquiring a large 70-frameL0mosaic covering the entire crater to assess its morphology, as well as stereo imaging on the western rim where more dark rocks are exposed. Another Mastcam mosaic will target the transition from the ejecta blocks back into typical rocks of the sulfate unit, and a fourth Mastcam mosaic will look toward the distant Gediz Vallis ridge, coordinated with a giant 20x1 ChemCam Long Distance RMI of the same area. Navcam will also be used to document the crater, to help generate high-resolution topography. This plan includes a drive on the second sol, during which Curiosity will take Full MAHLI Wheel Imaging — periodic documentation to assess the state and health of the rover wheels. The third sol contains untargeted science activities, including an autonomously selected ChemCam AEGIS target. Throughout the plan Curiosity will acquire observations to monitor the environment, including a Navcam dust devil movie, Mastcam twilight observation to search for iridescent ice clouds, and Navcam line-of-sight observations to assess dust content in the atmosphere.I was on shift as LTP today, and it was great to see this comprehensive plan come together to address a lot of different science areas. Looking forward to getting the results back and seeing what “Rustic Canyon” can tell us about Mars!Written by Lauren Edgar, Planetary Geologist at USGS Astrogeology Science Center

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TITLE: NASA's SPHEREx Observatory Completes First Cosmic Map Like No Other

6 min readNASA's SPHEREx Observatory Completes First Cosmic Map Like No Other

Jet Propulsion LaboratoryDec 18, 2025ArticleContentsSuperpowered telescopeMore about SPHERExThis panoramic view of SPHEREx's first all-sky map shows how the sky looks to the telescope. It transitions between observations of colors emitted by hot hydrogen gas (blue) and cosmic dust (red), and those primarily emitted by stars.Credit: NASA/JPL-CaltechLaunched in March, NASA's SPHEREx space telescope has completed its first infrared map of the entire sky in 102 colors. While not visible to the human eye, these 102 infrared wavelengths of light are prevalent in the cosmos, and observing the entire sky this way enables scientists to answer big questions, including how a dramatic event that occurred in the first billionth of a trillionth of a trillionth of a second after the big bang influenced the 3D distribution of hundreds of millions of galaxies in our universe. 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It will complete three additional all-sky scans during its two-year primary mission, and merging those maps together will increase the sensitivity of the measurements. The entire dataset isfreely available to scientistsand the public."SPHEREx is a mid-sized astrophysics mission delivering big science," said JPL Director Dave Gallagher. "It's a phenomenal example of how we turn bold ideas into reality, and in doing so, unlock enormous potential for discovery."NASA's SPHEREx has mapped the entire sky in 102 infrared colors, which are invisible to the human eye but can be used to reveal different features of the cosmos. This image features a selection of colors emitted primarily by stars (blue, green, and white), hot hydrogen gas (blue), and cosmic dust (red).NASA/JPL-CaltechThis SPHEREx image shows a selection of the infrared colors primarily emitted by stars and galaxies. The space telescope is observing hundreds of millions of distant galaxies across the sky. Its multiwavelength view will help astronomers measure the distance to those galaxies.NASA/JPL-CaltechThe infrared colors emitted primarily by dust (red) and hot gas (blue), key ingredients for forming new stars and planets, are seen in this SPHEREx image. Though these clouds of material cover a massive portion of the sky, they are invisible in most wavelengths of light, including those the human eye can detect.NASA/JPL-CaltechSuperpowered telescopeEach of the 102 colors detected by SPHEREx represents a wavelength of infrared light, and each wavelength provides unique information about the galaxies, stars, planet-forming regions, and other cosmic features therein. For example, dense clouds of dust in our galaxy where stars and planets form radiate brightly in certain wavelengths but emit no light (and are therefore totally invisible) in others. The process of separating the light from a source into its component wavelengths is called spectroscopy.And while a handful of previous missions has also mapped the entire sky, such as NASA'sWide-field Infrared Survey Explorer, none have done so in nearly as many colors as SPHEREx. By contrast, NASA'sJames Webb Space Telescopecan do spectroscopy with significantly more wavelengths of light than SPHEREx, but with a field of view thousands of times smaller. The combination of colors and such a wide field of view is why SPHEREx is so powerful."The superpower of SPHEREx is that it captures the whole sky in 102 colors about every six months. That's an amazing amount of information to gather in a short amount of time," said Beth Fabinsky, the SPHEREx project manager at JPL. "I think this makes us the mantis shrimp of telescopes, because we have an amazing multicolor visual detection system and we can also see a very wide swath of our surroundings."To accomplish this feat, SPHEREx uses six detectors, each paired with a specially designed filter that contains a gradient of 17 colors. That means every image taken with those six detectors contains 102 colors (six times 17). It also means that every all-sky map that SPHEREx produces is really 102 maps, each in a different color.The observatory will use those colors to measure the distance to hundreds of millions of galaxies. Though the positions of most of those galaxies have already been mapped in two dimensions by other observatories, SPHEREx's map will be in 3D, enabling scientists to measure subtle variations in the way galaxies are clustered and distributed across the universe.Each frame of this movie shows the entire sky in a different infrared wavelength, indicated by the color bar in the top right corner. Taken by NASA's SPHEREx observatory, the maps illustrate how viewing the universe in different wavelengths of light can reveal unique cosmic features.Credit: NASA/JPL-CaltechThose measurements will offer insights into an event that took place in the first billionth of a trillionth of a trillionth of a second after the big bang. In this moment, called inflation, the universe expanded by a trillion-trillionfold. Nothing like it has occurred in the universe since, and scientists want to understand it better. The SPHEREx mission's approach is one way to help in that effort.More about

SPHERExThe SPHEREx mission is managed by JPL for NASA's Astrophysics Division within the Science Mission Directorate in Washington. The telescope and the spacecraft bus were built by BAE Systems. The science analysis of the SPHEREx data is being conducted by a team of scientists at 10 institutions across the U.S., and in South Korea and Taiwan. Data is processed and archived atIPACat Caltech in Pasadena, which manages JPL for NASA. The mission's principal investigator is based at Caltech with a joint JPL appointment. The SPHEREx dataset is publicly available.For more information about the SPHEREx mission visit:https://science.nasa.gov/mission/spherex/News Media ContactsCalla CofieldJet Propulsion Laboratory, Pasadena, Calif.626-808-2469calla.e.cofield@jpl.nasa.gov2025-144

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For more information about the SPHEREx mission visit:

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Humans to Mars

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TITLE: Últimos preparativos para la primera misión tripulada a la Luna con la campaña Artemis de la NASA

11 min readÚltimos preparativos para la primera misión tripulada a la Luna con la campaña Artemis de la NASA

Tiernan P. DoyleJan 12, 2026ArticleEl cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) de la NASA, visto en la nave High Bay 3 del Edificio de Ensamblaje de Vehículos mientras los equipos t esperan la llegada de los tripulantes de Artemis II para abordar su nave espacial Orion en la parte superior del cohete como parte de la prueba de demostración de la cuenta atrás de Artemis II, el sábado 20 de diciembre de 2025, en el

Centro Espacial Kennedy de la NASA en Florida.NASA/Joel KowskyRead this article in Englishhere.Conforme la NASA se acerca al lanzamiento del vuelo de prueba Artemis II, la agencia pronto llevará por primera vez su cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) y la nave espacial Orion a la plataforma de lanzamiento en el Centro Espacial Kennedy de la agencia en Florida para comenzar la integración final, las pruebas y los ensayos para el lanzamiento.La NASA tiene como objetivo comenzar desde el sábado 17 de enero su traslado desde el Edificio de Ensamblaje de Vehículos hasta la Plataforma de Lanzamientos 39B, lo que tardará varias horas. El viaje de casi 6,5 kilómetros (cuatro millas) en el vehículo transportador oruga 2 podría durar hasta 12 horas. Los equipos técnicos están trabajando día y noche para dar por terminadas todas las tareas antes del transporte del cohete. Sin embargo, esta fecha objetivo está sujeta a cambios si fuera necesario tiempo adicional para los preparativos técnicos o debido a las condiciones meteorológicas."Nos estamos acercando a la misión Artemis II, y tenemos su lanzamiento a la vuelta de la esquina", dijo Lori Glaze, administradora asociada interina para la Dirección de Misiones de Desarrollo de Sistemas de Exploración de la NASA. "Nos quedan pasos importantes en nuestro camino hacia el lanzamiento, y la seguridad de la tripulación seguirá siendo nuestra principal prioridad en todo momento, a medida que nos acercamos al regreso de la humanidad a la Luna".Al igual que con todos los nuevos desarrollos de sistemas complejos, los ingenieros han estado solucionando varios problemas en los últimos días y semanas. Durante las comprobaciones finales antes del traslado, los técnicos detectaron que un cable relacionado con el sistema de terminación de vuelo estaba doblado en contra de las especificaciones. El personal técnico lo está reemplazando y hará pruebas con el nuevo cable durante el fin de semana. Además, una válvula relacionada con la presurización de la escotilla de Orion presentó problemas que hicieron necesario llevar a cabo pruebas dedemostración de la cuenta regresivael 20 de diciembre pasado. El 5 de enero, el equipo reemplazó la válvula e hizo pruebas de su funcionamiento que resultaron exitosas. Los ingenieros también trabajaron para resolver fugas en el hardware de soporte en tierra que es necesario para cargar oxígeno gaseoso en Orion a fin de proporcionar aire respirable.TrasladoUna vez que el cohete y la nave espacial integrados lleguen a la plataforma de lanzamiento, la NASA comenzará inmediatamente una larga lista de verificación para los preparativos en la plataforma de lanzamiento, incluyendo la conexión de equipos mecánicos de apoyo en tierra, como líneas eléctricas, conductos del sistema de control ambiental de combustible y tomas de surtido de combustible criogénico. Los equipos de personal técnico encenderán todos los sistemas integrados en la plataforma por primera vez para garantizar que los componentes del hardware de vuelo funcionen correctamente entre sí, con el lanzador móvil y con los sistemas de infraestructura terrestre.Una vez que esté todo completado, los astronautas de Artemis II, Reid Wiseman, Victor Glover y Christina Koch de la NASA, y el astronauta de la CSA (Agencia Espacial Canadiense) Jeremy Hansen, llevarán a cabo una caminata final en la plataforma.Ensayo general con circulación de combustible y llenado de tanquesA finales de enero, la NASA llevará a cabo un ensayo general con circulación de combustible, el cual es una prueba previa al lanzamiento para llenar los tanques de combustible en el cohete. Durante este ensayo general, el personal técnico hace una demostración de la capacidad de cargar más de 700.000 galones de combustible criogénico en el cohete, lleva a cabo una cuenta regresiva para el lanzamiento y practica la extracción segura del combustible del cohete sin tripulación presente en el sitio.Durante el lanzamiento,el equipo de cierre, o de tareas finales, será responsable de asegurar a los astronautas en Orion y cerrar sus escotillas. El personal de cierre también utilizará este ensayo para practicar sus procedimientos de forma segura sin tener tripulación a bordo de la nave espacial.El ensayo general con circulación de combustible incluirá varios “encendidos”, o pruebas de funcionamiento, para demostrar la capacidad del equipo de lanzamiento para detener, reanudar y reiniciar operaciones en varios momentos diferentes de los últimos 10 minutos de la cuenta regresiva, conocida como conteo terminal.La ejecución del primer encendido comenzará aproximadamente en las 49 horas antes del lanzamiento, cuando los equipos encargados de lanzamiento son llamados a sus estaciones, hasta 1 minuto y 30 segundos antes del lanzamiento, seguido de una pausa planificada de tres minutos y luego la reanudación de la cuenta regresiva hasta 33 segundos antes del lanzamiento, el punto en el que el secuenciador de lanzamiento automático del cohete controlará los últimos segundos de la cuenta regresiva. Luego, los equipos técnicos volverán a reiniciar a T-10 minutos y detendrán el conteo, y luego reanudarán los procedimientos hasta 30 segundos antes del lanzamiento como parte de una segunda ejecución.Si bien la NASA ha integrado las lecciones aprendidas con Artemis I en los procedimientos de la cuenta regresiva para el lanzamiento, la agencia hará una pausa para abordar cualquier problema durante la prueba o en cualquier otro momento si surgen retos técnicos. Los ingenieros vigilarán de cerca la carga de combustible de hidrógeno líquido y oxígeno líquido en el cohete, después de los desafíos que se encontraron con la carga de hidrógeno líquido durante losensayos generales con circulación de combustible de Artemis I. Los equipos técnicos también prestarán mucha atención a la efectividad de los procedimientos recientemente actualizados para limitar la cantidad de nitrógeno gaseoso que se acumula en el espacio que está entre el módulo de tripulación de Orion y las escotillas del sistema de cancelación de lanzamiento, lo que podría representar un problema para el personal de cierre.Es posible que se requieran ensayos generales con circulación de combustible adicionales para garantizar que el vehículo esté completamente revisado y apto para el vuelo.De ser necesario, la NASA podría trasladar al cohete SLS y la nave Orion de vuelta al Edificio de Ensamblaje de Vehículos para realizar trabajos adicionales antes del lanzamiento después del ensayo general con circulación de combustible.Próximos pasos para el lanzamientoDespués de un exitoso ensayo general con circulación de combustible, la NASA convocará una revisión de aptitud para el vuelo en la cual el equipo de gestión de la misión evaluará la aptitud de todos los sistemas, incluyendo el hardware de vuelo, la infraestructura y el personal de lanzamiento, vuelo y recuperación antes de comprometerse con una fecha de lanzamiento.Aunque la ventana para el lanzamiento de Artemis II se podría iniciar tan pronto como el viernes 6 de febrero, el equipo de gestión de la misión evaluará la aptitud para el vuelo después del ensayo general con toda la nave espacial, la infraestructura de lanzamiento, y la tripulación y el personal de operaciones antes de seleccionar una fecha para el lanzamiento.A fin de determinar las posibles fechas de lanzamiento, los ingenieros identificaron las restricciones clave necesarias para cumplir la misión y mantener a salvo a la tripulación dentro de Orion. Los períodos de lanzamiento resultantes son los días o las semanas en los que la nave espacial y el cohete pueden cumplir los objetivos de la misión. Estos períodos de lanzamiento explican la compleja mecánica orbital relacionada con el lanzamiento en una trayectoria precisa hacia la Luna mientras la Tierra rota sobre su eje y la Luna orbita la Tierra cada mes en su ciclo lunar. Esto da como resultado un patrón de alrededor de una semana de oportunidades de lanzamiento, seguido de tres semanas sin oportunidades de lanzamiento.Existen varios parámetros principales que establecen la disponibilidad del lanzamiento dentro de estos períodos. Debido a su trayectoria única en relación con las misiones de alunizaje posteriores, estas limitaciones clave son exclusivas del vuelo de prueba de la misión Artemis II.El día y la hora de lanzamiento deben permitir que SLS pueda llevar a Orion a una órbita terrestre alta, donde la tripulación y los equipos técnicos en tierra evaluarán los sistemas de soporte vital de la nave espacial antes de que la tripulación emprenda su viaje con rumbo a la Luna.Orion también debe estar en la alineación adecuada con la Tierra y la Luna en el momento del encendido de motores con inyección translunar. El encendido de motores con inyección translunar de Artemis II pone a Orion en rumbo de sobrevolar la Luna, y también lo pone en una trayectoria de retorno libre, en la cual la nave espacial utiliza la gravedad de la Luna para enviar la nave espacial de regreso a la Tierra sin maniobras adicionales importantes de propulsión.La trayectoria para un día determinado debe garantizar que Orion no esté en la oscuridad durante más de 90 minutos a la vez para que las alas de los paneles solares puedan recibir y convertir la luz solar en electricidad, y la nave espacial pueda mantener un rango de temperatura óptimo. Los planificadores de la misión eliminan las posibles fechas de lanzamiento que llevarían a Orion a eclipses prolongados durante el vuelo.La fecha de lanzamiento debe sustentar una trayectoria que permita el perfil de entrada adecuado planificado durante el regreso de Orion a la Tierra.Los períodos a continuación muestran la disponibilidad de llevar a cabo el lanzamientohasta abril de 2026. Los planificadores de la misión mejoran estos períodos en función de un análisis actualizado más o menos dos meses antes de que estos comiencen, y ellos están sujetos a cambios.Período de lanzamientodel 31 de enero al 14 de febreroOportunidades de lanzamiento los días 6, 7, 8, 10 y 11 de febreroPeríodo de lanzamientodel 28 de febrero al 13 de marzoOportunidades de lanzamiento los días 6, 7, 8, 9, 11 de marzoPeríodo de lanzamientodel 27 de marzo al 10 de abrilOportunidades de lanzamiento los días 1, 3, 4, 5, 6 de abrilAdemás de las oportunidades de lanzamiento basadas en la mecánica orbital y los requisitos de desempeño, también existen restricciones sobre qué días dentro de un período de lanzamiento pueden ser viables en función de la reposición de productos básicos, las condiciones meteorológicas y las operaciones de otros usuarios en el cronograma del Área Este. Como regla general, se pueden hacer hasta cuatro intentos de lanzamiento dentro de la semana aproximada de oportunidades que existen dentro de un período de lanzamiento.Mientras la agencia se prepara para su primera misión tripulada más allá de la órbita terrestre en más de 50 años, la NASA espera aprender durante los procesos, tanto en tierra como en vuelo, y dejará que la aptitud y el desempeño de sus sistemas indiquen el momento en que la agencia está lista para el lanzamiento.Como parte de una edad de oro de innovación y exploración, el vuelo de prueba de Artemis II, el cual tendrá una duración aproximada de 10 días, es el primer vuelo tripulado para la campaña Artemis de la NASA. Este es otro paso hacia nuevas misiones tripuladas de Estados Unidos en la superficie de la Luna, lo que llevará a una presencia sostenida en la Luna que ayudará a la agencia a prepararse para enviar a los primeros astronautas estadounidenses a Marte.Encuentra más información sobre la campaña Artemis de la NASA en el siguiente sitio web (en inglés):https://www.nasa.gov/artemis

Tiernan P. DoyleJan 12, 2026Article

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El cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) de la NASA, visto en la nave High Bay 3 del Edificio de Ensamblaje de Vehículos mientras los equipos t esperan la llegada de los tripulantes de Artemis II para abordar su nave espacial Orion en la parte superior del cohete como parte de la prueba de demostración de la cuenta atrás de Artemis II, el sábado 20 de diciembre de 2025, en el Centro Espacial Kennedy de la NASA en Florida.NASA/Joel KowskyRead this article in Englishhere.Conforme la NASA se acerca al lanzamiento del vuelo de prueba Artemis II, la agencia pronto llevará por primera vez su cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) y la nave espacial Orion a la plataforma de lanzamiento en el Centro Espacial Kennedy de la agencia en Florida para comenzar la integración final, las pruebas y los ensayos para el lanzamiento.La NASA tiene como objetivo comenzar desde el sábado 17 de enero su traslado desde el Edificio de Ensamblaje de Vehículos hasta la Plataforma de Lanzamientos 39B, lo que tardará varias horas. 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El personal técnico lo está reemplazando y hará pruebas con el nuevo cable durante el fin de semana. Además, una válvula relacionada con la presurización de la escotilla de Orion presentó problemas que hicieron necesario llevar a cabo pruebas dedemostración de la cuenta regresivael 20 de diciembre pasado. El 5 de enero, el equipo reemplazó la válvula e hizo pruebas de su funcionamiento que resultaron exitosas. Los ingenieros también trabajaron para resolver fugas en el hardware de soporte en tierra que es necesario para cargar oxígeno gaseoso en Orion a fin de proporcionar aire respirable.TrasladoUna vez que el cohete y la nave espacial integrados lleguen a la plataforma de lanzamiento, la NASA comenzará inmediatamente una larga lista de verificación para los preparativos en la plataforma de lanzamiento, incluyendo la conexión de equipos mecánicos de apoyo en tierra, como líneas eléctricas, conductos del sistema de control ambiental de combustible y tomas de surtido de combustible criogénico. Los equipos de personal técnico encenderán todos los sistemas integrados en la plataforma por primera vez para garantizar que los componentes del hardware de vuelo funcionen correctamente entre sí, con el lanzador móvil y con los sistemas de infraestructura terrestre.Una vez que esté todo completado, los astronautas de Artemis II, Reid Wiseman, Victor Glover y Christina Koch de la NASA, y el astronauta de la CSA (Agencia Espacial Canadiense) Jeremy Hansen, llevarán a cabo una caminata final en la plataforma.Ensayo general con circulación de combustible y llenado de tanquesA finales de enero, la NASA llevará a cabo un ensayo general con circulación de combustible, el cual es una prueba previa al lanzamiento para llenar los tanques de combustible en el cohete. Durante este ensayo general, el personal técnico hace una demostración de la capacidad de cargar más de 700.000 galones de combustible criogénico en el cohete, lleva a cabo una cuenta regresiva para el lanzamiento y practica la extracción segura del combustible del cohete sin tripulación presente en el sitio.Durante el lanzamiento,el equipo de cierre, o de tareas finales, será responsable de asegurar a los astronautas en Orion y cerrar sus escotillas. El personal de cierre también utilizará este ensayo para practicar sus procedimientos de forma segura sin tener tripulación a bordo de la nave espacial.El ensayo general con circulación de combustible incluirá varios "encendidos", o pruebas de funcionamiento, para demostrar la capacidad del equipo de lanzamiento para detener, reanudar y reiniciar operaciones en varios momentos diferentes de los últimos 10 minutos de la cuenta regresiva, conocida como conteo terminal.La ejecución del primer encendido comenzará aproximadamente en las 49 horas antes del lanzamiento, cuando los equipos encargados de lanzamiento son llamados a sus estaciones, hasta 1 minuto y 30 segundos antes del lanzamiento, seguido de una pausa planificada de tres minutos y luego la reanudación de la cuenta regresiva hasta 33 segundos antes del lanzamiento, el punto en el que el secuenciador de lanzamiento automático del cohete controlará los últimos segundos de la cuenta regresiva. Luego, los equipos técnicos volverán a reiniciar a T-10 minutos y detendrán el conteo, y luego reanudarán los procedimientos hasta 30 segundos antes del lanzamiento como parte de una segunda ejecución.Si bien la NASA ha integrado las lecciones aprendidas con Artemis I en los procedimientos de la cuenta regresiva para el lanzamiento, la agencia hará una pausa para abordar cualquier problema durante la prueba o en cualquier otro momento si surgen retos técnicos. Los ingenieros vigilarán de cerca la carga de combustible de hidrógeno líquido y oxígeno líquido en el cohete, después de los desafíos que se encontraron con la carga de hidrógeno líquido durante losensayos generales con circulación de combustible de Artemis I. Los equipos técnicos también prestarán mucha atención a la efectividad de los procedimientos recientemente actualizados para limitar la cantidad de nitrógeno gaseoso que se acumula en el espacio que está entre el módulo de tripulación de Orion y las escotillas del sistema de cancelación de lanzamiento, lo que podría representar un problema para el personal de cierre.Es posible que se requieran ensayos generales con circulación de combustible adicionales para garantizar que el vehículo esté completamente revisado y apto para el vuelo.De ser necesario, la NASA podría trasladar al cohete SLS y la nave Orion de vuelta al Edificio de Ensamblaje de Vehículos para realizar trabajos adicionales antes del lanzamiento después del ensayo general con circulación de combustible.Próximos pasos para el lanzamientoDespués de un exitoso ensayo general con circulación de combustible, la NASA convocará una revisión de aptitud para el vuelo en la cual el equipo de gestión de la misión evaluará la aptitud de todos los sistemas, incluyendo el hardware de vuelo, la infraestructura y el personal de lanzamiento, vuelo y recuperación antes de comprometerse con una fecha de lanzamiento.Aunque la ventana para el lanzamiento de Artemis II se podría iniciar tan pronto como el viernes 6 de febrero, el equipo de gestión de la misión evaluará la aptitud para el vuelo después del ensayo general con toda la nave espacial, la infraestructura de lanzamiento, y la tripulación y el personal de operaciones antes de seleccionar una fecha para el lanzamiento.A fin de determinar las posibles fechas de lanzamiento, los ingenieros identificaron las restricciones clave necesarias para cumplir la misión y mantener a salvo a la tripulación dentro de Orion. Los períodos de lanzamiento resultantes son los días o las semanas en los que la nave espacial y el cohete pueden cumplir los objetivos de la misión. Estos períodos de lanzamiento explican la compleja mecánica orbital relacionada con el lanzamiento en una trayectoria precisa hacia la Luna mientras la Tierra rota sobre su eje y la Luna orbita la Tierra cada mes en su ciclo lunar. Esto da como resultado un patrón de alrededor de una semana de oportunidades de lanzamiento, seguido de tres semanas sin oportunidades de lanzamiento.Existen varios parámetros principales que establecen la disponibilidad del lanzamiento dentro de estos períodos. Debido a su trayectoria única en relación con las misiones de alunizaje posteriores, estas limitaciones clave son exclusivas del vuelo de prueba de la misión Artemis II.El día y la hora de lanzamiento deben permitir que SLS pueda llevar a Orion a una órbita terrestre alta, donde la tripulación y los equipos técnicos en tierra evaluarán los sistemas de soporte vital de la nave espacial antes de que la tripulación emprenda su viaje con rumbo a la Luna.Orion también debe estar en la alineación adecuada con la Tierra y la Luna en el momento del encendido de motores con inyección translunar. El encendido de motores con inyección translunar de Artemis II pone a Orion en rumbo de sobrevolar la Luna, y también lo pone en una trayectoria de retorno libre, en la cual la nave espacial utiliza la gravedad de la Luna para enviar la nave espacial de regreso a la Tierra sin maniobras adicionales importantes de propulsión.La trayectoria para un día determinado debe garantizar que Orion no esté en la oscuridad durante más de 90 minutos a la vez para que las alas de los paneles solares puedan recibir y convertir la luz solar en electricidad, y la nave espacial pueda mantener un rango de temperatura óptimo. Los planificadores de la misión eliminan las posibles fechas de lanzamiento que llevarían a Orion a eclipses prolongados durante el vuelo.La fecha de lanzamiento debe sustentar una trayectoria que permita el perfil de entrada adecuado planificado durante el regreso de Orion a la Tierra.Los períodos a continuación muestran la disponibilidad de llevar a cabo el lanzamientohasta abril de 2026. Los planificadores de la misión mejoran estos períodos en función de un análisis actualizado más o menos dos meses antes de que estos comiencen, y ellos están sujetos a cambios.Período de lanzamientodel 31 de enero al 14 de febreroOportunidades de lanzamiento los días 6, 7, 8, 10 y 11 de febreroPeríodo de lanzamientodel 28 de febrero al 13 de marzoOportunidades de lanzamiento los días 6, 7, 8, 9, 11 de marzoPeríodo de lanzamientodel 27 de marzo al 10 de abrilOportunidades de lanzamiento los días 1, 3, 4, 5, 6 de abrilAdemás de las oportunidades de lanzamiento basadas en la mecánica orbital y los requisitos de desempeño, también existen restricciones sobre qué días dentro de un período de lanzamiento pueden ser viables en función de la reposición de productos básicos, las condiciones meteorológicas y las operaciones de otros usuarios en el cronograma del Área Este. Como regla general, se pueden hacer hasta cuatro intentos de lanzamiento dentro de la semana aproximada de oportunidades que existen dentro de un período de lanzamiento.Mientras la agencia se prepara para su primera misión tripulada más allá de la órbita terrestre en más de 50 años, la NASA espera aprender durante los procesos, tanto en tierra como en vuelo, y dejará que la aptitud y el desempeño de sus sistemas indiquen el momento en que la agencia está lista para el lanzamiento.Como parte de una edad de oro de innovación y exploración, el vuelo de prueba de Artemis II, el cual tendrá una duración aproximada de 10 días, es el primer vuelo tripulado para la campaña Artemis de la NASA. Este es otro paso hacia nuevas misiones tripuladas de Estados Unidos en la superficie de la Luna, lo que llevará a una presencia sostenida en la Luna que ayudará a la agencia a prepararse para enviar a los primeros astronautas estadounidenses a Marte.Encuentra más información sobre la campaña Artemis de la NASA en el siguiente sitio web (en inglés):https://www.nasa.gov/artemis

El cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) de la NASA, visto en la nave High Bay 3 del Edificio de Ensamblaje de Vehículos mientras los equipos t esperan la llegada de los tripulantes de Artemis II para abordar su nave espacial Orion en la parte superior del cohete como parte de la prueba de demostración de la cuenta atrás de Artemis II, el sábado 20 de diciembre de 2025, en el Centro Espacial Kennedy de la NASA en Florida.NASA/Joel Kowsky

El cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) de la NASA, visto en la nave High Bay 3 del Edificio de Ensamblaje de Vehículos mientras los equipos t esperan la llegada de los tripulantes de Artemis II para abordar su nave espacial Orion en la parte superior del cohete como parte de la prueba de demostración de la cuenta atrás de Artemis II, el sábado 20 de diciembre de 2025, en el Centro Espacial Kennedy de la NASA en Florida.

NASA/Joel Kowsky

Read this article in Englishhere.

Conforme la NASA se acerca al lanzamiento del vuelo de prueba Artemis II, la agencia pronto llevará por primera vez su cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) y la nave espacial Orion a la plataforma de lanzamiento en el Centro Espacial Kennedy de la agencia en Florida para comenzar la integración final, las pruebas y los ensayos para el lanzamiento.

La NASA tiene como objetivo comenzar desde el sábado 17 de enero su traslado desde el Edificio de Ensamblaje de Vehículos hasta la Plataforma de Lanzamientos 39B, lo que tardará varias horas. El viaje de casi 6,5 kilómetros (cuatro millas) en el vehículo transportador oruga 2 podría durar hasta 12 horas. Los equipos técnicos están trabajando día y noche para dar por terminadas todas las tareas antes del transporte del cohete. Sin embargo, esta fecha objetivo está sujeta a cambios si fuera necesario tiempo adicional para los preparativos técnicos o debido a las condiciones meteorológicas.

"Nos estamos acercando a la misión Artemis II, y tenemos su lanzamiento a la vuelta de la esquina", dijo Lori Glaze, administradora asociada interina para la Dirección de Misiones de Desarrollo de Sistemas de Exploración de la NASA. "Nos quedan pasos importantes en nuestro camino hacia el lanzamiento, y la seguridad de la tripulación seguirá siendo nuestra principal prioridad en todo momento, a medida que nos acercamos al regreso de la humanidad a la Luna".

Al igual que con todos los nuevos desarrollos de sistemas complejos, los ingenieros han estado solucionando varios problemas en los últimos días y semanas. Durante las comprobaciones finales antes del traslado, los técnicos detectaron que un cable relacionado con el sistema de terminación de vuelo estaba doblado en contra de las especificaciones. El personal técnico lo está reemplazando y hará pruebas con el nuevo cable durante el fin de semana. Además, una válvula relacionada con la presurización de la escotilla de Orion presentó problemas que hicieron necesario llevar a cabo pruebas dedemostración de la cuenta regresivael 20 de diciembre pasado. El 5 de enero, el equipo reemplazó la válvula e hizo pruebas de su funcionamiento que resultaron exitosas. Los ingenieros también trabajaron para resolver fugas en el hardware de soporte en tierra que es necesario para cargar oxígeno gaseoso en Orion a fin de proporcionar aire respirable.

Traslado

Una vez que el cohete y la nave espacial integrados lleguen a la plataforma de lanzamiento, la NASA comenzará inmediatamente una larga lista de verificación para los preparativos en la plataforma de lanzamiento, incluyendo la conexión de equipos mecánicos de apoyo en tierra, como líneas eléctricas, conductos del sistema de control ambiental de combustible y tomas de surtido de combustible criogénico. Los equipos de personal técnico encenderán todos los sistemas integrados en la plataforma por primera vez para garantizar que los componentes del hardware de vuelo funcionen correctamente entre sí, con el lanzador móvil y con los sistemas de infraestructura terrestre.

Una vez que esté todo completado, los astronautas de Artemis II, Reid Wiseman, Victor Glover y Christina Koch de la NASA, y el astronauta de la CSA (Agencia Espacial Canadiense) Jeremy Hansen, llevarán a cabo una caminata final en la plataforma.

Ensayo general con circulación de combustible y llenado de tanques

A finales de enero, la NASA llevará a cabo un ensayo general con circulación de combustible, el cual es una prueba previa al lanzamiento para llenar los tanques de combustible en el cohete. Durante este ensayo general, el personal técnico hace una demostración de la capacidad de cargar más de 700.000 galones de combustible criogénico en el cohete, lleva a cabo una cuenta regresiva para el lanzamiento y practica la extracción segura del combustible del cohete sin tripulación presente en el sitio.

Durante el lanzamiento,el equipo de cierre, o de tareas finales, será responsable de asegurar a los astronautas en Orion y cerrar sus escotillas. El personal de cierre también utilizará este ensayo para practicar sus procedimientos de forma segura sin tener tripulación a bordo de la nave espacial.

El ensayo general con circulación de combustible incluirá varios “encendidos”, o pruebas de funcionamiento, para demostrar la capacidad del equipo de lanzamiento para detener, reanudar y reiniciar operaciones en varios momentos diferentes de los últimos 10 minutos de la cuenta regresiva, conocida como conteo terminal.

La ejecución del primer encendido comenzará aproximadamente en las 49 horas antes del lanzamiento, cuando los equipos encargados de lanzamiento son llamados a sus estaciones, hasta 1 minuto y 30 segundos antes del lanzamiento, seguido de una pausa planificada de tres minutos y luego la reanudación de la cuenta regresiva hasta 33 segundos antes del lanzamiento, el punto en el que el secuenciador de lanzamiento automático del cohete controlará los últimos segundos de la cuenta regresiva. Luego, los equipos técnicos volverán a reiniciar a T-10 minutos y detendrán el conteo, y luego reanudarán los procedimientos hasta 30 segundos antes del lanzamiento como parte de una segunda ejecución.

Si bien la NASA ha integrado las lecciones aprendidas con Artemis I en los procedimientos de la cuenta regresiva para el lanzamiento, la agencia hará una pausa para abordar cualquier problema durante la prueba o en cualquier otro momento si surgen retos técnicos. Los ingenieros vigilarán de cerca la carga de combustible de hidrógeno líquido y oxígeno líquido en el cohete, después de los desafíos que se encontraron con la carga de hidrógeno líquido durante losensayos generales con circulación de combustible de Artemis I. Los equipos técnicos también prestarán mucha atención a la efectividad de los procedimientos recientemente actualizados para limitar la cantidad de nitrógeno gaseoso que se acumula en el espacio que está entre el módulo de tripulación de Orion y las escotillas del sistema de cancelación de lanzamiento, lo que podría representar un problema para el personal de cierre.

Es posible que se requieran ensayos generales con circulación de combustible adicionales para garantizar que el vehículo esté completamente revisado y apto para el vuelo.

De ser necesario, la NASA podría trasladar al cohete SLS y la nave Orion de vuelta al Edificio de Ensamblaje de Vehículos para realizar trabajos adicionales antes del lanzamiento después del ensayo general con circulación de combustible.

Próximos pasos para el lanzamiento

Después de un exitoso ensayo general con circulación de combustible, la NASA convocará una revisión de aptitud para el vuelo en la cual el equipo de gestión de la misión evaluará la aptitud de todos los sistemas, incluyendo el hardware de vuelo, la infraestructura y el personal de lanzamiento, vuelo y recuperación antes de comprometerse con una fecha de lanzamiento.

Aunque la ventana para el lanzamiento de Artemis II se podría iniciar tan pronto como el viernes 6 de febrero, el equipo de gestión de la misión evaluará la aptitud para el vuelo después del ensayo general con toda la nave espacial, la infraestructura de lanzamiento, y la tripulación y el personal de operaciones antes de seleccionar una fecha para el lanzamiento.

A fin de determinar las posibles fechas de lanzamiento, los ingenieros identificaron las restricciones clave necesarias para cumplir la misión y mantener a salvo a la tripulación dentro de Orion. Los períodos de lanzamiento resultantes son los días o las semanas en los que la nave espacial y el cohete pueden cumplir los objetivos de la misión. Estos períodos de lanzamiento explican la compleja mecánica orbital relacionada con el lanzamiento en una trayectoria precisa hacia la Luna mientras la Tierra rota sobre su eje y la Luna orbita la Tierra cada mes en su ciclo lunar. Esto da como resultado un patrón de alrededor de una semana de oportunidades de lanzamiento, seguido de tres semanas sin oportunidades de lanzamiento.

Existen varios parámetros principales que establecen la disponibilidad del lanzamiento dentro de estos períodos. Debido a su trayectoria única en relación con las misiones de alunizaje posteriores, estas limitaciones clave son exclusivas del vuelo de prueba de la misión Artemis II.

El día y la hora de lanzamiento deben permitir que SLS pueda llevar a Orion a una órbita terrestre alta, donde la tripulación y los equipos técnicos en tierra evaluarán los sistemas de soporte vital de la nave espacial antes de que la tripulación emprenda su viaje con rumbo a la Luna.

Orion también debe estar en la alineación adecuada con la Tierra y la Luna en el momento del encendido de motores con inyección translunar. El encendido de motores con inyección translunar de Artemis II pone a Orion en rumbo de sobrevolar la Luna, y también lo pone en una trayectoria de retorno libre, en la cual la nave espacial utiliza la gravedad de la Luna para enviar la nave espacial de regreso a la Tierra sin maniobras adicionales importantes de propulsión.

La trayectoria para un día determinado debe garantizar que Orion no esté en la oscuridad durante más de 90 minutos a la vez para que las alas de los paneles solares puedan recibir y convertir la luz solar en electricidad, y la nave espacial pueda mantener un rango de temperatura óptimo. Los planificadores de la misión eliminan las posibles fechas de lanzamiento que llevarían a Orion a eclipses prolongados durante el vuelo.

La fecha de lanzamiento debe sustentar una trayectoria que permita el perfil de entrada adecuado planificado durante el regreso de Orion a la Tierra.

Los períodos a continuación muestran la disponibilidad de llevar a cabo el lanzamientohasta abril de 2026. Los planificadores de la misión mejoran estos períodos en función de un análisis actualizado más o menos dos meses antes de que estos comiencen, y ellos están sujetos a cambios.

Período de lanzamientodel 31 de enero al 14 de febrero

Oportunidades de lanzamiento los días 6, 7, 8, 10 y 11 de febrero

Período de lanzamientodel 28 de febrero al 13 de marzo

Oportunidades de lanzamiento los días 6, 7, 8, 9, 11 de marzo

Período de lanzamientodel 27 de marzo al 10 de abril

Oportunidades de lanzamiento los días 1, 3, 4, 5, 6 de abril

Además de las oportunidades de lanzamiento basadas en la mecánica orbital y los requisitos de desempeño, también existen restricciones sobre qué días dentro de un período de lanzamiento pueden ser viables en función de la reposición de productos básicos, las condiciones meteorológicas y las operaciones de otros usuarios en el cronograma del Área Este. Como regla general, se pueden hacer hasta cuatro intentos de lanzamiento dentro de la semana aproximada de oportunidades que existen dentro de un período de lanzamiento.

Mientras la agencia se prepara para su primera misión tripulada más allá de la órbita terrestre en más de 50 años, la NASA espera aprender durante los procesos, tanto en tierra como en vuelo, y dejará que la aptitud y el desempeño de sus sistemas indiquen el momento en que la agencia está lista para el lanzamiento.

Como parte de una edad de oro de innovación y exploración, el vuelo de prueba de Artemis II, el cual tendrá una duración aproximada de 10 días, es el primer vuelo tripulado para la campaña Artemis de la NASA. Este es otro paso hacia nuevas misiones tripuladas de Estados Unidos en la superficie de la Luna, lo que llevará a una presencia sostenida en la Luna que ayudará a la agencia a prepararse para enviar a los primeros astronautas estadounidenses a Marte.

Encuentra más información sobre la campaña Artemis de la NASA en el siguiente sitio web (en inglés):

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First All-Sky Panorama from SPHEREx

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NASA’s Fermi Spots Young Star Cluster Blowing Gamma-Ray BubblesFor the first time, astronomers using NASA’s Fermi Gamma-ray Space Telescope have traced a budding outflow of gas from a cluster of young stars in our galaxy — insights that help us understand how the universe has evolved.Learn Moreabout NASA’s Fermi Spots Young Star Cluster Blowing Gamma-Ray Bubbles

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LOCATION:Bldg. 220 – HERA Habitat

SUBJECT: Photographic Support for Flight Analogs HERA Campaign 5 Mission 1 Crew exiting the HERA Habitat – with Sara Edwards, Roderick Borgie, Ian Porter and Dustin Wallace

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This project offers multiple challenges reaching a broad audience of middle and high schools, colleges, and universities across the nation.Learn MoreNASA Human Exploration Rover ChallengeHERC is a rigorous and continuously evolving activity which engages middle school, high school, and college/university level students in hands-on engineering design correlated to NASA's Artemis missions.Learn MoreAct NowLaunch Your Name Around Moon in 2026 on NASA's Artemis II MissionThe collected names will be put on an SD card loaded aboard Orion before launch.NASA is inviting the public to join the agency'sArtemis IItest flight as four astronauts venture around the Moon and back to test systems and hardware needed for deep space exploration. As part of the agency's "Send Your Name with Artemis II" effort, anyone can claim their spot by signing up before Jan. 21. Participants can download a boarding pass with their name on it as a collectable.Send Your Name Around the Moonabout Launch Your Name Around Moon in 2026 on NASA's Artemis II MissionBoarding passes will carry participants' names on NASA's Artemis II mission in 2026.Credit: NASAAttend a Virtual EventLaunches & LandingsVirtual Guest ProgramHeliophysics WebinarsFlight Opportunities WebinarsSmall Spacecraft WebinarsLaunches & LandingsTune in to scheduled mission updates that you can watch live.Read MoreNASA's PACE (Plankton, Aerosol, Cloud, ocean Ecosystem) spacecraft, atop a SpaceX Falcon 9 rocket, successfully lifts off from Space Launch Complex 40 at Cape Canaveral Space Force Station in Florida at 1:33 a.m. EST Thursday, Feb. 8. 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Boarding passes will carry participants’ names on NASA’s Artemis II mission in 2026.Credit: NASA

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URL: https://www.nasa.gov/humans-in-space/commercial-space/

TITLE: Commercial Space

Humans in SpaceCommercial SpaceNASA is enabling commercial industry to build, own, and operate space systems with the agency purchasing services for its science and research needs. Through public-private partnerships, NASA is opening space to more science, people, and opportunities.Low Earth Orbit EconomyAs the International Space Station nears the end of operations, NASA plans to transition to a new low Earth orbit model to continue leveraging microgravity benefits. Through commercial partnerships, NASA aims to maintain its leadership in microgravity research and ensure continued benefits for humanity.Learn Moreabout Low Earth Orbit EconomyPrivate Astronaut MissionsPrivate astronaut missions to the International Space Station help pave the way toward commercial space stations as part of NASA's efforts to develop a thriving low Earth orbit ecosystem and marketplace.Learn Moreabout Private Astronaut MissionsThe Axiom Mission 4, or Ax-4, crew will launch aboard a SpaceX Dragon spacecraft to the International Space Station from NASA's Kennedy Space Center in Florida. From left to right: ISRO (Indian Space Research Organization) astronaut Shubhanshu Shukla, former NASA astronaut Peggy Whitson, ESA (European Space Agency) astronaut Sławosz Uznański-Wiśniewski of Poland, and Tibor Kapu of Hungary.Credit: Axiom SpaceCommercial Space StationsNASA is committed to maintaining a continuous human presence in low Earth orbit as the agency transitions from the International Space Station to commercial space stations.Learn Moreabout Commercial Space StationsThis long-duration photograph highlights the city lights of Mexico's Yucatan peninsula, Earth's atmospheric glow, and star trails above taken from the International Space Station as it orbited 258 miles above.Credit: NASAGrowing the Lunar EconomyNASA is leading Artemis, humanity's return to the Moon. With international partners and U.S. industry providers, the future at the Moon holds promise for a robust lunar marketplace.Learn Moreabout Growing the Lunar EconomyHuman Landing SystemsNASA's commercial providers, SpaceX and Blue Origin, are building the human landing systems that will carry Artemis astronauts to the lunar surface and back to lunar orbit for their ride home to Earth aboard the Orion spacecraft.Learn Moreabout Human Landing SystemsSide-by-side illustrations of the SpaceX Starship lunar lander and the Blue Origin Blue Moon lunar lander.Credit: SpaceX/Blue OriginCommercial Lunar Payload ServicesNASA is working with several American companies to deliver science and technology to the lunar surface through the Commercial Lunar Payload Services initiative. These companies, ranging in size, bid on delivering payloads for NASA. Under Artemis, commercial deliveries are performing science experiments, testing technologies, and demonstrating capabilities to help NASA explore the Moon.Learn Moreabout Commercial Lunar Payload ServicesThe moon's surface taken by Lunar Orbiter 3.Credit: NASACommercial Crew ProgramNASA's Commercial Crew Program is delivering on its goal of safe, reliable, and cost-effective human transportation to and from the International Space Station through a partnership with American private industry.Learn Moreabout Commercial Crew ProgramLaunch Services ProgramNASA's Launch Service Program is responsible for launching commercial rockets delivering spacecraft that observe the Earth, visit other planets, and explore the universe, ranging from weather satellites and telescopes to Mars Rovers.Learn Moreabout Launch Services ProgramA SpaceX Falcon 9 rocket, carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) observatory and PUNCH (Polarimeter to Unify the Corona and Heliosphere) satellites, launches from Space Launch Complex 4 East at Vandenberg Space Force Base in California.Credit: SpaceXLatest Commercial Space NewsMore NASA NewsBlog3 Min ReadCrew Studies Cardiac Research, Artificial Intelligence as Dragon Boosts Station's OrbitArticle6 Min ReadNASA Kennedy Top 20 Stories of 2025Blog4 Min ReadSpace Station First: All Docking Ports Fully Occupied, 8 Spacecraft on OrbitBlog3 Min ReadCrew Studies Space Orientation, DNA Data Storage as Dragon Reboosts StationKeep ExploringDiscover More Topics From NASALow Earth Orbit EconomyCommercial Lunar Payload ServicesCommercial Space StationsGrowing the Lunar Economy

Commercial SpaceNASA is enabling commercial industry to build, own, and operate space systems with the agency purchasing services for its science and research needs. Through public-private partnerships, NASA is opening space to more science, people, and opportunities.

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Private Astronaut MissionsPrivate astronaut missions to the International Space Station help pave the way toward commercial space stations as part of NASA’s efforts to develop a thriving low Earth orbit ecosystem and marketplace.Learn Moreabout Private Astronaut MissionsThe Axiom Mission 4, or Ax-4, crew will launch aboard a SpaceX Dragon spacecraft to the International Space Station from NASA’s Kennedy Space Center in Florida. From left to right: ISRO (Indian Space Research Organization) astronaut Shubhanshu Shukla, former NASA astronaut Peggy Whitson, ESA (European Space Agency) astronaut Sławosz Uznański-Wiśniewski of Poland, and Tibor Kapu of Hungary.Credit: Axiom Space

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Credit: Axiom Space

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Human Landing SystemsNASA's commercial providers, SpaceX and Blue Origin, are building the human landing systems that will carry Artemis astronauts to the lunar surface and back to lunar orbit for their ride home to Earth aboard the Orion spacecraft.Learn Moreabout Human Landing SystemsSide-by-side illustrations of the SpaceX Starship lunar lander and the Blue Origin Blue Moon lunar lander.Credit: SpaceX/Blue Origin

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The moon’s surface taken by Lunar Orbiter 3.Credit: NASA

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URL: https://science.nasa.gov/biological-physical/programs/physical-sciences/

TITLE: Physical Sciences Program

Explore This SectionPhysical Sciences ProgramOverviewNASA’s Physical Science Research Program has made contributions in two distinct areas: first, fundamental research, which investigates physical phenomena in the absence of gravity and fundamental laws of the universe, and second, applied research, which contributes to the basic understanding underlying space exploration technologies. In completing these investigations, physical sciences provides basic scientific knowledge, results leading to societal benefit, and contributions to the basic understanding underlying space exploration technologies such as power generation and storage, space propulsion, life support systems, and environmental monitoring and control. All have led to improved space systems or new products on Earth.Our core objectives include:Investigate fundamental laws of physics, often using either microgravity or interplanetary distances as research toolsProvide mechanistic understanding of processes underlying space exploration technologies such as power generation and storage, space propulsion, life support systems, and environmental monitoring and controlSupport the transfer of knowledge and technology of space-based research to terrestrial systems to benefit life on EarthDeveloping cutting-edge technologies to facilitate spaceflight researchPromote open science through data sharingDisciplinesBiophysicsCombustion ScienceSoft MatterFluid PhysicsFundamental PhysicsMaterials ScienceThe International Space Station provides the highly desired condition of long-duration microgravity, allowing continuous and interactive research similar to Earth-based laboratories, even providing statistical validity when required. The program also has benefited from research collaborations with the International Space Station partners (Russia, Europe, Japan, Canada) and individual foreign governments with space programs, such as France, Germany and Italy.NASA's physical science research is organized into six disciplines – Biophysics, Combustion Science, Complex Fluids, Fluid Physics, Fundamental Physics and Materials Science. Conducted in a nearly weightless environment, experiments in these disciplines reveal how physical systems respond to the near absence of buoyancy-driven convection, sedimentation or sagging. They also reveal how other forces, such as capillary forces, which are small compared to gravity, can dominate the system behavior in space. The data acquired from these investigations is stored in NASA's Physical Sciences Informatics System (PSI) and is available to the public.If you are a researcher and interested in learning more about NASA's Physical Sciences Program, please contact Dan Walsh.The following information is a summary of the six disciplines and the PSI database.Physical Sciences Informatics System (database)In fulfillment of the new Open Science model, we are pleased to announce thePhysical Science Informatics (PSI)data repository for physical science experiments performed on the International Space Station (ISS). The PSI system is accessible and open to the public. This provides the opportunity for researchers to data mine results from prior flight investigations, expanding on the research performed. This approach will allow numerous ground-based investigations to be conducted from one flight experiment's data, exponentially increasing our body of knowledge. PSI also meets the President's Open Data policy.Fundamental PhysicsCommercially Enabled Rapid Space ScienceOpen Science

Physical Sciences Program

NASA's Physical Science Research Program has made contributions in two distinct areas: first, fundamental research, which investigates physical phenomena in the absence of gravity and fundamental laws of the universe, and second, applied research, which contributes to the basic understanding underlying space exploration technologies. In completing these investigations, physical sciences provides basic scientific knowledge, results leading to societal benefit, and contributions to the basic understanding underlying space exploration technologies such as power generation and storage, space propulsion, life support systems, and environmental monitoring and control. All have led to improved space systems or new products on Earth.

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Investigate fundamental laws of physics, often using either microgravity or interplanetary distances as research tools

Provide mechanistic understanding of processes underlying space exploration technologies such as power generation and storage, space propulsion, life support systems, and environmental monitoring and control

Support the transfer of knowledge and technology of space-based research to terrestrial systems to benefit life on Earth

Developing cutting-edge technologies to facilitate spaceflight research

Promote open science through data sharing

Disciplines

Biophysics

Combustion Science

Soft Matter

Fluid Physics

Fundamental Physics

Materials Science

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URL: https://science.nasa.gov/biological-physical/programs/space-biology/

TITLE: Space Biology Program

Explore This SectionSpace Biology ProgramContentsOverviewAnimal BiologyCell and Molecular BiologyMicrobiologyPlant BiologyDevelopmental, Reproductive and Evolutionary BiologyOverviewThe main objective of Space Biology research is to build a better understanding of how spaceflight affects living systems in spacecraft such as the International Space Station (ISS), or in ground-based experiments that mimic aspects of spaceflight, and to prepare for future human exploration missions far from Earth. The experiments we conduct on these platforms examine how plants, microbes, and animals adjust or adapt to living in space. We examine processes of metabolism, growth, stress response , physiology, and development. We study how organisms repair cellular damage and protect themselves from infection and disease in conditions of microgravity while being exposed to space radiation. And we do it across the spectrum of biological organization, from molecules to cells, from tissues and organs, and from systems to whole organisms to communities of microorganisms.Our program’s core objectives include:Discovering how biological systems respond, acclimate and adapt to the space environmentDeveloping integrated physiological models for biology in spaceIdentifying the underlying mechanisms and networks that govern biological processes in the space environmentPromoting open science through theGeneLab Data SystemandLife Sciences Data ArchiveDeveloping cutting-edge biological technologies to facilitate spaceflight researchDeveloping mechanistic understanding to support human health in spaceEnabling the transfer of knowledge and technology to the understanding of life on EarthSpace Biology Science Plan (PDF)Search the Space Station Research Explorer for Space Biology experimentsIn addition to providing useful information on how living organisms adapt to spaceflight, the discoveries NASA researchers make in space have enormous implications for life on Earth. Space Biology's research into the virulence of pathogens in space, loss of bone density, and the changes in the growth of plants can impact the development of drugs that promote wound healing or tissue regeneration, treatments designed to counter osteoporosis on Earth, and high-tech fertilizers that increase crop yield.Sign up to receive the Space Biology newsletterAccess back issues of the Space Biology newsletterAnimal BiologyLife in space produces profound changes in biology. All organisms on Earth have adapted to perform under conditions of gravity, atmosphere, and cycles of light and darkness that have not changed in millions of years, conditions which are altered aboard spacecraft like the ISS. For example, while circling Earth at speeds of 17,130 miles an hour, crewmembers of the ISS experience sunrise and sunset 16 times a day! Simply put, terrestrial organisms are not designed for life in space.The goal of the Space Biology Program in the animal biology area is to understand the basic mechanisms that animals use to adapt and/or acclimate to spaceflight and alterations in gravity in general. Animals are frequently used to model human disease as well as how humans respond to stressful stimuli. The most commonly used model organisms for which genomics are now well defined include vertebrate species, e.g., rodents, both rats and mice, and a variety of invertebrate species, e.g., nematodes and insects. NASA has used these model organisms extensively to evaluate biological spaceflight hazards, elucidate the fundamental mechanisms life uses to adapt to microgravity, and apply such knowledge to advance human exploration, and for societal benefits on Earth.Read about a Behavior Analysis Study that shows mice adapt to spaceflight.As we examine the impacts of spaceflight on animal biology and physiology, we ask the following questions:Do the effects of spaceflight level off over time, get worse, or get better?Are the effects of spaceflight exposure permanent or do they decrease and/or vanish with time upon return?Can we prevent adverse effects before they appear, or at least lessen their impact?What conditions are required for animals to live in space for years, come back to Earth, and remain healthy?Cell and Molecular BiologyCell and molecular biology research cuts across all science disciplines in Space Biology, from understanding how single-celled organisms, such as protozoa, bacteria, and fungi respond to the conditions of spaceflight, to how all of the various cells in a complex tissue or organ work together to help an organism as a whole acclimate to such a foreign environment. The overarching goal of Space Biology Cell and Molecular Biology research at NASA is to determine how the stresses of the spaceflight environment affect living systems at the basic cellular and molecular levels, using contemporary cell and molecular biology techniques and measures. This includes characterizing and identifying changes in gene and protein expression, DNA function and structure, cellular structure and morphology, and cell-to-cell communication.As we examine the impacts of spaceflight on the cell biology and physiology, we ask the following:What are the underlying genetic and molecular mechanisms of the cells that are influenced by gravitational changes and the space environment?Does the space flight environment affect or influence cell and molecular functions causing tissue/organ dysfunction or disease states?Does the space environment affect cellular and molecular functions in a manner that impact tissue morphogenesis or development?MicrobiologyNo Matter Where We Go, We Take Our MicrobesEverywhere we go, we take microorganisms with us, whether we want to or not. This is true even aboard the “clean” environment of the ISS, which is a closed environment. With a crew of astronauts on board, living, breathing, exercising, and sweating, the ISS is a breeding ground for microbes.Microbes Can Be Both Friends and FoesThe effects of spaceflight on the biology of microorganisms and on microbial populations are largely unknown. In addition to posing a risk to astronaut health, bio-corrosive microorganisms that grow on metallic surfaces in spacecraft can damage both equipment and hardware. Unlike cruise ships on Earth which can be evacuated, emptied, and cleaned, problems that arise due to microbial contamination on long-duration spaceflight missions can only be resolved using tools and resources already present within the vessel. Understanding how microbial species grow and interact with each other in this environment is the first step in preparing for such a scenario.While the presence of certain microbes can be problematic, it is also important to remember that many microbial species play important roles in Earth’s various ecosystems. For example, the symbiotic relationships that have evolved between plants and certain bacteria are vital in ensuring plants receive the nutrients they require to grow and flourish. Oxygen-producing cyanobacteria that live in terrestrial bodies of water help replenish Earth’s oxygen. Even the bacteria living within our digestive tracts play an important role in the digestion of our food and the proper absorption of nutrients. The presence of certain microbes may be important for the proper growth of some plants in space and may even be critical for the production of future bio-regenerative life support systems. 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URL: https://osdr.nasa.gov/bio/

TITLE: Biological and Physical Sciences Data

Explore This SectionBiological and Physical Sciences DataThe NASA Open Science Data Repository (OSDR) and Physical Sciences Informatics (PSI) provide open access to biological and physical science datasets from spaceflight and ground studies, enabling data reuse for discovery and innovation. Together, they accelerate research, broaden participation, and advance NASA's open science mission.BPS Data and RepositoriesOSDRBiological Data Project - Learn more about OSDRPSIPhysical Sciences Data Project - Learn more about PSINBISCBiorepository of non-human samples-Learn more about NBISCSMCCRepository for collections of space microbes-Learn more about SMCCExplore and ContributeBiological DataSearch biological space-relevant datasetsExploreBiological Submission PortalSubmit your biological space-relevant dataContributePhysical Sciences DataSearch physical sciences datasetsExplorePhysical Sciences Submission PortalSubmit your physical sciences dataContributeNew Biological DataOSD-655Pick-and-eat space crop production flight testing on the International Space Station (Elemental Analysis, ORAC, Phenolic Content)OSD-949Skeletal muscle-specific inducible AMPKα1/α2 knockout mice develop muscle weakness, glycogen depletion, and fibrosis that persists during disuse atrophyOSD-890Planetary Protection Low Biomass Nanopore Pilot StudyNew Physical Sciences DataPSI-190Particle Engulfment and Pushing By Solidifying Interfaces(PEP)PSI-177Coupled Growth in Hypermonotectics(CGH)PSI-186Investigating the Structure of Paramagnetic Aggregates from Colloidal Emulsions - 4 (InSPACE-4)Working Groups and Data ManagementOSDR AWGsJoin OSDR Analysis WorkingBPS Data Management PolicyPolicy and guidelines for managing scientific DataMore Data toolsVisualization PortalsExplore OSDR Visualization PortalsBiospecimen Search ToolSearch and request biospecimensMicrogravity ToolExplore Microgravity Analysis PlatformSPLASHDatabase of spaceflight experimental hardwareLatest Content2 min readOSDR and PSI Unveil New Consolidated WebsiteArticle4 months ago3 min readJoe A. Adam Presents Ring Sheared Drop (RSD) Research at 2025 ISSRDCArticle4 months ago2 min readNew File Download Process for PSI Large RequestsArticle4 months agoResourcesTask BookThe NASA Task Book is an online database of research projects supported by NASA'sBiological & Physical Sciences (BPS) DivisionandHuman Research Program(HRP).NSLSLThe NSLSL seeks to consolidate global space life sciences literature into a single database.PSI Database TutorialA guide to search and navigate investigations in PSI.PSI Workspace TutorialA guide designed to help you navigate the PSI workspace.OSDR TutorialsHow To Guides for OSDR applications.

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URL: https://science.nasa.gov/citizen-science/osdr-awg/

TITLE: Open Science Data Repository Analysis Working Groups (OSDR AWG)

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Dozens of project groups are active at any time.The OSDR provides free and open data analysis/visualization tools for anyone to use, and free, open, on-demand trainings for youth, students, and the public to learn bioinformatics, artificial intelligence, and machine learning for space biology. Your ideas and collaborative research may enable the next breakthrough to help life thrive in deep space!Go to Project Websiteabout Open Science Data Repository Analysis Working Groups (OSDR AWG)project taskIndependent work with datadivisionBiological and Physical SciencewhereOnlinelaunched2018To view this video please enable JavaScript, and consider upgrading to a web browser thatsupports HTML5 videoWhat you'll doThe Analysis Working Groups (AWG) have two main activities:provide feedback on scientific standards that support reuse of data related to life in space;mine and reuse OSDR data to conduct scientific analysis to learn more from the valuable datasets in it.Become part of a global network of passionate scientists and volunteers working together to unlock the secrets to help terrestrial life thrive in space.RequirementsTime:Time: Varies. All AWGs have monthly calls. There are sub-group project meetings as well. Depending on interest and expertise, members likely do ~5 hours of activity (writing, coding, analyzing data) between meetingsEquipment:Internet-connected computerKnowledge:None required. Familiarity with data engineering, statistics, coding, space biology literature is very helpful. There are tutorials here:https://osdr-tutorials.readthedocs.io/en/latest/, and various open trainings available to AWG members (see below).Get started!Visit the project website.Read the AWG CharterSubmit your request to join the AWG!Not sure you're ready? Check out the training resources below.Learn MoreThere is a 'Forum-Space' for OSDR AWG members to collaborate, find projects, chat with one another, ask questions, and post/find opportunities. Here is theForum-Space Homepageand pinned top post is themaintained list of all current projects. TheAbout pageon this site has lots of information and links to resources.In addition to theOSDR tutorialsthat cover the basics of accessing and navigating the repository, there are some special resources for learners.GeneLab for High Schoolsis a summer intensive internship for high school students that introduces space biology and bioinformatics analysis methods. Visit the link for information on this year's offerings.Past teachers who have participated as interns in this course alongside students have created curricular materials — units, worksheets, and supporting activities — to support classroom participation in space biology using OSDR.Click here for the teacher-created lessons.GeneLab for Colleges and UniversitiesandArtificial Intelligence/Machine Learning (AI/ML) in Space Biologytrainings are also available as free, open-access, on-demand courses designed to support skill development.Sign up for the OSDR Newsletter!The current Analysis Working Groups are listed below. 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What you'll doThe Analysis Working Groups (AWG) have two main activities:provide feedback on scientific standards that support reuse of data related to life in space;mine and reuse OSDR data to conduct scientific analysis to learn more from the valuable datasets in it.Become part of a global network of passionate scientists and volunteers working together to unlock the secrets to help terrestrial life thrive in space.RequirementsTime:Time: Varies. All AWGs have monthly calls. There are sub-group project meetings as well. Depending on interest and expertise, members likely do ~5 hours of activity (writing, coding, analyzing data) between meetingsEquipment:Internet-connected computerKnowledge:None required. Familiarity with data engineering, statistics, coding, space biology literature is very helpful. There are tutorials here:https://osdr-tutorials.readthedocs.io/en/latest/, and various open trainings available to AWG members (see below).Get started!Visit the project website.Read the AWG CharterSubmit your request to join the AWG!Not sure you're ready? Check out the training resources below.Learn MoreThere is a 'Forum-Space' for OSDR AWG members to collaborate, find projects, chat with one another, ask questions, and post/find opportunities. Here is theForum-Space Homepageand pinned top post is themaintained list of all current projects. TheAbout pageon this site has lots of information and links to resources.In addition to theOSDR tutorialsthat cover the basics of accessing and navigating the repository, there are some special resources for learners.GeneLab for High Schoolsis a summer intensive internship for high school students that introduces space biology and bioinformatics analysis methods. Visit the link for information on this year's offerings.Past teachers who have participated as interns in this course alongside students have created curricular materials — units, worksheets, and supporting activities — to support classroom participation in space biology using OSDR.Click here for the teacher-created lessons.GeneLab for Colleges and UniversitiesandArtificial Intelligence/Machine Learning (AI/ML) in Space Biologytrainings are also available as free, open-access, on-demand courses designed to support skill development.Sign up for the OSDR Newsletter!

What you'll do

The Analysis Working Groups (AWG) have two main activities:

provide feedback on scientific standards that support reuse of data related to life in space;

mine and reuse OSDR data to conduct scientific analysis to learn more from the valuable datasets in it.

Become part of a global network of passionate scientists and volunteers working together to unlock the secrets to help terrestrial life thrive in space.

Requirements

Time:Time: Varies. All AWGs have monthly calls. There are sub-group project meetings as well. Depending on interest and expertise, members likely do ~5 hours of activity (writing, coding, analyzing data) between meetings

Equipment:Internet-connected computer

Knowledge:None required. Familiarity with data engineering, statistics, coding, space biology literature is very helpful. There are tutorials here:https://osdr-tutorials.readthedocs.io/en/latest/, and various open trainings available to AWG members (see below).

Get started!

Visit the project website.

Read the AWG Charter

Submit your request to join the AWG!

Not sure you’re ready? Check out the training resources below.

There is a ‘Forum-Space’ for OSDR AWG members to collaborate, find projects, chat with one another, ask questions, and post/find opportunities. Here is theForum-Space Homepageand pinned top post is themaintained list of all current projects. TheAbout pageon this site has lots of information and links to resources.

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High above Earth, an astronaut looks out into space and to all the places humans might travel in the future.Image credit: NASA

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Image credit: NASA

Get to know the people of OSDR Team!Ryan ScottScience Lead, NASA Open Science Data RepositoryMeet the TeamLauren M SandersChair of AI/ML AWGMeet the TeamAmber M. PaulChair of the Animal AWGMeet the TeamBegum MathykChair of the Female Reproduction and Health AWGMeet the TeamEliah OverbeyChair of the Human AWGMeet the TeamDaniela BezdanChair of the Microbes AWGMeet the TeamAfshin BeheshtiChair of the Multi-Omics AWGMeet the TeamKirill GrigorevChair of the RadLab AWGMeet the TeamRichard BarkerChair of the Plant AWGRead More

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URL: https://science.nasa.gov/science-research/biological-physical-sciences/osdr-and-psi-unveil-new-consolidated-website/

TITLE: OSDR and PSI Unveil New Consolidated Website

2 min readOSDR and PSI Unveil New Consolidated Website

Danielle LopezSep 30, 2025ArticleTheOpen Science Data Repository (OSDR)andPhysical Sciences Informatics (PSI)has a new home. As part of NASA’s website consolidation initiative, the OSDR and PSI site have officially transitioned to theBiological and Physical

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TheOpen Science Data Repository (OSDR)andPhysical Sciences Informatics (PSI)has a new home. As part of NASA's website consolidation initiative, the OSDR and PSI site have officially transitioned to theBiological and Physical Sciences (BPS) Datapage, accessible through the "Data" menu on the Science Mission Directorate's (SMD) website at science.nasa.gov. This strategic move reflects NASA's broader effort to streamline user access to resources, unify digital platforms, and provide a more consistent experience across the SMD divisions.

The OSDR and PSI consolidation brings together two powerful resources, giving researchers a single point of access to search both biological and physical sciences datasets. By integrating these repositories, NASA is expanding opportunities for cross-disciplinary research, enabling scientists to draw connections across fields and gain deeper insights into how biology and physical systems respond to spaceflight environments.

The redesigned OSDR website continues to serve as a hub for open access to space science data, offering a modernized layout, improved navigation, and direct pathways to explore datasets and analysis tools, and submit data through the submission portals enabled by OSDR and PSI. Whether you are a scientist seeking resources for new investigations, a student learning about space research, or a collaborator from another discipline, the updated platform makes accessing NASA's open science data easier than ever. Check out the newBPS DataandOSDR, andPSIwebsites now!

The launch of the new consolidated OSDR and PSI websites underscores NASA's commitment to open science and to advancing knowledge through transparent, accessible, and reusable data. By situating OSDR under the BPS data ecosystem and combining it with PSI, NASA is strengthening visibility, fostering collaboration, and ensuring that both biological and physical sciences research in space continues to thrive.

URL: https://science.nasa.gov/biological-physical/precision-health/

TITLE: Precision Health

Explore This SectionPrecision HealthLeveraging space to unlock the secretsof aging and diseaseBIOLOGICAL & PHYSICAL SCIENCESLeveraging space to unlock the secretsof aging and diseaseStressors encountered during space travel can affect human health, including bone and muscle loss, immune system function, microbes, and other biological responses. NASA research could provide vital information needed to help protect astronauts during future deep-space missions and advance the prevention and treatment of disease for people on Earth.Information SheetBPS Road MapImpact & BenefitsKey Investigation: AVATAROrgan ChipsAging & DiseaseSynthetic BiologyAcclimation and AdaptationInvestigationsBPS Scientific GoalsRelated ResourcesAbout BPSPrecision Health Information SheetSep 11, 2025PDF (421.77 KB)Biological & Physical Sciences Road MapSep 12, 2025PDF (1.60 MB)Impacts & BenefitsPrecision HealthNASA studies the physiological, cellular, and genetic alterations that can occur during space travel. 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Studying biological and physical phenomena under extreme conditions allows researchers to advance the fundamental scientific knowledge required to go farther and stay longer in space, while also benefitting life on Earth.

Precision HealthLeveraging space to unlock the secretsof aging and disease

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Organ-On-a-Chip(Organ Chips)

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3 Min ReadI Am Artemis: Dustin GohmertDustin Gohmert, Orion Crew Survival System (OCSS) manager, sits in the OCSS Lab at NASA's Johnson Space Center in Houston.Credits:NASA/Rad Sinyak

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Erika PetersJan 23, 2026ArticleListen to this audio excerpt from Dustin Gohmert, Orion Crew Survival System(OCSS) manager:0:00 / 0:00Your browser does not support the audio element.During NASA's Artemis II mission around the Moon, the astronauts inside the Orion spacecraft will be wearing specialized pressure suits designed to protect them throughout their journey. At NASA's Johnson Space Center in Houston, Dustin Gohmert leads the team responsible for these suits, known as the Orion Crew Survival System (OCSS)."We work with the crew to say, 'Here's this design concept we have. How does this really work in the spaceflight environment?'" Gohmert said. "As we evolve the design, we take the crew's input and we adapt the suit over time to take into account not only the desire we have for safety, but the real-world impacts that it has."The suits will protect astronauts on launch day, throughout high-risk parts of missions near the Moon, during the high-speed return to Earth, and in emergency situations if such events arise. The OCSS suits are engineered to sustain life for up to six days in the event of an emergency, and can provide the astronauts oxygen, hydration, food, and waste management needed on their way back to Earth.Dustin Gohmert, Orion Crew Survival System (OCSS) manager, sits in the OCSS Lab at NASA's Johnson Space Center in Houston.Credits: NASA/Rad Sinyak"In an emergency, you're essentially living in a personal spacecraft that's only an inch bigger than your body," Gohmert said. "That's the reality of survival in space."Gohmert's team in the Orion Crew Survival Systems Lab manages every phase of the suits, including processing, designing, qualifying, and testing them for the mission, as well as integrating them with the Orion spacecraft. Their work addresses engineering challenges, such as how much internal pressure the suit can safely maintain and for how long.The team custom-builds each suit to fit the anatomy of the astronauts. Crew members undergo detailed sizing and multiple fit checks to ensure precision, and their feedback is a key part of the design evolution and refinement of the suit.Orion Crew Survival System (OCSS) Manager Dustin Gohmert and his team perform a flight suit long duration fit check with Artemis II crew member Christina Koch in the OCSS Lab at NASA's Johnson Space Center in Houston.Credit: NASA/Josh ValcarcelAfter earning his bachelor's in mechanical engineering from the University of Texas at San Antonio and his master's in engineering from the University of Texas at Austin, Gohmert joined United Space Alliance before becoming a NASA civil servant. He worked through the end of the Space Shuttle Program and later transitioned to Orion. Working on the suit throughout his career has been both technically challenging and a deeply personal responsibility.The weight of it is incredible; knowing the ultimate responsibility you and the team share in the safety of the crew and the mission. Every thought we have, every piece of paper we write — crew is the number one priority.dustin GohmertOrion Crew Survival Systems (OCSS) ManagerAs NASA prepares to explore deep space with Artemis II, Gohmert's role will play a part in safely sending crew members around the Moon and returning them home."I was born after the last Moon landing," he said. "To actually be a part of the next round is kind of overwhelming. It's awe-inspiring in every possible way."

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Jessica TaveauJan 23, 2026MEDIA ADVISORYM26-008NASA HeadquartersJohnson Space CenterNASA's SpaceX Crew-12 crew, from left to right, is NASA astronauts Jessica Meir and Jack Hathaway, ESA (European Space Agency) astronaut Sophie Adenot, and Roscosmos cosmonaut Andrey Fedyaev.Credit: SpaceXNASA and its partners will discuss the upcoming crew rotation to the International Space Station during a pair of news conferences on Friday, Jan. 30, from the agency's Johnson Space Center in Houston.At 11 a.m. EST, mission leadership will discuss final launch and mission preparations in a news conference that will stream on the agency'sYouTubechannel.Next, the crew of NASA's SpaceX Crew-12 mission will participate in a virtual news conference from NASA Johnson crew quarters at 1 p.m., also on the agency'sYouTubechannel. Individual streams for each of the events will be available on that page. 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For more information about the mission, visit:

https://www.nasa.gov/commercialcrew

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TITLE: Quesst Mission Image Gallery

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URL: https://www.nasa.gov/aeronautics/nasas-sonic-boom-research-takes-shape/

TITLE: NASA’s Sonic Boom Research Takes “Shape”

4 min readNASA’s Sonic Boom Research Takes “Shape”

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Jim BankeManaging Editor/Senior WriterNov 06, 2013ArticleThis book tells the story of a strangely shaped aircraft that proved to researchers it was possible to lower the volume of sonic booms.NASA / Mike RyanIf an airplane flies overhead at supersonic speed and no one below can hear it, did it make a sonic boom?Answering that twist on the classic falling tree in the forest question gets to the heart of a new book published by NASA that explores the agency’s research into understanding how sonic booms work and how best to make them less of a public nuisance.“Quieting the Boom: The Shaped Sonic Boom Demonstrator and the Quest for Quiet Supersonic Flight” is available online at no cost as an e-book, while printed versions of the book may be purchased from NASA’s Information Center.The 388-page book written by Lawrence R. Benson begins with an overview of early supersonic flight that triggered the first sonic booms and sets the stage for telling the story of the Shaped Sonic Boom Demonstrator research aircraft, which is the primary focus of the book.“I think ‘Quieting the Boom’ shows the value of sustaining basic and applied research to solve difficult scientific and engineering challenges, even if success takes decades to achieve,” Benson said.This modified Northrop F-5E jet was used during 2003 for NASA’s Shaped Sonic Boom Demonstration program, a successful effort to show that an aircraft’s shape can be used to reduce the intensity of the sonic booms it creates while flying supersonic.NASADuring the 1950s and into the 1960s, sonic booms were echoing across the nation as the Air Force deployed more and more of the Mach-1-busting jets that were designed to keep the peace during the Cold War.But the public voice protesting the noise was just as forceful as the sonic booms themselves.In fact, as Benson points out in the book, between 1956 and 1968 there were 38,831 claims against the Air Force (14,006 were approved) to cover losses from sonic booms that ranged from broken glass and cracked plaster to assertions of pets dying and livestock going insane.It was within this difficult environment that NASA and the Federal Aviation Administration led a research effort in support of industry designing and building a commercial supersonic transport, which became known as the SST.One of many microphones arrayed under the path of the F-5E SSBE (Shaped Sonic Boom Experiment) aircraft to record sonic booms.NASA / Tony LandisHowever, even as NASA continued to research sonic booms in support of the SST the public and political pressure became too much. The SST program was cancelled in 1971 and the future for supersonic airliners, business jets and other aircraft became gloomy at best.Fortunately, the nation’s aeronautical innovators were not deterred. They believed that with additional research a way could be found to quiet the booms enough to satisfy all critics and make commercial supersonic travel possible across the country.To help them find that solution, NASA partnered with others to conduct an innovative flight research program using what was named the Shaped Sonic Boom Demonstrator (SSBD), a Northrop Grumman F-5E Tiger fighter jet modified with a larger nose."This modification, which made the front of the F-5E somewhat resemble a pelican's beak, was carefully shaped to change the pattern of shock waves it would generate while flying faster than the speed of sound," Benson said.The theory behind the nose job was that the increased airplane volume at the front of the SSBD would result in less intense sonic booms heard or felt on the ground below.The modified F-5E Shaped Sonic Boom Demonstration aircraft (center) flies off the wing of NASA's F-15B Research testbed aircraft, which flew in the supersonic shockwave of the F-5E. Following the two aircraft is an unmodified U.S. Navy F-5E used for baseline sonic boom measurements.NASA / Carla ThomasDuring a series of some 30 tests flown in August 2003 and January 2004, data gathered by airborne and ground sensors of the SSBD in flight proved the theory was – well – sound, and that additional research in the future could enable the long-sought realization of commercial supersonic flight across the United States.Indeed, that research continues as NASA and its research partners explore fresh ideas and design new tools for making supersonic flight quieter.Most importantly, in telling what could easily be a complex story involving high speed flight, atmospheric sciences, acoustics and bizarre math, Benson said he attempted to keep the technical jargon to a minimum."This book is intended to be a general history of sonic boom research, emphasizing the people and organizations that have contributed, and not a technical study of the science and engineering involved," Benson said.The American Institute of Aeronautics and Astronautics has recognized the book with its prestigious History Manuscript Award for 2014.Nearly ten years after SSBD, aircraft shapes that can reduce the level of sonic booms tend to look more like this. This overland supersonic aircraft concept is being used by NASA researchers in computer simulations and to create models for wind tunnel tests.NASAPublication of "Quieting the Boom" was sponsored and funded by the communications and education department of NASA's Aeronautics Research Mission Directorate."Quieting the Boom" E-Book

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Answering that twist on the classic falling tree in the forest question gets to the heart of a new book published by NASA that explores the agency's research into understanding how sonic booms work and how best to make them less of a public nuisance.

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“Quieting the Boom” E-Book

About the AuthorJim BankeManaging Editor/Senior WriterJim Banke is a veteran aviation and aerospace communicator with more than 40 years of experience as a writer, producer, consultant, and project manager based at Cape Canaveral, Florida. He is part of NASA Aeronautics' Strategic Communications Team and is Managing Editor for the Aeronautics topic on the NASA website.

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TITLE: Telescopio espacial Roman

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La demostración de la tecnología del instrumento coronógrafo generará imágenes de alto contraste y efectuará espectroscopia de los exoplanetas cercanos individualmente.La vida útil de la misión principal del telescopio espacial Roman, cuyo lanzamiento está programado a más tardar para mayo de 2027, será de cinco años, con una posible extensión de la misión de cinco años más.MISIÓN¿Por qué Roman?Aunque es tan sensible como las cámaras de Hubble, el instrumento de campo amplio de 300 megapíxeles del telescopio espacial Roman obtendrá imágenes de un área del cielo cien veces más grande. Esto significa que una sola imagen de Roman tendrá el detalle equivalente a cien imágenes de Hubble.Aprende másL. 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Descubre más en este episodio de Universo curioso de la NASA.Escucha el episodioEsta representación artística muestra cómo podría verse el sistema planetario TRAPPIST 1 desde un punto de observación cercano al planeta TRAPPIST 1f (a la derecha).NASA/JPL-Caltech¿Quién es Nancy Grace Roman?Tenemos el placer de dedicar este observatorio espacial, encargado de aumentar nuestra comprensión de la energía oscura, el universo y nuestro lugar en él, a la doctora Nancy

Grace Roman, pionera de la astronomía espacial moderna.Lee másabout ¿Quién es Nancy Grace Roman? Noticias relacionadasArtículo7 Min ReadCómo el telescopio espacial Roman retrocederá a los orígenes del universoArtículo6 Min ReadEl telescopio Roman buscará agujeros negros originarios6 minutos de lecturaTelescopio Roman de la NASA buscará acumulaciones de materia oscuraArtículo8 minutos de lecturaTelescopio Roman estudiará las luces titilantes de la Vía LácteaArtículo1 minutos de lecturaMira cómo se construye el próximo telescopio espacial de la NASAArtículoExplora más temasCiencia de la NASAUniversoSistema solarTelescopio espacial James WebbTelescopio espacial Hubble

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¿Qué son los exoplanetas?

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URL: https://www.nasa.gov/aeronautics/wind-over-its-wing-nasas-x-66-model-tests-airflow/

TITLE: Wind Over Its Wing: NASA’s X-66 Model Tests Airflow

2 min readWind Over Its Wing: NASA’s X-66 Model Tests Airflow

Sarah MannNASA Armstrong Public Affairs SpecialistFeb 05, 2025ArticleNASA’s Sustainable Flight Demonstrator project concluded wind tunnel testing in the fall of 2024. Tests on a Boeing-built X-66 model were completed at NASA’s Ames Research Center in California’s Silicon Valley in its 11-Foot Transonic Unitary Plan Facility. The model underwent tests representing expected flight conditions to obtain engineering information to influence design of the wing and provide data for flight simulators.NASA/Brandon Torres NavarreteNASA’s Sustainable Flight Demonstrator (SFD) project recently concluded wind tunnel tests of its X-66 semi-span model in partnership with Boeing. The model, designed to represent half the aircraft, allows the research team to generate high-quality data about the aerodynamic forces that would affect the actual X-66.Test results will help researchers identify areas where they can refine the X-66 design – potentially reducing drag, enhancing fuel efficiency, or adjusting the vehicle shape for better flying qualities.Tests on the Boeing-built X-66 semi-span model were completed at NASA’s Ames Research Center in California’s Silicon Valley in its 11-Foot Transonic Unitary Plan Facility. The model underwent tests representing expected flight conditions so the team could obtain engineering information to influence the design of the aircraft’s wing and provide data for flight simulators.NASA’s Sustainable Flight Demonstrator project concluded wind tunnel testing in the fall of 2024. Tests on a Boeing-built X-66 model were completed at NASA’s Ames Research Center in California’s Silicon Valley in its 11-Foot Transonic Unitary Plan Facility. Pressure points, which are drilled holes with data sensors attached, are installed along the edge of the wing and allow engineers to understand the characteristics of airflow and will influence the final design of the wing.NASA/Brandon Torres NavarreteSemi-span tests take advantage of symmetry. The forces and behaviors on a model of half an aircraft mirror those on the other half. By using a larger half of the model, engineers increase the number of surface pressure measurements. Various sensors were placed on the wing to measure forces and movements to calculate lift, drag, stability, and other important characteristics.The semi-span tests followearlier wind tunnel workat NASA's Langley Research Center in Hampton, Virginia, using a smaller model of the entire aircraft. Engineers will study the data from all of the X-66 wind tunnel tests to determine any design changes that should be made before fabrication begins on the wing that will be used on the X-66 itself.The SFD project is NASA's effort to develop more efficient aircraft configurations as the nation moves toward aviation that's more economically, societally, and environmentally sustainable. The project seeks to provide information to inform the next generation of single-aisle airliners, the most common aircraft in commercial aviation fleets around the world. Boeing and NASA are partnering to develop the X-66 experimental demonstrator aircraft.

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TITLE: Commercial Low Earth Orbit (LEO) Destination Contract (CLDC) – ON HOLD

JSC-OP-SubCommercial Low Earth Orbit (LEO) Destination Contract (CLDC) – ON HOLDThe CLDC acquisition is on hold. NASA has determined that, rather than moving forward at this time with a firm fixed price contract for CLD certification and services, NASA will continue to support U.S. industry's design and demonstration of CLDs with funded SAAs for the next phase. After the SAA phase is enabled, NASA anticipates establishing a follow-on certification and services phase to meet government requirements under a FAR based acquisition. Details regarding the future acquisition will be released in the future. NASA remains committed to enabling sustainable development of commercially owned and operated LEO destinations under the National Aeronautics and Space Act (Space Act), 51 U.S.C. §§ 20101-20164.The following link provides additional information regarding the SAA strategy and competition process:Commercial Destinations – Development and Demonstration Objectives (C3DO) Space Act Agreement (SAA) Website (linkhttps://www.nasa.gov/johnson/jsc-procurement/c3do)This CLDC website and related announcements in www.sam.gov will remain accessible for reference purposes. In addition, NASA will continue to provide technical documents in the CLDC Technical Library site to keep industry up to date on current NASA technical standards that may be relevant to a future acquisition. Please continue to monitor www.sam.gov postings and the Acquisition Website for updates on the release of any CLDC information.CLDC Technical Documents Library (https://sam.gov/workspace/contract/opp/3b5b0bcd82e1462998a122448dad4aa3/view)Please note, the documents/links on the CLDC Acquisition Website listed here are for information and planning purposes only. It does not constitute a Request for Proposal, Invitation for Bid, or Request for Quotation, and it is not to be construed as a commitment by the Government to enter a contract.No solicitation exists; therefore, do not request a copy of a solicitation. If a solicitation is released, it will be synopsized on www.sam.gov. 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Commercial Low Earth Orbit (LEO) Destination Contract (CLDC) – ON HOLD

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The following link provides additional information regarding the SAA strategy and competition process:

Commercial Destinations – Development and Demonstration Objectives (C3DO) Space Act Agreement (SAA) Website (linkhttps://www.nasa.gov/johnson/jsc-procurement/c3do)

This CLDC website and related announcements in www.sam.gov will remain accessible for reference purposes. In addition, NASA will continue to provide technical documents in the CLDC Technical Library site to keep industry up to date on current NASA technical standards that may be relevant to a future acquisition. Please continue to monitor www.sam.gov postings and the Acquisition Website for updates on the release of any CLDC information.

CLDC Technical Documents Library (https://sam.gov/workspace/contract/opp/3b5b0bcd82e1462998a122448dad4aa3/view)

Please note, the documents/links on the CLDC Acquisition Website listed here are for information and planning purposes only. It does not constitute a Request for Proposal, Invitation for Bid, or Request for Quotation, and it is not to be construed as a commitment by the Government to enter a contract.

No solicitation exists; therefore, do not request a copy of a solicitation. If a solicitation is released, it will be synopsized on www.sam.gov. It is each interested party's responsibility to monitor the www.sam.gov website for the release of any subsequent solicitation or synopsis.

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Name:Rogelio CurielTitle:Contracting OfficerPhone: 575.524.5191Email:JSC-CLDC@mail.nasa.gov

Name:Matthew WindemuthTitle:Contract SpecialistPhone: 281.244.6212Email:JSC-CLDC@mail.nasa.gov

Source Selection Authority (SSA): TBD

Electronic responses are through the sam.gov

Current Information

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MilestoneTBD

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Equipo de redacción de CienciaJan 05, 2026ArtículoÍndice de contenidosChispas estelares en retiradaMaravillas cósmicas fugacesEsta animación muestra un posible diseño del patrón de mosaico del Sondeo de Altas Latitudes en el Dominio del Tiempo del telescopio espacial Nancy Grace Roman de la NASA. Este programa de observación será diseñado mediante un proceso comunitario, pero se espera que cubra cinco grados cuadrados —una región del cielo tan grande como 25 lunas llenas— y que penetre en el espacio, hasta el momento en que el universo tenía unos 500 millones de años, o menos del cuatro por ciento de su edad actual de 13.800 millones de años.Centro de Vuelo Espacial Goddard de la NASAEl telescopio espacial Nancy Grace Roman de la NASA trabajará a la par con otros observatorios espaciales que poseen un amplio campo de visión para descubrir el cosmos y su dinámica de maneras que nunca antes habían sido posibles."Roman trabajará en conjunto con observatorios de la NASA como el telescopio espacial James Webb y el Observatorio de rayos X Chandra, los cuales están diseñados para observar de cerca objetos transitorios singulares, una vez que estos han sido identificados; pero rara vez o nunca los descubren", dijo Julie McEnery, científica principal del proyecto del telescopio Roman en el Centro de Vuelo Espacial Goddard de la NASA en Greenbelt, Maryland. "El campo de visión mucho más grande de Roman revelará muchos de estos objetos que antes eran desconocidos. Y como nunca antes habíamos tenido un observatorio como este para explorar el cosmos, incluso podríamos encontrar clases completamente nuevas de objetos y fenómenos".La misión Sondeo de Altas Latitudes en el Dominio del Tiempo está bien diseñada para descubrir un tipo particular de estrellas que han hecho explosión, las cuales pueden ser usadas por los astrónomos para rastrear la evolución del universo e investigar posibles explicaciones para su expansión acelerada. Y dado que este estudio observará repetidamente la misma gran vista del espacio, los científicos también verán eventos esporádicos como el choque de cadáveres estelares y estrellas que son arrastradas hacia agujeros negros.El sondeo mirará más allá de nuestra galaxia para observar el mismo sector del cielo cada unos cinco días durante dos años. Al unir estas observaciones para hacer imágenes como animaciones cuadro por cuadro, se crearán videos que revelarán una gran cantidad de fenómenos transitorios.Chispas estelares en retiradaLos astrónomos buscarán a través de todos estos datos un tipo especial de estrella en explosión llamada supernova de tipo Ia. Estos fenómenos se originan en ciertos sistemas estelares binarios que contienen al menos una enana blanca: son los restos del núcleo pequeño y caliente de una estrella similar al Sol. En algunos casos, esta enana puede vaciar el material de su compañera. Esto desencadena una reacción nuclear fuera de control que finalmente hacer detonar a la estrella "ladrona". Los astrónomos también han hallado evidencia que respalda otro escenario, relacionado con dos enanas blancas que giran en espiral una hacia la otra hasta que se fusionan. Si su masa combinada es lo suficientemente grande, también pueden producir una supernova de tipo Ia.Dado que estas explosiones alcanzan su punto máximo con un brillo intrínseco conocido y similar, los astrónomos pueden usarlas para determinar a qué distancia están simplemente midiendo qué tan intenso es su brillo aparente. Los astrónomos usarán el telescopio Roman para estudiar el espectro de luz de estas supernovas para averiguar a qué velocidad parecen alejarse de nosotros debido a la expansión del espacio.Al comparar la velocidad con la que están retrocediendo las supernovas de tipo Ia a diferentes distancias, los científicos rastrearán la expansión cósmica a lo largo del tiempo. Esto nos ayudará a comprender si la energía oscura —la presión inexplicable que se cree que está acelerando la expansión del universo— ha cambiado a lo largo del tiempo y cómo lo ha hecho. El uso de estas y otras mediciones obtenidas por Roman también debería ayudar a aclarar las mediciones inconsistentes de la constante de Hubble, la cual es la tasa de expansión actual del universo.“Roman pintará una imagen más vívida del pasado y presente de nuestro universo, dándonos nuevas pistas sobre su posible destino”, dijo Rebekah Hounsell, científica investigadora de la Universidad de Maryland en el condado de Baltimore y del centro Goddard, quien está explorando formas de optimizar el Sondeo de Altas Latitudes en el Dominio del Tiempo de Roman. “Sus hallazgos podrían reconfigurar nuestra comprensión del cosmos”.Esta secuencia fotográfica de la supernova 2018gv en la galaxia NGC 2525 comprime casi un año de observaciones del telescopio espacial Hubble de la NASA en unos pocos segundos. Al inicio, la supernova eclipsa a las estrellas más brillantes de la galaxia antes de desvanecerse en la oscuridad. El telescopio espacial Nancy Grace Roman de la NASA podría captar tales eventos de principio a fin y alertar a otros telescopios, como Hubble y James Webb, para obtener datos aún más detallados.Crédito: NASA, ESA y A. Riess (STScI/JHU) y el equipo de SH0ES; Agradecimiento: M. Zamani (ESA/Hubble)Maravillas cósmicas fugacesDebido a la forma en que este estudio observará el cosmos, también detectará otros fenómenos poco comunes. Gracias a Roman, seremos testigos del nacimiento de nuevos agujeros negros que se forman cuando se fusionan las estrellas de neutrones, esto es, los núcleos de estrellas que han explotado y que no fueron lo suficientemente masivas como para colapsar y formar agujeros negros por sí solas. Estos eventos titánicos crean ondas en el tejido del espacio-tiempo y brillantes explosiones de kilonovas.También se espera que la misión descubra varias decenas de eventos de disrupción de mareas, los cuales ocurren cuando una estrella que se aventura demasiado cerca de un agujero negro es destrozada por la extrema gravedad del agujero negro. La metralla estelar produce una gran cantidad de luz a medida que se acelera en su camino hacia el agujero negro. Roman captará estas erupciones de energía para aprender cómo los agujeros negros afectan su entorno.El sondeo también permitirá a los astrónomos explorar objetos variables, como las galaxias activas cuyos núcleos albergan un cuásar extremadamente brillante. Un cuásar es un faro brillante de luz intensa alimentado por un agujero negro supermasivo. El agujero negro se alimenta vorazmente de la materia que cae dentro de él y desata un torrente de radiación. 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Esta transición marcó el comienzo del “amanecer cósmico”, ya que el universo pasó de ser en su mayor parte opaco a transparente, permitiendo que la luz visible y ultravioleta viajara libremente.“Este estudio de Roman descubrirá un tesoro oculto de datos para que los astrónomos lo examinen detalladamente, permitiendo una exploración cósmica con preguntas más abiertas de lo que normalmente es posible”, dijo McEnery. “Es posible que descubramos fortuitamente cosas completamente nuevas que aún no sabemos que debemos buscar”.El telescopio espacial Nancy Grace Roman es administrado desde el Centro de Vuelo Espacial Goddard de la NASA en Greenbelt, Maryland, con la participación del Laboratorio de Propulsión a Chorro (JPL, por sus siglas en inglés) de la NASA y Caltech/IPAC en el sur de California, el Instituto de Ciencias del Telescopio Espacial en Baltimore y un equipo integrado por científicos de diversas instituciones de investigación. Los principales socios de la industria son Ball Aerospace and Technologies Corporation en Boulder, Colorado; L3Harris Technologies en Melbourne, Florida; y Teledyne Scientific & Imaging en Thousand Oaks, California.Por Ashley BalzerCentro de Vuelo Espacial Goddard de la NASA, Greenbelt, MarylandRead this story in Englishhere.Más sobre Roman

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Chispas estelares en retirada

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Los principales socios de la industria son Ball Aerospace and Technologies Corporation en Boulder, Colorado; L3Harris Technologies en Melbourne, Florida; y Teledyne Scientific & Imaging en Thousand Oaks, California.Por Ashley BalzerCentro de Vuelo Espacial Goddard de la NASA, Greenbelt, MarylandRead this story in Englishhere.Más sobre Roman

Esta animación muestra un posible diseño del patrón de mosaico del Sondeo de Altas Latitudes en el Dominio del Tiempo del telescopio espacial Nancy Grace Roman de la NASA. Este programa de observación será diseñado mediante un proceso comunitario, pero se espera que cubra cinco grados cuadrados —una región del cielo tan grande como 25 lunas llenas— y que penetre en el espacio, hasta el momento en que el universo tenía unos 500 millones de años, o menos del cuatro por ciento de su edad actual de 13.800 millones de años.Centro de Vuelo Espacial Goddard de la NASA

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El telescopio espacial Nancy Grace Roman de la NASA trabajará a la par con otros observatorios espaciales que poseen un amplio campo de visión para descubrir el cosmos y su dinámica de maneras que nunca antes habían sido posibles.

“Roman trabajará en conjunto con observatorios de la NASA como el telescopio espacial James Webb y el Observatorio de rayos X Chandra, los cuales están diseñados para observar de cerca objetos transitorios singulares, una vez que estos han sido identificados; pero rara vez o nunca los descubren”, dijo Julie McEnery, científica principal del proyecto del telescopio Roman en el Centro de Vuelo Espacial Goddard de la NASA en Greenbelt, Maryland. “El campo de visión mucho más grande de Roman revelará muchos de estos objetos que antes eran desconocidos. Y como nunca antes habíamos tenido un observatorio como este para explorar el cosmos, incluso podríamos encontrar clases completamente nuevas de objetos y fenómenos".

La misión Sondeo de Altas Latitudes en el Dominio del Tiempo está bien diseñada para descubrir un tipo particular de estrellas que han hecho explosión, las cuales pueden ser usadas por los astrónomos para rastrear la evolución del universo e investigar posibles explicaciones para su expansión acelerada. Y dado que este estudio observará repetidamente la misma gran vista del espacio, los científicos también verán eventos esporádicos como el choque de cadáveres estelares y estrellas que son arrastradas hacia agujeros negros.

El sondeo mirará más allá de nuestra galaxia para observar el mismo sector del cielo cada unos cinco días durante dos años. Al unir estas observaciones para hacer imágenes como animaciones cuadro por cuadro, se crearán videos que revelarán una gran cantidad de fenómenos transitorios.

Los astrónomos buscarán a través de todos estos datos un tipo especial de estrella en explosión llamada supernova de tipo Ia. Estos fenómenos se originan en ciertos sistemas estelares binarios que contienen al menos una enana blanca: son los restos del núcleo pequeño y caliente de una estrella similar al Sol. En algunos casos, esta enana puede vaciar el material de su compañera. Esto desencadena una reacción nuclear fuera de control que finalmente hacer detonar a la estrella "ladrona". Los astrónomos también han hallado evidencia que respalda otro escenario, relacionado con dos enanas blancas que giran en espiral una hacia la otra hasta que se fusionan. Si su masa combinada es lo suficientemente grande, también pueden producir una supernova de tipo Ia.

Dado que estas explosiones alcanzan su punto máximo con un brillo intrínseco conocido y similar, los astrónomos pueden usarlas para determinar a qué distancia están simplemente midiendo qué tan intenso es su brillo aparente. Los astrónomos usarán el telescopio Roman para estudiar el espectro de luz de estas supernovas para averiguar a qué velocidad parecen alejarse de nosotros debido a la expansión del espacio.

Al comparar la velocidad con la que están retrocediendo las supernovas de tipo Ia a diferentes distancias, los científicos rastrearán la expansión cósmica a lo largo del tiempo. Esto nos ayudará a comprender si la energía oscura —la presión inexplicable que se cree que está acelerando la expansión del universo— ha cambiado a lo largo del tiempo y cómo lo ha hecho. El uso de estas y otras mediciones obtenidas por Roman también debería ayudar a aclarar las mediciones inconsistentes de la constante de Hubble, la cual es la tasa de expansión actual del universo.

“Roman pintará una imagen más vívida del pasado y presente de nuestro universo, dándonos nuevas pistas sobre su posible destino”, dijo Rebekah Hounsell, científica investigadora de la Universidad de Maryland en el condado de Baltimore y del centro Goddard, quien está explorando formas de optimizar el Sondeo de Altas Latitudes en el Dominio del Tiempo de Roman. “Sus hallazgos podrían reconfigurar nuestra comprensión del cosmos”.

Debido a la forma en que este estudio observará el cosmos, también detectará otros fenómenos poco comunes. Gracias a Roman, seremos testigos del nacimiento de nuevos agujeros negros que se forman cuando se fusionan las estrellas de neutrones, esto es, los núcleos de estrellas que han explotado y que no fueron lo suficientemente masivas como para colapsar y formar agujeros negros por sí solas. Estos eventos titánicos crean ondas en el tejido del espacio-tiempo y brillantes explosiones de kilonovas.

También se espera que la misión descubra varias decenas de eventos de disrupción de mareas, los cuales ocurren cuando una estrella que se aventura demasiado cerca de un agujero negro es destrozada por la extrema gravedad del agujero negro. La metralla estelar produce una gran cantidad de luz a medida que se acelera en su camino hacia el agujero negro. Roman captará estas erupciones de energía para aprender cómo los agujeros negros afectan su entorno.

El sondeo también permitirá a los astrónomos explorar objetos variables, como las galaxias activas cuyos núcleos albergan un cuásar extremadamente brillante. Un cuásar es un faro brillante de luz intensa alimentado por un agujero negro supermasivo. El agujero negro se alimenta vorazmente de la materia que cae dentro de él y desata un torrente de radiación. La mirada constante de Roman ayudará a los astrónomos a estudiar cómo y por qué fluctúa el brillo de estos estallidos.

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3 min readNASA's Universe of Learning Unveils Fresh Facilitator Guides Inspired by Community Feedback

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The image reveals how stars recycle their material back into the cosmos, seeding future generations of stars and planets, as NASA explores the secrets of the universe and our place in it.Image: Helix Nebula (NIRCam)This new image of a portion of the Helix Nebula from NASA's James Webb Space Telescope highlights comet-like knots, fierce stellar winds, and layers of gas shed off by a dying star interacting with its surrounding environment.Image: NASA, ESA, CSA, STScI; Image Processing: Alyssa Pagan (STScI)In the image from Webb's NIRCam (Near-Infrared Camera), pillars that look like comets with extended tails trace the circumference of the inner region of an expanding shell of gas. Here, blistering winds of fast-moving hot gas from the dying star are crashing into slower moving colder shells of dust and gas that were shed earlier in its life, sculpting the nebula's remarkable structure.The iconic Helix Nebula has been imaged by many ground- and space-based observatories over the nearly two centuries since it was discovered. Webb's near-infrared view of the target brings these knots to the forefront compared to theethereal image from NASA's Hubble Space Telescope, while its increased resolution sharpens focus fromNASA's retired Spitzer Space Telescope's snapshot. Additionally, the new near-infrared look shows the stark transition between the hottest gas to the coolest gas as the shell expands out from the central white dwarf.Image: Helix Nebula Context (VISTA and Webb)This image of the Helix Nebula from the ground-based Visible and Infrared Telescope for Astronomy (left) shows the full view of the planetary nebula, with a box highlighting Webb's field of view (right).Image: ESO, VISTA, NASA, ESA, CSA, STScI, J. Emerson (ESO); Acknowledgment: CASUA blazing white dwarf, the leftover core of the dying star, lies right at the heart of the nebula, out of the frame of the Webb image. Its intense radiation lights up the surrounding gas, creating a rainbow of features: hot ionized gas closest to the white dwarf, cooler molecular hydrogen farther out, and protective pockets where more complex molecules can begin to form within dust clouds. This interaction is vital, as it's the raw material from which new planets may one day form in other star systems.In Webb's image of the Helix Nebula, color represents the temperature and chemistry. A touch of a blue hue marks the hottest gas in this field, energized by intense ultraviolet light from the white dwarf. Farther out, the gas cools into the yellow regions where hydrogen atoms join into molecules. At the outer edges, the reddish tones trace the coolest material, where gas begins to thin and dust can take shape. Together, the colors show the star's final breath transforming into the raw ingredients for new worlds, adding to thewealth of knowledge gained from Webb about the origin of planets.Spitzer’s studies of the Helix Nebula hinted at the formation of more complex molecules, but Webb’s resolution shows how they form in shielded zones of the scene. In the Webb image, look for dark pockets of space amid the glowing orange and red.Video: Observatory Comparison (Hubble/Spitzer/Webb)To view this video please enable JavaScript, and consider upgrading to a web browser thatsupports HTML5 videoThis video compares images of the Helix Nebula from three NASA observatories: Hubble’s image in visible light, Spitzer’s infrared view, and Webb’s high-resolution near-infrared look.Video: NASA, ESA, CSA, STScI, Alyssa Pagan (STScI); Acknowledgment: NASA/JPL-Caltech, ESO, VISTA, CASU, Joseph Hora (CfA), J. Emerson (ESO)The Helix Nebula is located 650 light-years away from Earth in the constellation Aquarius. It remains a favorite among stargazers and professional astronomers alike due to its relative proximity to Earth, and its similar appearance to the “Eye of Sauron.”The James Webb Space Telescope is the world’s premier space science observatory. Webb is solving mysteries in our solar system, looking beyond to distant worlds around other stars, and probing the mysterious structures and origins of our universe and our place in it. Webb is an international program led by NASA with its partners, ESA (European Space Agency) and CSA (Canadian Space Agency).To learn more about Webb, visit:https://nasa.gov/webbDownloads & Related InformationThe following sections contain links to download this article's images and videos in all available resolutions followed by related information links, media contacts, and if available, research paper and Spanish translation links.

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This image of the Helix Nebula from the ground-based Visible and Infrared Telescope for Astronomy (left) shows the full view of the planetary nebula, with a box highlighting Webb's field of view (right).

Image: ESO, VISTA, NASA, ESA, CSA, STScI, J. Emerson (ESO); Acknowledgment: CASU

A blazing white dwarf, the leftover core of the dying star, lies right at the heart of the nebula, out of the frame of the Webb image. Its intense radiation lights up the surrounding gas, creating a rainbow of features: hot ionized gas closest to the white dwarf, cooler molecular hydrogen farther out, and protective pockets where more complex molecules can begin to form within dust clouds. This interaction is vital, as it's the raw material from which new planets may one day form in other star systems.

In Webb's image of the Helix Nebula, color represents the temperature and chemistry. A touch of a blue hue marks the hottest gas in this field, energized by intense ultraviolet light from the white dwarf. Farther out, the gas cools into the yellow regions where hydrogen atoms join into molecules. At the outer edges, the reddish tones trace the coolest material, where gas begins to thin and dust can take shape. Together, the colors show the star's final breath transforming into the raw ingredients for new worlds, adding to thewealth of knowledge gained from Webb about the origin of planets.

Spitzer's studies of the Helix Nebula hinted at the formation of more complex molecules, but Webb's resolution shows how they form in shielded zones of the scene. In the Webb image, look for dark pockets of space amid the glowing orange and red.

To view this video please enable JavaScript, and consider upgrading to a web browser thatsupports HTML5 videoThis video compares images of the Helix Nebula from three NASA observatories: Hubble's image in visible light, Spitzer's infrared view, and Webb's high-resolution near-infrared look.Video: NASA, ESA, CSA, STScI, Alyssa Pagan (STScI); Acknowledgment: NASA/JPL-Caltech, ESO, VISTA, CASU, Joseph Hora (CfA), J. Emerson (ESO)

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The Helix Nebula is located 650 light-years away from Earth in the constellation Aquarius. It remains a favorite among stargazers and professional astronomers alike due to its relative proximity to Earth, and its similar appearance to the "Eye of Sauron."

The James Webb Space Telescope is the world's premier space science observatory. Webb is solving mysteries in our solar system, looking beyond to distant worlds around other stars, and probing the mysterious structures and origins of our universe and our place in it. Webb is an international program led by NASA with its partners, ESA (European Space Agency) and CSA (Canadian Space Agency).

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NASA Hubble Mission TeamGoddard Space Flight CenterDec 23, 2025ArticleAstronomers using NASA'sHubble Space Telescopehave imaged the largest protoplanetary disk ever observed circling a young star. For the first time in visible light, Hubble has revealed the disk is unexpectedly chaotic and turbulent, with wisps of material stretching much farther above and below the disk than astronomers have seen in any similar system. Strangely, more extended filaments are only visible on one side of the disk. The findings, which published Tuesday in The Astrophysical Journal, mark a new milestone for Hubble and shed light on how planets may form in extreme environments, as NASA's missions lead humanity's exploration of the universe and our place in it.Located roughly 1,000 light-years from Earth, IRAS 23077+6707, nicknamed "Dracula's Chivito," spans nearly 400 billion miles — 40 times the diameter of our solar system to the outer edge of the Kuiper Belt of cometary bodies. The disk obscures the young star within it, which scientists believe may be either a hot, massive star, or a pair of stars. And the enormous disk is not only the largest known planet-forming disk; it's also shaping up to be one of the most unusual."The level of detail we're seeing is rare in protoplanetary disk imaging, and these new Hubble images show that planet nurseries can be much more active and chaotic than we expected," said lead author Kristina Monsch of the Center for Astrophysics | Harvard & Smithsonian (CfA). "We're seeing this disk nearly edge-on and its wispy upper layers and asymmetric features are especially striking. Both Hubble and NASA'sJames Webb Space Telescopehave glimpsed similar structures in other disks, but IRAS 23077+6707 provides us with an exceptional perspective — allowing us to trace its substructures in visible light at an unprecedented level of detail. This makes the system a unique, new laboratory for studying planet formation and the environments where it happens."The nickname "Dracula's Chivito" playfully reflects the heritage of its researchers—one from Transylvania and another from Uruguay, where the national dish is a sandwich called a chivito. The edge-on disk resembles a hamburger, with a dark central lane flanked by glowing top and bottom layers of dust and gas.This Hubble Space Telescope image shows the largest planet-forming disk ever observed around a young star. It spans nearly 400 billion miles — 40 times the diameter of our solar system.Image: NASA, ESA, STScI, Kristina Monsch (CfA); Image Processing: Joseph DePasquale (STScI)Puzzling asymmetryThe impressive height of these features wasn't the only thing that captured the attention of scientists. The new images revealed that vertically imposing filament-like features appear on just one side of the disk, while the other side appears to have a sharp edge and no visible filaments. This peculiar, lopsided structure suggests that dynamic processes, like the recent infall of dust and gas, or interactions with its surroundings, are shaping the disk."We were stunned to see how asymmetric this disk is," said co-investigator Joshua Bennett Lovell, also an astronomer at the CfA. "Hubble has given us a front row seat to the chaotic processes that are shaping disks as they build new planets — processes that we don't yet fully understand but can now study in a whole new way."All planetary systems form from disks of gas and dust encircling young stars. Over time, the gas accretes onto the star, and planets emerge from the remaining material. IRAS 23077+6707 may represent a scaled-up version of our early solar system, with a disk mass estimated at 10 to 30 times that of Jupiter — ample material for forming multiple gas giants. This, plus the new findings, makes it an exceptional case for studying the birth of planetary systems."In theory, IRAS 23077+6707 could host a vast planetary system," said Monsch. "While planet formation may differ in such massive environments, the underlying processes are likely similar. Right now, we have more questions than answers, but these new images are a starting point for understanding how planets form over time and in different environments."Credit: NASA's Goddard Space Flight Center; Lead Producer: Paul MorrisThe Hubble Space Telescope has been operating for over three decades and continues to make ground-breaking discoveries that shape our fundamental understanding of the universe. Hubble is a project of international cooperation between NASA and ESA (European Space Agency). NASA's Goddard Space Flight Center in Greenbelt, Maryland, manages the telescope and mission operations. Lockheed Martin Space, based in Denver, also supports mission operations at Goddard. The Space Telescope Science Institute in Baltimore, which is operated by the Association of Universities for Research in Astronomy, conducts Hubble science operations for NASA.Facebook logo@NASAHubble@NASAHubbleInstagram logo@NASAHubble

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Image: NASA, ESA, STScI, Kristina Monsch (CfA); Image Processing: Joseph DePasquale (STScI)

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NASA Today

Looking Forward

A History of Innovation

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NASA Missions

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Administrator Jared IsaacmanJared “Rook” Isaacman is the 15th Administrator of the National Aeronautics and Space Administration. An entrepreneur and skilled pilot, Isaacman commanded Inspiration4, the first all-civilian orbital spaceflight aboard the Dragon spacecraft. The mission was a major milestone in commercial spaceflight.Full BiographyAssociate Administrator Amit KshatriyaAmit Kshatriya is NASA’s associate administrator. Kshatriya serves as the highest-ranking civil servant at the agency and as a senior advisor to the administrator. Kshatriya leads the agency’s 10 center directors, as well as the mission directorate associate administrators at NASA Headquarters in Washington.Full BiographyDeputy Associate Administrator Casey SwailsCasey Swails is NASA’s deputy associate administrator. Swails is deputy and principal advisor to the associate administrator for day-to-day operations and long-term strategic direction across the agency's 10 field centers, mission directorates, and staff organizations.Full Biography

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Facebook:NASA Artemis

Instagram:@nasaartemis

X:@NASAArtemis

NASA Space Alerts

Facebook:@NASASpaceAlerts

X:@NASASpaceAlerts

Chandra X-Ray Observatory

Facebook:Chandra X-ray Observatory

Flickr:Chandra X-ray Observatory

Instagram:@nasachandraxray

X:@chandraxray

YouTube:Chandra X-ray Observatory

Facebook:NASA Earth

Instagram:@nasaearth

X:@NASAEarth

Facebook:NASA’s Hubble Space Telescope

Flickr:NASA Hubble Space Telescope

Instagram:@nasahubble

X:@NASAHubble

Facebook:International Space Station

Instagram:iss

X:@Space_Station

Facebook:NASA’s James Webb Space Telescope

Flickr:NASA’s James Webb Space Telescope

Instagram:@nasawebb

X:@NASAWebb

YouTube:James Webb Space Telescope (JWST)

Mars Exploration

Facebook:NASA Mars

X:@NASAMars

Facebook:Nancy Grace Roman Space Telescope

X:@NASARoman

Facebook:NASA Solar System Exploration

Instagram:@nasasolarsystem

X:@NASASolarSystem

YouTube:NASASolarSystem

Universe Exploration

Facebook:NASA Universe

Instagram:@nasauniverse

X:@NASAUniverse

NASA LeadershipThe official accounts of current NASA leadership.

Instagram:@nasaadmin

X:@NASAAdmin

Press Secretary Bethany Stevens

X:@NASASpox

Official NASA Astronaut Social Media Accounts

Read thebiographiesof NASA astronauts eligible for flight assignment – including those who are not active on social media.

Ayers, Nichole

Instagram:@astro_ayers

X:@Astro_Ayers

Barron, Kayla

Facebook:NASA Astronaut Kayla Barron

Instagram:@astro_kayla

Berríos, Marcos|en español

Instagram:@astro_berrios

X:@astro_berrios

Birch, Christina

Instagram: @astro_birch

X: @astro_birch

Boe, Eric A.

X:@Astro_Boe

Bresnik, Randolph J.

Facebook:NASA Astronaut Randy “Komrade” Bresnik

Instagram:@astrokomrade

X:@AstroKomrade

Burnham, Deniz

Instagram:@astronaut_deniz

X:@astro_deniz

Cardman, Zena

Instagram:@zenanaut

Reddit:u/zenacardman

X:@zenanaut

Chari, Raja

Facebook:NASA Astronaut Raja Chari

Instagram:@astro_raja

X:@Astro_Raja

Douglas, Andre

Instagram:@astro_andred

X:@Astro_AndreD

Fincke, E. Michael

X:@AstroIronMike

Glover, Victor J.

Instagram:@astrovicglover

X:@AstroVicGlover

Hathaway, Jack

Instagram:@astro_hathaway

X:@astro_hathaway

Hines, Bob

Instagram:@astro_farmerbo

X:@Astro_FarmerBob

Hoburg, Warren

Instagram:astro.woody

Reddit:u/warrenhoburg

X:@Astro_Woody

Kim, Jonny

Facebook:NASA Astronaut Jonny Kim

Instagram:@jonnykimusa

X:@JonnyKimUSA

Koch, Christina H.

Instagram:@astro_christina

X:@Astro_Christina

Lindgren, Kjell N.

Instagram:@astro_kjell

X:@astro_kjell

Mann, Nicole Aunapu

X:@AstroDuke

McClain, Anne C.

Facebook:NASA Astronaut Anne McClain

Instagram:@astro_annimal

X:@AstroAnnimal

Meir, Jessica U.

Instagram:@astro_jessica

X:@Astro_Jessica

Menon, Anil

Instagram:@astro_anil

X:@astro_anil

Moghbeli, Jasmin

Facebook:NASA Astronaut Jasmin Moghbeli

Instagram:@astrojaws

Morgan, Andrew R.

Facebook:NASA Astronaut Andrew “Drew” Morgan

Instagram:@astrodrewmorgan

X:@AstroDrewMorgan

O’Hara, Loral

Instagram:@loralohara

X:@lunarloral

Pettit, Donald R.

X:@astro_Pettit

Rubio, Frank

Facebook:Astronaut Frank Rubio

Instagram:@astro_frankrubio

Tingle, Scott D.

Instagram:@astro_maker

Vande Hei, Mark T.

X:@Astro_Sabot

Watkins, Jessica

Instagram:@astro_watkins

X:@astro_watkins

Wheelock, Douglas H.

X:@Astro_Wheels

Wilson, Stephanie D.

X:@Astro_Stephanie

Williams, Christopher L.

X:@Astro_ChrisW

Williams, Sunita L.

X:@Astro_Suni

Wiseman, G. Reid

Instagram:@astro_reid

X:@astro_reid

Wittner, Jessica

Instagram:@astro_wittner

X:@Astro_Wittner

NASA Astronaut Candidates

The term astronaut candidate refers to individuals who have been selected by NASA as candidates for astronaut corps and are currently undergoing a candidacy training program at NASA's Johnson Space Center. Learn more about theastronaut selection program. Read thebiographiesof the most recent class of NASA astronaut candidates – including those who are not active on social media.

Bailey, Ben

Instagram:@astro_benbailey

X:@astro_benbailey

Edgar, Lauren

Facebook:NASA-Astronaut-Lauren-Edgar

Instagram:@astronautedgar

Fuhrmann, Adam

Instagram:@astro_fuhrmann

X:@astro_fuhrmann

Jones, Cameron

Facebook:NASA Astronaut Cameron Jones

Instagram:@astro_camjones

X:@astro_camjones

Kubo, Yuri

Facebook:NASA Astronaut Yuri Kubo

Instagram:@astro_kubo

Lawler, Rebecca

Instagram:@astro_lawler

X:@astro_lawler

Menon, Anna

Instagram:@astroannamenon

Muller, Imelda

Instagram:@astro_imelda

Overcash, Erin

Facebook:NASA Astronaut Erin Overcash

Instagram:@astro_overcash

Spies, Katherine

Instagram:@spiesinspace

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URL: https://science.nasa.gov/dark-energy/

URL: https://science.nasa.gov/universe/search-for-life/

TITLE: Are We Alone?

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TITLE: Centennial Challenges

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Grade Levels

Grades K-4

Grades 5-8

Grades 9-12

Higher Education

Informal Education

Subject

Art

Computer Science

Earth ScienceGeographyGeologyHuman Interactions and EnvironmentPlanet EarthWeather and Climate

Geography

Geology

Human Interactions and Environment

Planet Earth

Education

Engineering Design

HistoryBiographiesFlightHistorical OverviewsPlanetary ExplorationScientists and InventorsTimelines

History

Biographies

Flight

Historical Overviews

Planetary Exploration

Scientists and Inventors

Timelines

Human Space ExplorationPastPresent and Future

Human Space Exploration

Present and Future

Life ScienceAnimals and PlantsHealth and NutritionHumansLife SupportLiving and Working in SpaceSpace Biology

Life Science

Animals and Plants

Health and Nutrition

Humans

Life Support

Living and Working in Space

Space Biology

MathematicsAlgebraCalculusFractions and DecimalsGeometryMeasurement and Data AnalysisNumbers and OperationsProbability and StatisticsProblem SolvingTrigonometry

Mathematics

Algebra

Calculus

Fractions and Decimals

Geometry

Measurement and Data Analysis

Numbers and Operations

Probability and Statistics

Problem Solving

Trigonometry

Physical ScienceChemistryEnergyFlight and AeronauticsForces and MotionGravityMatter and Its PropertiesMicrogravityOpticsPhysicsWavesLightSound

Physical Science

Chemistry

Energy

Flight and Aeronautics

Forces and Motion

Gravity

Matter and Its Properties

Microgravity

Optics

Physics

WavesLightSound

Waves

Light

Sound

Space ScienceAstrobiologyAstronomyEclipsesExoplanetsMissions to Planets and MoonsPlanetary GeologyRobotic ExplorersSolar System and PlanetsAsteroids Comets MeteoritesDwarf PlanetsEarthEarth's MoonJupiterMarsMercuryNeptunePlanetary MoonsSaturnSunUranusVenusSolar System and StarsSpace RadiationStarsTelescopesUniverse

Space Science

Astronomy

Missions to Planets and Moons

Planetary Geology

Robotic Explorers

Solar System and PlanetsAsteroids Comets MeteoritesDwarf PlanetsEarthEarth's MoonJupiterMarsMercuryNeptunePlanetary MoonsSaturnSunUranusVenus

Solar System and Planets

Asteroids Comets Meteorites

Dwarf Planets

Planetary Moons

Solar System and Stars

Space Radiation

Telescopes

Spanish

TechnologyCommunicationModelsPhotography and ImagingRemote SensingRoboticsRocketrySatellitesSpace StationSpace VehiclesSpinoffs (Benefits to You)Virtual Reality

Communication

Models

Photography and Imaging

Remote Sensing

Rocketry

Satellites

Space Station

Space Vehicles

Spinoffs (Benefits to You)

Virtual Reality

Type

Bookmarks

Data Sets

Educator Guides

Hands-on Activities

Image Galleries

Interactive Multimedia

Lesson Plans / Activities

Mobile Apps

Other Multimedia

Play and LearnColoring and PuzzlesGames

Play and Learn

Coloring and Puzzles

Posters

Program Brochures

Read About It

STEM Resource Collections

Websites

URL: https://www.nasa.gov/get-involved/#NASA-Classroom

URL: https://science.nasa.gov/uranus/

TITLE: Uranus

Explore This SectionUranusUranus is the seventh planet from the Sun, and the third largest planet in our solar system. It appears to spin sideways.FactsExplorationMoonsResourcesAbout Planet UranusUranus is the seventh planet from the Sun, and the third largest planet in our solar system – about four times wider than Earth.Uranus is very cold and windy. It is surrounded by faint rings, and more than two dozen small moons. It rotates at a nearly 90-degree angle from the plane of its orbit. This unique tilt makes Uranus appear to spin on its side.Get the FactsThis image of Uranus from the James Webb Telescope's Near-Infrared Camera (NIRCam) shows Uranus's seasonal north polar cap and dim inner and outer rings. This Webb image also shows 9 of the planet's 27 moons.NASA, ESA, CSA, STScIPlanet Uranus ExplorationOnly one spacecraft has explored planet Uranus.In January 1986,Voyager 2 made a close approach to Uranus, snapping images of the planet and some its moons.Read MoreNASA's Voyager 2 lifts off on Aug. 20, 1977, from Kennedy Space Center in Florida.NASAPlanet Uranus

MoonsMeet the Literary MoonsUranus has28 known moons, including five major moons: Miranda, Ariel, Umbriel, Titania, and Oberon. The moons are sometimes called the "literary moons" because they are named for Shakespearean characters, along with a couple of the moons being named for characters from the works of Alexander Pope.Explore Moons of Planet UranusThis image of Uranus from NASA's James Webb Space Telescope shows the planet and distant background galaxies. This image also includes 14 of the planet's 27 moons.NASA, ESA, CSA, STScIIce Giant ResourcesResources for Uranus and Neptune.Explore a curated collection of resources about ice giants, including activities that can be done at home, as well as videos and animations, images, and posters.Read MoreNASA/JPL-CaltechVoyager 240th Anniversary of Uranus FlybyThe first human-made object to fly past Uranus, Voyager 2 made its closest approach on Jan. 24, 1986, and during its visit discovered 10 new moons, two new rings, and a magnetic field stronger than Saturn's, while returning images and data that still fuel discoveries today. Voyager 2 finished its encounter in February, enroute to a rendezvous with Neptune in 1989.Learn More About Voyager 2's Uranus EncounterEven decades after Voyager 2 became the first, and so far only, spacecraft to visit Uranus, its data has led to revelations about the ice giant.Read 'Mining Old Data From NASA's Voyager 2 Solves Several Uranus Mysteries'Eyes on the Solar System: UranusEyes of the Solar System uses data and images from NASA missions to give you a simulated view of Uranus.Solar System NewsArticle4 Min ReadNASA Finds Lunar Regolith Limits Meteorites as Source of Earth's WaterBlog4 Min ReadNASA's SunRISE SmallSats Ace Tests, Moving Closer to LaunchBlog1 Min ReadStrong Flare Erupts From Sun25 Min ReadArtemis II: What NASA Learned From Launching Artemis ILearn MoreFor Kids: All About Planet UranusFacts about Planet Uranus for kids.Ice Giant ResourcesA curated collection of resources about ice giants, including activities that can be done at home, as well as videos and animations, images, and posters.NASA Photojournal: Planet UranusImages of planet Uranus from NASA spacecraft.Keep ExploringDiscover More Topics From NASANeptuneSunMercurySolar System Exploration

UranusUranus is the seventh planet from the Sun, and the third largest planet in our solar system. It appears to spin sideways.

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Moons

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Planet Uranus Moons

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Voyager 2

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Eyes on the Solar System: Uranus

Eyes of the Solar System uses data and images from NASA missions to give you a simulated view of Uranus.

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TITLE: NASA Policy on the Release of Information to News and Information Media

NASA Policy on the Release of Information to News and Information MediaEncyclopediaUpdated Dec 31, 2025

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Agency's communications program, adhering to Agency policies, procedures, and priorities, and coordinating their activities with Headquarters (and others where appropriate). They are responsible for the quality of public information prepared by Center public affairs officers. They also are responsible for the day-to-day production of public information covering their respective Center activities, which includes obtaining the necessary Center concurrences and coordinating, as necessary, with the appropriate Headquarters public affairs officers.(e) Center Directors have ultimate responsibility for the accuracy of public information that does not require the concurrence of Headquarters. (See "Public information coordination and concurrence," section (d).)(f) All NASA employees are required to coordinate, in a timely manner, with the appropriate public affairs officers prior to releasing information that has the potential to generate significant media, or public interest or inquiry.(g) All NASA public affairs officers are required to notify the appropriate Headquarters public affairs officers in a timely manner about activities or events that have the potential to generate significant media or public interest or inquiry.(h) All NASA public affairs employees are expected to adhere to the following code of conduct:Be honest and accurate in all communications.Honor publication embargoes.Respond promptly to media requests and respect media deadlines.Act promptly to correct mistakes or erroneous information, either internally or externally.Promote the free flow of scientific and technical information.Protect non-public information.(i) All NASA employees are responsible for adhering to plans (including schedules) for activities established by public affairs offices and senior management for the coordinated release of public information.(j) All NASA-funded missions will have a public affairs plan, approved by the Assistant Administrator for Public Affairs, which will be managed by Headquarters and/or a designated NASA Center.(k) Public affairs activities for NASA-funded missions will not be managed by non-NASA institutions, unless authorized by the Assistant Administrator for Public Affairs.(l) The requirements of this directive do not apply to the Office of Inspector General regarding its activities.Public Information Coordination and Concurrence(a) General. All NASA employees involved in preparing and issuing NASA public information are responsible for proper coordination among Headquarters, Center, and Mission Directorate offices to include review and clearance by appropriate officials prior to issuance. Such coordination will be accomplished through procedures developed and published by the NASA Assistant Administrator for Public Affairs.(b) Coordination. To ensure timely release of public information, Headquarters and Center public affairs officers are required to coordinate to obtain review and clearance by appropriate officials, keep each other informed of changes, delays, or cancellation of releases, and provide advance notification of the actual release.(c) All public information shall be coordinated through the appropriate Headquarters offices, including review by the appropriate Mission Directorate Associate Administrator and mission support office head, or their designees, to ensure scientific, technical, and programmatic accuracy, and review by the Assistant Administrator of Public Affairs or his/her designee to ensure that public information products are well written and appropriate for the intended audience.(d) Centers may, however, without the full coordination of Headquarters, issue public information that is institutional in nature, of local interest, or has been deemed not to be a Headquarters release. (The Assistant Administrator for Public Affairs or his/her designee will determine which public information will be issued nationwide by NASA Headquarters.) These releases must be coordinated through the appropriate Center offices and approved by the Center Director and Center Public Affairs Director. The Center Public Affairs Director is required to provide proper notification to the NASA Office of Public Affairs, Headquarters, prior to release. (The Assistant Administrator for Public Affairs shall publish guidelines for the release of public information that may be issued by Centers without clearance from Headquarters' offices.)(e) Dispute Resolution. Any dispute arising from a decision to proceed or not proceed with the issuance of a news release or other type of public information will be addressed and resolved by the Assistant Administrator for Public Affairs with the appropriate Mission Directorate Associate Administrator, mission support office head, Center Director, and others, such as Center Public Affairs Directors, as necessary. However, the appropriate Mission Directorate Associate Administrator shall be the arbiter of disputes about the accuracy or characterization of programmatic, technical, or scientific information. Additional appeals may be made to the Chief of Strategic Communications and to the Office of the Administrator. When requested by a Center Public Affairs Director, an explanation of the resolution will be provided in writing to all interested Agency parties.Interviews(a) Only spokespersons designated by the Assistant Administrator for Public Affairs, or his/her designee, are authorized to speak for the Agency in an official capacity regarding NASA policy, programmatic, and budget issues.(b) In response to media interview requests, NASA will offer articulate and knowledgeable spokespersons who can best serve the needs of the media and the American public. However, journalists may have access to the NASA officials they seek to interview, provided those NASA officials agree to be interviewed.(c) NASA employees may speak to the media and the public about their work. When doing so, employees shall notify their immediate supervisor and coordinate with their public affairs office in advance of interviews whenever possible, or immediately thereafter, and are encouraged, to the maximum extent practicable, to have a public affairs officer present during interviews. If public affairs officers are present, their role will be to attest to the content of the interview, support the interviewee, and provide post-interview follow up with the media as necessary.(d) NASA, as an Agency, does not take a position on any scientific conclusions. That is the role of the broad scientific community and the nature of the scientific process. NASA scientists may draw conclusions and may, consistent with this policy, communicate those conclusions to the media. However, NASA employees who present personal views outside their official area of expertise or responsibility must make clear that they are presenting their individual views – not the views of the Agency – and ask that they be sourced as such.(e) Appropriated funds may only be used to support Agency missions and objectives consistent with legislative or presidential direction. Government funds shall not be used for media interviews or other communication activities that go beyond the scope of Agency responsibilities and/or an employee's official area of expertise or responsibility.(f) Media interviews will be "on-the-record" and attributable to the person making the remarks, unless authorized to do otherwise by the Assistant Administrator for Public Affairs or Center Public Affairs Director, or their designees. Any NASA employee providing material to the press will identify himself/herself as the source.(g) Audio recordings may be made by NASA with consent of the interviewee.(h) NASA employees are not required to speak to the media.(i) Public information volunteered by a NASA official will not be considered exclusive to any one media source and will be made available to other sources, if requested.Preventing Release of Classified Information to the Media(a) Release of classified information in any form (e.g., documents, through interviews, audio/visual, etc.) to the news media is prohibited. The disclosure of classified information to unauthorized individuals may be cause for prosecution and/or disciplinary action against the NASA employee involved. Ignorance of NASA policy and procedures regarding classified information does not release a NASA employee from responsibility for preventing any unauthorized release. See NPR 1600.1, Chapter 5, Section 5.23 for internal NASA guidance on management of classified information. For further guidance that applies to all agencies, see Executive Order 12958, as amended, "Classified National Security Information" and its implementing directive at 32 CFR Parts 2001 and 2004.(b) Any attempt by news media representatives to obtain classified information will be reported through the Headquarters Office of Public Affairs or Installation Public Affairs Office to the Installation Security Office and Office of Security and Program Protection.(c) For classified operations and/or programs managed under the auspices of a DD Form 254, "Contract Security Classification Specification," all inquiries concerning this activity will be responded to by the appropriate PAO official designated in Item 12 on the DD Form 254.(d) For classified operations and/or information owned by other Government agencies (e.g., DOD, DOE, etc.), all inquiries will be referred to the appropriate Agency public affairs officer as established in written agreements.Preventing Unauthorized Release of Sensitive but Unclassified (SBU) Information and/or Material to the News Media(a) All NASA SBU information requires accountability and approval for release. Release of SBU information to unauthorized personnel is prohibited. Unauthorized release of SBU information may result in prosecution and/or disciplinary action. Ignorance of NASA policy and procedures regarding SBU information does not release a NASA employee from responsibility for unauthorized release. See NPR 1600.1, Chapter 5, Section

5.24 for guidance on identification, marking, accountability and release of NASA SBU information.(b) Examples of SBU information include: proprietary information of others provided to NASA under nondisclosure or confidentiality agreement; source selection and bid and proposal information; information subject to export control under the International Traffic in Arms Regulations (ITAR) or the Export Administration Regulations (EAR); information subject to the Privacy Act of 1974; predecisional materials such as national space policy not yet publicly released; pending reorganization plans or sensitive travel itineraries; and information that could constitute an indicator of U.S. government intentions, capabilities, operations, or activities or otherwise threaten operations security.(c) Upon request for access to information/material deemed SBU, coordination must be made with the information/material owner to determine if the information/material may be released. Other organizations that play a part in SBU information identification, accountability and release (e.g., General Counsel, External Relations, Procurement, etc.) must be consulted for assistance and/or concurrence prior to release.(d) Requests for SBU information from other Government agencies must be referred to the respective Agency public affairs officer.Multimedia Materials(a) NASA's multimedia material, from all sources, will be made available to the information media, the public, and to all Agency Centers and contractor installations utilizing contemporary delivery methods and emerging digital technology.(b) Centers will provide the media, the public, and as necessary, NASA Headquarters with:Selected prints and original or duplicate files of news-oriented imagery and other digital multimedia material generated within their respective areas.Selected video material in the highest quality format practical, which, in the opinion of the installations, would be appropriate for use as news feed material or features in pre-produced programs and other presentations.Audio and/or video files of significant news developments and other events of historic or public interest.Interactive multimedia features that can be incorporated into the Agency's Internet portal for use by internal and external audiences, including the media and the general public.News Releases Concerning International Activities(a) Releases of information involving NASA activities, views, programs, or projects involving another country or an international organization require prior coordination and approval by the Headquarters offices of External Relations and Public Affairs.(b) NASA Centers and Headquarters offices will report all visits proposed by representatives of foreign news media to the public affairs officer for the Office of External Relations for appropriate handling consistent with all NASA policies and procedures.

and/or Material to the News MediaMultimedia MaterialsNews Releases Concerning International Activities

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Not included under this definition are scientific and technical reports, web postings designed for technical or scientific interchange, and technical information presented at professional meetings or in professional journals.Applicability(a)This policy applies to NASA Headquarters, NASA Centers, and Component Facilities.(b) In the event of any conflict between this policy and any other NASA policy, directive, or regulation, this policy shall govern and supersede any previous issuance or directive.Principles(a) NASA, a scientific and technical agency, is committed to a culture of openness with the media and public that values the free exchange of ideas, data, and information as part of scientific and technical inquiry. 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However, NASA employees who present personal views outside their official area of expertise or responsibility must make clear that they are presenting their individual views – not the views of the Agency – and ask that they be sourced as such.(e) Appropriated funds may only be used to support Agency missions and objectives consistent with legislative or presidential direction. Government funds shall not be used for media interviews or other communication activities that go beyond the scope of Agency responsibilities and/or an employee's official area of expertise or responsibility.(f) Media interviews will be "on-the-record" and attributable to the person making the remarks, unless authorized to do otherwise by the Assistant Administrator for Public Affairs or Center Public Affairs Director, or their designees. Any NASA employee providing material to the press will identify himself/herself as the source.(g) Audio recordings may be made by NASA with consent of the interviewee.(h) NASA employees are not required to speak to the media.(i) Public information volunteered by a NASA official will not be considered exclusive to any one media source and will be made available to other sources, if requested.Preventing Release of Classified Information to the Media(a) Release of classified information in any form (e.g., documents, through interviews, audio/visual, etc.) to the news media is prohibited. The disclosure of classified information to unauthorized individuals may be cause for prosecution and/or disciplinary action against the NASA employee involved. Ignorance of NASA policy and procedures regarding classified information does not release a NASA employee from responsibility for preventing any unauthorized release. 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Other organizations that play a part in SBU information identification, accountability and release (e.g., General Counsel, External Relations, Procurement, etc.) must be consulted for assistance and/or concurrence prior to release.(d) Requests for SBU information from other Government agencies must be referred to the respective Agency public affairs officer.Multimedia Materials(a) NASA's multimedia material, from all sources, will be made available to the information media, the public, and to all Agency Centers and contractor installations utilizing contemporary delivery methods and emerging digital technology.(b) Centers will provide the media, the public, and as necessary, NASA Headquarters with:Selected prints and original or duplicate files of news-oriented imagery and other digital multimedia material generated within their respective areas.Selected video material in the highest quality format practical, which, in the opinion of the installations, would be appropriate for use as news feed material or features in pre-produced programs and other presentations.Audio and/or video files of significant news developments and other events of historic or public interest.Interactive multimedia features that can be incorporated into the Agency's Internet portal for use by internal and external audiences, including the media and the general public.News Releases Concerning International Activities(a) Releases of information involving NASA activities, views, programs, or projects involving another country or an international organization require prior coordination and approval by the Headquarters offices of External Relations and Public Affairs.(b) NASA Centers and Headquarters offices will report all visits proposed by representatives of foreign news media to the public affairs officer for the Office of External Relations for appropriate handling consistent with all NASA policies and procedures.

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(b) In the event of any conflict between this policy and any other NASA policy, directive, or regulation, this policy shall govern and supersede any previous issuance or directive.

(a) NASA, a scientific and technical agency, is committed to a culture of openness with the media and public that values the free exchange of ideas, data, and information as part of scientific and technical inquiry. Scientific and technical information from or about Agency programs and projects will be accurate and unfiltered.

(b) Consistent with NASA statutory responsibility, NASA will "provide for the widest practicable and appropriate dissemination of information concerning its activities and the results thereof." Release of public information concerning NASA activities and the results of NASA activities will be made promptly, factually, and completely.

(c) To ensure timely release of information, NASA will endeavor to ensure cooperation and coordination among the Agency's scientific, engineering, and public affairs communities.

(d) In keeping with the desire for a culture of openness, NASA employees may, consistent with this policy, speak to the press and the public about their work.

(e) This policy does not authorize or require disclosure of information that is exempt from disclosure under the Freedom of Information Act (5 U.S.C. § 552) or otherwise restricted by statute, regulation, Executive Order, or other Executive Branch policy or NASA policy (e.g., OMB Circulars, NASA Policy Directives). Examples of information not releasable under this policy include, without limitation, information that is, or is marked as, classified information, procurement sensitive information, information subject to the Privacy Act, other sensitive but unclassified information, and information subject to privilege, such as pre-decisional information or attorney-client communications.

(a) The Assistant Administrator for Public Affairs is responsible for developing and administering an integrated Agency-wide communications program, establishing Agency public affairs policies and priorities, and coordinating and reviewing the performance of all Agency public affairs activities. The Assistant Administrator will develop criteria to identify which news releases and other types of public information will be issued nationwide by NASA Headquarters. Decisions to release public information nationwide by NASA Headquarters will be made by the Assistant Administrator for Public Affairs or his/her designee.

(b) NASA's Mission Directorate Associate Administrators and Mission Support Office heads have ultimate responsibility for the technical, scientific, and programmatic accuracy of all information that is related to their respective programs and released by NASA.

(c) Under the direction of the Assistant Administrator for Public Affairs, public affairs officers assigned to Mission Directorates are responsible for the timely and efficient coordination of public information covering their respective programs. This coordination includes review by appropriate Mission Directorate officials. It also includes editing by public affairs staff to ensure that public information products are well written and appropriate for the intended audience. However, such editing shall not change scientific or technical data, or the meaning of programmatic content.

(d) Center Public Affairs Directors are responsible for implementing their portion of the Agency's communications program, adhering to Agency policies, procedures, and priorities, and coordinating their activities with Headquarters (and others where appropriate). They are responsible for the quality of public information prepared by Center public affairs officers. They also are responsible for the day-to-day production of public information covering their respective Center activities, which includes obtaining the necessary Center concurrences and coordinating, as necessary, with the appropriate Headquarters public affairs officers.

(e) Center Directors have ultimate responsibility for the accuracy of public information that does not require the concurrence of Headquarters. (See "Public information coordination and concurrence," section (d).)

(f) All NASA employees are required to coordinate, in a timely manner, with the appropriate public affairs officers prior to releasing information that has the potential to generate significant media, or public interest or inquiry.

(g) All NASA public affairs officers are required to notify the appropriate Headquarters public affairs officers in a timely manner about activities or events that have the potential to generate significant media or public interest or inquiry.

(h) All NASA public affairs employees are expected to adhere to the following code of conduct:

Be honest and accurate in all communications.

Honor publication embargoes.

Respond promptly to media requests and respect media deadlines.

Act promptly to correct mistakes or erroneous information, either internally or externally.

Promote the free flow of scientific and technical information.

Protect non-public information.

(i) All NASA employees are responsible for adhering to plans (including schedules) for activities established by public affairs offices and senior management for the coordinated release of public information.

(j) All NASA-funded missions will have a public affairs plan, approved by the Assistant Administrator for Public Affairs, which will be managed by Headquarters and/or a designated NASA Center.

(k) Public affairs activities for NASA-funded missions will not be managed by non-NASA institutions, unless authorized by the Assistant Administrator for Public Affairs.

(l) The requirements of this directive do not apply to the Office of Inspector General regarding its activities.

(a) General. All NASA employees involved in preparing and issuing NASA public information are responsible for proper coordination among Headquarters, Center, and Mission Directorate offices to include review and clearance by appropriate officials prior to issuance. Such coordination will be accomplished through procedures developed and published by the NASA Assistant Administrator for Public Affairs.

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(c) All public information shall be coordinated through the appropriate Headquarters offices, including review by the appropriate Mission Directorate Associate Administrator and mission support office head, or their designees, to ensure scientific, technical, and programmatic accuracy, and review by the Assistant Administrator of Public Affairs or his/her designee to ensure that public information products are well written and appropriate for the intended audience.

(d) Centers may, however, without the full coordination of Headquarters, issue public information that is institutional in nature, of local interest, or has been deemed not to be a Headquarters release. (The Assistant Administrator for Public Affairs or his/her designee will determine which public information will be issued nationwide by NASA Headquarters.) These releases must be coordinated through the appropriate Center offices and approved by the Center Director and Center Public Affairs Director. The Center Public Affairs Director is required to provide proper notification to the NASA Office of Public Affairs, Headquarters, prior to release. (The Assistant Administrator for Public Affairs shall publish guidelines for the release of public information that may be issued by Centers without clearance from Headquarters' offices.)

(e) Dispute Resolution. Any dispute arising from a decision to proceed or not proceed with the issuance of a news release or other type of public information will be addressed and resolved by the Assistant Administrator for Public Affairs with the appropriate Mission Directorate Associate Administrator, mission support office head, Center Director, and others, such as Center Public Affairs Directors, as necessary. However, the appropriate Mission Directorate Associate Administrator shall be the arbiter of disputes about the accuracy or characterization of programmatic, technical, or scientific information. Additional appeals may be made to the Chief of Strategic Communications and to the Office of the Administrator. When requested by a Center Public Affairs Director, an explanation of the resolution will be provided in writing to all interested Agency parties.

(a) Only spokespersons designated by the Assistant Administrator for Public Affairs, or his/her designee, are authorized to speak for the Agency in an official capacity regarding NASA policy, programmatic, and budget issues.

(b) In response to media interview requests, NASA will offer articulate and knowledgeable spokespersons who can best serve the needs of the media and the American public. However, journalists may have access to the NASA officials they seek to interview, provided those NASA officials agree to be interviewed.

(c) NASA employees may speak to the media and the public about their work. When doing so, employees shall notify their immediate supervisor and coordinate with their public affairs office in advance of interviews whenever possible, or immediately thereafter, and are encouraged, to the maximum extent practicable, to have a public affairs officer present during interviews. If public affairs officers are present, their role will be to attest to the content of the interview, support the interviewee, and provide post-interview follow up with the media as necessary.

(d) NASA, as an Agency, does not take a position on any scientific conclusions. That is the role of the broad scientific community and the nature of the scientific process. NASA scientists may draw conclusions and may, consistent with this policy, communicate those conclusions to the media. However, NASA employees who present personal views outside their official area of expertise or responsibility must make clear that they are presenting their individual views – not the views of the Agency – and ask that they be sourced as such.

(e) Appropriated funds may only be used to support Agency missions and objectives consistent with legislative or presidential direction. Government funds shall not be used for media interviews or other communication activities that go beyond the scope of Agency responsibilities and/or an employee’s official area of expertise or responsibility.

(f) Media interviews will be “on-the-record” and attributable to the person making the remarks, unless authorized to do otherwise by the Assistant Administrator for Public Affairs or Center Public Affairs Director, or their designees. Any NASA employee providing material to the press will identify himself/herself as the source.

(g) Audio recordings may be made by NASA with consent of the interviewee.

(h) NASA employees are not required to speak to the media.

(i) Public information volunteered by a NASA official will not be considered exclusive to any one media source and will be made available to other sources, if requested.

(a) Release of classified information in any form (e.g., documents, through interviews, audio/visual, etc.) to the news media is prohibited. The disclosure of classified information to unauthorized individuals may be cause for prosecution and/or disciplinary action against the NASA employee involved. Ignorance of NASA policy and procedures regarding classified information does not release a NASA employee from responsibility for preventing any unauthorized release. See NPR 1600.1, Chapter 5, Section 5.23 for internal NASA guidance on management of classified information. For further guidance that applies to all agencies, see Executive Order 12958, as amended, "Classified National Security Information" and its implementing directive at 32 CFR Parts 2001 and 2004.

(b) Any attempt by news media representatives to obtain classified information will be reported through the Headquarters Office of Public Affairs or Installation Public Affairs Office to the Installation Security Office and Office of Security and Program Protection.

(c) For classified operations and/or programs managed under the auspices of a DD Form 254, "Contract Security Classification Specification," all inquiries concerning this activity will be responded to by the appropriate PAO official designated in Item 12 on the DD Form 254.

(d) For classified operations and/or information owned by other Government agencies (e.g., DOD, DOE, etc.), all inquiries will be referred to the appropriate Agency public affairs officer as established in written agreements.

(a) All NASA SBU information requires accountability and approval for release. Release of SBU information to unauthorized personnel is prohibited. Unauthorized release of SBU information may result in prosecution and/or disciplinary action. Ignorance of NASA policy and procedures regarding SBU information does not release a NASA employee from responsibility for unauthorized release. See NPR 1600.1, Chapter 5, Section 5.24 for guidance on identification, marking, accountability and release of NASA SBU information.

(b) Examples of SBU information include: proprietary information of others provided to NASA under nondisclosure or confidentiality agreement; source selection and bid and proposal information; information subject to export control under the International Traffic in Arms Regulations (ITAR) or the Export Administration Regulations (EAR); information subject to the Privacy Act of 1974; predecisional materials such as national space policy not yet publicly released; pending reorganization plans or sensitive travel itineraries; and information that could constitute an indicator of U.S. government intentions, capabilities, operations, or activities or otherwise threaten operations security.

(c) Upon request for access to information/material deemed SBU, coordination must be made with the information/material owner to determine if the information/material may be released. Other organizations that play a part in SBU information identification, accountability and release (e.g., General Counsel, External Relations, Procurement, etc.) must be consulted for assistance and/or concurrence prior to release.

(d) Requests for SBU information from other Government agencies must be referred to the respective Agency public affairs officer.

(a) NASA’s multimedia material, from all sources, will be made available to the information media, the public, and to all Agency Centers and contractor installations utilizing contemporary delivery methods and emerging digital technology.

(b) Centers will provide the media, the public, and as necessary, NASA Headquarters with:

Selected prints and original or duplicate files of news-oriented imagery and other digital multimedia material generated within their respective areas.

Selected video material in the highest quality format practical, which, in the opinion of the installations, would be appropriate for use as news feed material or features in pre-produced programs and other presentations.

Audio and/or video files of significant news developments and other events of historic or public interest.

Interactive multimedia features that can be incorporated into the Agency’s Internet portal for use by internal and external audiences, including the media and the general public.

(a) Releases of information involving NASA activities, views, programs, or projects involving another country or an international organization require prior coordination and approval by the Headquarters offices of External Relations and Public Affairs.

(b) NASA Centers and Headquarters offices will report all visits proposed by representatives of foreign news media to the public affairs officer for the Office of External Relations for appropriate handling consistent with all NASA policies and procedures.

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Elyna Niles-CarnesDec 22, 2025ArticleNASA’s Artemis II Orion spacecraft with its launch abort system is stacked atop the agency’s SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA’s Kennedy Space Center in Florida on Monday, Oct. 20, 2025. The spacecraft will carry NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen on a 10-day mission around the Moon and back in early 2026. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to NASA Kennedy’s Launch Complex 39B for the wet dress rehearsal.NASA/Kim ShiflettTeams at NASA’s Kennedy Space Center in Florida spent 2025 preparing the launch vehicle and its powerhouse SLS (Space Launch System) rocket to launch four astronauts around the Moon for Artemis II in early 2026. The center also celebrated milestones by conducting science experiments at the International Space Station to studying the Sun’s solar wind impacts on Earth to traveling to Mars in hopes of one day exploring the Red Planet in person.JANUARYNASA Kennedy Marks New Chapter for Florida Space IndustryKennedy Space Center Director Janet Petro and charter members of the Florida University Space Research Consortium sign a memorandum of understanding in research and development to assist with missions and contribute to NASA’s Moon to Mars exploration approach.From left: Jennifer Kunz, Associate Director, Technical, Kennedy Space Center; Kelvin Manning, Deputy Director, Kennedy Space Center; Dr. Kent Fuchs, Interim President, University of Florida; Janet Petro, Director, Kennedy Space Center; Jeanette Nuñez, Florida Lieutenant Governor; Dr. Alexander Cartwright, President, University of Central Florida; Dr. Barry Butler, President, Embry-Riddle Aeronautical University.NASA/Kim ShiffletFirefly Launches Blue Ghost Mission OneFirefly Aerospace launched Blue Ghost Mission One lunar lander with a suite of NASA scientific instruments on January 15, from Launch Complex 39A at NASA Kennedy. The lander and instruments landed March 2 on the Moon.Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Firefly Aerospace’s Blue Ghost Mission One lander soars upward after liftoff from Launch Complex 39A at NASA’s Kennedy Space Center in Florida on Wednesday, Jan. 15, 2025 as part of NASA’s CLPS (Commercial Lunar Payload Services) initiative.NASA/Cory S HustonFEBRUARYIntuitive Machines Launches to the MoonIntuitive Machines’ IM-2 Nova C lunar lander launched Feb. 26 from Launch Complex 39A, carrying NASA science and technology demonstrations to the Mons Mouton region of the Moon. IM-2 reached the surface of the Moon on March 6.Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Intuitive Machines’ Nova-C lunar lander (IM-2) soars upward after liftoff from Launch Complex 39A at NASA’s Kennedy Space Center in Florida at 7:16 p.m. EST Wednesday, Feb. 26, 2025, as part of NASA’s CLPS

(Commercial Lunar Payload Services) initiative.NASA/Frank MichauxMARCHNASA's SpaceX Crew-10 LaunchNASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov launched March 14 from Launch Complex 39A to the International Space Station for a five-month science mission.Members of NASA's SpaceX Crew-10, from left, Roscosmos cosmonaut Kirill Peskov, mission specialist; NASA astronauts Nichole Ayers, pilot and Anne McClain, commander; and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, mission specialist.SpaceXNASA's SPHEREx, PUNCH Missions LaunchA SpaceX Falcon 9 rocket launched on March 11, from Space Launch Complex 4 East at Vandenberg Space Force Base in California carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) and PUNCH (Polarimeter to Unify the Corona and Heliosphere) missions. NASA's Launch Services Program, based at NASA Kennedy managed the launch service for SPHEREx.NASA's SPHEREx observatory is installed in the Titan Thermal Vacuum (TVAC) test Chamber at BAE Systems in Boulder, Colorado, in June 2024. As part of the test setup, the spacecraft and photon shield are covered in multilayer insulation and blankets and surrounded by ground support equipment.Jet Propulsion LaboratoryNASA's SpaceX Crew-9 ReturnsNASA astronauts Nick Hague, Suni Williams, and Butch Wilmore were greeted by dolphins and recovery teams after their SpaceX Dragon spacecraft splashed down on March 18, off the coast of Tallahassee, Florida following their long-duration mission at the International Space Station.Support teams work around a SpaceX Dragon spacecraft shortly after it landed with NASA astronauts Nick Hague, Suni Williams, Butch Wilmore, and Roscosmos cosmonaut Aleksandr Gorbunov aboard in the water off the coast of Tallahassee, Florida, Tuesday, March 18, 2025. Hague, Gorbunov, Williams, and Wilmore are returning from a long-duration science expedition aboard the International Space Station.NASA/Keegan BarberNASA Causeway Bridge OpensThe Florida Department of Transportation opened the westbound portion of the NASA Causeway Bridge on March 19, completing construction in both directions spanning the Indian River Lagoon and connecting NASA Kennedy and Cape Canaveral Space Force Station to the mainland.Cars drive over the newly completed westbound portion (right side of photo) of the NASA Causeway Bridge leading away from NASA's Kennedy Space Center in Florida on Wednesday, March 19, 2025. The Florida Department of Transportation (FDOT) opened the span on Tuesday, March 18, 2025, alongside its twin on the eastbound side, which has accommodated traffic in both directions since FDOT opened it on June 9, 2023.NASA/Glenn BensonNASA Artemis Teams Complete URT-12Teams from NASA and the Department of War train during a week-long Underway Recovery Test-12 in March off the coast of California for Artemis II test flight crewmembers and the Orion spacecraft. The series of tests demonstrate and evaluate the processes, procedures, and hardware used in recovery operations for crewed lunar missions.Waves break inside USS Somerset as the Crew Module Test Article, a full scale mockup of the Orion spacecraft, is tethered during Underway Recovery Test-12 off the coast of California, Wednesday, March 26, 2025. During the test, NASA and Department of Defense teams are practicing to ensure recovery procedures are validated as NASA plans to send Artemis II astronauts around the Moon and splashdown in the Pacific Ocean.NASA/Joel KowskyAPRILNASA's SpaceX 32nd Commercial Resupply MissionA SpaceX Falcon 9 rocket and a Dragon spacecraft carrying nearly 6,700 pounds of scientific investigations, food, supplies, and equipment launched on April 21 from Launch Complex 39A to the International Space Station.The SpaceX Falcon 9 rocket carrying the Dragon spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Monday, April 21, on the company's 32nd commercial resupply services mission for the agency to the International Space Station.SpaceXJULYArtemis III Begins ProcessingNASA's Artemis III SLS engine section and boat-tail made the journey from the Space Systems Processing Facility at NASA Kennedy to the spaceport's Vehicle Assembly Building in July to complete integration and check-out testing. Beginning withthe Artemis III hardware, NASA moved certain operations to NASA Kennedy to streamline the manufacturing process and enable simultaneous production operations of two core stages.Teams from NASA's Kennedy Space Center in Florida integrate NASA's Artemis III SLS (Space Launch System) core stage engine section with its boat-tail inside the spaceport's Vehicle Assembly Building on Wednesday, July 30, 2025. The boat-tail is a fairing-like structure that protects the bottom end of the core stage, while the engine section is one the most complex and intricate parts of the rocket stage that will help power the Artemis missions to the Moon.NASA/Ronald BeardAUGUSTNASA's SpaceX Crew-11 LaunchesNASA astronauts Zena Cardman and Mike Fincke, JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonov launched aboard a SpaceX Dragon spacecraft and its Falcon 9 rocket on Aug. 1 from Launch Complex 39A bound for a long-duration mission to the International Space Station.NASA's SpaceX Crew-11 mission is the eleventh crew rotation mission of the SpaceX Dragon spacecraft and Falcon 9 rocket to the International Space Station as part of the agency's Commercial Crew Program.NASA/Joel KowskyNASA's SpaceX Crew-10 ReturnsNASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov became the first Commercial Crew to splash down in the Pacific Ocean off the coast of California on Aug. 9, completing their nearly five-month mission at the orbiting outpost as part of the agency's Commercial Crew Program.Roscosmos cosmonaut Kirill Peskov, left, NASA astronauts Nichole Ayers, Anne McClain, and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi returned after 147 days in space as part of Expedition 73 aboard the International Space Station.NASA/Keegan BarberNASA's SpaceX 33rd Commercial Resupply MissionA SpaceX Falcon 9 launched the company’s Dragon spacecraft carrying more than 5,000 pounds of food, crew supplies, science investigations, spacewalk equipment, and more to the space station on Aug. 24 from Launch Complex 39A.A SpaceX Dragon cargo spacecraft with its nosecone open and carrying over 5,000 pounds of science, supplies, and hardware for NASA’s SpaceX CRS-33 mission approaches the International Space Station for an automated docking to the Harmony module’s forward port.NASAOrion Tested, Stacked With HardwareTeams transported NASA’s Orion spacecraft from Kennedy’s Multi-Payload Processing Facility to the Launch Abort System Facility in August where crews integrated the 44-foot-tall launch abort system. The Orion spacecraft will send NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen around the Moon for the Artemis II mission in early 2026. The launch abort system is designed to carry the crew to safety in the event of an emergency atop the SLS.The launch abort tower on NASA’s Artemis II Orion spacecraft is pictured inside the Launch Abort System Facility at the agency’s Kennedy Space Center in Florida on Wednesday, Aug. 27, 2025, after teams with NASA’s Exploration Ground Systems Program installed the tower on Wednesday, Aug. 20, 2025. Positioned at the top of Orion, the 44-foot-tall launch abort system is designed to carry the crew to safety in the event of an emergency during launch or ascent, with its three solid rocket motors working together to propel Orion – and astronauts inside – away from the rocket for a safe landing in the ocean, or detach from the spacecraft when it is no longer needed. The final step to complete integration will be the installation of the ogive fairings, which are four protective panels that will shield the crew module from the severe vibrations and sounds it will experience during launch.NASA/Cory HustonSEPTEMBERNASA Launches IMAP MissionNASA’s IMAP (Interstellar Mapping and Acceleration Probe) launched from Launch Complex 39A on Sept. 24, to help researchers better understand the boundary of the heliosphere, a huge bubble created by the Sun surrounding and protecting our solar system.A SpaceX Falcon 9 rocket carrying NASA’s IMAP (Interstellar Mapping and Acceleration Probe), the agency’s Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration’s (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA’s Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.BAE Systems/Benjamin FryNASA’s Northrop Grumman Commercial Resupply MissionA Northrop Grumman Cygnus XL spacecraft atop a SpaceX Falcon 9 rocket lifted off from Launch Complex 39A to the International Space Station delivering NASA science investigations, supplies, and equipment as part of the agency’s partnership to resupply the orbiting laboratory.Northrop Grumman’s Cygnus XL cargo craft, carrying over 11,000 pounds of new science and supplies for the Expedition 73 crew, is pictured in the grips of the International Space Station's Canadarm2 robotic arm following its capture. Both spacecraft were orbiting 257 miles above Tanzania. Cygnus XL is Northrop Grumman's expanded version of its previous Cygnus cargo craft increasing its payload capacity and pressurized cargo volume.NASAOCTOBEROrion Integrated With SLS RocketTeams stacked NASA's Orion spacecraft with its launch abort system on the agency's SLS rocket in High Bay 3 of the Vehicle Assembly Building at NASA Kennedy on Oct. 20 for the agency's Artemis II mission. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to Launch Complex 39B for the wet dress rehearsal.NASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025.NASA/Kim ShiflettNOVEMBERNASA's ESCAPADE Begins Journey to MarsNASA's ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) twin spacecraft launched aboard Blue Origin's New Glenn rocket on Nov. 13 from Launch Complex 36 at Cape Canaveral Space Force Station. Its twin orbiters will take simultaneous observations from different locations around Mars to reveal how the solar wind interacts with Mars' magnetic environment and how this interaction drives the planet's atmospheric escape.Near Cape Canaveral Lighthouse, Blue Origin's New Glenn rocket carrying NASA's twin ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPADE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.Blue OriginNASA, European Partners Launch Sea SatelliteA SpaceX Falcon 9 rocket carrying the U.S.-European Sentinel-6B satellite launched at Nov. 16 from Space Launch Complex 4 East at Vandenberg Space Force Base in California. Sentinel-6B will observe Earth's ocean, measuring sea levels to improve weather forecasts and flood predictions, safeguard public safety, benefit commercial industry, and protect coastal infrastructure.A SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.SpaceXDECEMBERNASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA astronaut Jeremy Hansen participated in a dry dress rehearsal at NASA Kennedy on Dec. 20 to mimic launch day operations for the Artemis II launch. The crew donned their spacesuits, exited the Neil A. Operations and Checkout Building, and took the journey to the Vehicle Assembly Building, up the mobile launcher to the crew access arm, and entered the Orion spacecraft that will take them around the Moon and back to Earth.From right to left, NASA astronauts Christina Koch, mission specialist; Reid Wiseman, commander; Victor Glover, pilot; and CSA (Canadian Space Agency) astronaut Jeremy Hansen, mission specialist are seen as they depart the Neil A. Armstrong Operations and Checkout Building to board their Orion spacecraft atop NASA's Space Launch System rocket inside the Vehicle Assembly Building as part of the Artemis II countdown demonstration test, Saturday, Dec. 20, 2025, at NASA's Kennedy Space Center in Florida. For this operation, the Artemis II crew and launch teams are simulating the launch day timeline including suit-up, walkout, and spacecraft ingress and egress. Through the Artemis campaign, NASA will send astronauts to explore the Moon for scientific discovery, economic benefits, and to build the foundation for the first crewed missions to Mars, for the benefit of all.NASA/Aubrey Gemignani

Elyna Niles-CarnesDec 22, 2025Article

Elyna Niles-Carnes

NASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025. The spacecraft will carry NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen on a 10-day mission around the Moon and back in early 2026. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to NASA Kennedy's Launch Complex 39B for the wet dress rehearsal.NASA/Kim ShiflettTeams at NASA's Kennedy Space Center in Florida spent 2025 preparing the launch vehicle and its powerhouse SLS (Space Launch System) rocket to launch four astronauts around the Moon for Artemis II in early 2026. 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During the test, NASA and Department of Defense teams are practicing to ensure recovery procedures are validated as NASA plans to send Artemis II astronauts around the Moon and splashdown in the Pacific Ocean.NASA/Joel KowskyAPRILNASA's SpaceX 32nd Commercial Resupply MissionA SpaceX Falcon 9 rocket and a Dragon spacecraft carrying nearly 6,700 pounds of scientific investigations, food, supplies, and equipment launched on April 21 from Launch Complex 39A to the International Space Station.The SpaceX Falcon 9 rocket carrying the Dragon spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Monday, April 21, on the company's 32nd commercial resupply services mission for the agency to the International Space Station.SpaceXJULYArtemis III Begins ProcessingNASA's Artemis III SLS engine section and boat-tail made the journey from the Space Systems Processing Facility at NASA Kennedy to the spaceport's Vehicle Assembly Building in July to complete integration and check-out testing. Beginning withthe Artemis III hardware, NASA moved certain operations to NASA Kennedy to streamline the manufacturing process and enable simultaneous production operations of two core stages.Teams from NASA's Kennedy Space Center in Florida integrate NASA's Artemis III SLS (Space Launch System) core stage engine section with its boat-tail inside the spaceport's Vehicle Assembly Building on Wednesday, July 30, 2025. The boat-tail is a fairing-like structure that protects the bottom end of the core stage, while the engine section is one the most complex and intricate parts of the rocket stage that will help power the Artemis missions to the Moon.NASA/Ronald BeardAUGUSTNASA's SpaceX Crew-11 LaunchesNASA astronauts Zena Cardman and Mike Fincke, JAXA (Japan

Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonov launched aboard a SpaceX Dragon spacecraft and its Falcon 9 rocket on Aug. 1 from Launch Complex 39A bound for a long-duration mission to the International Space Station.NASA's SpaceX Crew-11 mission is the eleventh crew rotation mission of the SpaceX Dragon spacecraft and Falcon 9 rocket to the International Space Station as part of the agency's Commercial Crew Program.NASA/Joel KowskyNASA's SpaceX Crew-10 ReturnsNASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov became the first Commercial Crew to splash down in the Pacific Ocean off the coast of California on Aug. 9, completing their nearly five-month mission at the orbiting outpost as part of the agency's Commercial Crew Program.Roscosmos cosmonaut Kirill Peskov, left, NASA astronauts Nichole Ayers, Anne McClain, and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi returned after 147 days in space as part of Expedition 73 aboard the International Space Station.NASA/Keegan BarberNASA's SpaceX 33rd Commercial Resupply MissionA SpaceX Falcon 9 launched the company's Dragon spacecraft carrying more than 5,000 pounds of food, crew supplies, science investigations, spacewalk equipment, and more to the space station on Aug. 24 from Launch Complex 39A.A SpaceX Dragon cargo spacecraft with its nosecone open and carrying over 5,000 pounds of science, supplies, and hardware for NASA's SpaceX CRS-33 mission approaches the International Space Station for an automated docking to the Harmony module's forward port.NASAOrion Tested, Stacked With HardwareTeams transported NASA's Orion spacecraft from Kennedy's Multi-Payload Processing Facility to the Launch Abort System Facility in August where crews integrated the 44-foot-tall launch abort system. The Orion spacecraft will send NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen around the Moon for the Artemis II mission in early 2026. The launch abort system is designed to carry the crew to safety in the event of an emergency atop the SLS.The launch abort tower on NASA's Artemis II Orion spacecraft is pictured inside the Launch Abort System Facility at the agency's Kennedy Space Center in Florida on Wednesday, Aug. 27, 2025, after teams with NASA's Exploration Ground Systems Program installed the tower on Wednesday, Aug. 20, 2025. Positioned at the top of Orion, the 44-foot-tall launch abort system is designed to carry the crew to safety in the event of an emergency during launch or ascent, with its three solid rocket motors working together to propel Orion – and astronauts inside – away from the rocket for a safe landing in the ocean, or detach from the spacecraft when it is no longer needed. The final step to complete integration will be the installation of the ogive fairings, which are four protective panels that will shield the crew module from the severe vibrations and sounds it will experience during launch.NASA/Cory HustonSEPTEMBERNASA Launches IMAP MissionNASA's IMAP (Interstellar Mapping and Acceleration Probe) launched from Launch

Complex 39A on Sept. 24, to help researchers better understand the boundary of the heliosphere, a huge bubble created by the Sun surrounding and protecting our solar system.A SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.BAE Systems/Benjamin FryNASA's Northrop Grumman Commercial Resupply MissionA Northrop Grumman Cygnus XL spacecraft atop a SpaceX Falcon 9 rocket lifted off from Launch Complex 39A to the International Space Station delivering NASA science investigations, supplies, and equipment as part of the agency's partnership to resupply the orbiting laboratory.Northrop Grumman's Cygnus XL cargo craft, carrying over 11,000 pounds of new science and supplies for the Expedition 73 crew, is pictured in the grips of the International Space Station's Canadarm2 robotic arm following its capture. Both spacecraft were orbiting 257 miles above Tanzania. Cygnus XL is Northrop Grumman's expanded version of its previous Cygnus cargo craft increasing its payload capacity and pressurized cargo volume.NASAOCTOBEROrion Integrated With SLS RocketTeams stacked NASA's Orion spacecraft with its launch abort system on the agency's SLS rocket in High Bay 3 of the Vehicle Assembly Building at NASA Kennedy on Oct. 20 for the agency's Artemis II mission. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to Launch Complex 39B for the wet dress rehearsal.NASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025.NASA/Kim ShiflettNOVEMBERNASA's ESCAPADE Begins Journey to MarsNASA's ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) twin spacecraft launched aboard Blue Origin's New Glenn rocket on Nov. 13 from Launch Complex 36 at Cape Canaveral Space Force Station. Its twin orbiters will take simultaneous observations from different locations around Mars to reveal how the solar wind interacts with Mars' magnetic environment and how this interaction drives the planet's atmospheric escape.Near Cape Canaveral Lighthouse, Blue Origin's New Glenn rocket carrying NASA's twin ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPADE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.Blue OriginNASA, European Partners Launch Sea SatelliteA SpaceX Falcon 9 rocket carrying the U.S.-European Sentinel-6B satellite launched at Nov. 16 from Space Launch Complex 4 East at Vandenberg Space Force Base in California. Sentinel-6B will observe Earth's ocean, measuring sea levels to improve weather forecasts and flood predictions, safeguard public safety, benefit commercial industry, and protect coastal infrastructure.A SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.SpaceXDECEMBERNASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA astronaut Jeremy Hansen participated in a dry dress rehearsal at NASA Kennedy on Dec. 20 to mimic launch day operations for the Artemis II launch. The crew donned their spacesuits, exited the Neil A. Operations and Checkout Building, and took the journey to the Vehicle Assembly Building, up the mobile launcher to the crew access arm, and entered the Orion spacecraft that will take them around the Moon and back to Earth.From right to left, NASA astronauts Christina Koch, mission specialist; Reid Wiseman, commander; Victor Glover, pilot; and CSA (Canadian Space Agency) astronaut Jeremy Hansen, mission specialist are seen as they depart the Neil A. Armstrong Operations and Checkout Building to board their Orion spacecraft atop NASA's Space Launch System rocket inside the Vehicle Assembly Building as part of the Artemis II countdown demonstration test, Saturday, Dec. 20, 2025, at NASA's Kennedy Space Center in Florida. For this operation, the Artemis II crew and launch teams are simulating the launch day timeline including suit-up, walkout, and spacecraft ingress and egress. Through the Artemis campaign, NASA will send astronauts to explore the Moon for scientific discovery, economic benefits, and to build the foundation for the first crewed missions to Mars, for the benefit of all.NASA/Aubrey Gemignani

NASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025. The spacecraft will carry NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen on a 10-day mission around the Moon and back in early 2026. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to NASA Kennedy's Launch Complex 39B for the wet dress rehearsal.NASA/Kim Shiflett

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Teams at NASA's Kennedy Space Center in Florida spent 2025 preparing the launch vehicle and its powerhouse SLS (Space Launch System) rocket to launch four astronauts around the Moon for Artemis II in early 2026. The center also celebrated milestones by conducting science experiments at the International Space Station to studying the Sun's solar wind impacts on Earth to traveling to Mars in hopes of one day exploring the Red Planet in person.

JANUARYNASA Kennedy Marks New Chapter for Florida Space Industry

Kennedy Space Center Director Janet Petro and charter members of the Florida University Space Research Consortium sign a memorandum of understanding in research and development to assist with missions and contribute to NASA's Moon to Mars exploration approach.

From left: Jennifer Kunz, Associate Director, Technical, Kennedy Space Center; Kelvin Manning, Deputy Director, Kennedy Space Center; Dr. Kent Fuchs, Interim President, University of Florida; Janet Petro, Director, Kennedy Space Center; Jeanette Nuñez, Florida Lieutenant Governor; Dr. Alexander Cartwright, President, University of Central Florida; Dr. Barry Butler, President, Embry-Riddle Aeronautical University.NASA/Kim Shifflet

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NASA/Kim Shifflet

Firefly Launches Blue Ghost Mission One

Firefly Aerospace launched Blue Ghost Mission One lunar lander with a suite of NASA scientific instruments on January 15, from Launch Complex 39A at NASA Kennedy. The lander and instruments landed March 2 on the Moon.

Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Firefly Aerospace's Blue Ghost Mission One lander soars upward after liftoff from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Wednesday, Jan. 15, 2025 as part of NASA's CLPS (Commercial Lunar Payload Services) initiative.NASA/Cory S Huston

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NASA/Cory S Huston

FEBRUARYIntuitive Machines Launches to the Moon

Intuitive Machines' IM-2 Nova C lunar lander launched Feb. 26 from Launch Complex 39A, carrying NASA science and technology demonstrations to the Mons Mouton region of the Moon. IM-2 reached the surface of the Moon on March 6.

Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Intuitive Machines' Nova-C lunar lander (IM-2) soars upward after liftoff from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:16 p.m. EST Wednesday, Feb. 26, 2025, as part of NASA's CLPS (Commercial Lunar Payload Services) initiative.NASA/Frank Michaux

Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Intuitive Machines' Nova-C lunar lander (IM-2) soars upward after liftoff from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:16 p.m. EST Wednesday, Feb. 26, 2025, as part of NASA's CLPS (Commercial Lunar Payload Services) initiative.

MARCHNASA's SpaceX Crew-10 Launch

NASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov launched March 14 from Launch Complex 39A to the International Space Station for a five-month science mission.

Members of NASA's SpaceX Crew-10, from left, Roscosmos cosmonaut Kirill Peskov, mission specialist; NASA astronauts Nichole Ayers, pilot and Anne McClain, commander; and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, mission specialist.SpaceX

Members of NASA's SpaceX Crew-10, from left, Roscosmos cosmonaut Kirill Peskov, mission specialist; NASA astronauts Nichole Ayers, pilot and Anne McClain, commander; and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, mission specialist.

NASA's SPHEREx, PUNCH Missions Launch

A SpaceX Falcon 9 rocket launched on March 11, from Space Launch Complex 4 East at Vandenberg Space Force Base in California carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) and PUNCH (Polarimeter to Unify the Corona and Heliosphere) missions. NASA's Launch Services Program, based at NASA Kennedy managed the launch service for SPHEREx.

NASA's SPHEREx observatory is installed in the Titan Thermal Vacuum (TVAC) test Chamber at BAE Systems in Boulder, Colorado, in June 2024. As part of the test setup, the spacecraft and photon shield are covered in multilayer insulation and blankets and surrounded by ground support equipment.Jet Propulsion Laboratory

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NASA's SpaceX Crew-9 Returns

NASA astronauts Nick Hague, Suni Williams, and Butch Wilmore were greeted by dolphins and recovery teams after their SpaceX Dragon spacecraft splashed down on March 18, off the coast of Tallahassee, Florida following their long-duration mission at the International Space Station.

Support teams work around a SpaceX Dragon spacecraft shortly after it landed with NASA astronauts Nick Hague, Suni Williams, Butch Wilmore, and Roscosmos cosmonaut Aleksandr Gorbunov aboard in the water off the coast of Tallahassee, Florida, Tuesday, March 18, 2025. Hague, Gorbunov, Williams, and Wilmore are returning from a long-duration science expedition aboard the International Space Station.NASA/Keegan Barber

Support teams work around a SpaceX Dragon spacecraft shortly after it landed with NASA astronauts Nick Hague, Suni Williams, Butch Wilmore, and Roscosmos cosmonaut Aleksandr Gorbunov aboard in the water off the coast of Tallahassee, Florida, Tuesday, March 18, 2025. Hague, Gorbunov, Williams, and Wilmore are returning from a long-duration science expedition aboard the International Space Station.

NASA Causeway Bridge Opens

The Florida Department of Transportation opened the westbound portion of the NASA Causeway Bridge on March 19, completing construction in both directions spanning the Indian River Lagoon and connecting NASA Kennedy and Cape Canaveral Space Force Station to the mainland.

Cars drive over the newly completed westbound portion (right side of photo) of the NASA Causeway Bridge leading away from NASA's Kennedy Space Center in Florida on Wednesday, March 19, 2025. The Florida Department of Transportation (FDOT) opened the span on Tuesday, March 18, 2025, alongside its twin on the eastbound side, which has accommodated traffic in both directions since FDOT opened it on June 9, 2023.NASA/Glenn Benson

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NASA/Glenn Benson

NASA Artemis Teams Complete URT-12

Teams from NASA and the Department of War train during a week-long Underway Recovery Test-12 in March off the coast of California for Artemis II test flight crewmembers and the Orion spacecraft. The series of tests demonstrate and evaluate the processes, procedures, and hardware used in recovery operations for crewed lunar missions.

Waves break inside USS Somerset as the Crew Module Test Article, a full scale mockup of the Orion spacecraft, is tethered during Underway Recovery Test-12 off the coast of California, Wednesday, March 26, 2025. During the test, NASA and Department of Defense teams are practicing to ensure recovery procedures are validated as NASA plans to send Artemis II astronauts around the Moon and splashdown in the Pacific Ocean.NASA/Joel Kowsky

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APRILNASA's SpaceX 32nd Commercial Resupply Mission

A SpaceX Falcon 9 rocket and a Dragon spacecraft carrying nearly 6,700 pounds of scientific investigations, food, supplies, and equipment launched on April 21 from Launch Complex 39A to the International Space Station.

The SpaceX Falcon 9 rocket carrying the Dragon spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Monday, April 21, on the company's

32nd commercial resupply services mission for the agency to the International Space Station.SpaceX

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JULYArtemis III Begins Processing

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Teams from NASA's Kennedy Space Center in Florida integrate NASA's Artemis III SLS (Space Launch System) core stage engine section with its boat-tail inside the spaceport's Vehicle Assembly Building on Wednesday, July 30, 2025. The boat-tail is a fairing-like structure that protects the bottom end of the core stage, while the engine section is one the most complex and intricate parts of the rocket stage that will help power the Artemis missions to the Moon.NASA/Ronald Beard

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NASA/Ronald Beard

AUGUSTNASA's SpaceX Crew-11 Launches

NASA astronauts Zena Cardman and Mike Fincke, JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonov launched aboard a SpaceX Dragon spacecraft and its Falcon 9 rocket on Aug. 1 from Launch Complex 39A bound for a long-duration mission to the International Space Station.

NASA's SpaceX Crew-11 mission is the eleventh crew rotation mission of the SpaceX Dragon spacecraft and Falcon 9 rocket to the International Space Station as part of the agency's Commercial Crew Program.NASA/Joel Kowsky

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NASA’s SpaceX Crew-10 Returns

NASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov became the first Commercial Crew to splash down in the Pacific Ocean off the coast of California on Aug. 9, completing their nearly five-month mission at the orbiting outpost as part of the agency’s Commercial Crew Program.

Roscosmos cosmonaut Kirill Peskov, left, NASA astronauts Nichole Ayers, Anne McClain, and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi returned after 147 days in space as part of Expedition 73 aboard the International Space Station.NASA/Keegan Barber

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NASA’s SpaceX 33rd Commercial Resupply Mission

A SpaceX Falcon 9 launched the company’s Dragon spacecraft carrying more than 5,000 pounds of food, crew supplies, science investigations, spacewalk equipment, and more to the space station on Aug. 24 from Launch Complex 39A.

A SpaceX Dragon cargo spacecraft with its nosecone open and carrying over 5,000 pounds of science, supplies, and hardware for NASA’s SpaceX CRS-33 mission approaches the International Space Station for an automated docking to the Harmony module’s forward port.NASA

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Orion Tested, Stacked With Hardware

Teams transported NASA’s Orion spacecraft from Kennedy’s Multi-Payload Processing Facility to the Launch Abort System Facility in August where crews integrated the 44-foot-tall launch abort system. The Orion spacecraft will send NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen around the Moon for the Artemis II mission in early 2026. The launch abort system is designed to carry the crew to safety in the event of an emergency atop the SLS.

The launch abort tower on NASA's Artemis II Orion spacecraft is pictured inside the Launch Abort System Facility at the agency's Kennedy Space Center in Florida on Wednesday, Aug. 27, 2025, after teams with NASA's Exploration Ground Systems Program installed the tower on Wednesday, Aug. 20, 2025. Positioned at the top of Orion, the 44-foot-tall launch abort system is designed to carry the crew to safety in the event of an emergency during launch or ascent, with its three solid rocket motors working together to propel Orion – and astronauts inside – away from the rocket for a safe landing in the ocean, or detach from the spacecraft when it is no longer needed. The final step to complete integration will be the installation of the ogive fairings, which are four protective panels that will shield the crew module from the severe vibrations and sounds it will experience during launch.NASA/Cory Huston

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BAE Systems/Benjamin Fry

NASA's Northrop Grumman Commercial Resupply Mission

A Northrop Grumman Cygnus XL spacecraft atop a SpaceX Falcon 9 rocket lifted off from Launch Complex 39A to the International Space Station delivering NASA science investigations, supplies, and equipment as part of the agency's partnership to resupply the orbiting laboratory.

Northrop Grumman's Cygnus XL cargo craft, carrying over 11,000 pounds of new science and supplies for the Expedition 73 crew, is pictured in the grips of the International Space Station's Canadarm2 robotic arm following its capture. Both spacecraft were orbiting 257 miles above Tanzania. Cygnus XL is Northrop Grumman's expanded version of its previous Cygnus cargo craft increasing its payload capacity and pressurized cargo volume.NASA

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OCTOBEROrion Integrated With SLS Rocket

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NASA’s Artemis II Orion spacecraft with its launch abort system is stacked atop the agency’s SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA’s Kennedy Space Center in Florida on Monday, Oct. 20, 2025.NASA/Kim Shiflett

NASA’s Artemis II Orion spacecraft with its launch abort system is stacked atop the agency’s SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA’s Kennedy Space Center in Florida on Monday, Oct. 20, 2025.

NOVEMBERNASA’s ESCAPADE Begins Journey to Mars

NASA’s ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) twin spacecraft launched aboard Blue Origin’s New Glenn rocket on Nov. 13 from Launch Complex 36 at Cape Canaveral Space Force Station. Its twin orbiters will take simultaneous observations from different locations around Mars to reveal how the solar wind interacts with Mars’ magnetic environment and how this interaction drives the planet’s atmospheric escape.

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Blue Origin

NASA, European Partners Launch Sea Satellite

A SpaceX Falcon 9 rocket carrying the U.S.-European Sentinel-6B satellite launched at Nov. 16 from Space Launch Complex 4 East at Vandenberg Space Force Base in California. Sentinel-6B will observe Earth’s ocean, measuring sea levels to improve weather forecasts and flood predictions, safeguard public safety, benefit commercial industry, and protect coastal infrastructure.

A SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.SpaceX

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DECEMBER

NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA astronaut Jeremy Hansen participated in a dry dress rehearsal at NASA Kennedy on Dec. 20 to mimic launch day operations for the Artemis II launch. The crew donned their spacesuits, exited the Neil A. Operations and Checkout Building, and took the journey to the Vehicle Assembly Building, up the mobile launcher to the crew access arm, and entered the Orion spacecraft that will take them around the Moon and back to Earth.

From right to left, NASA astronauts Christina Koch, mission specialist; Reid Wiseman, commander; Victor Glover, pilot; and CSA (Canadian Space Agency) astronaut Jeremy Hansen, mission specialist are seen as they depart the Neil A. Armstrong Operations and Checkout Building to board their Orion spacecraft atop NASA's Space Launch System rocket inside the Vehicle Assembly Building as part of the Artemis II countdown demonstration test, Saturday, Dec. 20, 2025, at NASA's Kennedy Space Center in Florida. For this operation, the Artemis II crew and launch teams are simulating the launch day timeline including suit-up, walkout, and spacecraft ingress and egress. Through the Artemis campaign, NASA will send astronauts to explore the Moon for scientific discovery, economic benefits, and to build the foundation for the first crewed missions to Mars, for the benefit of all.NASA/Aubrey Gemignani

From right to left, NASA astronauts Christina Koch, mission specialist; Reid Wiseman, commander; Victor Glover, pilot; and CSA (Canadian Space Agency) astronaut Jeremy Hansen, mission specialist are seen as they depart the Neil A. Armstrong Operations and Checkout Building to board their Orion spacecraft atop NASA's Space Launch System rocket inside the Vehicle Assembly Building as part of the Artemis II countdown demonstration test, Saturday, Dec. 20, 2025, at NASA's Kennedy Space Center in Florida. For this operation, the Artemis II crew and launch teams are simulating the launch day timeline including suit-up, walkout, and spacecraft ingress and egress. Through the Artemis campaign, NASA will send astronauts to explore the Moon for scientific discovery, economic benefits, and to build the foundation for the first crewed missions to Mars, for the benefit of all.

NASA/Aubrey Gemignani

URL: https://www.nasa.gov/kennedy/launch-services-program/

TITLE: Launch Services Program

Launch Services ProgramLaunch Services ProgramNASA's Launch Services Program (LSP) is responsible for launching rockets delivering spacecraft that observe the Earth, visit other planets and explore the universe – from weather satellites to telescopes to Mars rovers and more.LSP functions as a broker, matching spacecraft with the best-suited rockets, managing the launch process, providing support from pre-mission planning to post-launch. LSP helps implement NASA's policy of a mixed-fleet launch strategy, which uses both existing and emerging domestic launch capabilities to assure access to space.Learn Moreabout Launch Services ProgramLatest NewsMore LSP NewsBlog2 Min ReadNASA's Pandora Satellite Acquires SignalBlog1 Min ReadNASA's Exoplanet Observing Satellite Separated From RocketBlog1 Min ReadLiftoff of NASA's Newest Planet-Observing SatelliteBlog3 Min ReadNASA's Pandora Mission, CubeSats Ready for Flight3 Min ReadWelcome to Launch Day for NASA's Pandora Mission, CubeSatsBlog6 Min ReadNASA Kennedy Top 20 Stories of 2025Article4 Min ReadNASA Cost-Saving Technology Demo is Ready for LaunchBlog5 Min ReadNASA, SpaceX Launch US-European Satellite to Monitor Earth's OceansNews ReleaseRocketsFirefly Aerospace AlphaUnited Launch Alliance Atlas VRocket Lab ElectronSpaceX Falcon 9SpaceX Falcon HeavyBlue Origin New GlennNorthrop Grumman PegasusXLUnited Launch Alliance VulcanFirefly Aerospace AlphaFirefly's small-lift rocket, Alpha, is equipped to launch more than 1,000 kg to low Earth orbit. Alpha can be launched at Firefly's launch facilities at the Vandenberg Space Force Base in California and new launch capabilities coming soon at the Mid-Atlantic Regional Spaceport (MARS) on Wallops Island, Virginia as early as 2025.Read MoreFirefly Aerospace's "Noise of Summer" launched eight CubeSats for NASA on its Venture-Class Launch Services Demonstration 2 contract in July of 2024.Firefly Aerospace/Trevor MahlmannUnited Launch Alliance Atlas VAtlas V uses a standard common core booster, up to five solid rocket boosters (SRBs), a Centaur upper stage in a single- or dual-engine configuration, and one of several sizes of payload fairings. Atlas V is on NASA's Launch Services II contract.Read MoreA United Launch Alliance Atlas V 541 rocket lifts off from

Space Launch Complex 41 at Cape Canaveral Air Force Station in Florida on July 30, 2020, at 7:50 a.m. EDT, carrying NASA's Mars Perseverance rover and Ingenuity helicopter.NASA/Tony Gray and Tim PowersRocket Lab ElectronRocket Lab's Electron is a reusable orbital-class small rocket. Capturing and reflying Electron's first stage enables higher launch frequency without expanding production and lowers launch costs. Rocket Lab has three launch pads at two launch sites, including two launch pads at a private orbital launch site located in New Zealand and a third pad at NASA's Wallops Flight Facility in Virginia.Read MoreA Rocket Lab Electron rocket lifts off Launch Complex 1, Pad B, in Māhia, New Zealand on May 8 at 1 p.m. New Zealand time (May 7 at 9 p.m. EDT), carrying two NASA CubeSats designed to study tropical cyclones, including hurricanes and typhoons. NASA's Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) CubeSats will provide data on temperature, precipitation, water vapor, and clouds by measuring microwave frequencies, providing insight into storm formation and intensification.Rocket LabSpaceX Falcon 9Falcon 9 is a reusable, two-stage rocket designed and manufactured by SpaceX for the reliable and safe transport of people and payloads into Earth orbit and beyond. Falcon 9 is on NASA's Launch Services II contract.Read MoreNASA's TRACERS (Tandem Reconnection and Cusp Electrodynamics Reconnaissance Satellites) mission launches at 11:13 a.m. PDT (2:13 p.m. EDT) on Wednesday, July 23, 2025, atop a SpaceX Falcon 9 rocket at Space Launch Complex 4 East at Vandenberg Space Force Base in California. The TRACERS mission will study magnetic reconnection around Earth — a process in which electrically charged plasmas exchange energy in the atmosphere — to understand how the Sun's solar wind interacts with the magnetosphere, Earth's protective magnetic shield.SpaceXSpaceX Falcon HeavyFalcon Heavy is composed of three reusable Falcon 9 nine-engine cores whose 27 Merlin engines together generate more than 5 million pounds of thrust at liftoff, equal to approximately eighteen 747 aircraft. As one of the world's most powerful operational rockets, Falcon Heavy can lift nearly 64 metric tons (141,000 lbs.) to orbit. Falcon Heavy is on NASA's Launch Services II contract.Read MoreA SpaceX Falcon Heavy rocket carrying NASA's Europa Clipper spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 12:06 p.m. EDT on Monday, Oct. 14, 2024. After launch, the spacecraft plans to fly by Mars in February 2025, then back by Earth in December 2026, using the gravity of each planet to increase its momentum. With help of these "gravity assists," Europa Clipper will achieve the velocity needed to reach Jupiter in April 2030.SpaceXBlue Origin New GlennNamed after John Glenn, the first American astronaut to orbit Earth, New Glenn is a single-configuration, heavy-lift orbital launch vehicle with a first stage designed for a minimum of 25 flights. Powered by seven Blue Engine 4 (BE-4) engines, it can carry more than 13 metric tons to geostationary transfer orbit and 45 metric tons to low Earth orbit. New Glenn is on NASA's Launch Services II contract.Read MoreBlue Origin's New Glenn rocket carrying NASA's twin ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. 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The rocket is attached beneath the company's L-1011 Stargazer aircraft which launches the rocket once in the air.NASAUnited Launch Alliance VulcanVulcan is United Launch Alliance's next generation rocket available in four standard offering configurations including zero, two, four and six solid rocket booster variants. ULA will launch Vulcan from both Cape Canaveral Space Force Station in Florida and Vandenberg Space Force Base in California. Vulcan is on NASA's Launch Services II contract.Read MoreUnited Launch Alliance's Vulcan rocket is rolled out of the Vertical Integration Facility at Space Launch Complex 41 on Cape Canaveral Space Force Station in Florida in preparation for a commercial launch in January 2024.NASA/Cory S HustonLaunch SitesKennedy Space CenterCape CanaveralVandenbergWallopsKwajalein AtollAdditional SitesKennedy Space CenterOne of two primary launch sites for NASA's Launch Vehicles.Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit.View SiteA SpaceX Falcon 9 rocket with NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft atop stands vertical at Launch Complex 39A as the sun sets on Monday, Sept. 22, 2025, at the agency's Kennedy Space Center in Florida. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.BAE Systems/Benjamin FryCape Canaveral Space Force StationOne of two primary launch sites for NASA's Launch Vehicles.Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit.View SiteA United Launch Alliance Atlas V rocket with the Lucy spacecraft aboard is seen in this 2 minute and 30 second exposure photograph as it launches from Space Launch Complex 41, Saturday, Oct. 16, 2021, at Cape Canaveral Space Force Station in Florida. Lucy will be the first spacecraft to study Jupiter's Trojan Asteroids. Like the mission's namesake – the fossilized human ancestor, "Lucy," whose skeleton provided unique insight into humanity's evolution – Lucy will revolutionize our knowledge of planetary origins and the formation of the solar system.NASA/Bill IngallsVandenberg Space Force BaseOne of two primary launch sites for NASA's Launch Vehicles.Located along California's central coast between Los Angeles and San Francisco, Vandenberg is preferred for spacecraft requiring a north-south, or polar, orbit.Visit SiteA SpaceX Falcon 9 rocket carrying NASA's R5-S7 (Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company's Transporter-15 mission lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 10:44 a.m. PST Friday, Nov. 28, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users.SpaceXWallops Flight FacilityLocated along the Atlantic Coast on Wallops Island, Virginia.Operated by Goddard Space Flight Center in Greenbelt, Maryland, Wallops is NASA's principal facility for suborbital research programs.View SiteA Northrop Grumman Antares rocket carrying a Cygnus resupply spacecraft is seen on the Mid-Atlantic Regional Spaceport's Pad-0A, Thursday, October 1, 2020, at NASA's Wallops Flight Facility in Virginia. Northrop Grumman's 14th contracted cargo resupply mission with NASA to the International Space Station will deliver nearly 8,000 pounds of science and research, crew supplies and vehicle hardware to the orbital laboratory and its crew. The CRS-14 Cygnus spacecraft is named after the first female astronaut of Indian descent, Kalpana Chawla.NASA/Patrick BlackReagan Test Site, Kwajalein AtollAn additional Expendable Launch Vehicle launch location in the Republic of the Marshall Islands in the North Pacific.The Kwajalein site is located between Hawaii and Australia and is used for missions requiring equatorial orbits and low inclinations.Visit SiteOrbital Science Corp.'s L-1011 aircraft "Stargazer" flies over the runway on Kwajalein Atoll with the company's Pegasus rocket slung underneath. Kwajalein is part of the Marshall Islands chain in the Pacific Ocean. It's also part of the Reagan Test Site and used for launches of NASA, commercial and military missions.NASAAdditional Launch SitesAdditional launch sites specific to LSP's VADR launches include Rocket Lab's Launch Complex 1 in Mahia, New Zealand; and SpaceX's Starbase in Boca Chica, Texas. Kodiak Island, located in Alaska, is considered one of the best locations in the world for polar launch operations, providing a wide launch azimuth and unobstructed downrange flight path.Read MoreA wet dress rehearsal is underway for Rocket Lab's Electron rocket at Launch Complex 1 in Mahia, New Zealand on April 28, 2023. NASA's Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) CubeSats are secured in the payload fairing atop the rocket.Rocket LabRomanThe Nancy Grace Roman Space Telescope will be able to block starlight to directly see exoplanets and planet-forming disks, complete a statistical census of planetary systems in our galaxy, and settle essential questions in the areas of dark energy, exoplanets, and infrared astrophysics.Learn Moreabout RomanUpcoming MissionsLearn more about future missions managed by NASA's Launch Services Program.More LSP MissionsPandoraPandora is a small satellite designed to characterize exoplanet atmospheres and their host stars. It is slated to observe at least 20 different planets during its one year of science operations.Read MoreGatewayAstronauts will explore the scientific mysteries of deep space with Gateway, humanity's first space station around the Moon, and chart a path for the first human missions to Mars and beyond.Read MoreDragonflyDragonfly, the first-of-its-kind rotorcraft to explore another world, will fly to various locations on Saturn's moon Titan. The Dragonfly rotorcraft will break the barriers for exploration of other planetary bodies.Read MoreLearn More About Launch Services ProgramLaunch Services Program AwardsView the latest contract award news from Launch Services Program.Venture-Class Acquisition of Dedicated and RideshareFind out how NASA is enabling greater access to space for science and technology missions through the agency's VADR (Venture-Class Acquisition of Dedicated and Rideshare) contract.LSP Launch ArchiveCheck out detailed information about the spacecraft and launch vehicles that have lofted important scientific missions into Earth's orbit and to the outer reaches of our solar system.LSP YouTubeWatch videos of past Launch Services Program missions.Media ResourcesLearn more about Launch Services Program.AboutCubeSat Launch InitiativeSince its inception, NASA's CubeSat Launch Initiative has launched over 150 CubeSats on more than 40 Educational Launch of Nanosatellites (ELaNa) missions.NASA's CubeSat Launch initiative (CSLI) provides low-cost access to space for U.S. educational institutions, informal educational institutions such as museums and science centers, non-profits with an education/outreach component, and NASA centers for early career workforce development. The initiative's intent is to inspire and develop the next generation of scientists, engineers, and technologists by offering a unique opportunity to conduct scientific research and develop/demonstrate novel technologies in space.Learn Moreabout CubeSat Launch InitiativeOnce Launch Services Program pairs CubeSats to a launch, an Educational Launch of Nanosatellites (ELaNa) mission number is assigned. Pictured is ELaNa 19 / Venture Class CubeSats (CHOMPTT) inside Rocket Lab's facility, located at Huntington Beach California.Image credit: NASAA SpaceX Falcon 9 rocket carrying NASA's R5-S7

(Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company's Transporter-15 mission stands vertical on the launch pad of Space Launch Complex 4 East at Vandenberg Space Force Base in California on Thursday, Nov. 27, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users.SpaceXA long exposure photo shows two streaks – the SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifting off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025, and the rocket's first stage returning minutes later to land at Vandenberg's Landing Zone 4 East. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.SpaceXBlue Origin's New Glenn rocket carrying NASA's twin ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPADE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.Blue OriginA SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.SpaceXNASA's TRACERS (Tandem Reconnection and Cusp Electrodynamics Reconnaissance Satellites) mission launches at 11:13 a.m. PDT (2:13 p.m. EDT) on Wednesday, July 23, 2025, atop a SpaceX Falcon 9 rocket at Space Launch Complex 4 East at Vandenberg Space Force Base in California. The TRACERS mission will study magnetic reconnection around Earth — a process in which electrically charged plasmas exchange energy in the atmosphere — to understand how the Sun's solar wind interacts with the magnetosphere, Earth's protective magnetic shield.SpaceXA SpaceX Falcon 9 rocket, carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) observatory and PUNCH (Polarimeter to Unify the Corona and Heliosphere) satellites, launches from Space Launch

Complex 4 East at Vandenberg Space Force Base in California on Tuesday, March 11, 2025. SPHEREx will use its telescope to provide an all-sky spectral survey, creating a 3D map of the entire sky to help scientists investigate the origins of our universe. PUNCH will study origins of the Sun's outflow of material, or the solar wind, capturing continuous 3D images of the Sun's corona and the solar wind's journey into the solar system.SpaceXA reflection in the water shows NASA's Europa Clipper spacecraft atop SpaceX's Falcon Heavy rocket at Launch Pad 39A on Sunday, Oct. 13, 2024, at the agency's Kennedy Space Center in Florida ahead of launch to Jupiter's icy moon, Europa. The spacecraft will complete nearly 50 flybys of Europa to determine if there are conditions suitable for life beyond Earth. Launch is targeting 12:06 p.m. EDT on Monday, Oct. 14, from Launch Complex 39A at Kennedy Space Center in Florida.SpaceXA United Launch Alliance Atlas V 541 rocket, carrying the National Oceanic and Atmospheric Administration's (NOAA) Geostationary Operational Environmental Satellite-T (GOES-T), lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station in Florida on March 1, 2022. Liftoff was at 4:38 p.m. EST. GOES-T is the third satellite in the GOES-R series that will continue to help meteorologists observe and predict local weather events that affect public safety. GOES-T will be renamed GOES-18 once it reaches geostationary orbit. GOES-18 will go into operational service as GOES West to provide critical data for the U.S. West Coast, Alaska, Hawaii, Mexico, Central America, and the Pacific Ocean. The launch was managed by NASA's Launch Services Program based at Kennedy Space Center in Florida, America's multi-user spaceport.NASA/Kevin O'Connell and Kevin DavisA Rocket Lab Electron rocket that will carry the Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment (CAPSTONE) spacecraft is seen at Rocket Lab's Launch Complex 1 on the Mahia Peninsula of New Zealand.A SpaceX Falcon Heavy rocket with the Psyche spacecraft onboard is seen as it is rolled to the launch pad at Launch Complex 39A as preparations continue for the Psyche mission, Tuesday, Oct. 10, 2023, at NASA's Kennedy Space Center in Florida. NASA's Psyche spacecraft will travel to a metal-rich asteroid by the same name orbiting the Sun between Mars and Jupiter to study it's composition. The spacecraft also carries the agency's Deep Space Optical Communications technology demonstration, which will test laser communications beyond the Moon. Photo Credit: (NASA/Aubrey Gemignani)NASA's PACE (Plankton, Aerosol, Cloud, ocean Ecosystem) spacecraft, atop a SpaceX Falcon 9 rocket, successfully lifts off from Space Launch Complex 40 at Cape Canaveral Space Force Station in Florida at 1:33 a.m. EST Thursday, Feb. 8. PACE is NASA's newest earth-observing satellite that will help increase our understanding of Earth's oceans, atmosphere, and climate by delivering hyperspectral observations of microscopic marine organisms called phytoplankton, as well new data on clouds and aerosols.SpaceXMeet Our TeamMore BiographiesAlbert SierraAlbert Sierra is the program manager of the Launch Services Program at Kennedy.Read MoreJennifer W.

LyonsJenny Lyons is the deputy program manager of the Launch Services Program at Kennedy.Read MoreTim DunnTim Dunn is senior launch director of the Launch Services Program at Kennedy.Read MoreFacebook logo@NASAKennedy@NASA@NASAKennedy@NASAInstagram logo@NASA@NASAKennedyLinkedin logo@company/NASAKeep ExploringDiscover More Topics From NASAMissionsHumans in SpaceEarth & ClimateSolar System

Launch Services ProgramNASA's Launch Services Program (LSP) is responsible for launching rockets delivering spacecraft that observe the Earth, visit other planets and explore the universe – from weather satellites to telescopes to Mars rovers and more.LSP functions as a broker, matching spacecraft with the best-suited rockets, managing the launch process, providing support from pre-mission planning to post-launch. LSP helps implement NASA's policy of a mixed-fleet launch strategy, which uses both existing and emerging domestic launch capabilities to assure access to space.Learn Moreabout Launch Services Program

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Rockets

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Launch Sites

Kennedy Space CenterCape CanaveralVandenbergWallopsKwajalein AtollAdditional SitesKennedy Space CenterOne of two primary launch sites for NASA's Launch Vehicles.Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit.View SiteA SpaceX Falcon 9 rocket with NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft atop stands vertical at Launch Complex 39A as the sun sets on Monday, Sept. 22, 2025, at the agency's Kennedy Space Center in Florida. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.BAE Systems/Benjamin FryCape Canaveral Space Force StationOne of two primary launch sites for NASA's Launch Vehicles.Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit.View SiteA United Launch Alliance Atlas V rocket with the Lucy spacecraft aboard is seen in this 2 minute and 30 second exposure photograph as it launches from Space Launch Complex 41, Saturday, Oct. 16, 2021, at Cape Canaveral Space Force Station in Florida. Lucy will be the first spacecraft to study Jupiter's Trojan Asteroids. Like the mission's namesake – the fossilized human ancestor, "Lucy," whose skeleton provided unique insight into humanity's evolution – Lucy will revolutionize our knowledge of planetary origins and the formation of the solar system.NASA/Bill IngallsVandenberg Space Force BaseOne of two primary launch sites for NASA's Launch Vehicles.Located along California's central coast between Los Angeles and San Francisco, Vandenberg is preferred for spacecraft requiring a north-south, or polar, orbit.Visit SiteA SpaceX Falcon 9 rocket carrying NASA's R5-S7 (Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company's Transporter-15 mission lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 10:44 a.m. PST Friday, Nov. 28, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users.SpaceXWallops Flight FacilityLocated along the Atlantic Coast on Wallops Island, Virginia.Operated by Goddard Space Flight Center in Greenbelt, Maryland, Wallops is NASA's principal facility for suborbital research programs.View SiteA Northrop Grumman Antares rocket carrying a Cygnus resupply spacecraft is seen on the Mid-Atlantic Regional Spaceport's Pad-0A, Thursday, October 1, 2020, at NASA's Wallops Flight Facility in Virginia. Northrop Grumman's 14th contracted cargo resupply mission with NASA to the International Space Station will deliver nearly 8,000 pounds of science and research, crew supplies and vehicle hardware to the orbital laboratory and its crew. The CRS-14 Cygnus spacecraft is named after the first female astronaut of Indian descent, Kalpana Chawla.NASA/Patrick BlackReagan Test Site, Kwajalein AtollAn additional Expendable Launch Vehicle launch location in the Republic of the Marshall Islands in the North Pacific.The Kwajalein site is located between Hawaii and Australia and is used for missions requiring equatorial orbits and low inclinations.Visit SiteOrbital Science Corp.'s L-1011 aircraft "Stargazer" flies over the runway on Kwajalein Atoll with the company's Pegasus rocket slung underneath. Kwajalein is part of the Marshall Islands chain in the Pacific Ocean. It's also part of the Reagan Test Site and used for launches of NASA, commercial and military missions.NASAAdditional Launch SitesAdditional launch sites specific to LSP's VADR launches include Rocket Lab's Launch Complex 1 in Mahia, New Zealand; and SpaceX's Starbase in Boca Chica, Texas. Kodiak Island, located in Alaska, is considered one of the best locations in the world for polar launch operations, providing a wide launch azimuth and unobstructed downrange flight path.Read MoreA wet dress rehearsal is underway for Rocket Lab's Electron rocket at Launch Complex 1 in Mahia, New Zealand on April 28, 2023. NASA's Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) CubeSats are secured in the payload fairing atop the rocket.Rocket Lab

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Kwajalein Atoll

Additional Sites

Kennedy Space CenterOne of two primary launch sites for NASA's Launch Vehicles.Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit.View SiteA SpaceX Falcon 9 rocket with NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft atop stands vertical at Launch Complex 39A as the sun sets on Monday, Sept. 22, 2025, at the agency's Kennedy Space Center in Florida. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.BAE Systems/Benjamin FryCape Canaveral Space Force StationOne of two primary launch sites for NASA's Launch Vehicles.Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit.View SiteA United Launch Alliance Atlas V rocket with the Lucy spacecraft aboard is seen in this 2 minute and 30 second exposure photograph as it launches from Space Launch Complex 41, Saturday, Oct. 16, 2021, at Cape Canaveral Space Force Station in Florida. Lucy will be the first spacecraft to study Jupiter's Trojan Asteroids. Like the mission's namesake – the fossilized human ancestor, "Lucy," whose skeleton provided unique insight into humanity's evolution – Lucy will revolutionize our knowledge of planetary origins and the formation of the solar system.NASA/Bill IngallsVandenberg Space Force BaseOne of two primary launch sites for NASA's Launch Vehicles.Located along California's central coast between Los Angeles and San Francisco, Vandenberg is preferred for spacecraft requiring a north-south, or polar, orbit.Visit SiteA SpaceX Falcon 9 rocket carrying NASA's R5-S7 (Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company's Transporter-15 mission lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 10:44 a.m. PST Friday, Nov. 28, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users.SpaceXWallops Flight FacilityLocated along the Atlantic Coast on Wallops Island, Virginia.Operated by Goddard Space Flight Center in Greenbelt, Maryland, Wallops is NASA's principal facility for suborbital research programs.View SiteA Northrop Grumman Antares rocket carrying a Cygnus resupply spacecraft is seen on the Mid-Atlantic Regional Spaceport's Pad-0A, Thursday, October 1, 2020, at NASA's Wallops Flight Facility in Virginia. Northrop Grumman's 14th contracted cargo resupply mission with NASA to the International Space Station will deliver nearly 8,000 pounds of science and research, crew supplies and vehicle hardware to the orbital laboratory and its crew. The CRS-14 Cygnus spacecraft is named after the first female astronaut of Indian descent, Kalpana Chawla.NASA/Patrick BlackReagan Test Site, Kwajalein AtollAn additional Expendable Launch Vehicle launch location in the Republic of the Marshall Islands in the North Pacific.The Kwajalein site is located between Hawaii and Australia and is used for missions requiring equatorial orbits and low inclinations.Visit SiteOrbital Science Corp.'s L-1011 aircraft "Stargazer" flies over the runway on Kwajalein Atoll with the company's Pegasus rocket slung underneath. Kwajalein is part of the Marshall Islands chain in the Pacific Ocean. It's also part of the Reagan Test Site and used for launches of NASA, commercial and military missions.NASAAdditional Launch SitesAdditional launch sites specific to LSP's VADR launches include Rocket Lab's Launch Complex 1 in Mahia, New Zealand; and SpaceX's Starbase in Boca Chica, Texas. Kodiak Island, located in Alaska, is considered one of the best locations in the world for polar launch operations, providing a wide launch azimuth and unobstructed downrange flight path.Read MoreA wet dress rehearsal is underway for Rocket Lab's Electron rocket at Launch Complex 1 in Mahia, New Zealand on April 28, 2023. NASA's Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) CubeSats are secured in the payload fairing atop the rocket.Rocket Lab

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Reagan Test Site, Kwajalein AtollAn additional Expendable Launch Vehicle launch location in the Republic of the Marshall Islands in the North Pacific.The Kwajalein site is located between Hawaii and Australia and is used for missions requiring equatorial orbits and low inclinations.Visit SiteOrbital Science Corp.’s L-1011 aircraft “Stargazer” flies over the runway on Kwajalein Atoll with the company’s Pegasus rocket slung underneath. Kwajalein is part of the Marshall Islands chain in the Pacific Ocean. It’s also part of the Reagan Test Site and used for launches of NASA, commercial and military missions.NASA

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AboutCubeSat Launch InitiativeSince its inception, NASA's CubeSat Launch Initiative has launched over 150 CubeSats on more than 40 Educational Launch of Nanosatellites (ELaNa) missions.NASA’s CubeSat Launch initiative (CSLI) provides low-cost access to space for U.S. educational institutions, informal educational institutions such as museums and science centers, non-profits with an education/outreach component, and NASA centers for early career workforce development. The initiative’s intent is to inspire and develop the next generation of scientists, engineers, and technologists by offering a unique opportunity to conduct scientific research and develop/demonstrate novel technologies in space.Learn Moreabout CubeSat Launch InitiativeOnce Launch Services Program pairs CubeSats to a launch, an Educational Launch of Nanosatellites (ELaNa) mission number is assigned. Pictured is ELaNa 19 / Venture Class CubeSats (CHOMPTT) inside Rocket Lab’s facility, located at Huntington Beach California.Image credit: NASA

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A SpaceX Falcon 9 rocket carrying NASA's R5-S7 (Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company's Transporter-15 mission stands vertical on the launch pad of Space Launch Complex 4 East at Vandenberg Space Force Base in California on Thursday, Nov. 27, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users.SpaceXA long exposure photo shows two streaks – the SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifting off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025, and the rocket's first stage returning minutes later to land at Vandenberg's Landing Zone 4 East. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.SpaceXBlue Origin's New Glenn rocket carrying NASA's twin ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPADE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.Blue OriginA SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.SpaceXNASA's TRACERS (Tandem Reconnection and Cusp Electrodynamics Reconnaissance Satellites) mission launches at 11:13 a.m. PDT (2:13 p.m. EDT) on Wednesday, July 23, 2025, atop a SpaceX Falcon 9 rocket at Space Launch Complex 4 East at Vandenberg Space Force Base in California. The TRACERS mission will study magnetic reconnection around Earth — a process in which electrically charged plasmas exchange energy in the atmosphere — to understand how the Sun's solar wind interacts with the magnetosphere, Earth's protective magnetic shield.SpaceXA SpaceX Falcon 9 rocket, carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) observatory and PUNCH (Polarimeter to Unify the Corona and Heliosphere) satellites, launches from Space Launch Complex 4 East at Vandenberg Space Force Base in California on Tuesday, March 11, 2025. SPHEREx will use its telescope to provide an all-sky spectral survey, creating a 3D map of the entire sky to help scientists investigate the origins of our universe. PUNCH will study origins of the Sun's outflow of material, or the solar wind, capturing continuous 3D images of the Sun's corona and the solar wind's journey into the solar system.SpaceXA reflection in the water shows NASA's Europa Clipper spacecraft atop SpaceX's Falcon Heavy rocket at Launch Pad 39A on Sunday, Oct. 13, 2024, at the agency's Kennedy Space Center in Florida ahead of launch to Jupiter's icy moon, Europa. The spacecraft will complete nearly 50 flybys of Europa to determine if there are conditions suitable for life beyond Earth. Launch is targeting 12:06 p.m. EDT on Monday, Oct. 14, from Launch Complex 39A at Kennedy Space Center in Florida.SpaceXA United Launch Alliance Atlas V 541 rocket, carrying the National Oceanic and Atmospheric Administration's (NOAA) Geostationary Operational Environmental Satellite-T (GOES-T), lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station in Florida on March 1, 2022. Liftoff was at 4:38 p.m. EST. GOES-T is the third satellite in the GOES-R series that will continue to help meteorologists observe and predict local weather events that affect public safety. GOES-T will be renamed GOES-18 once it reaches geostationary orbit. GOES-18 will go into operational service as GOES West to provide critical data for the U.S. West Coast, Alaska, Hawaii, Mexico, Central America, and the Pacific Ocean. The launch was managed by NASA’s Launch Services Program based at Kennedy Space Center in Florida, America’s multi-user spaceport.NASA/Kevin O’Connell and Kevin DavisA Rocket Lab Electron rocket that will carry the Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment (CAPSTONE) spacecraft is seen at Rocket Lab’s Launch Complex 1 on the Mahia Peninsula of New Zealand.A SpaceX Falcon Heavy rocket with the Psyche spacecraft onboard is seen as it is rolled to the launch pad at Launch Complex 39A as preparations continue for the Psyche mission, Tuesday, Oct. 10, 2023, at NASA’s Kennedy Space Center in Florida. NASA’s Psyche spacecraft will travel to a metal-rich asteroid by the same name orbiting the Sun between Mars and Jupiter to study it’s composition. The spacecraft also carries the agency’s Deep Space Optical Communications technology demonstration, which will test laser communications beyond the Moon. Photo Credit: (NASA/Aubrey Gemignani)NASA’s PACE (Plankton, Aerosol, Cloud, ocean Ecosystem) spacecraft, atop a SpaceX Falcon 9 rocket, successfully lifts off from Space Launch Complex 40 at Cape Canaveral Space Force Station in Florida at 1:33 a.m. EST Thursday, Feb. 8. PACE is NASA’s newest earth-observing satellite that will help increase our understanding of Earth’s oceans, atmosphere, and climate by delivering hyperspectral observations of microscopic marine organisms called phytoplankton, as well new data on clouds and aerosols.SpaceX

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A SpaceX Falcon 9 rocket carrying NASA’s R5-S7 (Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company’s Transporter-15 mission stands vertical on the launch pad of Space Launch Complex 4 East at Vandenberg Space Force Base in California on Thursday, Nov. 27, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users.

A long exposure photo shows two streaks – the SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifting off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025, and the rocket's first stage returning minutes later to land at Vandenberg's Landing Zone 4 East. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.SpaceX

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A SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.SpaceX

A SpaceX Falcon 9 rocket, carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) observatory and PUNCH (Polarimeter to Unify the Corona and Heliosphere) satellites, launches from Space Launch Complex 4 East at Vandenberg Space Force Base in California on Tuesday, March 11, 2025. SPHEREx will use its telescope to provide an all-sky spectral survey, creating a 3D map of the entire sky to help scientists investigate the origins of our universe. PUNCH will study origins of the Sun's outflow of material, or the solar wind, capturing continuous 3D images of the Sun's corona and the solar wind's journey into the solar system.SpaceX

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A reflection in the water shows NASA's Europa Clipper spacecraft atop SpaceX's Falcon Heavy rocket at Launch Pad 39A on Sunday, Oct. 13, 2024, at the agency's Kennedy Space Center in Florida ahead of launch to Jupiter's icy moon, Europa. The spacecraft will complete nearly 50 flybys of Europa to determine if there are conditions suitable for life beyond Earth. Launch is targeting 12:06 p.m. EDT on Monday, Oct. 14, from Launch Complex 39A at Kennedy Space Center in Florida.SpaceX

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A United Launch Alliance Atlas V 541 rocket, carrying the National Oceanic and Atmospheric Administration's (NOAA) Geostationary Operational Environmental Satellite-T (GOES-T), lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station in Florida on March 1, 2022. Liftoff was at 4:38 p.m. EST. GOES-T is the third satellite in the GOES-R series that will continue to help meteorologists observe and predict local weather events that affect public safety. GOES-T will be renamed GOES-18 once it reaches geostationary orbit. GOES-18 will go into operational service as GOES West to provide critical data for the U.S. West Coast, Alaska, Hawaii, Mexico, Central America, and the Pacific Ocean. The launch was managed by NASA's Launch Services Program based at Kennedy Space Center in Florida, America's multi-user spaceport.NASA/Kevin O'Connell and Kevin Davis

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NASA/Kevin O'Connell and Kevin Davis

A Rocket Lab Electron rocket that will carry the Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment (CAPSTONE) spacecraft is seen at Rocket Lab's Launch Complex 1 on the Mahia Peninsula of New Zealand.

A SpaceX Falcon Heavy rocket with the Psyche spacecraft onboard is seen as it is rolled to the launch pad at Launch Complex 39A as preparations continue for the Psyche mission, Tuesday, Oct. 10, 2023, at NASA's Kennedy Space Center in Florida. NASA's Psyche spacecraft will travel to a metal-rich asteroid by the same name orbiting the Sun between Mars and Jupiter to study it's composition. The spacecraft also carries the agency's Deep Space Optical Communications technology demonstration, which will test laser communications beyond the Moon. Photo Credit: (NASA/Aubrey Gemignani)

NASA's PACE (Plankton, Aerosol, Cloud, ocean Ecosystem) spacecraft, atop a SpaceX Falcon 9 rocket, successfully lifts off from Space Launch Complex 40 at Cape Canaveral Space Force Station in Florida at 1:33 a.m. EST Thursday, Feb. 8. PACE is NASA's newest earth-observing satellite that will help increase our understanding of Earth's oceans, atmosphere, and climate by delivering hyperspectral observations of microscopic marine organisms called phytoplankton, as well new data on clouds and aerosols.SpaceX

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TITLE: Earth — A Photo-Essay

Earth ObservatoryEarth — A Photo-EssayOf all celestial bodies within reach or view, as far as we can see, out to the edge, the most wonderful and marvelous and mysterious is turning out to be our own planet earth. There is nothing to match it anywhere, not yet anyway. —Lewis ThomasA photo-essay from NASA's Earth Science DivisionDownloadEarthinPDF,MOBI(Kindle), orePubformats.ForewordSixty years ago, with the launch of Explorer 1, NASA made its first observations of Earth from space. Fifty years ago, astronauts left Earth orbit for the first time and looked back at our "blue marble." All of these years later, as we send spacecraft and point our telescopes past the outer edges of the solar system, as we study our planetary neighbors and our Sun in exquisite detail, there remains much to see and explore at home.We are still just getting to know Earth through the tools of science. For centuries, painters, poets, philosophers, and photographers have sought to teach us something about our home through their art.This book stands at an intersection of science and art. From its origins, NASA has studied our planet in novel ways, using ingenious tools to study physical processes at work—from beneath the crust to the edge of the atmosphere. We look at it in macrocosm and microcosm, from the flow of one mountain stream to the flow of jet streams. Most of all, we look at Earth as a system, examining the cycles and processes—the water cycle, the carbon cycle, ocean circulation, the movement of heat—that interact and influence each other in a complex, dynamic dance across seasons and decades.We measure particles, gases, energy, and fluids moving in, on, and around Earth. And like artists, we study the light—how it bounces, reflects, refracts, and gets absorbed and changed. 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He is a baseball player and fan, a longtime singer and guitarist, and the proud father of three science and engineering majors.Lawrence Friedldirects the Applied Sciences Program in the Earth Science Division of NASA's Science Mission Directorate. He works to enable innovative and practical uses of data from Earth-observing satellites. He has worked at the U.S. Environmental Protection Agency and as a Space Shuttle flight controller in NASA's Mission Control Center. He and his wife have three children, and he enjoys ultimate frisbee and hiking.Kevin Wardmanages NASA's Earth Observatory Group, including the Earth Observatory, Visible Earth, NASA Earth Observations (NEO), and EONET. He holds a master's degree in library and information science and has spent more than 20 years developing Web-accessible resources in support of NASA Earth science communications. 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Earth — A Photo-EssayOf all celestial bodies within reach or view, as far as we can see, out to the edge, the most wonderful and marvelous and mysterious is turning out to be our own planet earth. There is nothing to match it anywhere, not yet anyway. —Lewis Thomas

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Foreword

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Michael Carlowiczis managing editor of the NASA Earth Observatory. He has written about Earth science and geophysics since 1991 for several NASA divisions, the American Geophysical Union, the Woods Hole Oceanographic Institution, and in three popular science books. He is a baseball player and fan, a longtime singer and guitarist, and the proud father of three science and engineering majors.

Lawrence Friedldirects the Applied Sciences Program in the Earth Science Division of NASA's Science Mission Directorate. He works to enable innovative and practical uses of data from Earth-observing satellites. He has worked at the U.S. Environmental Protection Agency and as a Space Shuttle flight controller in NASA's Mission Control Center. He and his wife have three children, and he enjoys ultimate frisbee and hiking.

Kevin Wardmanages NASA's Earth Observatory Group, including the Earth Observatory, Visible Earth, NASA Earth Observations (NEO), and EONET. He holds a master's degree in library and information science and has spent more than 20 years developing Web-accessible resources in support of NASA Earth science communications. He and his wife have a son and a deep love of music.

Kathy Carrollsupports the Earth Science Division in the Science Mission Directorate at NASA Headquarters. She previously worked as a manager and organizer at for-profit and non-profit organizations and on political campaigns. She is a diehard baseball and hockey fan, and she volunteers with animal rescue organizations.

Stephen Schaeberleis a graphic designer with the Communications Support Services Center at NASA Headquarters. He holds a bachelor of fine arts from the Pratt Institute, and he has received numerous awards and honors for his work and designs. He enjoys boating and fishing on the Chesapeake Bay.

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NASA conceives, builds, and flies cutting-edge satellites and instruments; harvests data from those tools and from commercial partners; combines it with expert insights and the world's most advanced models — all to advance solutions for monitoring, prediction, and management.Why it mattersUsing the power of space-based observations, innovative technologies, engineering, science, and partnerships, NASA can offer a comprehensive, digital view of Earth as a system. NASA works across commercial and industrial sectors and federal and international agencies to:generate the scientific data necessary to inform decisions from the local to state to federal to international levelleverage innovation from academia and the private sector to develop novel applications of Earth science informationcreate products and analyses that directly address societal needslink technology, research & development, and actionable science to ensure rapid transfer from innovation to impact.advance America's space exploration goals, using our Earth tools and insights to better prepare us for a sustainable presence on the Moon and Mars.Featured VideoNASA Earth: Powering America's FutureDid you know the very same cutting-edge technology NASA uses to study other planets is exactly what we use to study our own?From the ground, air, and skies, NASA gathers information crucial to life here on Earth, and we build instruments and technologies and programs that equip the US to meet critical, real-world challenges.Watch on YouTubeEarth Science at Work,a white paperBy 2030, the potential economic value of EO (Earth observations) could exceed $700 billion globally...better data can inform decision-making, optimize solutions and enable innovation.Amplifying the Global Value of Earth ObservationWorld Economic Forum reportWhat We DoNASA's research and data provide a competitive advantage to American businesses and support the U.S. economy.Mappingrare Earth mineralsfor use in domestic, technology-dominant industries such as telecommunications, energy generation, and aerospace.AidingU.S. agriculturewith continuous measurements of water resources, crop health, and global production that inform business decisions, improve farm and ranch efficiency, and reduce input costs, resulting in a positive impact from the commodity-trading floor to the grocery store.In FY24, NASA's Earth science data was downloaded by more than8.3 million distinct usersfrom a wide variety of fields and industries.NASA, BLM, USGS/Yolanda Fong-SamNASA Earth supports national security with valuable information and tools.Gathering data and establishing parameters that underpin theposition, navigation, and timing accuracy of satellites and GPS.Providing data needed to forecast extreme events like floods -- through the Global HydroIntelligence System.Enhancingsituational awareness ofice cover and other conditions around Arctic seas.Supplyingland surface monitoring capabilitiesthrough theLand Information System.Providingsea level dataapplicable to U.S. coastal military installations and operations around the world.Satellite observations at night indicate that oil, gas, and mining have expanded across Earth's northern latitudes.Data is the key. If we can measure it, we can quantify it, we can solve it.Dwane RothFarmer and OpenET Data User, Finney County, KansasNASA is improving the nation’s ability to forecast and respond to severe storms, droughts, wildland fires, and other natural hazards.Offeringnear real-time data products, early warning tools, and disaster mapping portals.Developing an advancedlandslide forecastmodel.Improving our understanding of hurricanes and sharing those insights and tools with NOAA’s National Hurricane Center and the Joint Typhoon Warning CenterWorking with the U.S. Forest Service and other agencies to provide key observations of conditions before, during, and afterwildland fires- such as estimating fuel loads and soil moisture.A selection of four images showing a volcano erupting, flooded streets in Vermont, a hurricane seen from space, and fires burning a hillside.NASA Earth Science DivisionOur data supports decision-making that affects human health and quality of life.Satellite measurements of temperature, humidity, and precipitation help predictoutbreaks of mosquito-borne illnesses.Satellite observations and models helpassess freshwater resourcesfrom the surface through the root zone to the deep aquifer.Observations of the amount of freshwater stored on land and underground (left) contribute to our understanding of how water availability puts a strain on farmers and communities, potentially leading to famine, conflicts, poverty, and an increased risk of disease, including those transmitted by mosquitoes (right).NASA Earth Observatory, CDCNASA's consistent sea level rise data is the best way and the best tool for us to make sure that we spend our money carefully and wisely so that the future generations get to enjoy our beaches and bays and rivers just as much as I did.Casi CallawayChief Resilience Officer, City of Mobile, AlabamaNASA is uniquely positioned to take scientific and societal questions from conception to public utility. We develop and fund technological leaps, build and launch the satellites and instruments, gather and disseminate the data, analyze the findings, and work with commercial, industry, academic, and government partners to serve societal needs.NASA Earth Science DivisionStudying Earth from SpaceSince the 1960s, we have been observing our own planet because it is the one we need to know best. NASA applies ingenuity and expertise gained from decades of planetary and deep-space exploration to the study of our home planet.From Headquarters and multiple NASA centers and labs, the Earth Science Division operates more than 20 satellites in orbit, sponsors hundreds of research programs and studies, and funds opportunities to put data to use for societal needs.We develop new ways to observe the oceans, land, ice, atmosphere, and life, and we measure how changes in one drive changes in others over the short and long term.While listening to and collaborating with industry leaders, international partners, academic institutions, and other users of our data, we deepen knowledge of our planet, drive innovations, and deliver science to help inform decisions that benefit the nation and the world.NASA Research and Data in ActionArticle5 min readOpenET Study Helps Water Managers and Farmers Put NASA Data to WorkArticle4 min readNASA and Forest Service

Use Balloon to Help Firefighters CommunicateArticle6 min readNASA Flights Map Critical Minerals from Skies Above Western USArticle4 min readNASA Data Helps Protect US Embassy Staff from Polluted AirArticle5 min readNASA Laser Reflecting Instruments to Help Pinpoint Earth MeasurementsArticle5 min readNASA, Pacific Disaster Center Increase Landslide Hazard AwarenessArticle4 min readNASA Data Helps Bangladeshi Farmers Save Water, Money, EnergyArticle5 min readCutting-Edge Satellite Tracks Lake Water Levels in Ohio River BasinLearn moreEarth Science at Work,a white paperEarth Science to Action StrategyWithin a decade, we will advance and integrate Earth science knowledge to empower humanity to create a more resilient world.

NASA Earth Sciencehelps Americans respond to challenges and societal needs— such as wildland fires, hurricanes, and water supplies for farming — by putting actionable satellite information into the hands of decisionmakers. NASA conceives, builds, and flies cutting-edge satellites and instruments; harvests data from those tools and from commercial partners; combines it with expert insights and the world's most advanced models — all to advance solutions for monitoring, prediction, and management.

Why it matters

Using the power of space-based observations, innovative technologies, engineering, science, and partnerships, NASA can offer a comprehensive, digital view of Earth as a system. NASA works across commercial and industrial sectors and federal and international agencies to:

generate the scientific data necessary to inform decisions from the local to state to federal to international level

leverage innovation from academia and the private sector to develop novel applications of Earth science information

create products and analyses that directly address societal needs

link technology, research & development, and actionable science to ensure rapid transfer from innovation to impact.

advance America's space exploration goals, using our Earth tools and insights to better prepare us for a sustainable presence on the Moon and Mars.

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Earth Science at Work,a white paper

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NASA's research and data provide a competitive advantage to American businesses and support the U.S. economy.Mappingrare Earth mineralsfor use in domestic, technology-dominant industries such as telecommunications, energy generation, and aerospace.AidingU.S. agriculturewith continuous measurements of water resources, crop health, and global production that inform business decisions, improve farm and ranch efficiency, and reduce input costs, resulting in a positive impact from the commodity-trading floor to the grocery store.In FY24, NASA’s Earth science data was downloaded by more than8.3 million distinct usersfrom a wide variety of fields and industries.NASA, BLM, USGS/Yolanda Fong-SamNASA Earth supports national security with valuable information and tools.Gathering data and establishing parameters that underpin theposition, navigation, and timing accuracy of satellites and GPS.Providing data needed to forecast extreme events like floods -- through the Global HydroIntelligence System.Enhancingsituational awareness ofice cover and other conditions around Arctic seas.Supplyingland surface monitoring capabilitiesthrough theLand Information System.Providingsea level dataapplicable to U.S. coastal military installations and operations around the world.Satellite observations at night indicate that oil, gas, and mining have expanded across Earth’s northern latitudes.

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NASA's consistent sea level rise data is the best way and the best tool for us to make sure that we spend our money carefully and wisely so that the future generations get to enjoy our beaches and bays and rivers just as much as I did.Casi CallawayChief Resilience Officer, City of Mobile, Alabama

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Studying Earth from Space

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We develop new ways to observe the oceans, land, ice, atmosphere, and life, and we measure how changes in one drive changes in others over the short and long term.

While listening to and collaborating with industry leaders, international partners, academic institutions, and other users of our data, we deepen knowledge of our planet, drive innovations, and deliver science to help inform decisions that benefit the nation and the world.

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5 Min ReadNASA Webb Finds Young Sun-Like Star Forging, Spewing Common CrystalsNASA's James Webb Space Telescope's 2024 NIRCam image shows protostar EC 53 circled. Researchers using new data from Webb's MIRI proved that crystalline silicates form in the hottest part of the disk of gas and dust surrounding the star — and may be shot to the system's edges.Credits:Image: NASA, ESA, CSA, STScI, Klaus Pontoppidan (NASA-JPL), Joel Green (STScI); Image Processing: Alyssa Pagan (STScI)

5 Min ReadNASA Webb Finds Young Sun-Like Star Forging, Spewing Common Crystals

NASA's James Webb Space Telescope's 2024 NIRCam image shows protostar EC 53 circled. Researchers using new data from Webb's MIRI proved that crystalline silicates form in the hottest part of the disk of gas and dust surrounding the star — and may be shot to the system's edges.Credits:Image: NASA, ESA, CSA, STScI, Klaus Pontoppidan (NASA-JPL), Joel Green (STScI); Image Processing: Alyssa Pagan (STScI)

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NASA Webb Mission TeamGoddard Space Flight CenterJan 21, 2026ArticleContentsImage: Protostar EC 53 in the Serpens Nebula (NIRCam Image)Image: Silicate Crystallization and Movement Near Protostar EC 53 (Illustration)Look aheadDownloads & Related InformationRelated LinksAstronomers have long sought evidence to explain whycometsat the outskirts of our ownsolar systemcontain crystalline silicates, since crystals require intense heat to form and these "dirty snowballs" spend most of their time in the ultracoldKuiper BeltandOort Cloud. Now, looking outside our solar system, NASA's James Webb Space Telescope has returned the first conclusive evidence that links how those conditions are possible. The telescope clearly showed for the first time that the hot, inner part of the disk of gas and dust surrounding a very young, actively forming star is where crystalline silicates are forged. Webb also revealed a strong outflow that is capable of carrying the crystals to the outer edges of this disk. Compared to our own fully formed, mostly dust-cleared solar system, the crystals would be forming approximately between the Sun and Earth.Webb's sensitive mid-infrared observations of the protostar, cataloged EC 53, also show that the powerful winds from the star's disk are likely catapulting these crystals into distant locales, like the incredibly cold edge of its protoplanetary disk where comets may eventually form."EC 53's layered outflows may lift up these newly formed crystalline silicates and transfer them outward, like they're on a cosmic highway," said Jeong-Eun Lee, the lead author of anew paper in Natureand a professor at Seoul National University in South Korea. "Webb not only showed us exactly which types of silicates are in the dust near the star, but also where they are both before and during a burst."Image: Protostar EC 53 in the Serpens Nebula (NIRCam Image)NASA's James Webb Space Telescope's 2024 NIRCam image shows protostar EC 53 circled. Researchers using new data from Webb's MIRI proved that crystalline silicates form in the hottest part of the disk of gas and dust surrounding the star — and may be shot to the system's edges.Image: NASA, ESA, CSA, STScI, Klaus Pontoppidan (NASA-JPL), Joel Green (STScI); Image Processing: Alyssa Pagan (STScI)The team used Webb'sMIRI (Mid-Infrared Instrument)to collect two sets of highly detailedspectrato identify specific elements and molecules, and determine their structures. Next, they precisely mapped where everything is, both when EC 53 is "quiet" (but still gradually "nibbling" at its disk) and when it's more active (what's known as an outburst phase).This star, which has been studied by this team and others for decades, is highly predictable. (Other young stars have erratic outbursts, or their outbursts last for hundreds of years.) About every 18 months, EC 53 begins a 100-day, bombastic burst phase, kicking up the pace and absolutely devouring nearby gas and dust, while ejecting some of its intake as powerful jets and outflows. These expulsions may fling some of the newly formed crystals into the outskirts of the star's protoplanetary disk."Even as a scientist, it is amazing to me that we can find specific silicates in space, including forsterite and enstatite near EC 53," said Doug Johnstone, a co-author and a principal research officer at the National Research Council of Canada. "These are common minerals on Earth. The main ingredient of our planet is silicate." For decades, research has also identified crystalline silicates not only on comets in our solar system, but also in distant protoplanetary disks around other, slightly older stars — but couldn't pinpoint how they got there. With Webb's new data, researchers now better understand how these conditions might be possible."It's incredibly impressive that Webb can not only show us so much, but also where everything is," said Joel Green, a co-author and an instrument scientist at the Space Telescope Science Institute in Baltimore, Maryland. "Our research team mapped how the crystals move throughout the system. We've effectively shown how the star creates and distributes these superfine particles, which are each significantly smaller than a grain of sand."Webb's MIRI data also clearly shows the star's narrow, high-velocity jets of hot gas near its poles, and the slightly cooler and slower outflows that stem from the innermost and hottest area of the disk that feeds the star. The image above, which was taken by another Webb instrument, NIRCam (Near-Infrared Camera), shows one set of winds and scattered light from EC 53's disk as a white semi-circle angled toward the right. Its winds also flow in the opposite direction, roughly behind the star, but in near-infrared light, this region appears dark. Its jets are too tiny to pick out.Image: Silicate Crystallization and Movement Near Protostar EC 53 (Illustration)This illustration represents half the disk of gas and dust surrounding the protostar EC 53. Stellar outbursts periodically form crystalline silicates, which are launched up and out to the edges of the system, where comets and other icy rocky bodies may eventually form.Illustration: NASA, ESA, CSA, Elizabeth Wheatley (STScI)Look aheadEC 53 is still "wrapped" in dust and may be for another 100,000 years. Over millions of years, while a young star's disk is heavily populated with teeny grains of dust and pebbles, an untold number of collisions will occur that may slowly build up a range of larger rocks, eventually leading to the formation of terrestrial and gas giant planets. As the disk settles, both the star itself and any rocky planets will finish forming, the dust will largely clear (no longer obscuring the view), and a Sun-like star will remain at the center of a cleared planetary system, with crystalline silicates "littered" throughout.EC 53 is part of the Serpens Nebula, which lies 1,300 light-years from Earth and is brimming with actively forming stars.The James Webb Space Telescope is the world's premier space science observatory. Webb is solving mysteries in our solar system, looking beyond to distant worlds around other stars, and probing the mysterious structures and origins of our universe and our place in it. Webb is an international program led by NASA with its partners, ESA (European Space Agency) and CSA (Canadian Space Agency).To learn more about Webb, visit:https://science.nasa.gov/webbDownloads & Related InformationThe following sections contain links to download this article's images and videos in all available resolutions followed by related information links, media contacts, and if available, research paper and Spanish translation links.

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Compared to our own fully formed, mostly dust-cleared solar system, the crystals would be forming approximately between the Sun and Earth.Webb’s sensitive mid-infrared observations of the protostar, cataloged EC 53, also show that the powerful winds from the star’s disk are likely catapulting these crystals into distant locales, like the incredibly cold edge of its protoplanetary disk where comets may eventually form.“EC 53’s layered outflows may lift up these newly formed crystalline silicates and transfer them outward, like they’re on a cosmic highway,” said Jeong-Eun Lee, the lead author of anew paper in Natureand a professor at Seoul National University in South Korea. “Webb not only showed us exactly which types of silicates are in the dust near the star, but also where they are both before and during a burst.”Image: Protostar EC 53 in the Serpens Nebula (NIRCam Image)NASA’s James Webb Space Telescope’s 2024 NIRCam image shows protostar EC 53 circled. Researchers using new data from Webb’s MIRI proved that crystalline silicates form in the hottest part of the disk of gas and dust surrounding the star — and may be shot to the system's edges.Image: NASA, ESA, CSA, STScI, Klaus Pontoppidan (NASA-JPL), Joel Green (STScI); Image Processing: Alyssa Pagan (STScI)The team used Webb'sMIRI (Mid-Infrared Instrument)to collect two sets of highly detailedspectrato identify specific elements and molecules, and determine their structures. Next, they precisely mapped where everything is, both when EC 53 is "quiet" (but still gradually "nibbling" at its disk) and when it's more active (what's known as an outburst phase).This star, which has been studied by this team and others for decades, is highly predictable. (Other young stars have erratic outbursts, or their outbursts last for hundreds of years.) About every 18 months, EC 53 begins a 100-day, bombastic burst phase, kicking up the pace and absolutely devouring nearby gas and dust, while ejecting some of its intake as powerful jets and outflows. These expulsions may fling some of the newly formed crystals into the outskirts of the star's protoplanetary disk."Even as a scientist, it is amazing to me that we can find specific silicates in space, including forsterite and enstatite near EC 53," said Doug Johnstone, a co-author and a principal research officer at the National Research Council of Canada. "These are common minerals on Earth. The main ingredient of our planet is silicate." For decades, research has also identified crystalline silicates not only on comets in our solar system, but also in distant protoplanetary disks around other, slightly older stars — but couldn't pinpoint how they got there. With Webb's new data, researchers now better understand how these conditions might be possible."It's incredibly impressive that Webb can not only show us so much, but also where everything is," said Joel Green, a co-author and an instrument scientist at the Space Telescope Science Institute in Baltimore, Maryland. "Our research team mapped how the crystals move throughout the system. We've effectively shown how the star creates and distributes these superfine particles, which are each significantly smaller than a grain of sand."Webb's MIRI data also clearly shows the star's narrow, high-velocity jets of hot gas near its poles, and the slightly cooler and slower outflows that stem from the innermost and hottest area of the disk that feeds the star. The image above, which was taken by another Webb instrument, NIRCam (Near-Infrared Camera), shows one set of winds and scattered light from EC 53's disk as a white semi-circle angled toward the right. Its winds also flow in the opposite direction, roughly behind the star, but in near-infrared light, this region appears dark. Its jets are too tiny to pick out.Image: Silicate Crystallization and Movement Near Protostar EC 53 (Illustration)This illustration represents half the disk of gas and dust surrounding the protostar EC 53. Stellar outbursts periodically form crystalline silicates, which are launched up and out to the edges of the system, where comets and other icy rocky bodies may eventually form.Illustration: NASA, ESA, CSA, Elizabeth Wheatley (STScI)Look aheadEC 53 is still "wrapped" in dust and may be for another 100,000 years. Over millions of years, while a young star's disk is heavily populated with teeny grains of dust and pebbles, an untold number of collisions will occur that may slowly build up a range of larger rocks, eventually leading to the formation of terrestrial and gas giant planets. As the disk settles, both the star itself and any rocky planets will finish forming, the dust will largely clear (no longer obscuring the view), and a Sun-like star will remain at the center of a cleared planetary system, with crystalline silicates “littered” throughout.EC 53 is part of the Serpens Nebula, which lies 1,300 light-years from Earth and is brimming with actively forming stars.The James Webb Space Telescope is the world’s premier space science observatory. Webb is solving mysteries in our solar system, looking beyond to distant worlds around other stars, and probing the mysterious structures and origins of our universe and our place in it. 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Why go to space?Human space exploration addresses fundamental questions about our place in the universe and the history of our solar system. NASA’s exploration vision is anchored in providing value for humanity by answering some of the most fundamental questions: Why are we here? How did it all begin? Are we alone? What comes next?Learn Moreabout Why go to space?

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NASA astronaut Victor Glover and JAXA astronaut Soichi Noguchi inside the SpaceX Dragon spacecraft at the International Space Station.Credits: NASA

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ISSExpedition 72Mission TypeHumans in Spacemeet the crewSuni Williams, Butch Wilmore, NIck Hague, Don Pettit, Alexey Ovchinin, Ivan Vagner, Aleksandr GorbunovstartSep. 23, 2024LandingApril 19, 2025The official portrait of the International Space Station’s Expedition 72 crew. At the top (from left) are, Roscosmos cosmonaut and

Flight Engineer Alexey Ovchinin, NASA astronaut and space station Commander Suni Williams, and NASA astronaut and Flight Engineer Butch Wilmore. In the middle row are, Roscosmos cosmonaut and Flight Engineer Ivan Vagner and NASA astronaut and Flight Engineer Don Pettit. In the bottom row are, Roscosmos cosmonaut and Flight Engineer Aleksandr Gorbunov and NASA astronaut and Flight Engineer Nick Hague.NASA/Bill Stafford/Robert MarkowitzCrew MembersSuni Williams, CommanderButch Wilmore, Flight EngineerNick Hague, Flight EngineerDon Pettit, Flight EngineerAlexey Ovchinin, Flight EngineerIvan Vagner, Flight EngineerAleksandr Gorbunov, Flight EngineerCrew and Cargo Missions4/19/2025 – Soyuz MS-26 Undock4/8/2025 – Soyuz MS-27 Dock4/8/2025 – Soyuz MS-27 Launch3/28/2025 – Cygnus NG-21 Release3/18/2025 – SpaceX Crew-9 Splashdown3/18/2025 – SpaceX Crew-9 Undock3/15/2025 – SpaceX Crew-10 Dock3/14/2025 – SpaceX Crew-10 Launch3/1/2025 – Progress 91 Dock2/27/2025 – Progress 91 Launch2/25/2025 – Progress 89 Undock12/16/2024 – SpaceX CRS-31 Undock11/23/2024 – Progress 90 Dock11/21/2024 – Progress 90 Launch11/19/2024 – Progress 88 Undock11/5/2024 – SpaceX CRS-31 Dock11/4/2024 – SpaceX CRS-31 Launch11/3/2024 – SpaceX Crew-9 Relocation10/25/2024 – SpaceX Crew-8 Splashdown10/23/2024 – SpaceX Crew-8 Undock9/29/2024 – SpaceX Crew-9 Dock9/28/2024 – SpaceX Crew-9 Launch9/23/2024 – Soyuz MS-25 Land9/23/2024 – Soyuz MS-25 UndockSpacewalksDate:Jan. 30, 2025Duration:5 hours, 26 minutesSpacewalkers:Suni Williams,Butch WilmoreDate:Jan. 16, 2025Duration:6 hoursSpacewalkers:Nick Hague,Suni WilliamsDate:Dec. 19, 2024Duration:7 hours, 17 minutesSpacewalkers:Alexey Ovchinin, Ivan VagnerExpedition 72 Image Gallery619 ImagesGo To GalleryGo To GalleryExpedition 72 ResourcesMission BlogsThe daily blog updates for Expedition 72 highlighting crew and cargo missions, spacewalks, and advanced space research.Mission ScienceLearn more about the range of advanced space research, technology demonstrations, and more the Expedition 72 crew is conducting.SpaceX Crew-9 ScienceThe SpaceX Crew-9 members will investigate blood cell development in space, X-rays emitted by neutron stars, taking a vitamin supplement to improve vision, space agriculture, and more.SpaceX Crew-10 ScienceThe SpaceX Crew-10 members are studying how microgravity affects production of protein by microalgae, testing a microscope to capture microbial activity, and studying genetic activity in biofilms.Mission ImageryThe Expedition 72 crew is taking images of the Earth, the orbiting station, their daily lives aboard, and the experiments they conduct during their stay, plus so much more. Check out some of the images in the Expedition 72 gallery.Space Station ImageryView imagery taken from the International Space Station since 1999 of daily crew activities, spacewalks, visiting vehicles, and the Earth below.Latest Expedition 72 NewsMore Expedition 72 NewsArticle3 Min ReadCity Lights Glow Along Moonlit WatersArticle4 Min ReadWelcome Home, Expedition 72 Crew!Blog1 Min ReadDragon

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Crew Members

Suni Williams, CommanderButch Wilmore, Flight EngineerNick Hague, Flight EngineerDon Pettit, Flight EngineerAlexey Ovchinin, Flight EngineerIvan Vagner, Flight EngineerAleksandr Gorbunov, Flight Engineer

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TITLE: Expedition 73

ISSExpedition 73Mission TypeHumans in Spacemeet the crewAlexey Zubritsky, Sergey Ryzhikov, Jonny Kim, Kimiya Yui, Oleg Platonov, Zena Cardman, Mike FinckestartApril 19, 2025LandingNov. 8, 2025Crew MembersAlexey Zubritsky, Flight EngineerSergey Ryzhikov, CommanderJonny Kim, Flight EngineerKimiya Yui, Flight EngineerOleg Platonov, Flight EngineerZena Cardman, Flight EngineerMike Fincke, Flight EngineerCrew and Cargo Missions12/8/2025 – Soyuz MS-27 Undock12/1/2025 – Cygnus NG-23

Reinstallation11/27/2025 – Soyuz MS-28 Dock11/27/2025 – Soyuz MS-28 Launch11/24/2025 – Cygnus NG-23 Uninstallation10/29/2025 – HTV-X1 Capture10/25/2025 – HTV-X1 Launch9/18/2025 – Cygnus NG-23 Capture/Installation9/14/2025 – Cygnus NG-23 Launch9/13/2025 – Progress 93 Dock9/11/2025 – Progress 93 Launch9/9/2025 – Progress 91 Undock8/25/2025 – SpaceX CRS-33 Dock8/24/2025 – SpaceX CRS-33 Launch8/9/2025 – SpaceX Crew-10 Splashdown8/8/2025 – SpaceX Crew-10 Undock8/2/2025 – SpaceX Crew-11 Dock8/1/2025 – SpaceX Crew-11 Launch7/15/2025 – Axiom Mission 4 Splashdown7/14/2025 – Axiom Mission 4 Undock7/5/2025 – Progress 92 Dock7/3/2025 – Progress 92 Launch7/1/2025 – Progress 90 Undock6/26/2025 – Axiom Mission 4 Dock6/25/2025 – Axiom Mission 4 Launch5/23/2025 – SpaceX CRS-32 Undock4/22/2025 – SpaceX CRS-32 Dock4/21/2025 – SpaceX CRS-32 Launch4/19/2025 – Soyuz MS-26 Land4/19/2025 – Soyuz MS-26 UndockSpacewalksDate:Oct. 28, 2025Duration:6 hours, 54 minutesSpacewalkers:Sergey Ryzhikov, Alexey ZubritskyDate:Oct. 16, 2025Duration:6 hours, 9 minutesSpacewalkers:Sergey Ryzhikov, Alexey ZubritskyDate:May 1, 2025Duration:6 hours, 35 minutesSpacewalkers:Anne McClain,Nichole AyersExpedition 73 Image Gallery727 ImagesGo To GalleryGo To GalleryExpedition 73 ResourcesMission BlogsThe daily blog updates for Expedition 73 highlighting crew and cargo missions, spacewalks, and advanced space research.Mission ScienceLearn more about the range of advanced space research, technology demonstrations, and more the Expedition 73 crew is conducting.SpaceX Crew-10 ScienceThe SpaceX Crew-10 members are studying how microgravity affects production of protein by microalgae, testing a microscope to capture microbial activity, and studying genetic activity in biofilms.Mission ImageryThe Expedition 73 crew is taking images of the Earth, the orbiting station, their daily lives aboard, and the experiments they conduct during their stay, plus so much more. Check out some of the images in the Expedition 73 gallery.Space Station ImageryView imagery taken from the International Space Station since 1999 of daily crew activities, spacewalks, visiting vehicles, and the Earth below.Latest Expedition 73 NewsMore Expedition 73 NewsBlog2 Min ReadStation Trio Back on Earth; Expedition 74 Keeps Up Science, Maintains SystemsBlog1 Min ReadSoyuz Crew Lands Ending Eight-Month Space Research JourneyBlog1 Min ReadLive on NASA+: Three Soyuz Crewmates Landing on Earth SoonBlog1 Min ReadSoyuz Undocks from Station, Crew Heads for Earth Return1 Min ReadLive on NASA+: Soyuz Crew Undocking from Station SoonBlog1 Min ReadHatches Closed, Crew Inside Soyuz Ready to DepartBlog2 Min ReadLive on NASA+: Three Crewmates Say Farewell, Set to Depart StationBlog3 Min ReadCrew Swaps Commanders on Sunday as Trio Packs for DepartureBlogKeep ExploringMore Humans in Space TopicsCommercial Crew Program OverviewSpace Station Research and TechnologyStation Benefits for HumanityArtemis

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startApril 19, 2025

LandingNov. 8, 2025

Nov. 8, 2025

Alexey Zubritsky, Flight EngineerSergey Ryzhikov, CommanderJonny Kim, Flight EngineerKimiya Yui, Flight EngineerOleg Platonov, Flight EngineerZena Cardman, Flight EngineerMike Fincke, Flight Engineer

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URL: https://www.nasa.gov/mission/expedition-70/

TITLE: Expedition 70

ISSExpedition 70Expedition 70 researched heart health, cancer treatments, space manufacturing techniques, and more during their long-duration stay in Earth orbit.Mission TypeISS ExpeditionMeet the crewAndreas Mogensen, Jasmin Moghbeli, Satoshi Furukawa, Loral O'Hara, Konstantin Borisov, Oleg Kononenko, Nikolai ChubstartSept. 27, 2023ENDApril 5, 2024iss070-s-002 (March 6, 2023) — The official Expedition 70 crew portrait with (top row from left) Roscosmos cosmonauts Nikolai Chub, Konstantin Borisov, and Oleg Kononenko; JAXA (Japan Aerospace Exploration Agency) astronaut Satoshi Furukawa; and NASA astronaut Loral O'Hara. In the front row are, ESA (European Space Agency) astronaut and Expedition 70 Commander Andreas Mogensen and NASA astronaut Jasmin Moghbeli.NASAThe official patch of the International Space Station's Expedition 70 crew. The Expedition 70 patch is designed around the central yin-yang symbol representing balance; first and foremost, the balance of our beautiful planet Earth that is encircled by the yin-yang symbol and which forms part of the Expedition number. In our exploration of space, we are reminded of the uniqueness of Earth; the further we push the boundaries of human existence, the stronger our longing for our home planet grows. As our understanding of the cosmos expands, so does our understanding of Earth, and although we live in an ever-changing world, we recognize the need for a planet in balance to ensure our future. Space exploration is also about creating the future of our dreams. The tentative first steps we take today will hopefully become a well-trodden path in the future. This is represented stylistically by the "retro-futuristic" design of the patch, which mimics the design of the posters depicting the future from the early days of the space age. It is also emphasized by the yellow, orange, and red colors suggesting a sunrise. Lastly, the dynamism in the depiction of the number 7 suggests not only the physical launch into space, but also humanity's progress towards the future.Crew MembersAndreas Mogensen, CommanderJasmin Moghbeli, Flight EngineerSatoshi Furukawa, Flight EngineerLoral O'Hara, Flight EngineerKonstantin Borisov, Flight EngineerOleg Kononenko, Flight EngineerNikolai Chub, Flight EngineerCrew and Cargo Missions4/6/2024 – Soyuz MS-24 Land4/5/2024 – Soyuz MS-24 Undock3/25/24 – Soyuz MS-25 Dock3/23/24 – Soyuz MS-25 Launch3/23/24 – SpaceX CRS-30 Dock3/21/2024 – SpaceX CRS-30 Launch3/12/2024 – SpaceX Crew-7 Splashdown3/11/2024 – SpaceX Crew-7 Undock3/5/2024 – SpaceX Crew-8 Dock3/3/2024 – SpaceX Crew-8 Launch2/17/24 – ISS Progress 87 Dock2/14/24 – ISS Progress 87 Launch2/7/2024 – Axiom Mission 3 Undock2/1/2024 – Cygnus CRS-20 Capture1/30/2024 – Cygnus CRS-20 Launch1/20/2024 – Axiom Mission 3 Dock1/18/2024 – Axiom Mission 3 Launch12/21/2023 – SpaceX CRS-29 Undock12/1/23 – ISS Progress 86 Launch/Dock11/11/23 – SpaceX CRS-29 Dock11/9/23 – SpaceX CRS-29 Launch9/27/23 – Soyuz MS-23 Undock/Land9/15/23 – Soyuz MS-24 Launch/Dock8/27/23 – SpaceX Crew-7 Dock8/26/23 – SpaceX Crew-7 LaunchSpacewalksDate:Nov. 1, 2023Duration:6 hours, 42 minutesSpacewalkers:Jasmin Moghbeli,Loral O'HaraDate:Oct. 25, 2023Duration:7 hours, 41 minutesSpacewalkers:Oleg Kononenko, Nikolai ChubExpedition 70 Image Gallery493 ImagesGo To GalleryGo To GalleryExpedition 70 ResourcesMission BlogsThe daily blog updates for Expedition 70 highlighting crew and cargo missions, spacewalks, and advanced space research.Mission ScienceLearn more about the range of microgravity science encompassing human research, space phsyics, and more the Expedition 70 crew has conducted.SpaceX Crew-7 ScienceThe SpaceX Crew-7 members compared Earth and space sleep patterns, monitored numerous astronaut health parameters, examined disinfectants to clean spacecraft surfaces, and more.SpaceX Crew-8 ScienceThe SpaceX Crew-8 members explored the causes of neurodegenerative disorders, protecting plants from space radiation, space-caused changes in eye structure and vision, and more.Mission ImageryThe Expedition 70 crew is taking images of the Earth, the orbiting station, their daily lives aboard, and the experiments they conduct during their stay, plus so much more. Check out some of the images in the Expedition 70 gallery.Space Station ImageryView imagery taken from the International Space Station since 1999 of daily crew activities, spacewalks, visiting vehicles, and the Earth below.Expedition 70 NewsMore Expedition 70 NewsBlog1 Min ReadNASA, SpaceX Adjust Crew-8 Undocking DateArticle3 Min ReadWelcome Back to Planet Earth, Expedition 70 Crew!Blog2 Min ReadSoyuz Lands Returning O’Hara, Two Crewmates Back to EarthBlog1 Min ReadLoral O’Hara, Two Crewmates Returning to Earth on NASA TV2 Min ReadSoyuz Spacecraft Undocks to Return Three Crewmates to EarthBlog2 Min ReadThree Crew Members Departing Station Live on NASA TVBlog2 Min ReadTrio Boards Soyuz Crew Ship and Closes HatchBlog3 Min ReadTrio Finalizes Packing, Science Activities Before Friday DepartureBlogKeep ExploringKeep ExploringCommercial Crew Program OverviewSpace Station Research and TechnologyStation Benefits for HumanityHumans In Space

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Meet the crew

Andreas Mogensen, Jasmin Moghbeli, Satoshi Furukawa, Loral O'Hara, Konstantin Borisov, Oleg Kononenko, Nikolai Chub

startSept. 27, 2023ENDApril 5, 2024

startSept. 27, 2023

Sept. 27, 2023

ENDApril 5, 2024

END

April 5, 2024

iss070-s-002 (March 6, 2023) — The official Expedition 70 crew portrait with (top row from left) Roscosmos cosmonauts Nikolai Chub, Konstantin Borisov, and Oleg Kononenko; JAXA (Japan Aerospace Exploration Agency) astronaut Satoshi Furukawa; and NASA astronaut Loral O'Hara. In the front row are, ESA (European Space Agency) astronaut and Expedition 70 Commander Andreas Mogensen and NASA astronaut Jasmin Moghbeli.NASAThe official patch of the International Space Station's Expedition 70 crew.   The Expedition 70 patch is designed around the central yin-yang symbol representing balance; first and foremost, the balance of our beautiful planet Earth that is encircled by the yin-yang symbol and which forms part of the Expedition number. In our exploration of space, we are reminded of the uniqueness of Earth; the further we push the boundaries of human existence, the stronger our longing for our home planet grows. As our understanding of the cosmos expands, so does our understanding of Earth, and although we live in an ever-changing world, we recognize the need for a planet in balance to ensure our future. Space exploration is also about creating the future of our dreams. The tentative first steps we take today will hopefully become a well-trodden path in the future. This is represented stylistically by the "retro-futuristic" design of the patch, which mimics the design of the posters depicting the future from the early days of the space age. It is also emphasized by the yellow, orange, and red colors suggesting a sunrise. Lastly, the dynamism in the depiction of the number 7 suggests not only the physical launch into space, but also humanity’s progress towards the future.

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The official patch of the International Space Station’s Expedition 70 crew.   The Expedition 70 patch is designed around the central yin-yang symbol representing balance; first and foremost, the balance of our beautiful planet Earth that is encircled by the yin-yang symbol and which forms part of the Expedition number. In our exploration of space, we are reminded of the uniqueness of Earth; the further we push the boundaries of human existence, the stronger our longing for our home planet grows. As our understanding of the cosmos expands, so does our understanding of Earth, and although we live in an ever-changing world, we recognize the need for a planet in balance to ensure our future. Space exploration is also about creating the future of our dreams. The tentative first steps we take today will hopefully become a well-trodden path in the future. This is represented stylistically by the “retro-futuristic” design of the patch, which mimics the design of the posters depicting the future from the early days of the space age. It is also emphasized by the yellow, orange, and red colors suggesting a sunrise. Lastly, the dynamism in the depiction of the number 7 suggests not only the physical launch into space, but also humanity’s progress towards the future.

Andreas Mogensen, CommanderJasmin Moghbeli, Flight EngineerSatoshi Furukawa, Flight EngineerLoral O’Hara, Flight EngineerKonstantin Borisov, Flight EngineerOleg Kononenko, Flight EngineerNikolai Chub, Flight Engineer

4/6/2024 – Soyuz MS-24 Land4/5/2024 – Soyuz MS-24 Undock3/25/24 – Soyuz MS-25 Dock3/23/24 – Soyuz MS-25 Launch3/23/24 – SpaceX CRS-30 Dock3/21/2024 – SpaceX CRS-30 Launch3/12/2024 – SpaceX Crew-7 Splashdown3/11/2024 – SpaceX Crew-7 Undock3/5/2024 – SpaceX Crew-8 Dock3/3/2024 – SpaceX Crew-8 Launch2/17/24 – ISS Progress 87 Dock2/14/24 – ISS Progress 87 Launch2/7/2024 – Axiom Mission 3

Undock2/1/2024 – Cygnus CRS-20 Capture1/30/2024 – Cygnus CRS-20 Launch1/20/2024 – Axiom Mission 3 Dock1/18/2024 – Axiom Mission 3 Launch12/21/2023 – SpaceX CRS-29 Undock12/1/23 – ISS Progress 86 Launch/Dock11/11/23 – SpaceX CRS-29 Dock11/9/23 – SpaceX CRS-29 Launch9/27/23 – Soyuz MS-23 Undock/Land9/15/23 – Soyuz MS-24 Launch/Dock8/27/23 – SpaceX Crew-7 Dock8/26/23 – SpaceX Crew-7 Launch

Date:Nov. 1, 2023Duration:6 hours, 42 minutesSpacewalkers:Jasmin Moghbeli,Loral O'HaraDate:Oct. 25, 2023Duration:7 hours, 41 minutesSpacewalkers:Oleg Kononenko, Nikolai Chub

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Expedition 70 ResourcesMission BlogsThe daily blog updates for Expedition 70 highlighting crew and cargo missions, spacewalks, and advanced space research.Mission ScienceLearn more about the range of microgravity science encompassing human research, space phsyics, and more the Expedition 70 crew has conducted.SpaceX Crew-7 ScienceThe SpaceX Crew-7 members compared Earth and space sleep patterns, monitored numerous astronaut health parameters, examined disinfectants to clean spacecraft surfaces, and more.SpaceX Crew-8 ScienceThe SpaceX Crew-8 members explored the causes of neurodegenerative disorders, protecting plants from space radiation, space-caused changes in eye structure and vision, and more.Mission ImageryThe Expedition 70 crew is taking images of the Earth, the orbiting station, their daily lives aboard, and the experiments they conduct during their stay, plus so much more. Check out some of the images in the Expedition 70 gallery.Space Station ImageryView imagery taken from the International Space Station since 1999 of daily crew activities, spacewalks, visiting vehicles, and the Earth below.

Expedition 70 Resources

Mission BlogsThe daily blog updates for Expedition 70 highlighting crew and cargo missions, spacewalks, and advanced space research.

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Mission ScienceLearn more about the range of microgravity science encompassing human research, space phsyics, and more the Expedition 70 crew has conducted.

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SpaceX Crew-7 ScienceThe SpaceX Crew-7 members compared Earth and space sleep patterns, monitored numerous astronaut health parameters, examined disinfectants to clean spacecraft surfaces, and more.

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The SpaceX Crew-7 members compared Earth and space sleep patterns, monitored numerous astronaut health parameters, examined disinfectants to clean spacecraft surfaces, and more.

SpaceX Crew-8 ScienceThe SpaceX Crew-8 members explored the causes of neurodegenerative disorders, protecting plants from space radiation, space-caused changes in eye structure and vision, and more.

SpaceX Crew-8 Science

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Mission ImageryThe Expedition 70 crew is taking images of the Earth, the orbiting station, their daily lives aboard, and the experiments they conduct during their stay, plus so much more. Check out some of the images in the Expedition 70 gallery.

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Expedition 70 NewsMore Expedition 70 NewsBlog1 Min ReadNASA, SpaceX Adjust Crew-8 Undocking DateArticle3 Min ReadWelcome Back to Planet Earth, Expedition 70 Crew!Blog2 Min ReadSoyuz Lands Returning O'Hara, Two Crewmates Back to EarthBlog1 Min ReadLoral O'Hara, Two Crewmates Returning to Earth on NASA TV2 Min ReadSoyuz Spacecraft Undocks to Return Three Crewmates to EarthBlog2 Min ReadThree Crew Members Departing Station Live on NASA TVBlog2 Min ReadTrio Boards Soyuz Crew Ship and Closes HatchBlog3 Min ReadTrio Finalizes Packing, Science Activities Before Friday DepartureBlog

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TITLE: A to Z Topics Listing

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Safety Advisory PanelAIM (Aeronomy of Ice in the Mesosphere)Air Mobility Pathfinders ProjectAir Traffic Control LabsAir Traffic SolutionsAirborne ScienceAirMOSS (Airborne Microwave Observatory of Subcanopy and Subsurface)AirSTARAJAX (Alpha Jet Atmospheric eXperiment)Akatsuki/Venus Climate Orbiter mission (PLANET-C)Alan B. Shepard Jr.Alan G. PoindexterAlan L. BeanAlbert Sacco Jr.Alfred M. WordenAnalog MissionsAndre DouglasAndrew J. FeustelAndrew M. AllenAndrew R. MorganAndrew S. W. ThomasAnil MenonAnimal BiologyAnna L. FisherAnne C. McClainAntarctic StationsAntarctic Station NSFAnthony W. EnglandAPH (Advanced Plant Habitat)APIApollo ProgramApollo 1Apollo 7Apollo 8Apollo 9Apollo 10Apollo 11Apollo 12Apollo 13Apollo 14Apollo 15Apollo 15 SubsatelliteApollo 16Apollo 16 SubsatelliteApollo 17Apollo-Soyuz Test ProjectApophisApplied Sciences ProgramApply for FundingAppsAquaAquariusARCTAS (Arctic Research of the Composition of the Troposphere from Aircraft and Satellites)ARIEL (Atmospheric Remote-sensing Infrared Exoplanet Large-survey)ARISE 2014Armstrong Flight Research CenterArmstrong Test Facility (Plum Brook)ArrokothArtemisArtemis 1Artemis 2Artemis 3Artemis 4Artemis Campaign Development DivisionASCA (Advanced Satellite for Cosmology and Astrophysics)Asteroid InitiativeAsian American and Native Hawaiian Pacific Islander HeritageAsteroidsASTHROS (Astrophysics Stratospheric Telescope for High Spectral Resolution Observations at Submillimeter-wavelengths)Astro Observatory 1Astro Observatory 2AstrobiologyAstrogramAstromaterialsAstronautsAstronaut CandidatesAstrophysicsAstronomyAstrophysics DivisionAstrophysics Explorers ProgramAstrophysics PioneersAstrophysics Research ProgramAtmospheric CompositionATLAS (Atmospheric Laboratory of Applications and Science)Atmospheric Tomography Mission (ATom)ATS (Applications Technology Satellite Program)ATTREX (Airborne Tropical TRopopause Experiment)AudioAuraAurorasAwarded AbstractsAWE (Atmospheric Waves Experiment)BB200B-377 Super GuppyBarbara R. MorganBARREL (Balloon Array for Radiation-belt Relativistic Electron Losses)Barry E. WilmoreBBXRT (Broad Band X-ray Telescope)BECCAL (Bose Einstein Condensates & Cold Atoms Lab)Become an AstronautBenefits Back on EarthBenefits to HumanityBenefits to ScienceBenjamin Alvin DrewBennuBepiColomboBeresheetBernard A. Harris Jr.BetelgeuseBill NelsonBinary StarsBiological & Physical SciencesBiological Experiment-01 (BioExpt-01)BION / BiocosmosBiophysicsBlack History MonthBlack HolesBlazarsBlogsBlue GhostBob HinesBoeing StarlinerBonnie J. DunbarBorrellyBrand Guidelines CenterBrent W. JettBrewster H. Shaw Jr.Brian DuffyBrian T. O'LearyBRIC (Biological Research in Canisters)Brown DwarfsBruce E. MelnickBruce McCandlessBryan D. O'ConnorBudget & Annual ReportsBuzz AldrinByron K. LichtenbergCC-130C-20AC-9Caldwell CatalogCALIPSO (Cloud-Aerosol Lidar and Infrared Pathfinder Satellite)Callie First Graphic NovelCallistoCandidate AstronautsCarbon CycleCareersCarl E. WalzCarl J.

MeadeCarlos I. NoriegaCarruthers Geocorona Observatory (GLIDE)CARVE (Carbon in Arctic Reservoirs Vulnerability Experiment)Cassini-HuygensCatherine G. ColemanCATS (Cloud-Aerosol Transport System)Cell and Molecular BiologyCentennial ChallengesCentennial Challenges NewsCenter Innovation Fund Center of Excellence for Collaborative Innovation (CoECI)CERES (Clouds and the Earth's Radiant Energy System)CeREs (Compact Radiation Belt Explorer)Ceres Dwarf PlanetCessna C206CGRO (Compton Gamma Ray Observatory)CHAMP (Challenging Mini-satellite Payload)Chandra X-ray ObservatoryChandrayaan-1Chandrayaan-2Chandrayaan-3Charles A. Bassett IICharles CamardaCharles Conrad Jr.Charles D. GemarCharles D. WalkerCharles E. Brady Jr.Charles F. Bolden Jr.Charles G. FullertonCharles J. PrecourtCharles Lacy VeachCharles M. Duke Jr.Charles O. HobaughCharonChiaki MukaiCHIPS (Cosmic Hot Interstellar Plasma Spectrometer)Christina BirchChristina H. KochChristopher J. FergusonChristopher J. LoriaChristopher J. CassidyChristopher L. WilliamsCiencia e InvestigaciónCINDI (Coupled Ion Neutral Dynamic Investigation)CIPHER (Complement of Integrated Protocols for Human Exploration Research)Cirrus SR-22Citizen ScienceClayton C. AndersonClementineClifton C. Williams Jr.Climate ChangeClimate ScienceCloudsatCLPS (Commercial Lunar Payload Services)Cluster IICOBE (Cosmic Background Explorer)Color it!Columbia 300Combustion ScienceCometsCommercial CrewCommercial Crew RocketsCommercial Crew SpacecraftCommercial Lunar Payload Services (CLPS)Commercial SpaceCommercial Space ProgramsCommercial Supersonic TechnologyCommunicating and Navigating with MissionsCommunity of Practice WebinarCommunity PartnersComplex FluidsConjunction Assessment Risk Analysis Team (CARA)ConstellationsCONTOUR (COmet Nucleus TOUR)Convergent Aeronautics SolutionsCORAL (COral Reef Airborne Observatory)Coronal Mass EjectionsCOSI (Compton Spectrometer and Imager)Cosmic Origins ProgramCourses & Curriculums for ProfessionalsCOVID-19 ResponseCrab NebulaCrew Health and Performance Exploration Analog (CHAPEA)Crew Vehicle Systems Research FacilityCross Program IntegrationCRRES (Combined Release and Radiation Effects Satellite)Cryogenic Fluid Management (CFM)Cryogenic Propellant HandlingCryosphereCubeSat Launch InitiativeCubeSat Or Microsat Probabilistic and Analogies Cost Tool (COMPACT)CuPID (Cusp Plasma Imaging Detector)Curiosity (Rover)Curtis L. Brown Jr.CUSP (CubeSat Mission to Study Solar Particles)Cycles & Focus AreasCYGNSS (Cyclone Global Navigation Satellite System)DDale A. GardnerDaniel C. BrandensteinDaniel C. BurbankDaniel M. TaniDaniel T. BarryDaniel W. BurschDark Matter & Dark EnergyDART (Double Asteroid Redirection Test)Data ResourcesDavid A. WolfDavid C. HilmersDavid C. LeestmaDavid M. BrownDavid M. WalkerDavid R. ScottDAVINCI (Deep Atmosphere Venus Investigation of Noble gases, Chemistry, and Imaging)DawnDC-8Deep Impact / EPOXI (Extrasolar Planet Observation and Deep Impact Extended Investigation)Deep Space 1Deep Space Atomic ClockDeep

Space Network (DSN)Deep Space Optical Communications (DSOC)DeimosDeniz BurnhamDesert RATS (Research and Technology Studies)Developmental, Reproductive and Evolutionary Biology(DECLIC) DEvice for the study of Critical LIquids and CrystallizationDHCDidymos & DimorphosDiffuse X-Ray Spectrometer (DXS)DISCOVER AQ (Deriving Information on Surface Conditions from COlumn and VERtically Resolved Observations Relevant to Air Quality)Discovery ProgramDoing Business with NASADominic A. AntonelliDominic L. GorieDon L. LindDonald A. ThomasDonald E. WilliamsDonald H. PetersonDonald K. SlaytonDonald L. HolmquestDonald R. McMonagleDonald R. PettitDonn F. EiseleDorothy M. Metcalf-LindenburgerDouglas G. HurleyDouglas H. WheelockDragon Crew SpacecraftDragonflyDraper SERIES-2Drones & YouDroughtsDSCOVR (Deep Space Climate Observatory)Duane G. CareyDuane GravelineDust StormsDwarf PlanetsDynamics Explorer (DE)E:envihabEarly Career FacultyEarthEarth DayEarth Information CenterEarth ObservatoryEarth Observing-1 (EO-1)Earth ScienceEarth Science Data Systems ProgramEarth Science DivisionEarth Science Technology OfficeEarth Surface & InteriorEarth System Observatory (ESO)EarthquakesEarth’s AtmosphereEarth’s Magnetic FieldEarth’s MoonEarth’s Vital SignsE-BooksEclipsesEcostress (ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station)Edgar D. MitchellEdward G. GibsonEdward G. Givens Jr.Edward H. White IIEdward M. FinckeEdward T. LuEileen M. CollinsELaNa (Educational Launch of Nanosatellites)Electrified Powertrain Flight DemoElectroHydroDynamic (EHD) Test ChamberElectromagnetic SpectrumELFIN (Electron Losses & Fields Investigation)Ellen OchoaEllen S. BakerEllington FieldElliot M. See Jr.Elliptical GalaxiesEllison S. OnizukaEmergency & Pandemic ResponseEMIT (Earth Surface Mineral Dust Source Investigation)EnceladusEnvironmental Management Division (EMD)Environmental Science Services Administration (ESSA)EnVisionEquator-SER-2ERBS (Earth Radiation Budget Satellite)Eric A. BoeErisErosEscaPADE (Escape and Plasma Acceleration and Dynamics Explorers)EspañolEstablished Program to Stimulate Competitive Research (EPSCoR)Eta AquaridsE-TBEx (Enhanced Tandem Beacon Experiment)EuclidEugene A. CernanEugene H. TrinhEuropaEuropa ClipperEUVE (Extreme-Ultraviolet Explorer)EventsEveryday Extraordinary StoriesExoMarsExoplanet CatalogExoplanet DiscoveriesExoplanet Exploration ProgramExoplanet ScienceExoplanet Travel BureauExoplanetsExosphereExpedition 1Expedition 2Expedition 3Expedition 4Expedition 5Expedition 6Expedition 7Expedition 8Expedition 9Expedition 10Expedition 11Expedition 12Expedition 13Expedition 14Expedition 15Expedition 16Expedition 17Expedition 18Expedition 19Expedition 20Expedition 21Expedition 22Expedition 23Expedition 24Expedition 25Expedition 26Expedition 27Expedition 28Expedition 29Expedition 30Expedition 31Expedition 32Expedition 33Expedition 34Expedition 35Expedition 36Expedition 37Expedition 38Expedition 39Expedition 40Expedition 41Expedition

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4 for EducatorsGRAIL (Gravity Recovery And Interior Laboratory)Grants & OpportunitiesGravity Probe B (GP-B)Green Aviation TechGreenhouse EffectGreenhouse GassesGregory C. JohnsonGregory E. ChamitoffGregory H. JohnsonGregory J. HarbaughGregory JarvisGregory T. LinterisGriffin (lander)Ground Facilities for ISSGuion S. Bluford Jr.Gulfstream C-20AGulfstream G-IIIGulfstream G-VGulfstream IVGuy S. GardnerHH-135HALCA (Highly Advanced Laboratory for Communications and Astronomy)Hale-BoppHalley's CometHaloSatHarrison H. SchmittHaumeaHayabusaHayabusa 2Heidemarie M. Stefanyshyn-PiperHeliophysicsHeliophysics DivisionHeliophysics Research ProgramHelios 1Helios 2HeliosphereHelioSwarmHelix NebulaHenry W. Hartsfield Jr.HERA (Human Exploration Research Analog)HERMES (Heliophysics Environmental and Radiation Measurement Experiment Suite)Herschel Space ObservatoryHETE-1 (High Energy Transient Explorer 1)HETE-2 (High Energy Transient Explorer 2)High-Tech ComputingHinode (Solar B)HipparcosHi-Rate Composite Aircraft ManufacturingHispanic Heritage MonthHistory: NASAHiten / HagomoroHitomi (ASTRO-H)Horsehead NebulaHS3 Hurricane Mission (Hurricane and Severe Storm Sentinel)HU-25Hubble Space TelescopeHuman DimensionsHuman Exploration of Space: Why We GoHuman Health and PerformanceHuman Landing System ProgramHuman Research ProgramHuman Space Travel ResearchHuman Spaceflight Capabilities DivisionHumans in SpaceHurricanesHybrid Thermally Efficient CoreHyperionHypersonic TechnologyHyspIRI (Hyperspectral Infrared Imager)IIBEX (Interstellar Boundary Explorer)Ice GiantsIceBridgeICESat (Ice, Clouds and Land Elevation Satellite)ICESat-2 (Ice, Cloud and land Elevation Satellite-2)ICON (Ionospheric Connection Explorer)IdaIEH-3 (International Extreme Ultraviolet Hitchhiker)Ilan RamonIMAGE (Imager for Magnetopause-to-Aurora Global Exploration)ImagesImages & Media GuidelinesIMAP (Interstellar Mapping and Acceleration Probe)IMP-8 (Interplanetary Monitoring Platform 8)In-flight Education DownlinksINCUS (Investigation of Convective Updrafts)Infrared Space Observatory (ISO)Ingenuity (Helicopter)InSight (Interior Exploration using Seismic Investigations, Geodesy and Heat Transport)Institutional FundingIntegrated Aviation Systems ProgramInteractivesInternational Gamma-ray Astrophysics Laboratory (INTEGRAL)International Space Station (ISS)International Space Station DivisionInternshipsISRU (In-Situ Resource Utilization)IoIonosphereIRIS (Interface Region Imaging Spectrograph)Irregular GalaxiesISEE-3/ICE (International Sun-Earth Explorer-3 / International Cometary Explorer)ISERV (ISS SERVIR Environmental Research and Visualization System)ISS-RapidScat (Rapid Scatterometer)ISS ResearchIUE (International Ultraviolet Explorer)IXPE (Imaging X-ray Polarimetry Explorer)JJack D. FischerJack HathawayJack R. LousmaJake GarnJames A. Lovell Jr.James A. McDivittJames A. PawelczykJames B. IrwinJames C. AdamsonJames D. A. Van HoftenJames D. Halsell

Jr.James D. WetherbeeJames F. BuchliJames F. ReillyJames H. NewmanJames M. KellyJames P. BagianJames P. DuttonJames S. VossJames Webb Space Telescope (JWST)Janet L. KavandiJanice E. VossJanusJasmin MoghbeliJason-1Jason-3Jason-CS (Continuity of Service) / Sentinel-6Jay Clark BuckeyJeanette J. EppsJean-Jacques FavierJeffrey A. HoffmanJeffrey N. WilliamsJeffrey S. AshbyJerome AptJerry L. RossJerry M. LinengerJessica U. MeirJessica WatkinsJessica WittnerJet Propulsion LaboratoryJoan E. HigginbothamJoe F. Edwards Jr.Joe H. EngleJohn A. LlewellynJohn B. HerringtonJohn D. OlivasJohn E. BlahaJohn H. CasperJohn H. Glenn Jr.John L. PhillipsJohn L. Swigert Jr.John M. FabianJohn M. GrunsfeldJohn M. LoungeJohn O. CreightonJohn S. BullJohn W. YoungJohnson Flight OperationsJohnson Space CenterJohnson’s Mission Control CenterJoint Agency Satellite DivisionJon A. McBrideJonny KimJosé M. HernándezJoseph M. AcabaJoseph P. AllenJoseph P. KerwinJoseph R. TannerJosh A. CassadaJPSS (Joint Polar Satellite System)JPSS-3JPSS-4Judith A. ResnikJUICE (Jupiter Icy Moons Explorer)JunoJupiterJupiter MoonsJWST (James Webb Space Telescope)KK. Megan McArthurKaguyaKalpana ChawlaKaren L. NybergKarl G. HenizeKarol J. BobkoKatherine Johnson IV & V FacilityKathleen RubinsKathryn C. ThorntonKathryn D. SullivanKathryn P. HireKayla BarronKeck Interferometer (KI)Kennedy Space CenterKenneth D. BowersoxKenneth D. CameronKenneth D. CockrellKenneth S. Reightler Jr.Kenneth T. HamKent V. RomingerKepler / K2Kepler 22BKevin A. FordKevin P. ChiltonKevin R. KregelKjell N. LindgrenKnowledge Inventory For ProfessionalsKuiper Airborne Observatory (KAO)LL. Blaine Hammond Jr.L. Gordon Cooper Jr.La Tierra y calentamiento globalLADEE (Lunar Atmosphere Dust Environment Explorer)LAGEOS (LAser GEOdynamics Satellite)LandsatLandsat 7Landsat 8 / LDCM (Landsat Data Continuity Mission)Landsat 9Langley Research CenterLaser Communications RelayLaunch Services OfficeLaunch Services ProgramLaurel B. ClarkLawrence J. DelucasLBTI (Large Binocular Telescope Interferometer)LCROSS (Lunar Crater Observation and Sensing Satellite)Learning ResourcesLee J. ArchambaultLee M. MorinLEIA (Lunar Explorer Instrument for space biology Applications)Leland D. MelvinLenticular GalaxiesLeonid K. KadenyukLeonid MAC (Multi-instrument Aircraft Campaign)LeonidsLeroy ChiaoLessons Learned for ProfessionalsLGBTQ PrideLiberaLife at NASALIS (Lightning Imaging Sensor)Linda M. GodwinLISA (Laser Interferometer Space Antenna)Lisa M. NowakLodewijk Van Den BergLOFTID (Low-Earth Orbit Flight Test of an Inflatable Decelerator)Logistics Management Division (LMD)Logo & Media UsageLoral O’HaraLoren J. ShriverLoren W. ActonLow Boom Flight DemonstratorLow-Density Supersonic DeceleratorLow-Earth Orbit EconomyLRO (Lunar Reconnaissance Orbiter)LSII ConsortiumLucyLuke DelanyLunaH-Map (Lunar Polar Hydrogen Mapper)Lunar Discovery & Exploration ProgramLunar EclipsesLunar Lander SimulationLunar OrbiterLunar Orbiter 1Lunar Orbiter 2Lunar Orbiter 3Lunar Orbiter 4Lunar Orbiter 5Lunar ProspectorLunar ScienceLunar Surface Innovation ConsortiumLunar

Surface Innovation InitiativeLunar Surface Technology ResearchLunar TrailblazerLyridsMM. Scott CarpenterMae C. JemisonMagellanMagnetarsMagnetosphereMAIA (Multi-Angle Imager for Aerosols)Management AstronautsManley Lanier Carter Jr.Manufacturing, Materials, 3-D PrintingMarCO (Mars Cube One)Marcos BerriosMargaret Rhea SeddonMarinerMariner 1Mariner 2Mariner 3Mariner 4Mariner 5Mariner 6Mariner 7Mariner 8Mariner 9Mariner 10Mario Runco Jr.Mark C. LeeMark E. KellyMark L. PolanskyMark N. BrownMark T. Vande HeiMarsMars 2020Mars Campaign Development DivisionMars Climate OrbiterMars Exploration ProgramMars Exploration Rovers (MER)Mars ExpressMars Global Surveyor (MGS)Mars MoonsMars ObserverMars OdysseyMars Orbiter Mission (MOM)Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE)Mars PathfinderMars PhoenixMars Polar Lander / Deep Space 2Mars Reconnaissance Orbiter (MRO)Mars Sample Return (MSR)Mars Science Laboratory (MSL)Marsha IvinsMarshall Space Flight CenterMarshall Test Facility and Support InfrastructureMartian Moon Exploration (MMX)Martin J. FettmanMary E. WeberMary L. CleaveMaterials ScienceMaterials Science Research Rack (MSRR)Matthew DominickMAVEN (Mars Atmosphere and Volatile EvolutioN)Media ResourcesMegan McArthurMerchandise ApprovalsMercuryMesosphereMESSENGER (MErcury Surface, Space ENvironment, GEochemistry, and Ranging)Messier CatalogMeteor ShowersMeteors & MeteoritesMetOp (Meteorological Operational satellite)Michael A. BakerMichael CollinsMichael FinckeMichael E. FossumMichael E. Lopez-AlegriaMichael J. BloomfieldMichael J. ForemanMichael J. MassiminoMichael J. McCulleyMichael J. SmithMichael L. CoatsMichael L. GernhardtMichael P. AndersonMichael R. BarrattMichael R. CliffordMichael S. HopkinsMichael T. GoodMichel F. CurtisMichoud Assembly FacilityMicro-10Microbial Tracking (MT)MicrobiologyMillie Hughes-FulfordMini-RF InstrumentMinXSS (Miniature X-ray Solar Spectrometer)MirandaMission EquityMission Support DirectorateMissionsMMS (Magnetospheric Multiscale)Modeling, Analysis and Prediction (MAP) ProgramModerate Resolution Imaging Spectroradiometer (MODIS)Modern FiguresModular Optoelectric Multispectral Scanner (MOMS)MoonsMOS (Marine Observation Satellite)MSI ExchangeMSL InstrumentationMulti-Spectral Fluorescence Imaging System (Spectrum)MultimediaMUREPMUSE (Multi-slit Solar Explorer)Museum AllianceNN. Jan DavisNAAMES (North Atlantic Aerosols and Marine Ecosystems Study)Nancy Grace Roman Space TelescopeNancy J. Currie-GreggNanoSail-DNASA Advisory Council (NAC)NASA Aeronautics CommitteeNASA AircraftNASA AppsNASA Centers & FacilitiesNASA DirectoratesNASA e-BooksNASA Educator Professional Development CenterNASA en españolNASA Engineering and Safety CenterNASA Engineering & Safety Center AcademyNASA HeadquartersNASA HistoryNASA History OfficeNASA Home & CityNASA ImagesNASA Innovative Advanced Concepts (NIAC) ProgramNASA InteractivesNASA iTech ProgramNASA Knowledge ProgramNASA Open

Source SoftwareNASA PlusNASA PodcastsNASA Safety CenterNASA Safety Center Professional DevelopmentNASA Shared Services CenterNASA Social Media ContactsNASA Socials ProgramNASA STEMNASA STEM ProjectsNASA TVNASA WorldwindNational Aeronautics Research InstituteNational Space Council Users' Advisory Group (NSpC UAG)Native American Heritage MonthNatural DisastersNEA Scout (Near Earth Asteroid Scout)NEAR ShoemakerNear Space NetworkNear-Earth Object (NEO)Nebulae GameNEEMO (NASA Extreme Environment Mission Operations)Neil A. ArmstrongNeil Gehrels Swift ObservatoryNeil W. Woodward IIINEK/SIRIUS (Nezemnyy Eksperimental'nyy Kompleks/Scientific International Research In a Unique terrestrial Station)NEO Surveyor (Near-Earth Object Surveyor Space Telescope)NEOWISENeptuneNeptune MoonsNeptune-Like ExoplanetsNeutral Buoyancy LabNeutron StarsNew Frontiers ProgramNew HorizonsNewslettersNExIS (NASA's Exploration & In-space Services)Next Gen STEMNIAC StudiesNIAC SymposiumNICER (Neutron star Interior Composition Explorer)Nicholas J. M. PatrickNichole AyersNicole A. MannNicole P. StottNight Sky NetworkNimbusNISAR (NASA-ISRO Synthetic Aperture Radar)NOAA (National Oceanic and Atmospheric Administration)NOAA-20 (JPSS-1)NOAA-21 (JPSS-2)NOAA-N PrimeNorman E. ThagardNorthrop Grumman Commercial ResupplyNova-C (Lander)NozomiNuSTAR (Nuclear Spectroscopic Telescope Array)OO / OREOS (Organism / Organic Exposure to Orbital Stresses)Ocean Surface Topography Mission (OSTM) / Jason-2Ocean WorldsOceanographyOceansOceans Melting Greenland (OMG)OCO (Orbiting Carbon Observatory)OCO-2 (Orbiting Carbon Observatory 2)OCO-3 (Orbiting Carbon Observatory 3)Office of International and Interagency Relations (OIIR)Office of Legislative and Intergovernmental Affairs (OLIA)Office of Small Business Programs (OSBP)Office of Strategic Infrastructure (OSI)Office of Technology, Policy and Strategy (OTPS)Office of the Chief Engineer (OCE)Office of the Chief Financial Officer (OCFO)Office of the Chief Health and Medical Officer (OCHMO)Office of the Chief Information Officer (OCIO)Office of the Chief Scientist (OCS)Office of the General Counsel (OGC)One Year CrewOpportunities For Educators to Get InvolvedOpportunities For International Participants to Get InvolvedOpportunities For Researchers to Get InvolvedOpportunities For Students to Get InvolvedOpportunities For U.S. Citizens to Get InvolvedOpportunities to Contribute to NASA Missions & Get InvolvedOpportunity (Rover)ORACLES (ObseRvations of Aerosols above CLouds and their IntEractionS)Orbiting Astronomical Observatory (OAO)ORFEUS (Orbiting and Retrievable Far and Extreme Ultraviolet Spectrometer)-SPAS IIOrganizationsOrigin & Evolution of the UniverseOrion Multi-Purpose Crew VehicleOrion NebulaOrion SpacecraftOrionids OSAM-1 (On-Orbit Servicing, Assembly, and Manufacturing 1)OSIRIS-REx (Origins, Spectral Interpretation, Resource Identification, Security-Regolith Explorer)OumuamuaOuter Planets & Ocean Worlds ProgramOutside the ClassroomOV-10Owen K. GarriottOzone LayerOzone Mapping and Profiler Suite (OMPS)PP-

3PACE (Plankton, Aerosol, Cloud, Ocean Ecosystem)Pamela A. MelroyPandoraParker Solar Probe (PSP)Partner AstronautsPartner with NASA STEMPartner With UsPatricia C. RobertsonPatrick G. ForresterPaul D. Scully-PowerPaul J. WeitzPaul S. LockhartPaul W. RichardsPeggy A. WhitsonPeople of NASAPeregrine Lunar LanderPerseidsPerseverance (Rover)Peter J. K. WisoffPFMI (Pore Formation and Mobility Investigation)PharmaSatPhilae (Lander)Philip K. ChapmanPhobosPhoebePhysical SciencesPhysical Sciences ProgramPI ResourcesPierre J. ThuotPiers J. SellersPioneerPioneer 0 / Able 1Pioneer 1 / Able 2Pioneer 2Pioneer 3Pioneer 4Pioneer 5Pioneer 6Pioneer 7Pioneer 8Pioneer 9Pioneer 10Pioneer 11Pioneer P-3 / Able 4BPioneer P-30 / Able 5APioneer P-31 / Able 5BPioneer VenusPioneer Venus 1Pioneer Venus 2Pioneer-EPlanckPlanet XPlanetary DefensePlanetary Defense Coordination OfficePlanetary Environments & AtmospheresPlanetary Geosciences & GeophysicsPlanetary SciencePlanetary Science DivisionPlanetary TransitsPlanetsPlant BiologyPlasma Kristall 4 (PK-4)PlayPlay it!Plume-Surface Interaction (PSI) ProjectPlutoPluto MoonsPodcastsPolarPopular HistoryPost Doc ProgramPotentially Hazardous Asteroid (PHA)PredatorPREFIRE (Polar Radiant Energy in the Far-InfraRed Experiment)PRESat (PharmaSat Risk Evaluation Satellite)Private Astronaut MissionsPrizes, Challenges & CrowdsourcingPrizes, Challenges & Crowdsourcing NewsProject Management & Systems EngineeringProject MercuryProtostarsProxima Centauri bPsyche AsteroidPsyche MissionPulsarsPUNCH (Polarimeter to Unify the Corona and Heliosphere)QQ-PACE (CubeSat Particle Aggregation and Collision Experiment)QuadrantidsQuasarsQuesst (X-59)Quesst: The FlightsQuesst: The MissionQuesst: The ScienceQuesst: The TeamQuesst: The VehicleQuikSCAT (Quick Scatterometer)RRADARSAT-1 (Radar Satellite 1)RAD-SEEDRaja ChariRandolph J. BresnikRangerRanger 1Ranger 2Ranger 3Ranger 4Ranger 5Ranger 6Ranger 7Ranger 8Ranger 9Read it!Red DwarfsRed GiantsRedstone 3 (Freedom 7)Redstone 4 (Liberty Bell 7)Research & Analytics ProgramResearch ArchiveResearch Opportunities in Space and Earth Sciences (ROSES)Revolutionary Vertical Lift TechnologyRex J. WalheimRheaRHESSI (Reuven Ramaty High Energy Solar Spectroscopic Imager)Richard A. MastracchioRichard A. SearfossRichard F. Gordon Jr.Richard H. TrulyRichard J. HiebRichard M. LinnehanRichard M. MullaneRichard N. RichardsRichard O. CoveyRichard R. ArnoldRick D. HusbandRings of JupiterRings of NeptuneRings of SaturnRing-Sheared Drop (RSD)RLL (Robotic Lunar Lander)Robb KulinRobert A. R. ParkerRobert C. SpringerRobert D. CabanaRobert F. OvermyerRobert J. CenkerRobert L. BehnkenRobert L. CrippenRobert L. Curbeam Jr.Robert L. GibsonRobert L. Satcher Jr.Robert L. StewartRobert Shane KimbroughRobotic Lunar Exploration ProgramRoboticsRodent Research (RR)Roger B. ChaffeeRoger K. CrouchRonald A. PariseRonald E. EvansRonald E. McNairRonald J. Garan Jr.Ronald J. GrabeRonald M. SegaRosalind Franklin (Rover)ROSAT (ROentgen SATellite)RosettaRoy D. Bridges Jr.RS-25Russell L. SchweickartRXTE (Rossi X-ray Timing Explorer)RyuguSS. Christa McAuliffeS.

David GriggsS3VISAC-B (Satélite de Aplicaciones Científicas-B)SAGE III (Stratospheric Aerosol and Gas Experiment)SAGE-III Meteor-3M (Stratospheric Aerosol and Gas Experiment III on Meteor-3M)SakigakeSally K. RideSAMPEX (Solar Anomalous and Magnetospheric Particle Explorer)Samuel T. DurranceSandra H. MagnusSaturnSaturn MoonsSBIRSBIR / STTRScience & ResearchScience ActivationScience for EveryoneScience in the AirScience InstrumentsScience LeadershipScience Mission DirectorateScience of Space ExplorationScience SolicitationsScience-enabling TechnologyScientific BalloonsScott D. AltmanScott D. TingleScott E. ParazynskiScott J. HorowitzScott J. KellySea IceSEAC4RS (Studies of Emissions, Atmospheric Composition, Clouds and Climate Coupling by Regional Surveys)SeasatSeaWiFS (Sea-viewing Wide Field-of-view Sensor)SeaWindsSentinel-6 Michael Freilich SatelliteSentinel-6BSerena M. Auñón-ChancellorSeres Humanos en el spacioSERVIR (Regional Visualization and Monitoring System)Seyfert GalaxiesShannon W. LucidShannon WalkerSherwood C. SpringShoemaker-Levy 9Shuttle-MirSidney M. GutierrezSierra UAS #2SimLabsSIR-C/X-SAR (Shuttle Imaging Radar-C / X-Band Synthetic Aperture Radar)Sistema solarSkylabSkywatchingSkywatching TipsSmall Bodies of the Solar SystemSmall Business Innovation Research / Small BusinessSmall Satellite MissionsSmall Spacecraft Technology ProgramSmallSats ProgramSMAP (Soil Moisture Active Passive)SMART-1 (Small Missions for Advanced Research in Technology 1)SNOE (Student Nitric Oxide Explorer)Social MediaSOFIA (Stratospheric Observatory for Infrared Astronomy) / 747-SPSoft Matter Dynamics (SMD) / FOAMSoftware to LicenseSOHO (Solar and Heliospheric Observatory)Solar CruiserSolar Dynamics Observatory (SDO)Solar EclipsesSolar Electric Propulsion (SEP)Solar FlaresSolar OrbiterSolar Sail – Beyond Plum BrookSolar System AmbassadorsSolar System Exploration Research Virtual InstituteSolar Terrestrial Probes ProgramSolar WindSolid Fuel Ignition and Extinction (SoFIE)Solidification Using a Baffle in Sealed Ampoules (SUBSA)SORCE (Solar Radiation and Climate Experiment)SORTIE (Scintillation Observations & Response of the Ionosphere to Electrodynamics)Sounding RocketsSounding Rockets ProgramSounds & RingtonesSouthern Delta AquariidsSpace BiologySpace Biology ProgramSpace Biosciences at AmesSpace Communications & Navigation Program (SCaN)Space Communications TechnologySpace Environment Testbeds (SET-1)Space Environments Testing Management Office (SETMO)Space Flight Awareness (SFA)Space GrantSLS (Space Launch System)Space Life & Physical Sciences Research & Applications DivisionSpace Nuclear Propulsion (SNP)Space Operations Directorate Resource Management OfficeSpace Operations Mission DirectorateSpace Samples for Your ClassroomSpace ShuttleSpace Station Research OpportunitiesSpace Tech Graduate ResearchSpace Technology 5Space Technology 6Space Technology 7 / Disturbance Reduction System (DRS)Space Technology GrantsSpace Technology Mission DirectorateSpace Technology Research GrantsSpace Vehicle Mockup FacilitySpace

WeatherSpaceships and RocketsSpacesuitsSpaceX Commercial ResupplySpartanSpectrum ManagementSPHEREx (Spectro-Photometer for the History of the Universe and Ices Explorer)SpinoffsSpiral GalaxiesSpirit (Rover)Spitzer Space TelescopeSpot the International Space StationSRTM (Shuttle Radar Topography Mission)ST5Stanley G. LoveStardust / Stardust NExTStarlinerStarsSTEM Engagement at NASAStennis Space CenterStennis Test Facility and Support InfrastructureStephanie D. WilsonStephen D. ThorneStephen G. BowenStephen K. RobinsonStephen N. FrickStephen S. OswaldSTEREO (Solar TErrestrial RElations Observatory)Steven A. HawleySteven L. SmithSteven R. NagelSteven R. SwansonSteven W. LindseyStory MusgraveSTPStrange New WorldsStrategic Integration and Management DivisionStrategic Partnerships GuidelinesStratosphereSTS- 1STS- 2STS- 3STS- 4STS- 5STS- 6STS- 7STS- 8STS- 9STS- 26STS- 27STS- 28STS- 29STS- 30STS- 31STS- 32STS- 33STS- 34STS- 35STS- 36STS- 37STS- 38STS- 39STS- 40STS- 41STS- 41BSTS- 41CSTS- 41DSTS- 41GSTS- 42STS- 43STS- 44STS- 45STS- 46STS- 47STS- 48STS- 49STS- 50STS- 51STS- 51ASTS- 51BSTS- 51CSTS- 51DSTS- 51FSTS- 51GSTS- 51ISTS- 51JSTS- 51LSTS- 52STS- 53STS- 54STS- 55STS- 56STS- 57STS- 58STS- 59STS- 60STS- 61STS- 61ASTS- 61BSTS- 61CSTS- 62STS- 63STS- 64STS- 65STS- 66STS- 67STS- 68STS- 69STS- 70STS- 71STS- 72STS- 73STS- 74STS- 75STS- 76STS- 77STS- 78STS- 79STS- 80STS- 81STS- 82STS- 83STS- 84STS- 85STS-86STS- 87STS- 88STS- 89STS- 90STS- 91STS- 92STS- 93STS- 94STS- 95STS- 96STS- 97STS- 98STS- 99STS- 100STS- 101STS- 102STS- 103STS- 104STS- 105STS- 106STS- 107STS- 108STS- 109STS- 110STS- 111STS- 112STS- 113STS- 114STS- 115STS- 116STS- 117STS- 118STS- 119STS- 120STS- 121STS- 122STS- 123STS- 124STS- 125STS- 126STS- 127STS- 128STS- 129STS- 130STS- 131STS- 132STS- 133STS- 134STS- 135Stuart A. RoosaStudying ExoplanetsSubmillimeter Wave Astronomy Satellite (SWAS)Sunita L. WilliamsSunRISE (Sun Radio Interferometer Space Experiment)SunspotsSuomi NPP (Suomi National Polar-orbiting Partnership)Super King AirSuper-Earth ExoplanetsSupermoonsSupernovaeSupersonic FlightSurveyorSurveyor 1Surveyor 2Surveyor 3Surveyor 4Surveyor 5Surveyor 6Surveyor 7Surveyor Model 1Surveyor Model 2Surveyor Model SD-1Susan J. HelmsSusan L. KilrainSustainability at Kennedy Space CenterSustainable Flight DemonstratorSustainable Flight National PartnershipSuzakuSWOT (Surface Water and Ocean Topography)Synchronous Meteorological Satellite (SMS)System-Wide SafetyTT-34CT-38 Astronaut TrainerTamara E. JerniganTaylor WangTech Demo MissionsTech Dev ProjectsTech PortfolioTechnologyTechnology DemonstrationTechnology Demonstration Missions ProgramTechnology for Living in SpaceTechnology for Space TravelTechnology HighlightsTechnology ResearchTechnology Transfer & SpinoffsTecnologíaTeleroboticsTempel 1Terence T. HenricksTerraTerrence W. WilcuttTerrestrial ExoplanetsTerrestrial PlanetsTerriers (Tomographic Experiment Using Radiative Recombinative Ionospheric EUV and Radio Sources)Terry J. HartTerry W. Virts

Jr.TESS (Transiting Exoplanet Survey Satellite)Tethered Satellite System (TSS)TG-14The Big BangThe Future of Commercial SpaceThe Great Red SpotThe Habitable ZoneThe Human Body in SpaceThe Kuiper BeltThe Milky WayThe North StarThe Oort CloudThe Search for LifeThe Solar SystemThe SunThe UniverseTHEMIS (Time History of Events and Macroscale Interactions During Substorms)THEMIS-ARTEMIS (Time History of Events and Macroscale Interactions During Substorms – Acceleration, Reconnection, Turbulence and Electrodynamics of Moon’s Interaction with the Sun)Theodore C. FreemanThermosphereThomas D. AkersThomas D. JonesThomas H. MarshburnThomas J. HennenThomas K. Mattingly IIThomas P. StaffordTIMED (Thermospere Ionosphere Mesosphere Energetics and Dynamics Mission)Timothy J. CreamerTimothy L. KopraTIROS (Television Infrared Observation Satellite Program)TitanTOMS-EP (Total Ozone Mapping Spectrometer – Earth Probe)TOPEX / Poseidon (ocean TOPography EXperiment)TornadoesTotal and Spectral Solar Irradiance Sensor – 1 (TSIS-1)Total and Spectral Solar Irradiance Sensor – 2 (TSIS-2)TP / ACOTRACE (Transition Region and Coronal Explorer)Trace Gas Orbiter (TGO)Tracking Aerosol Convection Interactions Experiment (TRACER)Tracking and Data Relay Satellite (TDRS)Tracy Caldwell DysonTransformational ToolsTechnologiesTransformative Aeronautics Concepts ProgramTritonTrojan AsteroidsTropical Rainfall Measuring Mission (TRMM)TROPICS (Time-Resolved Observations of Precipitation Structure and Storm Intensity with a Constellation of Smallsats)TroposphereTropospheric Emissions: Monitoring of Pollution (TEMPO)TsunamisTWINS (Two Wide-Angle Imaging Neutral-Atom Spectrometers)Twins StudyTyler N. HagueUUAS in the NASUAS Traffic ManagementUH-1HUH-IIUlyssesUnderway Recovery Test 10Universe of MonstersUniversity InnovationUniversity Leadership InitiativeUniversity Student Research ChallengeUniversoUpper Atmosphere Research Satellite (UARS)UranusUranus MoonsUrban Air Mobility SimulationUSGS (United States Geological Survey)VValles MarinerisVan Allen ProbesVance D. BrandVegetable Production System (VEGGIE)VenusVenus ExpressVERITAS (Venus Emissivity, Radio Science, InSAR, Topography & Spectroscopy)Vertical Motion SimulatorVestaVictor J. GloverVideoVikingViking 1Viking 2Viking 400VIPER (Volatiles Investigating Polar Exploration Rover)Virgil I. GrissomVirtual Guest ProgramVisible Infrared Imaging Radiometer Suite (VIIRS)VolcanoesVoyager 1Voyager 2Voyager ProgramWWallops Flight FacilityWalter CunninghamWalter M. Schirra Jr.Warren HoburgWater & Energy CycleWater on EarthWB-57Weather and Atmospheric DynamicsWendy B. LawrenceWetLab-2What We DoWhite DwarfsWhite Sands Test FacilityWhite Sands Test Facility and Support InfrastructureWhy Go To SpaceWild 2WildfiresWilkinson Microwave Anisotropy Probe (WMAP)William A. AndersWilliam A. OefeleinWilliam A. PailesWilliam B. LenoirWilliam C. McCoolWilliam E. ThorntonWilliam F. FisherWilliam F. ReaddyWilliam G. GregoryWilliam M. ShepherdWilliam R. PogueWilliam S. McArthur Jr.Wind MissionWinston E. ScottWIRE (Wide-Field Infrared

Explorer)WISE (Wide-field Infrared Survey Explorer)Women at NASAWomen’s History MonthXXelene Lunar Lander (XL-1)xEVA & Human Surface MobilityXMM-Newton (X-ray Multi-Mirror Newton)XRISM (X-Ray Imaging and Spectroscopy Mission)YYO-3AYohkohYvonne Darlene CagleZZena CardmanZero Boil-Off Tank (ZBOT-NC)

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2023 Annular Eclipse

3D Tissue Chips

67P/Churyumov-Gerasimenko

AAM Drones and You

Accessibility at NASA

Accessing Flight Tests

ACE (Advanced Composition Explorer)

Acerca de NASA

ACME (Advanced Combustion via Microgravity Experiments)

AcrimSat (Active Cavity Irradiance Monitor Satellite)

ACT (Atmospheric Carbon and Transport) – America

ACTE (Adaptive Compliant Trailing Edge)

Active Galaxies

ADEOS (Advanced Earth Observing Satellite) / MIDORI

Advanced Air Vehicles Program

Advanced Composites

Advanced Studies & Concepts (PACE)

Advanced Spacesuits

Aerospace Safety Advisory Panel

AIM (Aeronomy of Ice in the Mesosphere)

Air Mobility Pathfinders Project

Air Traffic Control Labs

AirMOSS (Airborne Microwave Observatory of Subcanopy and Subsurface)

AirSTAR

AJAX (Alpha Jet Atmospheric eXperiment)

Akatsuki/Venus Climate Orbiter mission (PLANET-C)

Alan G. Poindexter

Alan L. Bean

Albert Sacco Jr.

Alfred M. Worden

Analog Missions

Andre Douglas

Andrew J. Feustel

Andrew M. Allen

Andrew R. Morgan

Andrew S. W. Thomas

Anil Menon

Anna L. Fisher

Anne C. McClain

Antarctic Stations

Antarctic Station NSF

Anthony W. England

APH (Advanced Plant Habitat)

API

Apollo Program

Apollo 1

Apollo 7

Apollo 9

Apollo 10

Apollo 12

Apollo 13

Apollo 14

Apollo 15

Apollo 15 Subsatellite

Apollo 16

Apollo 16 Subsatellite

Apollo 17

Applied Sciences Program

Apply for Funding

Apps

Aqua

Aquarius

ARCTAS (Arctic Research of the Composition of the Troposphere from Aircraft and Satellites)

ARIEL (Atmospheric Remote-sensing Infrared Exoplanet Large-survey)

ARISE 2014

Armstrong Test Facility (Plum Brook)

Artemis 4

Artemis Campaign Development Division

ASCA (Advanced Satellite for Cosmology and Astrophysics)

Asteroid Initiative

ASTHROS (Astrophysics Stratospheric Telescope for High Spectral Resolution Observations at Submillimeter-wavelengths)

Astro Observatory 1

Astro Observatory 2

Astrogram

Astronaut Candidates

Astrophysics Explorers Program

Astrophysics Pioneers

Astrophysics Research Program

Atmospheric Composition

ATLAS (Atmospheric Laboratory of Applications and Science)

Atmospheric Tomography Mission (ATom)

ATS (Applications Technology Satellite Program)

ATTREX (Airborne Tropical TRopopause Experiment)

Aura

Awarded Abstracts

AWE (Atmospheric Waves Experiment)

B-377 Super Guppy

Barbara R. Morgan

BARREL (Balloon Array for Radiation-belt Relativistic Electron Losses)

Barry E. Wilmore

BBXRT (Broad Band X-ray Telescope)

BECCAL (Bose Einstein Condensates & Cold Atoms Lab)

Become an Astronaut

Benefits to Humanity

Benefits to Science

Benjamin Alvin Drew

BepiColombo

Beresheet

Bernard A. Harris Jr.

Betelgeuse

Bill Nelson

Binary Stars

Biological Experiment-01 (BioExpt-01)

BION / Biocosmos

Black History Month

Blazars

Blue Ghost

Boeing Starliner

Bonnie J. Dunbar

Borrelly

Brand Guidelines Center

Brent W. Jett

Brewster H. Shaw Jr.

Brian Duffy

Brian T. O'Leary

BRIC (Biological Research in Canisters)

Brown Dwarfs

Bruce E. Melnick

Bruce McCandless

Bryan D. O'Connor

Byron K. Lichtenberg

C-130

C-20A

C-9

Caldwell Catalog

CALIPSO (Cloud-Aerosol Lidar and Infrared Pathfinder Satellite)

Callie First Graphic Novel

Callisto

Candidate Astronauts

Carl E. Walz

Carl J. Meade

Carlos I. Noriega

Carruthers Geocorona Observatory (GLIDE)

CARVE (Carbon in Arctic Reservoirs Vulnerability Experiment)

Cassini-Huygens

Catherine G. Coleman

CATS (Cloud-Aerosol Transport System)

Centennial Challenges News

Center Innovation Fund Center of Excellence for Collaborative Innovation (CoECI)

CERES (Clouds and the Earth's Radiant Energy System)

CeREs (Compact Radiation Belt Explorer)

Ceres Dwarf Planet

Cessna C206

CGRO (Compton Gamma Ray Observatory)

CHAMP (Challenging Mini-satellite Payload)

Chandra X-ray Observatory

Chandrayaan-1

Chandrayaan-2

Chandrayaan-3

Charles A. Bassett II

Charles Camarda

Charles Conrad Jr.

Charles D. Gemar

Charles D. Walker

Charles E. Brady Jr.

Charles F. Bolden Jr.

Charles G. Fullerton

Charles J. Precourt

Charles Lacy Veach

Charles M. Duke Jr.

Charles O. Hobaugh

Charon

Chiaki Mukai

CHIPS (Cosmic Hot Interstellar Plasma Spectrometer)

Christina Birch

Christina H. Koch

Christopher J. Ferguson

Christopher J. Loria

Christopher J. Cassidy

Christopher L. Williams

CINDI (Coupled Ion Neutral Dynamic Investigation)

CIPHER (Complement of Integrated Protocols for Human Exploration Research)

Cirrus SR-22

Clayton C. Anderson

Clementine

Clifton C. Williams Jr.

Cloudsat

CLPS (Commercial Lunar Payload Services)

Cluster II

COBE (Cosmic Background Explorer)

Columbia 300

Commercial Crew Rockets

Commercial Crew Spacecraft

Commercial Lunar Payload Services (CLPS)

Community of Practice Webinar

Complex Fluids

Conjunction Assessment Risk Analysis Team (CARA)

Constellations

CONTOUR (COmet Nucleus TOUR)

Convergent Aeronautics Solutions

CORAL (COral Reef Airborne Observatory)

Coronal Mass Ejections

COSI (Compton Spectrometer and Imager)

Cosmic Origins Program

Crab Nebula

Crew Health and Performance Exploration Analog (CHAPEA)

Crew Vehicle Systems Research Facility

Cross Program Integration

CRRES (Combined Release and Radiation Effects Satellite)

Cryogenic Fluid Management (CFM)

Cryogenic Propellant Handling

CubeSat Or Microsat Probabilistic and Analogies Cost Tool (COMPACT)

CuPID (Cusp Plasma Imaging Detector)

Curtis L. Brown Jr.

CUSP (CubeSat Mission to Study Solar Particles)

Cycles & Focus Areas

CYGNSS (Cyclone Global Navigation Satellite System)

Dale A. Gardner

Daniel C. Brandenstein

Daniel C. Burbank

Daniel M. Tani

Daniel T. Barry

Daniel W. Bursch

David A. Wolf

David C. Hilmers

David C. Leestma

David M. Brown

David M. Walker

David R. Scott

Dawn

DC-8

Deep Impact / EPOXI (Extrasolar Planet Observation and Deep Impact Extended Investigation)

Deep Space 1

Deep Space Network (DSN)

Deep Space Optical Communications (DSOC)

Deimos

Deniz Burnham

Desert RATS (Research and Technology Studies)

(DECLIC) DEvice for the study of Critical LIquids and Crystallization

DHC

Didymos & Dimorphos

Diffuse X-Ray Spectrometer (DXS)

DISCOVER AQ (Deriving Information on Surface Conditions from COlumn and VERtically Resolved Observations Relevant to Air Quality)

Discovery Program

Dominic A. Antonelli

Dominic L. Gorie

Don L. Lind

Donald A. Thomas

Donald E. Williams

Donald H. Peterson

Donald K. Slayton

Donald L. Holmquest

Donald R. McMonagle

Donald R. Pettit

Donn F. Eisele

Dorothy M. Metcalf-Lindenburger

Douglas G. Hurley

Douglas H. Wheelock

Dragon Crew Spacecraft

Draper SERIES-2

Droughts

DSCOVR (Deep Space Climate Observatory)

Duane G. Carey

Duane Graveline

Dynamics Explorer (DE)

:envihab

Early Career Faculty

Earth Information Center

Earth Observing-1 (EO-1)

Earth Science Data Systems Program

Earth Science Technology Office

Earth System Observatory (ESO)

Earthquakes

Earth’s Atmosphere

Earth’s Magnetic Field

Earth’s Vital Signs

Ecostress (ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station)

Edgar D. Mitchell

Edward G. Gibson

Edward G. Givens Jr.

Edward H. White II

Edward M. Fincke

Edward T. Lu

Eileen M. Collins

Electrified Powertrain Flight Demo

ElectroHydroDynamic (EHD) Test Chamber

Electromagnetic Spectrum

ELFIN (Electron Losses & Fields Investigation)

Ellen Ochoa

Ellen S. Baker

Ellington Field

Elliot M. See Jr.

Elliptical Galaxies

Ellison S. Onizuka

Emergency & Pandemic Response

EMIT (Earth Surface Mineral Dust Source Investigation)

Enceladus

Environmental Management Division (EMD)

Environmental Science Services Administration (ESSA)

EnVision

Equator-S

ERBS (Earth Radiation Budget Satellite)

Eric A. Boe

Eros

EscaPADE (Escape and Plasma Acceleration and Dynamics Explorers)

Established Program to Stimulate Competitive Research (EPSCoR)

Eta Aquarids

E-TBEx (Enhanced Tandem Beacon Experiment)

Euclid

Eugene A. Cernan

Eugene H. Trinh

EUVE (Extreme-Ultraviolet Explorer)

Everyday Extraordinary Stories

Exosphere

Expedition 1

Expedition 2

Expedition 3

Expedition 4

Expedition 5

Expedition 6

Expedition 7

Expedition 8

Expedition 9

Expedition 10

Expedition 11

Expedition 12

Expedition 13

Expedition 14

Expedition 15

Expedition 16

Expedition 17

Expedition 18

Expedition 19

Expedition 20

Expedition 21

Expedition 22

Expedition 23

Expedition 24

Expedition 25

Expedition 26

Expedition 27

Expedition 28

Expedition 29

Expedition 30

Expedition 31

Expedition 32

Expedition 33

Expedition 34

Expedition 35

Expedition 41

Expedition 58

Expedition 60

Expedition 62

Expedition 63

Expedition 66

Expeditions

Exploration Capabilities Division

EFT 1 (Exploration Flight Test 1)

Exploration Operations Division

Explorer

Explorer 33

Explorer 35

Explorer 49

Explorers Program

Extraterrestrial Life

Extreme Ultraviolet High-Throughput Spectroscopic Telescope (EUVST) Epsilon

Extreme Weather Events

EZIE (Electrojet Zeeman Imaging Explorer)

F / A-18

F. Drew Gaffney

F-18

FAST (Fast Auroral SnapshoT Explorer)

FASTSat (Fast, Affordable, Science and Technology Satellite)

FBCE (Flow Boiling and Condensation Experiment)

Fermi Gamma-Ray Space Telescope

Fernando Caldeiro

Film & Documentary Guidelines

First Woman Graphic Novel

Fission Surface Power (FSP)

FLARE (Flammability Limits At Reduced-g Experiment)

Flight Program

Flight Providers

Flight Simulation

Francis R. Scobee

Franco Malerba

Frank Borman

Frank L. Culbertson Jr.

Franklin R. Chang-Díaz

Fred W. Haise Jr.

Fred W. Leslie

Frederick D. Gregory

Frederick H. Hauck

Frederick W. Sturckow

Fruit Fly Lab (FFL)

FUSE (Far Ultraviolet Spectroscopic Explorer)

FutureFlight Central

G. David Low

G. Reid Wiseman

Gaia

Galaxies, Stars, & Black Holes Research

GALEX (Galaxy Evolution Explorer)

Galileo

Galileo Jupiter Atmospheric Probe

Gamma Rays

Gamma-Ray Bursts

Garrett E. Reisman

Gary E. Payton

Gas Giants

Gateway Program

GDC (Geospace Dynamics Constellation)

GEDI (Global Ecosystem Dynamics Investigation)

Gemini I

Gemini II

Gemini III

Gemini IV

Gemini V

Gemini VI

Gemini VII

Gemini VIII

Gemini IX

Gemini X

Gemini XI

Gemini XII

Geminids

Genesis

GRIP (Genesis and Rapid Intensification Processes)

General George D. Nelson

George D. Zamka

Geosat (U.S. Navy GEOdetic SATellite)

Geospace Dynamics Constellation (GDC)

Geotail

Gerald Carr

Get Social

GII-STA

Giotto

Glaciers

GLIMR (Geosynchronous Littoral Imaging and Monitoring Radiometer)

Glory

GOES-N

GOES-R

GOLD (Global-scale Observations of the Limb and Disk)

GPIM (Green Propellant Infusion Mission)

GPM (Global Precipitation Global Precipitation Measurement)

GRACE (Gravity Recovery And Climate Experiment)

GRACE-FO (Gravity Recovery and Climate Experiment Follow-on)

Grades 5 – 8

Grades 5 – 8 for Educators

Grades 9 – 12

Grades K – 4

Grades K – 4 for Educators

GRAIL (Gravity Recovery And Interior Laboratory)

Gravity Probe B (GP-B)

Greenhouse Effect

Greenhouse Gasses

Gregory C. Johnson

Gregory E. Chamitoff

Gregory H. Johnson

Gregory J. Harbaugh

Gregory Jarvis

Gregory T. Linteris

Griffin (lander)

Ground Facilities for ISS

Guion S. Bluford Jr.

Gulfstream C-20A

Gulfstream G-III

Gulfstream G-V

Gulfstream IV

Guy S. Gardner

H-135

HALCA (Highly Advanced Laboratory for Communications and Astronomy)

Hale-Bopp

Halley's Comet

HaloSat

Harrison H. Schmitt

Hayabusa

Hayabusa 2

Heidemarie M. Stefanyshyn-Piper

Heliophysics Research Program

Helios 1

Helios 2

Heliosphere

HelioSwarm

Helix Nebula

Henry W. Hartsfield Jr.

HERA (Human Exploration Research Analog)

HERMES (Heliophysics Environmental and Radiation Measurement Experiment Suite)

Herschel Space Observatory

HETE-1 (High Energy Transient Explorer 1)

HETE-2 (High Energy Transient Explorer 2)

Hinode (Solar B)

Hipparcos

Hi-Rate Composite Aircraft Manufacturing

History: NASA

Hiten / Hagomoro

Hitomi (ASTRO-H)

Horsehead Nebula

HU-25

Human Exploration of Space: Why We Go

Human Health and Performance

Human Spaceflight Capabilities Division

Hurricanes

Hybrid Thermally Efficient Core

Hyperion

Hypersonic Technology

HyspIRI (Hyperspectral Infrared Imager)

IBEX (Interstellar Boundary Explorer)

ICESat-2 (Ice, Cloud and land Elevation Satellite-2)

ICON (Ionospheric Connection Explorer)

Ida

IEH-3 (International Extreme Ultraviolet Hitchhiker)

Ilan Ramon

IMAGE (Imager for Magnetopause-to-Aurora Global Exploration)

Images & Media Guidelines

IMAP (Interstellar Mapping and Acceleration Probe)

IMP-8 (Interplanetary Monitoring Platform 8)

INCUS (Investigation of Convective Updrafts)

Infrared Space Observatory (ISO)

Institutional Funding

International Gamma-ray Astrophysics Laboratory (INTEGRAL)

International Space Station Division

ISRU (In-Situ Resource Utilization)

Io

Ionosphere

IRIS (Interface Region Imaging Spectrograph)

Irregular Galaxies

ISEE-3/ICE (International Sun-Earth Explorer-3 / International Cometary Explorer)

ISERV (ISS SERVIR Environmental Research and Visualization System)

ISS-RapidScat (Rapid Scatterometer)

IUE (International Ultraviolet Explorer)

IXPE (Imaging X-ray Polarimetry Explorer)

Jack D. Fischer

Jack Hathaway

Jack R. Lousma

Jake Garn

James A. Lovell Jr.

James A. McDivitt

James A. Pawelczyk

James B. Irwin

James C. Adamson

James D. A. Van Hoften

James D. Halsell Jr.

James D. Wetherbee

James F. Buchli

James F. Reilly

James H. Newman

James M. Kelly

James P. Bagian

James P. Dutton

James S. Voss

Janet L. Kavandi

Janice E. Voss

Janus

Jason-1

Jason-CS (Continuity of Service) / Sentinel-6

Jay Clark Buckey

Jean-Jacques Favier

Jeffrey A. Hoffman

Jeffrey N. Williams

Jeffrey S. Ashby

Jerome Apt

Jerry L. Ross

Jerry M. Linenger

Jessica U. Meir

Jessica Wittner

Joan E. Higginbotham

Joe F. Edwards Jr.

Joe H. Engle

John A. Llewellyn

John B. Herrington

John D. Olivas

John E. Blaha

John H. Casper

John H. Glenn Jr.

John L. Phillips

John L. Swigert Jr.

John M. Fabian

John M. Grunsfeld

John M. Lounge

John O. Creighton

John S. Bull

John W. Young

Johnson Flight Operations

Johnson's Mission Control Center

Joint Agency Satellite Division

Jon A. McBride

Jonny Kim

José M. Hernández

Joseph M. Acaba

Joseph P. Allen

Joseph P. Kerwin

Joseph R. Tanner

Josh A. Cassada

JPSS (Joint Polar Satellite System)

JPSS-3

JPSS-4

Judith A. Resnik

JUICE (Jupiter Icy Moons Explorer)

JWST (James Webb Space Telescope)

Kaguya

Kalpana Chawla

Karen L. Nyberg

Karl G. Henize

Karol J. Bobko

Katherine Johnson IV & V Facility

Kathleen Rubins

Kathryn C. Thornton

Kathryn D. Sullivan

Kathryn P. Hire

Kayla Barron

Keck Interferometer (KI)

Kenneth D. Bowersox

Kenneth D. Cameron

Kenneth D. Cockrell

Kenneth S. Reightler Jr.

Kenneth T. Ham

Kent V. Rominger

Kepler / K2

Kepler 22B

Kevin A. Ford

Kevin P. Chilton

Kevin R. Kregel

Knowledge Inventory For Professionals

Kuiper Airborne Observatory (KAO)

L. Blaine Hammond Jr.

L. Gordon Cooper Jr.

La Tierra y calentamiento global

LAGEOS (LAser GEOdynamics Satellite)

Landsat 7

Landsat 8 / LDCM (Landsat Data Continuity Mission)

Landsat 9

Laser Communications Relay

Launch Services Office

Laurel B. Clark

Lawrence J. Delucas

LBTI (Large Binocular Telescope Interferometer)

LCROSS (Lunar Crater Observation and Sensing Satellite)

Lee J. Archambault

Lee M. Morin

LEIA (Lunar Explorer Instrument for space biology Applications)

Leland D. Melvin

Lenticular Galaxies

Leonid K. Kadenyuk

Leonid MAC (Multi-instrument Aircraft Campaign)

Leonids

Leroy Chiao

Lessons Learned for Professionals

LGBTQ Pride

Libera

LIS (Lightning Imaging Sensor)

Linda M. Godwin

LISA (Laser Interferometer Space Antenna)

Lisa M. Nowak

Lodewijk Van Den Berg

LOFTID (Low-Earth Orbit Flight Test of an Inflatable Decelerator)

Logistics Management Division (LMD)

Logo & Media Usage

Loral O’Hara

Loren J. Shriver

Loren W. Acton

Low-Density Supersonic Decelerator

Low-Earth Orbit Economy

LRO (Lunar Reconnaissance Orbiter)

LSII Consortium

Luke Delany

LunaH-Map (Lunar Polar Hydrogen Mapper)

Lunar Discovery & Exploration Program

Lunar Lander Simulation

Lunar Orbiter

Lunar Orbiter 1

Lunar Orbiter 2

Lunar Orbiter 3

Lunar Orbiter 4

Lunar Orbiter 5

Lunar Prospector

Lunar Surface Technology Research

Lunar Trailblazer

Lyrids

M. Scott Carpenter

Mae C. Jemison

Magellan

Magnetars

Magnetosphere

MAIA (Multi-Angle Imager for Aerosols)

Management Astronauts

Manley Lanier Carter Jr.

Marcos Berrios

Margaret Rhea Seddon

Mariner 1

Mariner 2

Mariner 3

Mariner 4

Mariner 5

Mariner 6

Mariner 7

Mariner 8

Mariner 9

Mariner 10

Mario Runco Jr.

Mark C. Lee

Mark E. Kelly

Mark L. Polansky

Mark N. Brown

Mark T. Vande Hei

Mars Campaign Development Division

Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE)

Marsha Ivins

Marshall Test Facility and Support Infrastructure

Martian Moon Exploration (MMX)

Martin J. Fettman

Mary E. Weber

Mary L. Cleave

Materials Science Research Rack (MSRR)

Matthew Dominick

Megan McArthur

Merchandise Approvals

Mesosphere

Messier Catalog

Meteor Showers

MetOp (Meteorological Operational satellite)

Michael A. Baker

Michael Collins

Michael Fincke

Michael E. Fossum

Michael E. Lopez-Alegria

Michael J. Bloomfield

Michael J. Foreman

Michael J. Massimino

Michael J. McCulley

Michael J. Smith

Michael L. Coats

Michael L. Gernhardt

Michael P. Anderson

Michael R. Barratt

Michael R. Clifford

Michael S. Hopkins

Michael T. Good

Michel F. Curtis

Micro-10

Microbial Tracking (MT)

Millie Hughes-Fulford

Mini-RF Instrument

MinXSS (Miniature X-ray Solar Spectrometer)

Miranda

Mission Equity

Mission Support Directorate

MMS (Magnetospheric Multiscale)

Modeling, Analysis and Prediction (MAP) Program

Moderate Resolution Imaging Spectroradiometer (MODIS)

Modern Figures

Modular Optoelectric Multispectral Scanner (MOMS)

MOS (Marine Observation Satellite)

MSL Instrumentation

Multi-Spectral Fluorescence Imaging System (Spectrum)

MUSE (Multi-slit Solar Explorer)

Museum Alliance

N. Jan Davis

NAAMES (North Atlantic Aerosols and Marine Ecosystems Study)

Nancy J. Currie-Gregg

NanoSail-D

NASA Aeronautics Committee

NASA Engineering and Safety Center

NASA History Office

NASA Home & City

NASA Interactives

NASA iTech Program

NASA Knowledge Program

NASA Open Source Software

NASA Plus

NASA Safety Center

NASA Safety Center Professional Development

NASA Shared Services Center

NASA Social Media Contacts

NASA Socials Program

NASA STEM

NASA Worldwind

National Aeronautics Research Institute

National Space Council Users' Advisory Group (NSpC UAG)

NEA Scout (Near Earth Asteroid Scout)

NEAR Shoemaker

Near-Earth Object (NEO)

Nebulae Game

NEEMO (NASA Extreme Environment Mission Operations)

Neil A. Armstrong

Neil Gehrels Swift Observatory

Neil W. Woodward III

NEK/SIRIUS (Nezemnyy Eksperimental’nyy Kompleks/Scientific International Research In a Unique terrestrial Station)

NEO Surveyor (Near-Earth Object Surveyor Space Telescope)

NEOWISE

Neptune Moons

Neutral Buoyancy Lab

Neutron Stars

New Frontiers Program

NExIS (NASA’s Exploration & In-space Services)

NIAC Studies

NIAC Symposium

NICER (Neutron star Interior Composition Explorer)

Nicholas J. M. Patrick

Nicole A. Mann

Nicole P. Stott

Nimbus

NISAR (NASA-ISRO Synthetic Aperture Radar)

NOAA-20 (JPSS-1)

NOAA-21 (JPSS-2)

NOAA-N Prime

Norman E. Thagard

Northrop Grumman Commercial Resupply

Nova-C (Lander)

Nozomi

NuSTAR (Nuclear Spectroscopic Telescope Array)

O / OREOS (Organism / Organic Exposure to Orbital Stresses)

Ocean Surface Topography Mission (OSTM) / Jason-2

Oceanography

Oceans Melting Greenland (OMG)

OCO (Orbiting Carbon Observatory)

OCO-2 (Orbiting Carbon Observatory 2)

OCO-3 (Orbiting Carbon Observatory 3)

Office of Legislative and Intergovernmental Affairs (OLIA)

Office of Strategic Infrastructure (OSI)

Office of Technology, Policy and Strategy (OTPS)

Office of the Chief Engineer (OCE)

Office of the Chief Financial Officer (OCFO)

Office of the Chief Health and Medical Officer (OCHMO)

Office of the Chief Information Officer (OCIO)

Office of the Chief Scientist (OCS)

Office of the General Counsel (OGC)

One Year Crew

ORACLES (ObseRvations of Aerosols above CLouds and their IntEractionS)

Orbiting Astronomical Observatory (OAO)

ORFEUS (Orbiting and Retrievable Far and Extreme Ultraviolet Spectrometer)-SPAS II

Orion Nebula

Orion Spacecraft

Orionids OSAM-1 (On-Orbit Servicing, Assembly, and Manufacturing 1)

Oumuamua

OV-10

Owen K. Garriott

Ozone Mapping and Profiler Suite (OMPS)

P-3

PACE (Plankton, Aerosol, Cloud, Ocean Ecosystem)

Pamela A. Melroy

Partner Astronauts

Patricia C. Robertson

Patrick G. Forrester

Paul D. Scully-Power

Paul J. Weitz

Paul S. Lockhart

Paul W. Richards

Peggy A. Whitson

Peregrine Lunar Lander

Perseids

Peter J. K. Wisoff

PFMI (Pore Formation and Mobility Investigation)

PharmaSat

Philae (Lander)

Philip K. Chapman

Phobos

Phoebe

Physical Sciences

PI Resources

Pierre J. Thuot

Piers J. Sellers

Pioneer

Pioneer 0 / Able 1

Pioneer 1 / Able 2

Pioneer 2

Pioneer 3

Pioneer 4

Pioneer 5

Pioneer 6

Pioneer 7

Pioneer 8

Pioneer 9

Pioneer 10

Pioneer 11

Pioneer P-3 / Able 4B

Pioneer P-30 / Able 5A

Pioneer P-31 / Able 5B

Pioneer Venus

Pioneer Venus 1

Pioneer Venus 2

Pioneer-E

Planck

Planetary Defense Coordination Office

Plasma Kristall 4 (PK-4)

Play

Play it!

Plume-Surface Interaction (PSI) Project

Pluto Moons

Polar

Predator

PREFIRE (Polar Radiant Energy in the Far-InfraRed Experiment)

PRESat (PharmaSat Risk Evaluation Satellite)

Prizes, Challenges & Crowdsourcing

Prizes, Challenges & Crowdsourcing News

Proxima Centauri b

Psyche Asteroid

Psyche Mission

Pulsars

PUNCH (Polarimeter to Unify the Corona and Heliosphere)

Q-PACE (CubeSat Particle Aggregation and Collision Experiment)

Quadrantids

Quasars

Quesst: The Flights

Quesst: The Vehicle

QuikSCAT (Quick Scatterometer)

RADARSAT-1 (Radar Satellite 1)

RAD-SEED

Raja Chari

Randolph J. Bresnik

Ranger

Ranger 1

Ranger 2

Ranger 3

Ranger 4

Ranger 5

Ranger 6

Ranger 7

Ranger 8

Ranger 9

Red Dwarfs

Red Giants

Redstone 3 (Freedom 7)

Redstone 4 (Liberty Bell 7)

Research & Analytics Program

Research Archive

Revolutionary Vertical Lift Technology

Rex J. Walheim

Rhea

RHESSI (Reuven Ramaty High Energy Solar Spectroscopic Imager)

Richard A. Mastracchio

Richard A. Searfoss

Richard F. Gordon Jr.

Richard H. Truly

Richard J. Hieb

Richard M. Linnehan

Richard M. Mullane

Richard N. Richards

Richard O. Covey

Richard R. Arnold

Rick D. Husband

Rings of Jupiter

Rings of Neptune

Rings of Saturn

Ring-Sheared Drop (RSD)

RLL (Robotic Lunar Lander)

Robb Kulin

Robert A. R. Parker

Robert C. Springer

Robert D. Cabana

Robert F. Overmyer

Robert J. Cenker

Robert L. Behnken

Robert L. Crippen

Robert L. Curbeam Jr.

Robert L. Gibson

Robert L. Satcher Jr.

Robert L. Stewart

Robert Shane Kimbrough

Robotic Lunar Exploration Program

Rodent Research (RR)

Roger B. Chaffee

Roger K. Crouch

Ronald A. Parise

Ronald E. Evans

Ronald E. McNair

Ronald J. Garan Jr.

Ronald J. Grabe

Ronald M. Sega

Rosalind Franklin (Rover)

ROSAT (ROentgen SATellite)

Rosetta

Roy D. Bridges Jr.

RS-25

Russell L. Schweickart

RXTE (Rossi X-ray Timing Explorer)

Ryugu

S. Christa McAuliffe

S. David Griggs

S3VI

SAC-B (Satélite de Aplicaciones Científicas-B)

SAGE III (Stratospheric Aerosol and Gas Experiment)

SAGE-III Meteor-3M (Stratospheric Aerosol and Gas Experiment III on Meteor-3M)

Sakigake

Sally K. Ride

SAMPEX (Solar Anomalous and Magnetospheric Particle Explorer)

Samuel T. Durrance

Sandra H. Magnus

SBIR

SBIR / STTR

Science Leadership

Science of Space Exploration

Science Solicitations

Scott D. Altman

Scott D. Tingle

Scott E. Parazynski

Scott J. Horowitz

Scott J. Kelly

SEAC4RS (Studies of Emissions, Atmospheric Composition, Clouds and Climate Coupling by Regional Surveys)

Seasat

SeaWiFS (Sea-viewing Wide Field-of-view Sensor)

SeaWinds

Sentinel-6 Michael Freilich Satellite

Sentinel-6B

Serena M. Auñón-Chancellor

Seres Humanos en el spacio

SERVIR (Regional Visualization and Monitoring System)

Seyfert Galaxies

Shannon W. Lucid

Shannon Walker

Sherwood C. Spring

Shoemaker-Levy 9

Shuttle-Mir

Sidney M. Gutierrez

Sierra UAS #2

SimLabs

SIR-C/X-SAR (Shuttle Imaging Radar-C / X-Band Synthetic Aperture Radar)

Skywatching Tips

Small Bodies of the Solar System

Small Business Innovation Research / Small Business

Small Spacecraft Technology Program

SMAP (Soil Moisture Active Passive)

SMART-1 (Small Missions for Advanced Research in Technology 1)

SNOE (Student Nitric Oxide Explorer)

Soft Matter Dynamics (SMD) / FOAM

Software to License

SOHO (Solar and Heliospheric Observatory)

Solar Cruiser

Solar Electric Propulsion (SEP)

Solar Flares

Solar Orbiter

Solar Sail – Beyond Plum Brook

Solar System Exploration Research Virtual Institute

Solar Terrestrial Probes Program

Solid Fuel Ignition and Extinction (SoFIE)

Solidification Using a Baffle in Sealed Ampoules (SUBSA)

SORCE (Solar Radiation and Climate Experiment)

SORTIE (Scintillation Observations & Response of the Ionosphere to Electrodynamics)

Sounds & Ringtones

Southern Delta Aquariids

Space Biosciences at Ames

Space Communications & Navigation Program (SCaN)

Space Environment Testbeds (SET-1)

Space Environments Testing Management Office (SETMO)

Space Flight Awareness (SFA)

SLS (Space Launch System)

Space Life & Physical Sciences Research & Applications Division

Space Nuclear Propulsion (SNP)

Space Operations Directorate Resource Management Office

Space Station Research Opportunities

Space Tech Graduate Research

Space Technology 5

Space Technology 6

Space Technology 7 / Disturbance Reduction System (DRS)

Space Vehicle Mockup Facility

Space Weather

Spartan

Spectrum Management

Spiral Galaxies

Spot the International Space Station

SRTM (Shuttle Radar Topography Mission)

ST5

Stanley G. Love

Stardust / Stardust NExT

Starliner

Stennis Test Facility and Support Infrastructure

Stephanie D. Wilson

Stephen D. Thorne

Stephen K. Robinson

Stephen N. Frick

Stephen S. Oswald

Steven A. Hawley

Steven L. Smith

Steven R. Nagel

Steven R. Swanson

Steven W. Lindsey

Story Musgrave

STP

Strategic Integration and Management Division

Strategic Partnerships Guidelines

Stratosphere

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Stuart A. Roosa

Submillimeter Wave Astronomy Satellite (SWAS)

SunRISE (Sun Radio Interferometer Space Experiment)

Sunspots

Suomi NPP (Suomi National Polar-orbiting Partnership)

Supermoons

Supernovae

Surveyor 1

Surveyor 2

Surveyor 3

Surveyor 4

Surveyor 5

Surveyor 6

Surveyor 7

Surveyor Model 1

Surveyor Model 2

Surveyor Model SD-1

Susan J. Helms

Susan L. Kilrain

Sustainable Flight National Partnership

Suzaku

Synchronous Meteorological Satellite (SMS)

T-38 Astronaut Trainer

Tamara E. Jernigan

Taylor Wang

Tech Demo Missions

Tech Dev Projects

Tech Portfolio

Technology Highlights

Tempel 1

Terence T. Henricks

Terra

Terrence W. Wilcutt

Terrestrial Planets

Terriers (Tomographic Experiment Using Radiative Recombinative Ionospheric EUV and Radio Sources)

Terry J. Hart

Terry W. Virts Jr.

TESS (Transiting Exoplanet Survey Satellite)

Tethered Satellite System (TSS)

TG-14

The Great Red Spot

The Habitable Zone

The Human Body in Space

The Milky Way

The North Star

THEMIS (Time History of Events and Macroscale Interactions During Substorms)

THEMIS-ARTEMIS (Time History of Events and Macroscale Interactions During Substorms – Acceleration, Reconnection, Turbulence and Electrodynamics of Moon's Interaction with the Sun)

Theodore C. Freeman

Thermosphere

Thomas D. Akers

Thomas D. Jones

Thomas H. Marshburn

Thomas J. Hennen

Thomas K. Mattingly II

Thomas P. Stafford

TIMED (Thermospere Ionosphere Mesosphere Energetics and Dynamics Mission)

Timothy J. Creamer

Timothy L. Kopra

TIROS (Television Infrared Observation Satellite Program)

TOMS-EP (Total Ozone Mapping Spectrometer – Earth Probe)

TOPEX / Poseidon (ocean TOPography EXperiment)

Tornadoes

Total and Spectral Solar Irradiance Sensor – 1 (TSIS-1)

Total and Spectral Solar Irradiance Sensor – 2 (TSIS-2)

TP / ACO

TRACE (Transition Region and Coronal Explorer)

Trace Gas Orbiter (TGO)

Tracking Aerosol Convection Interactions Experiment (TRACER)

Tracking and Data Relay Satellite (TDRS)

Tracy Caldwell Dyson

Transformational Tools

Technologies

Triton

Tropical Rainfall Measuring Mission (TRMM)

TROPICS (Time-Resolved Observations of Precipitation Structure and Storm Intensity with a Constellation of Smallsats)

Troposphere

Tsunamis

TWINS (Two Wide-Angle Imaging Neutral-Atom Spectrometers)

Twins Study

Tyler N. Hague

UAS in the NAS

UAS Traffic Management

UH-1H

UH-II

Ulysses

Underway Recovery Test 10

Universe of Monsters

Upper Atmosphere Research Satellite (UARS)

Uranus Moons

Urban Air Mobility Simulation

USGS (United States Geological Survey)

Valles Marineris

Van Allen Probes

Vance D. Brand

Venus Express

Vertical Motion Simulator

Victor J. Glover

Viking 400

Virgil I. Grissom

Visible Infrared Imaging Radiometer Suite (VIIRS)

Walter Cunningham

Walter M. Schirra Jr.

Wendy B. Lawrence

WetLab-2

White Dwarfs

White Sands Test Facility and Support Infrastructure

Wild 2

Wilkinson Microwave Anisotropy Probe (WMAP)

William A. Anders

William A. Oefelein

William A. Pailes

William B. Lenoir

William C. McCool

William E. Thornton

William F. Fisher

William F. Readdy

William G. Gregory

William M. Shepherd

William R. Pogue

William S. McArthur Jr.

Wind Mission

Winston E. Scott

WIRE (Wide-Field Infrared Explorer)

Women’s History Month

Xelene Lunar Lander (XL-1)

XMM-Newton (X-ray Multi-Mirror Newton)

XRISM (X-Ray Imaging and Spectroscopy Mission)

YO-3A

Yohkoh

Yvonne Darlene Cagle

Zena Cardman

Zero Boil-Off Tank (ZBOT-NC)

URL: https://www.nasa.gov/topics/moon-to-mars/

TITLE: Destinations

Humans in SpaceDestinationsNASA is taking a steppingstone approach to human exploration in space. Building on NASA’s 60 years of exploration experience and more than 20 years of continuous human presence on the International Space Station in low Earth orbit, we will extend humanity farther into space than ever before. The International Space Station has built the foundation to conduct complex operations in space, perform research in a microgravity environment, foster a growing space economy, and forge international partnerships toward a common goal. Artemis missions will establish our long-term presence at the Moon as astronauts explore more of the lunar surface than ever before to learn about the origins of the solar system and prepare for humanity’s next giant leap: human missions to Mars.Quick FactsThe space station has been continuously occupied since November 2000.Missions to the Moon are about 1,000 times farther from Earth than missions to the International Space Station.Thanks to nearly 20 years of continuous human habitation on the space station, future life support systems for Mars can be designed with a 36% reduction in mass.An international partnership of five space agencies from 15 countries operates the International Space Station.ExploreLearn more about NASA's destinations for human exploration from the orbiting laboratory in low-Earth orbit, to Artemis missions at the Moon, and leading to the boldest mission yet: sending humans to Mars.All Humans in Space TopicsInternational Space StationAstronauts on the orbiting laboratory are helping us prove many of the technologies and communications systems needed for human missions in deep space, including to the Moon and Mars. The space station also advances our understanding of how the body reacts and changes in space and how to protect astronaut health.Read MoreEarth's MoonArtemis missions will collaborate with commercial and international partners to demonstrate innovative technologies on the lunar surface and around the Moon for scientific discovery and to prepare for human missions to Mars.Read MoreMarsEngineers and scientists around the country are developing the technologies astronauts will use to one day live and work on Mars and safely return home from humanity's next giant leap. NASA is working to address the challenges of living farther from Earth, such as using existing resources, options for disposing of trash, and more.Read MoreCommercial Destinations in Low Earth OrbitNASA’s Commercial Low Earth Orbit Development Program is supporting the development of commercially owned and operated low Earth orbit destinations from which NASA, along with other customers, can purchase services and stimulate the growth of commercial activities close to Earth.Read MoreArchitectureMoon to Mars ArchitectureNASA's Moon to Mars architecture represents the hardware and operations needed for human missions to the Moon and Mars, and how they function together as a system.The architecture is not a mission, a manifest, or a set of requirements, but it does define the elements — rockets, spacecraft, rovers, spacesuits, communications relays, and more — that will be incrementally developed and delivered to the Moon and Mars for long-term, human-led scientific discovery in deep space.Learn Moreabout Moon to Mars ArchitectureIllustration of three astronaut silhouettesNASAThat's one small step for (a) man; one giant leap for mankind.Neil ArmstrongApollo astronautLatest NewsNews Release5 Min ReadNASA Sets Briefings for SpaceX Crew-12 Mission to Space StationArticle4 Min ReadNASA Selects Participants to Track Artemis II Mission5 Min ReadNASA Astronaut Suni Williams RetiresNews Release3 Min ReadNASA Unlocks Golden Age of Innovation, Exploration in Trump’s First YearNews Release25 Min ReadArtemis II: What NASA Learned From Launching Artemis IKeep ExploringDiscover More Topics From NASAInternational Space StationArtemisHumans to MarsLike the Moon, Mars is a rich destination for scientific discovery and a driver of technologies that will enable humans…Humans In Space

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Illustration of three astronaut silhouettesNASA

Illustration of three astronaut silhouettes

That's one small step for (a) man; one giant leap for mankind.Neil ArmstrongApollo astronaut

That's one small step for (a) man; one giant leap for mankind.

Neil ArmstrongApollo astronaut

Neil Armstrong

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URL: https://www.nasa.gov/humans-in-space/commercial-space/low-earth-orbit-economy/commercial-destinations-in-low-earth-orbit/

TITLE: CommercialSpace Stations

Humans in SpaceCommercialSpace StationsNASA is supporting the development of commercially owned and operated space stations in low Earth orbit from which the agency, along with other customers, can purchase services and stimulate the growth of commercial activities in microgravity. NASA is using a phased approach to support the design, development, and demonstration of commercial space stations followed by a phase to finalize certification and procure services.Read Moreabout CommercialSpace StationsFeatured VideoAmerica's Future in Low Earth OrbitNASA plans to transition to a new low Earth orbit model featuring commercial space stations to continue leveraging the unique environment and microgravity research benefits.Watch NowCommercial Space Station NewsArticle3 min readNASA Seeks Industry Input on Next Phase of Commercial Space StationsArticle2 min readNASA Sees Key Progress on Starlab Commercial Space StationArticle2 min readNASA Helps with Progress on Vast's Haven-1 Commercial Space StationArticle3 min readNASA Sees Progress on Blue Origin's Orbital Reef Design DevelopmentArticle3 min readNASA, Axiom Space Change Assembly Order of Commercial Space StationArticle3 min readNASA Sees Progress on Starlab Commercial Space Station DevelopmentArticle6 min readNASA Commercial Partners Make Progress on Low Earth Orbit ProjectsArticle3 min readNASA Quiet Space Fan Research to Benefit Commercial Space StationsArticle3 min readNASA Supports Burst Test for Orbital Reef Commercial Space StationArticle2 min readNASA Shares Use Requirements with Commercial Destination PartnersArticle3 min readNASA Shares Lessons of Human Systems Integration with IndustryArticle5 min readNASA Shares Medical Expertise with New Space Station PartnersArticle3 min readNASA Sees Progress on Blue Origin's Orbital Reef Life Support SystemArticle4 min readNASA Adjusts Agreements to Benefit Commercial Station DevelopmentArticle4 min readNASA's Commercial Partners Continue Progress on New Space StationsArticle3 min readNASA Partners Combine Efforts for Low Earth Orbit Commercial StationArticle4 min readNASA's Commercial Partners Pass Milestones for New Space StationsArticle5 min readHow NASA's Work Led to Commercial Spaceflight RevolutionArticle3 min readNASA's Commercial Partners Move Needle for Space Station DestinationsArticle4 min readNASA Seeks Input for Commercial Low Earth Orbit DestinationsArticle2 min readNASA Seeks Feedback on Commercial Destination CertificationKeep ExploringDiscover More Topics From NASALow Earth Orbit EconomyHumans In SpacePrivate Astronaut MissionsCommercial Space

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Exhibition view of NASA's Earth Information CenterNASA/Dan Goods

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Destiny DoranInternational Space Station Research Communications TeamJan 15, 2026ArticleContentsBolstering bone resilienceObserving Earth and beyondSpace catchTracking internal temperatureA new cargo vehicleMaking nutrients on demandCelebrating a historic milestoneNASA's SpaceX Crew-11 mission with agency astronauts Zena Cardman and Mike Fincke, JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonovreturned to Earthafter a long-duration mission aboard the International Space Station.During their stay, Cardman, Fincke, and Yui contributed more than 850 hours of research to help prepare humanity for the return to the Moon and future missions to Mars, while improving life back on Earth.Here's a glimpse into the science completed during the Crew-11 mission:Bolstering bone resilienceNASA astronaut Zena Cardman works with bone stem cells aboard the International Space Station to improve our understanding of how bone loss occurs during spaceflight. Studying bone cell activity in microgravity could help researchers learn how to control bone loss to protect astronauts' bone density during future long-duration space missions and inform treatments for diseases like osteoporosis on Earth.Learn more aboutMABL-B.Observing Earth and beyondJAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui photographs the Earth from the International Space Station's cupola. For more than 40 years, astronauts have used hand-held cameras to capture millions of images documenting Earth's geographic features, weather patterns, urban growth, changes to its surface, and the impacts of natural disasters such as hurricanes and floods.Astronauts also use the cupola and other viewports aboard the space station to gaze into the cosmos without Earth's atmospheric interference. Just as viewing Earth from 250 miles above provides a new perspective on our home planet, looking out into the stars from the orbiting laboratory offers a clearer view of our universe.Space catchNASA astronaut Mike Fincke poses aboard the International Space Station with a new device designed to test an inflatable capture bag's ability to open, close, and stay airtight in microgravity. This technology could be used to remove space debris from orbit, protecting future spacecraft and crew members. It also may enable trapping samples during exploration missions and support the capture and mining of small asteroids.Learn more aboutCapture Bag Demo.Tracking internal temperatureNASA astronaut Mike Fincke wears a temperature-monitoring headband that tracks how the human body regulates its core temperature during spaceflight. Adjusting to living and working aboard the International Space Station can influence human temperature regulation. This headband provides an easy, non-invasive way to collect temperature data while astronauts conduct their daily activities. The sensor is also being tested on Earth and may help prevent hyperthermia in people working in high-temperature environments.Learn more aboutT-Mini.A new cargo vehicleJAXA's (Japan Aerospace Exploration Agency) new cargo resupply spacecraft, HTV-X1, is shown afterbeing capturedby the International Space Station's Canadarm2 robotic arm during the Crew-11 mission. The spacecraft launched from Tanegashima Space Center on Oct. 26, 2025, delivering approximately 12,800 pounds of science, supplies, and hardware to the orbital complex. New cargo spacecraft expand the station's capability to support more research and receive critical supplies.Making nutrients on demandJAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui holds yogurt bags produced aboard the International Space Station that could provide important nutrients during missions far from Earth. Certain nutrients degrade when stored for long periods of time, and deficiency in even one can lead to illness. Researchers are building on previous experiments to develop a method for producing on-demand vitamins and nutrients in space using microorganisms.Learn more aboutBioNutrients-3.Celebrating a historic milestoneThe Expedition 73 crew poses for a portrait to commemorate25 yearsof continuous human presence aboard the International Space Station. In the front row from left, NASA astronaut Jonny Kim, Roscosmos cosmonaut Sergey Ryzhikov, and Roscosmos cosmonaut Alexey Zubritsky. In the back row, Roscosmos cosmonaut Oleg Platonov, NASA astronauts Mike Fincke and Zena Cardman, and JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui.A truly global endeavor, the space station has been visited by more than 290 people from 26 countries, along with a variety of international and commercial spacecraft. Since the first crew arrived, NASA and its partners have conducted thousands of research investigations and technology demonstrations to advance exploration of the Moon and Mars and benefit life on Earth.

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Jan 15, 2026

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NASA’s SpaceX Crew-11 mission with agency astronauts Zena Cardman and Mike Fincke, JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonovreturned to Earthafter a long-duration mission aboard the International Space Station.During their stay, Cardman, Fincke, and Yui contributed more than 850 hours of research to help prepare humanity for the return to the Moon and future missions to Mars, while improving life back on Earth.Here’s a glimpse into the science completed during the Crew-11 mission:Bolstering bone resilienceNASA astronaut Zena Cardman works with bone stem cells aboard the International Space Station to improve our understanding of how bone loss occurs during spaceflight. Studying bone cell activity in microgravity could help researchers learn how to control bone loss to protect astronauts’ bone density during future long-duration space missions and inform treatments for diseases like osteoporosis on Earth.Learn more aboutMABL-B.Observing Earth and beyondJAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui photographs the Earth from the International Space Station's cupola. For more than 40 years, astronauts have used hand-held cameras to capture millions of images documenting Earth's geographic features, weather patterns, urban growth, changes to its surface, and the impacts of natural disasters such as hurricanes and floods.Astronauts also use the cupola and other viewports aboard the space station to gaze into the cosmos without Earth's atmospheric interference. Just as viewing Earth from 250 miles above provides a new perspective on our home planet, looking out into the stars from the orbiting laboratory offers a clearer view of our universe.Space catchNASA astronaut Mike Fincke poses aboard the International Space Station with a new device designed to test an inflatable capture bag's ability to open, close, and stay airtight in microgravity. This technology could be used to remove space debris from orbit, protecting future spacecraft and crew members. It also may enable trapping samples during exploration missions and support the capture and mining of small asteroids.Learn more aboutCapture Bag Demo.Tracking internal temperatureNASA astronaut Mike Fincke wears a temperature-monitoring headband that tracks how the human body regulates its core temperature during spaceflight. Adjusting to living and working aboard the International Space Station can influence human temperature regulation. This headband provides an easy, non-invasive way to collect temperature data while astronauts conduct their daily activities. The sensor is also being tested on Earth and may help prevent hyperthermia in people working in high-temperature environments.Learn more aboutT-Mini.A new cargo vehicleJAXA's (Japan Aerospace Exploration Agency) new cargo resupply spacecraft, HTV-X1, is shown afterbeing capturedby the International Space Station's Canadarm2 robotic arm during the Crew-11 mission. The spacecraft launched from Tanegashima Space Center on Oct. 26, 2025, delivering approximately 12,800 pounds of science, supplies, and hardware to the orbital complex. New cargo spacecraft expand the station's capability to support more research and receive critical supplies.Making nutrients on demandJAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui holds yogurt bags produced aboard the International Space Station that could provide important nutrients during missions far from Earth. Certain nutrients degrade when stored for long periods of time, and deficiency in even one can lead to illness. Researchers are building on previous experiments to develop a method for producing on-demand vitamins and nutrients in space using microorganisms.Learn more aboutBioNutrients-3.Celebrating a historic milestoneThe Expedition 73 crew poses for a portrait to commemorate25 yearsof continuous human presence aboard the International Space Station. In the front row from left, NASA astronaut Jonny Kim, Roscosmos cosmonaut Sergey Ryzhikov, and Roscosmos cosmonaut Alexey

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Moon to Mars ArchitectureWe're designing a roadmap for long-term exploration of the lunar surface, our first steps on the Red Planet, and the journey beyond, working with our partners in industry, academia, and the international community.Moon to Mars Architecture – Deep DiveWhat is Architecture?When most think of architecture, they envision skyscrapers, cantilevered homes, or marbled museums. In this case, architecture isn't the built environment. It isn't a mission, a manifest, or a set of requirements. Instead, NASA's Moon to Mars Architecture defines the elements needed for long-term, human-led scientific discovery in deep space.NASA's architecture approach distills agency-developed objectives into operational capabilities and elements that support science and exploration goals. Working with experts across the agency, industry, academia, and the international community, NASA continuously evolves that blueprint for crewed exploration, setting humanity on a path to the Moon, Mars, and beyond.Dive deeper into the architecture effort using the navigation bar at top of this page. You can find historical details about exploration architectures, products from the agency's annual Architecture Concept Review, and information about getting involved in the architecture effort.2026 Architecture workshopsRequest to RegisterAre you a representative of a industry, academic, or international partner organization interested in getting involved in NASA's Moon to Mars Architecture? Request to register for the 2026 Moon to Mars Architecture workshops at the link below.Learn MoreArchitecture Documents2025 Architecture UpdateArchitecture Definition DocumentsArchitecture White PapersLunar ExplorationArchitecture and ArtemisWe are going.With Artemis missions, we are exploring the Moon for scientific discovery, technology advancement, and to learn how to live and work on another world as we prepare for human missions to Mars. The Artemis missions are a key steps in realizing NASA’s Moon to Mars Architecture and Objectives.Learn Moreabout Architecture and ArtemisArtemis II astronauts, from left, NASA astronaut Victor Glover (left), CSA (Canadian Space Agency) astronaut Jeremy Hansen, NASA astronauts Christina Koch and Reid Wiseman stand on the crew access arm of the mobile launcher at Launch Pad 39B as part of an integrated ground systems test at Kennedy Space Center in Florida on Wednesday, Sept. 20. The test ensures the ground systems team is ready to support the crew timeline on launch day. Photo Credit: NASA/Frank MichauxSubscribe to UpdatesDomestic EngagementInternational EngagementHear from us!Fill out the form linked below if you’d like to be added to the mailing list for updates on NASA’s Moon to Mars Architecture.Updates are intended for organizations seeking to engage with the architecture process or provide feedback on the architecture.Subscribe140 attendees representing 110 industry and academic organizations attended the Moon to Mars Architecture Workshop on Feb. 22. In this photo, attendees and NASA personnel gather for a photo in the great hall of the National Academy of Sciences in Washington.Work with us!Help define our shared future in space.If you are an authorized representative of a U.S.-based organization and wish to engage further on NASA’s Moon to Mars architecture, please contact us following review of the Architecture Definition Document. Please do not send any business proprietary or personally sensitive information.Contact UsAttendees of NASA’s Moon to Mars Architecture Workshop pose for a group picture, Tuesday, June 27, 2023, at the Gaylord National Resort and Convention Center in National Harbor, Md. Following the release of the 2022 Architecture Concept Review, NASA is conducting the workshop to engage the broader space community and collect feedback from U.S. industry and academia to inform the Moon to Mars mission architecture and operational delivery.NASA/Joel KowskyWork with us!Help define our shared future in space.If you are an authorized representative of an international space agency or government and wish to engage further on NASA’s Moon to Mars architecture, please contact us following review of the Architecture Definition Document. Please do not send any business proprietary or personally sensitive information.Contact UsThis graphic displays the flags of the nations that have signed the Artemis Accords against a background image of the Moon in the blackness of space. The graphic is titled “Artemis Accords.” The words, “United for Peaceful Exploration of Deep Space” appear on the bottom of the image

Moon to Mars ArchitectureWe’re designing a roadmap for long-term exploration of the lunar surface, our first steps on the Red Planet, and the journey beyond, working with our partners in industry, academia, and the international community.

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Moon to Mars Architecture – Deep Dive

What is Architecture?

When most think of architecture, they envision skyscrapers, cantilevered homes, or marbled museums. In this case, architecture isn’t the built environment. It isn’t a mission, a manifest, or a set of requirements. Instead, NASA’s Moon to Mars Architecture defines the elements needed for long-term, human-led scientific discovery in deep space.

NASA’s architecture approach distills agency-developed objectives into operational capabilities and elements that support science and exploration goals. Working with experts across the agency, industry, academia, and the international community, NASA continuously evolves that blueprint for crewed exploration, setting humanity on a path to the Moon, Mars, and beyond.

Dive deeper into the architecture effort using the navigation bar at top of this page. You can find historical details about exploration architectures, products from the agency’s annual Architecture Concept Review, and information about getting involved in the architecture effort.

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Lunar Exploration

Architecture and Artemis

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Eyes of the Solar System uses data and images from NASA missions to give you a simulated view of Mercury.

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Our astronauts also need technologies to make food and eat, exercise, sleep and maintain their overall wellbeing.Manufacturing and MaterialsAdvanced manufacturing technologies for both terrestrial and in-space purposes will make commercial and exploration missions more efficient and affordable. We're developing new materials with improved or combined properties, learning how to build infrastructure and innovating on manufacturing processes.RoboticsNASA uses robotic systems to explore other planets and objects in our solar system as precursors to crewed missions, assist astronauts on the International Space Station, study the universe, and much more. This also includes autonomous systems that enable robotics, spacecraft and aircraft to operate in a dynamic environment independent of external control.Science InstrumentsScience instruments are state-of-the-art tools for acquiring information about planetary geology, atmosphere, and environmental conditions. This includes technologies for remote sensing, mapping and observation capabilities, analyzing light from the universe, and even searching for signs of life.High-Tech ComputingSpace-qualified computing technology is designed to address the most computationally-intensive part of a mission. But for the future, NASA is going to need more power and more processing capabilities. We also work on modeling and simulation tech, software, and unique electronics and computing hardware for flight systems, whether in space or the atmosphere.Space CommunicationsReliable space communication systems are critical to every NASA mission. Spacecraft commands, never-before-seen images, and scientific data are sent and received daily by NASA's giant antennas on Earth. From the Voyager mission exploring beyond our solar system, to astronauts onboard the International Space Station, space communications provide the crucial connection to our home planet.StarlingNASA’s Starling mission is advancing the readiness of various technologies for cooperative groups of spacecraft – also known as distributed missions, clusters, or swarms. Starling will demonstrate technologies to enable multipoint science data collection by several small spacecraft flying in swarms. 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Space TravelThe path to the Moon, Mars, and beyond requires technologies to get us where we need to go quickly, safely and efficiently. Space travel includes launch and in-space propulsion systems, cryogenic fluid management, and thermal management, as well as navigation and landing systems to get our supplies, equipment, and robotic or human explorers to diverse surface destinations.

Space Travel

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Living in SpaceTechnologies for living in space enable our robotic and human explorers to achieve their mission goals. This means everything from having power to do their work, to the habitats or spacecraft they'll call home. Our astronauts also need technologies to make food and eat, exercise, sleep and maintain their overall wellbeing.

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Manufacturing and MaterialsAdvanced manufacturing technologies for both terrestrial and in-space purposes will make commercial and exploration missions more efficient and affordable. We're developing new materials with improved or combined properties, learning how to build infrastructure and innovating on manufacturing processes.

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RoboticsNASA uses robotic systems to explore other planets and objects in our solar system as precursors to crewed missions, assist astronauts on the International Space Station, study the universe, and much more. This also includes autonomous systems that enable robotics, spacecraft and aircraft to operate in a dynamic environment independent of external control.

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Science InstrumentsScience instruments are state-of-the-art tools for acquiring information about planetary geology, atmosphere, and environmental conditions. This includes technologies for remote sensing, mapping and observation capabilities, analyzing light from the universe, and even searching for signs of life.

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Space CommunicationsReliable space communication systems are critical to every NASA mission. Spacecraft commands, never-before-seen images, and scientific data are sent and received daily by NASA's giant antennas on Earth. From the Voyager mission exploring beyond our solar system, to astronauts onboard the International Space Station, space communications provide the crucial connection to our home planet.

Space Communications

StarlingNASA’s Starling mission is advancing the readiness of various technologies for cooperative groups of spacecraft – also known as distributed missions, clusters, or swarms. Starling will demonstrate technologies to enable multipoint science data collection by several small spacecraft flying in swarms. The six-month mission will use four CubeSats in low-Earth orbit to test four technologies that let spacecraft operate in a synchronized manner without resources from the ground.Visit the Starling Mission Pageabout Starling

Starling

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Station Technology Demonstration

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Rachel BarryJan 23, 2026ArticleA close-up view of a portion of a "relatively fresh" crater, looking southeast, as photographed during the third Apollo 15 lunar surface moonwalk.Credit: NASAA new NASA study of its Apollo lunar soils clarifies the Moon's record of meteorite impacts and timing of water delivery. These findings place upper bounds on how much water meteorites could have supplied later in Earth's history.Research has previously shown that meteorites may have been a significant source of Earth's water as they bombarded our planet early in the solar system's development. In a paper published Tuesday in the Proceedings to the National Academy of Sciences, researchers led by Tony Gargano, a postdoctoral fellow at NASA's Johnson Space Center and the Lunar and Planetary Institute (LPI), both in Houston, used a novel method for analyzing the dusty debris that covers the Moon's surface calledregolith. They learned that even under generous assumptions, meteorite delivery since about four billion years ago could only have supplied a small fraction of Earth's water.The Moon serves as an ancient archive of the impact history the Earth-Moon system has experienced over billions of years. Where Earth's dynamic crust and weather erase such records, lunar samples preserve them. The records don't come without challenge, though. Traditional methods of studying regolith have relied on analyzing metal-loving elements. These elements can get muddied by repeated impacts on the Moon, making it harder to untangle and reconstruct what the original meteoroids contained.Enter triple oxygen isotopes, high precision "fingerprints" that take advantage of the fact that oxygen, the dominant element by mass in rocks, is unaffected by impact or other external forces. The isotopes offer a clearer understanding of the composition of meteorites that impacted the Earth-Moon system. The oxygen-isotope measurements revealed that at least ~1% by mass of the regolith contained material from carbon-rich meteorites that were partially vaporized when they hit the Moon. Using the known properties of such meteorites allowed the team to calculate the amount of water that would have been carried within."The lunar regolith is one of the rare places we can still interpret a time-integrated record of what was hitting Earth's neighborhood for billions of years," said Gargano. "The oxygen-isotope fingerprint lets us pull an impactor signal out of a mixture that's been melted, vaporized, and reworked countless times."The findings have implications for our understanding of water sources on Earth and the Moon. When scaled up by roughly 20 times to account for the substantially higher rate of impacts on Earth, the cumulative water shown in the model made up only a small percent of the water in Earth's oceans. That makes it difficult to reconcile the hypothesis that late delivery of water-rich meteorites was the dominant source of Earth’s water.“Our results don’t say meteorites delivered no water,” added co-author Justin Simon, a planetary scientist at NASA Johnson ’ s Astromaterials Research and Exploration Science Division. “They say the Moon’s long-term record makes it very hard for late meteorite delivery to be the dominant source of Earth’s oceans.”For the Moon, the implied delivery since about 4 billion years ago is tiny on an Earth-ocean scale but is not insignificant for the Moon. The Moon’s accessible water inventory is concentrated in small, permanently shadowed regions at the North and South Poles. These are some of the coldest spots in the solar system and introduce unique opportunities for scientific discovery and potential resources for lunar exploration when NASA lands astronauts on the Moon through Artemis III and beyond.The samples analyzed for this study came from parts of the Moon near the equator on the side of the Moon facing Earth, where all sixApollo missions landed. The rocks and dust collected more than 50 years ago continue to reveal new insights but are constrained to a small portion of the Moon. Samples delivered through Artemis will open the door for a new generation of discoveries for decades to come.“I’m part of the next generation of Apollo scientists — people who didn’t fly the missions, but who were trained on the samples and the questions Apollo made possible,” said Gargano. “The value of the Moon is that it gives us ground truth: real, physical material we can measure in the lab and use to anchor what we infer from orbital data and telescopes. I can’t wait to see what the Artemis samples have to teach us and the next generation about our place in the solar system.”For more information on NASA’s Astromaterials Research and Exploration Science Division, visit:https://science.nasa.gov/astromaterialsKaren Fox / Molly WasserHeadquarters, Washington240-285-5155 / 240-419-1732karen.c.fox@nasa.gov/molly.l.wasser@nasa.govVictoria SegoviaNASA’s Johnson Space Center281-483-5111victoria.segovia@nasa.govRead MoreFacebook logo@NASASolarSystem@NASASolarSystemInstagram logo@NASASolarSystem

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A close-up view of a portion of a "relatively fresh" crater, looking southeast, as photographed during the third Apollo 15 lunar surface moonwalk.Credit: NASA

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For more information on NASA's Astromaterials Research and Exploration Science Division, visit:

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Jennifer M. DoorenJan 20, 2026RELEASE26-008NASA HeadquartersThe Space Shuttle Columbia Memorial is seen during a wreath laying ceremony that was part of NASA’s Day of Remembrance, Thursday, Jan. 23, 2025, at Arlington National Cemetery in Virginia. Wreaths were laid in memory of those men and women who lost their lives in the quest for space exploration.Credit: NASA/Bill IngallsNASA will observe its annual Day of Remembrance on Thursday, Jan. 22, which includes commemorating the crews of Apollo 1 and the space shuttles Challenger and Columbia. The event is traditionally held every year on the fourth Thursday of January, as all three astronaut accidents happened around the end of the month.“On NASA’s Day of Remembrance, we pause to honor the members of the NASA family who lost their lives while pushing the boundaries of exploration and discovery,” said NASA Administrator Jared Isaacman. “We remember them not to retreat from risk, but to respect it — to learn, to improve, and continue onward. Their sacrifice and the strength of their families will forever inspire us as we continue to reach for the stars and pursue the secrets of the universe.”Isaacman will lead an observance at 1 p.m. EST at Arlington National Cemetery in Virginia, which will begin with a wreath-laying ceremony at the Tomb of the Unknown Soldier, followed by observances for the Apollo 1, Challenger, and Columbia crews.Several agency centers also will hold observances:Johnson Space Center in HoustonNASA Johnson will hold a commemoration at 10 a.m. CST at the Astronaut Memorial Grove with remarks by Center Director Vanessa Wyche, NASA astronaut Jasmin Moghbeli, and Cheryl McNair, widow of Challenger astronaut Ronald McNair. The event will have a moment of silence, a NASA T-38 flyover, taps performed by the Texas A&M Squadron 17, and a procession placing flowers at Apollo I, Challenger, and Columbia memorial trees.Kennedy Space Center in FloridaNASA Kennedy and the

Astronauts Memorial Foundation will host a ceremony at the Space Shuttle Atlantis building at Kennedy's Visitor Complex at 11 a.m. EST. The event will include musical guests, a bell ringing commemoration, a moment of silence, and wreath-laying. Kelvin Manning, deputy director at NASA Kennedy, and Bob Cabana, former NASA associate administrator and Kennedy center director, will provide remarks during the ceremony, which will livestream on the center's Facebookpage.Ames Research Center in California's Silicon ValleyNASA Ames will hold a remembrance ceremony at 1 p.m. PST that includes remarks from Center Director Eugene Tu, a moment of silence, and bell ringing commemoration for each astronaut lost in service.Langley Research Center in Hampton, VirginiaNASA Langley will hold a remembrance ceremony at 1 p.m. EST with acting Center Director Trina Dyal, followed by placing flags at the Langley Workers Memorial.Marshall Space Flight Center in Huntsville, AlabamaNASA Marshall will hold a candle-lighting ceremony and wreath placement at 9:30 a.m. CST and include remarks from Rae Ann Meyer, Marshall's acting center director, and Bill Hill, director of Safety and Mission Assurance at Marshall.Stennis Space Flight Center in Bay St. Louis, MississippiNASA Stennis and the NASA Shared Services Center will hold a wreath-laying ceremony and moment of silence at 10:30 a.m. CST with remarks from Center Director John Bailey and Anita Harrell, NASA Shared Services Center executive director.The agency also is paying tribute to its fallen astronauts with special online content, updated on NASA's Day of Remembrance, at:https://www.nasa.gov/dor-end-Bethany Stevens / Elizabeth ShawHeadquarters, Washington202-358-1600bethany.c.stevens@nasa.gov/elizabeth.a.shaw@nasa.gov

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Jennifer M. Dooren

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The Space Shuttle Columbia Memorial is seen during a wreath laying ceremony that was part of NASA's Day of Remembrance, Thursday, Jan. 23, 2025, at Arlington National Cemetery in Virginia. Wreaths were laid in memory of those men and women who lost their lives in the quest for space exploration.Credit: NASA/Bill IngallsNASA will observe its annual Day of Remembrance on Thursday, Jan. 22, which includes commemorating the crews of Apollo 1 and the space shuttles Challenger and Columbia. The event is traditionally held every year on the fourth Thursday of January, as all three astronaut accidents happened around the end of the month."On NASA's Day of Remembrance, we pause to honor the members of the NASA family who lost their lives while pushing the boundaries of exploration and discovery," said NASA Administrator Jared Isaacman. "We remember them not to retreat from risk, but to respect it — to learn, to improve, and continue onward. Their sacrifice and the strength of their families will forever inspire us as we continue to reach for the stars and pursue the secrets of the universe."Isaacman will lead an observance at 1 p.m. EST at Arlington National Cemetery in Virginia, which will begin with a wreath-laying ceremony at the Tomb of the Unknown Soldier, followed by observances for the Apollo 1, Challenger, and Columbia crews.Several agency centers also will hold observances:Johnson Space Center in HoustonNASA Johnson will hold a commemoration at 10 a.m. CST at the Astronaut Memorial Grove with remarks by Center Director Vanessa Wyche, NASA astronaut Jasmin Moghbeli, and Cheryl McNair, widow of Challenger astronaut Ronald McNair. The event will have a moment of silence, a NASA T-38 flyover, taps performed by the Texas A&M Squadron 17, and a procession placing flowers at Apollo I, Challenger, and Columbia memorial trees.Kennedy Space Center in FloridaNASA Kennedy and the Astronauts Memorial Foundation will host a ceremony at the Space Shuttle Atlantis building at Kennedy's Visitor Complex at 11 a.m. EST. The event will include musical guests, a bell ringing commemoration, a moment of silence, and wreath-laying. Kelvin Manning, deputy director at NASA Kennedy, and Bob Cabana, former NASA associate administrator and Kennedy center director, will provide remarks during the ceremony, which will livestream on the center's Facebookpage.Ames Research Center in California's Silicon ValleyNASA Ames will hold a remembrance ceremony at 1 p.m. PST that includes remarks from Center Director Eugene Tu, a moment of silence, and bell ringing commemoration for each astronaut lost in service.Langley Research Center in Hampton, VirginiaNASA Langley will hold a remembrance ceremony at 1 p.m. EST with acting Center Director Trina Dyal, followed by placing flags at the Langley Workers Memorial.Marshall Space Flight Center in Huntsville, AlabamaNASA Marshall will hold a candle-lighting ceremony and wreath placement at 9:30 a.m. CST and include remarks from Rae Ann Meyer, Marshall's acting center director, and Bill Hill, director of Safety and Mission Assurance at Marshall.Stennis Space Flight Center in Bay St. Louis, MississippiNASA Stennis and the NASA Shared Services Center will hold a wreath-laying ceremony and moment of silence at 10:30 a.m. CST with remarks from Center Director John Bailey and Anita Harrell, NASA Shared Services Center executive director.The agency also is paying tribute to its fallen astronauts with special online content, updated on NASA's Day of Remembrance, at:https://www.nasa.gov/dor-end-Bethany Stevens / Elizabeth ShawHeadquarters, Washington202-358-1600bethany.c.stevens@nasa.gov/elizabeth.a.shaw@nasa.gov

The Space Shuttle Columbia Memorial is seen during a wreath laying ceremony that was part of NASA's Day of Remembrance, Thursday, Jan. 23, 2025, at Arlington National Cemetery in Virginia. Wreaths were laid in memory of those men and women who lost their lives in the quest for space exploration.Credit: NASA/Bill Ingalls

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Several agency centers also will hold observances:

Johnson Space Center in Houston

NASA Johnson will hold a commemoration at 10 a.m. CST at the Astronaut Memorial Grove with remarks by Center Director Vanessa Wyche, NASA astronaut Jasmin Moghbeli, and Cheryl McNair, widow of Challenger astronaut Ronald McNair. The event will have a moment of silence, a NASA T-38 flyover, taps performed by the Texas A&M Squadron 17, and a procession placing flowers at Apollo I, Challenger, and Columbia memorial trees.

Kennedy Space Center in Florida

NASA Kennedy and the Astronauts Memorial Foundation will host a ceremony at the Space Shuttle Atlantis building at Kennedy's Visitor Complex at 11 a.m. EST. The event will include musical guests, a bell ringing commemoration, a moment of silence, and wreath-laying. Kelvin Manning, deputy director at NASA Kennedy, and Bob Cabana, former NASA associate administrator and Kennedy center director, will provide remarks during the ceremony, which will livestream on the center's Facebookpage.

Ames Research Center in California's Silicon Valley

NASA Ames will hold a remembrance ceremony at 1 p.m. PST that includes remarks from Center Director Eugene Tu, a moment of silence, and bell ringing commemoration for each astronaut lost in service.

Langley Research Center in Hampton, Virginia

NASA Langley will hold a remembrance ceremony at 1 p.m. EST with acting Center Director Trina Dyal, followed by placing flags at the Langley Workers Memorial.

Marshall Space Flight Center in Huntsville, Alabama

NASA Marshall will hold a candle-lighting ceremony and wreath placement at 9:30 a.m. CST and include remarks from Rae Ann Meyer, Marshall's acting center director, and Bill Hill, director of Safety and Mission Assurance at Marshall.

Stennis Space Flight Center in Bay St. Louis, Mississippi

NASA Stennis and the NASA Shared Services Center will hold a wreath-laying ceremony and moment of silence at 10:30 a.m. CST with remarks from Center Director John Bailey and Anita Harrell, NASA Shared Services Center executive director.

The agency also is paying tribute to its fallen astronauts with special online content, updated on NASA's Day of Remembrance, at:

https://www.nasa.gov/dor

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Most Inspirational Award: Photonics OdysseyTeam Members: Manish D., Deeraj K., Prasanth G., Rajalingam N., Rashi M., Sakthi R.Challenge: Commercializing Low Earth Orbit (LEO)Country/Territory: IndiaPhotonics Odyssey reimagined satellite internet as a sovereign national infrastructure rather than a private service, proposing a phased-array antenna approach that reduces ground dependency and expands broadband access to remote regions of India. The concept aims to help connect more than 700 million people who lack access to broadband internet.Learn more about Photonics Odysseys' project

Best Use of Storytelling Award: HerCode SpaceTeam Members: Alice R., Joselyn R., Paula C., Pierina J.Challenge: Stellar Stories: Space Weather Through the Eyes of EarthlingsCountry/Territory: Universal EventHerCode Space combined NASA data and heliophysics concepts with powerful storytelling and vibrant illustrations to teach kids how space weather affects daily life and why it matters. HerCode Science hopes their story, "A Solar Tale," can bridge science and imagination, and bring heliophysics to life in classrooms, libraries, and outreach programs.Learn more about HerCode Spaces' project

Global Connection Award: Gaia+LEOTeam Members: Adam H., Katia L., Prajwal S., Upendra K.Challenge: Commercializing Low Earth Orbit (LEO)Country/Territory: United StatesTeam Gaia+LEO developed a mixed-integer optimization framework that co-designs orbital and terrestrial data-center networks to support large-scale AI training and climate modeling in orbit. Their goal is to reduce the power, and water demands of Earth-based systems and help accelerate the shift toward space-based, green computing within the emerging orbital economy.Learn more about Gaia+LEOs' project

Art & Technology Award: Zumorroda-XTeam Members: Alaa A., Esraa A., Malak S., Mennatulla E.Challenge: NASA Farm Navigators: Using NASA Data Exploration in AgricultureCountry/Territory: EgyptTeam Zumorroda-X created mini games that allow players to step into the shoes of a farmer who sets off on an epic journey around the world. Through this game, players can learn how farmers globally adapt to heat waves, flooding, and other environmental challenges.Learn more about Zumorroda-Xs' project

Local Impact Award: QUEÑARISTeam Members: Borax Q., Carlos Y., Marcelo S., Máximo S., Oscar M., Pamela P.Challenge: BloomWatch: An Earth Observation Application for Global Flowering PhenologyCountry/Territory: PeruTeam QUEÑARIS' project addresses critical water scarcity in Peru's second-largest city, Arequipa, caused by the degradation of queñua forests, which are vital for water retention. Their platform combines native microorganisms, NASA satellite data, drones, and artificial intelligence to accelerate tree growth, identify the best areas for reforestation, and monitor ecosystem health.Learn more about QUEÑARIS' project

Stay up to date with #SpaceApps by following these accounts:X:@SpaceAppsInstagram:@nasa_spaceappsFacebook:@spaceappschallengeYouTube:@NASASpaceAppsChallenge

NASA Space Apps is funded by NASA'sEarth Science Divisionthrough a contract withBooz Allen Hamilton,Mindgrub, andSecondMuse.

To learn more about what inspired these winning projects, visit:

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TITLE: NASA Launches Research Program for Students to Explore Big Ideas

NASA STEM TeamDec 16, 2025ArticleNASA is now accepting concepts for a new research challenge. The Opportunities in Research, Business, Innovation, and Technology (ORBIT) challenge is a multi-phase innovation competition designed to empower university and college students to develop next-generation solutions that benefit life on Earth and deep-space exploration.With up to $380,000 in total prize funding, NASA's ORBIT challenges student teams to bring their most forward-thinking concepts to the table, either utilizing NASA intellectual property or conceptualizing their own. Teams are tasked with conducting targeted research, designing early mockups or models, and performing feasibility analyses to refine their ideas. Finalists then advance to a live showcase where they present their work to a panel of expert judges, who evaluate the proposals and select winners based on the teams' final pitches and responses to questions.The ORBIT has two challenge tracks for teams to choose from. The ORBIT Earth track requires teams to select a NASA-owned patent and develop novel commercial or nonprofit applications addressing real-world problems. From adapting aerospace materials for disaster response and preparedness, to repurposing space-based sensors for healthcare, students must demonstrate clear pathways to public benefit.The ORBIT Space track asks teams to design new system concepts aligned with NASA's current and future missions, particularly supporting the Artemis program's goal of establishing a sustainable human presence on the Moon and preparing for eventual missions to Mars and beyond. Students will create technically feasible designs for everything from lunar habitats that could house future Artemis astronauts to deep space robotics that open more pathways to in-situ resource utilization. Teams that successfully integrate objectives from both tracks may qualify for an optional integration bonus.This challenge accelerates innovation in areas critical to NASA's future goals while cultivating a pipeline of interdisciplinary talent. By engaging the next generation in NASA's dual mission to explore space and improve life on Earth, ORBIT inspires students to join the agency's talent network while delivering tangible benefits to American communities and industries.Beyond monetary awards, participants stand to gain mentorship from NASA experts, access to agency facilities, and hands-on experience in systems design, entrepreneurship, and commercialization.For complete competition details, eligibility requirements, and official rules, visit:https://go.nasa.gov/4q2TS9uRegistration is open until Feb. 9, 2026, through theNASA STEM Gateway.

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ArtemisArtemis IIThe first crewed Artemis flight marks a key step toward long-term return to the Moon and future missions to Mars.Future MissionWHY we goScientific DiscoveryLUNAR DESTINATIONEarth's MoonMEET THE CREWOur Artemis AstronautsArtemis II builds on the success of the uncrewed Artemis I in 2022, and will demonstrate a broad range of capabilities needed on deep space missions. The Artemis II test flight will be NASA’s first mission with crew aboard the SLS (Space Launch System) rocket and Orion spacecraft.Mission TypeCrewed Lunar FlybyCREW SIZE4LaunchNo Earlier Than Feb. 6, 2026mission duration10 DaysArtemis II NewsMore Mission NewsArticle4 Min ReadNASA Selects Participants to Track Artemis II MissionArticle3 Min ReadI Am Artemis: Dustin Gohmert38 Min ReadArtemis II: RecoveryArticle5 Min ReadNASA’s Artemis II Mission to Fly Legacy Keepsakes with Astronaut Crew2 Min ReadLaunch Pad Preparations Progress Ahead of Artemis II Wet Dress RehearsalBlog1 Min ReadNASA’s Artemis II Rocket and Spacecraft Make Their Way to Launch PadImage Article3 Min ReadNASA Unlocks Golden Age of Innovation, Exploration in Trump’s First YearNews Release25 Min ReadArtemis II: What NASA Learned From Launching Artemis IOur Artemis II CrewMeet the astronauts who will venture around the Moon on Artemis II, the first crewed flight aboard NASA’s human deep space capabilities, paving the way for future lunar surface missions.Forging New Frontiersabout Our Artemis II CrewThe SpacecraftThe RocketThe Ground SystemsThe ScienceNASA’s Newest SpacecraftOrion is developed to be capable of sending astronauts to the Moon and is a crucial step toward eventually sending crews on to Mars.The Orion spacecraft will serve as the exploration vehicle that will carry and sustain the crew on Artemis missions to the Moon and return them safely to Earth. Orion will launch on NASA’s new heavy-lift rocket, the SLS (Space Launch System).About OrionTechnicians with NASA’s Exploration Ground Systems team use a crane to lift and secure NASA’s Orion spacecraft on top of the SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA’s Kennedy Space Center in Florida on

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Engineers and technicians with NASA’s Exploration Ground Systems Program prepare to lift the left center booster segment shown with the iconic NASA “worm” insignia for the agency’s SLS (Space Launch System) rocket to the Vehicle Assembly Building at NASA’s Kennedy Space Center in Florida on Friday, Jan. 24, 2025.

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Artemis II Flight PathThis visualization depicts the Artemis II mission’s 10-day "figure-eight" trajectory, featuring a high-altitude lunar flyby that will send four astronauts further into deep space than any humans in history by 2026.Visualizing Artemis IIabout Artemis II Flight Path

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Artemis II Boarding PassSend Your Name Around the MoonArtemis II will test NASA’s deep space capabilities as humans fly on the Space Launch System rocket and Orion spacecraft for the first time. Join the mission by launching your name around the moon alongside NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and Canadian Space Agency astronaut Jeremy Hansen.Claim Your Passabout Send Your Name Around the MoonArtemis II crew posterNASA

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Info for Parents, Teachers, and CaregiversNASA provides a safe place for children to play games as they learn about NASA and its missions. NASA encourages you to visit NASA Kids' Club and use its games and activities to inspire the next generation. On this site, you will find games of various skill levels for children pre-K through grade 4. These games support national education standards in science, technology, engineering and mathematics.Learn More

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TITLE: Air Traffic Solutions

AeronauticsAir Traffic SolutionsThe skies are getting busier every day, with new types of aircraft aiming to provide innovative air travel services that seek to improve all of our lives, whether that means delivering cargo to your front door or providing personalized taxi service across town. NASA is working with its government and industry partners to keep the sky lanes open, efficient, and safer than ever before. Follow their work here.Latest NewsMore Air Traffic Solutions NewsArticle3 Min ReadNASA Develops Blockchain Technology to Enhance Air Travel Safety and SecurityArticle3 Min ReadNASA Works with Boeing, Other Collaborators Toward More Efficient Global FlightsArticle3 Min ReadNASA, Partners Push Forward with Remotely Piloted Airspace IntegrationArticle4 Min ReadNASA Tests 5G-Based Aviation Network to Boost Air Taxi Connectivity3 Min ReadNASA Tests Mixed Reality Pilot Simulation in Vertical Motion SimulatorArticle23 Min ReadSmall Steps, Giant Leaps: Episode 154: NASA's Air Traffic Management-eXploration Project3 Min ReadNASA's Moffett Federal Airfield Hosts Boeing Digital Taxi TestsArticle5 Min ReadNASA Demonstrates New Wildland Fire Airspace Management SystemArticleThe air traffic solutions NASA is working on today build from a foundation of research that goes back decades. Three recently concluded projects in particular are helping to develop new ways of doing things in the 21st century. The two features presented here help tell those project's stories.A Story for Axel: Big Ideas for Better Air Mobility with Drones of all SizesIn this essay from 2021 — which was originally presented as a three-part web feature — NASA Aeronautics' senior writer Jim Banke tells his newborn grandson, Axel, the story of how two recently concluded major air traffic management research projects by NASA will help achieve a long-held vision for the future of aviation.Read this Story About NASA's Air Traffic SolutionsAs NASA Aeronautics senior writer Jim Banke tells his grandson, Axel, about two recently concluded research projects involving drones large and small, Axel appears to be imagining what the major elements of that future as told to him by his grandpa might be.NASA/ Jim Banke, Lillian GipsonNASA Delivers on Making Gate to Gate Flights More EfficientIn this story from March 2022, a summary of NASA's three-part Airspace Technology Demonstration (ATD) project is presented. It details how a team of NASA's aeronautical innovators working with government and industry partners spent fives years developing and demonstrating new tools to help you fly from here to there in the most time-efficient, fuel-saving, and cost-effective manner possible.Read this Story About ATDAt NASA's air traffic management laboratory near the Dallas/Ft. Worth International Airport in Texas, researchers Al Capps (seated) and Paul Borchers demonstrate Airspace Technology

Demonstration 2 (ATD-2) tools that air traffic managers began testing in 2017.NASA / Jim BankeAir traffic controllers test NASA-developed concepts in FutureFlight Central, an ultra-realistic, 360-degree, simulated control tower at NASA Ames Research Center.NASA / Dominic HartAir Traffic Solutions Gallery7 ImagesGo To GalleryGo To GalleryHere is Some Other Cool NASA Aeronautics Stuff to Check OutAeronautics @ HomeA collection of activities and lessons to learn more about NASA Aeronautics and the science of flight that you can do at home.STEM Learning ResourcesExplore hands-on activities, interactive lesson plans, educator guides, and other downloadable content related to aeronautics.Aeronautics BackgroundsTake video calls from your home or office in the cockpit of the X-59 or from inside a wind tunnel! And other aviation themes.Free E-Books to DownloadAdd these books covering NASA Aeronautics and other aviation and space topics to your electronic library.NASA X Video FilesThis binge worthy page collects aeronautics-themed topics from the popular NASA X video series.Facebook logo@NASA@NASAaero@NASAes@NASA@NASAaero@NASA_esInstagram logo@NASA@NASAaero@NASA_esLinkedin logo@NASAOther Aeronautics topicsEven More AeronauticsSupersonic FlightResearch about flying faster than the speed of sound.Flight InnovationStories about pioneering the frontiers of 21st century flight.Drones & YouAll about new ways to get from here to there in the air.Ultra-Efficient AviationLearn about research to make aviation more sustainable.Air Traffic SolutionsRead about how NASA is opening up the sky for all.Keep ExploringDiscover More Topics From NASAMissionsHumans In SpaceSolar System ExplorationEyes on the Solar SystemExplore NASA's History

Air Traffic SolutionsThe skies are getting busier every day, with new types of aircraft aiming to provide innovative air travel services that seek to improve all of our lives, whether that means delivering cargo to your front door or providing personalized taxi service across town. NASA is working with its government and industry partners to keep the sky lanes open, efficient, and safer than ever before. Follow their work here.

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MultimediaNASA ImagesNASA has always told its story through its images, a few of which have become icons of human history.TodayImage Of The DayRed, Green Light ShowA green and red aurora streams across Earth's horizon above the city lights of Europe in this Jan. 19, 2026, photograph, which looks north across Italy toward Germany. The International Space Station was orbiting 262 miles above the Mediterranean Sea at approximately 10:02 p.m. local time when the image was captured.Browse Image ArchiveTodayImage Of The DayView imageNASA HistoryFlickr Commons has more than 3,000 photos from NASA's history. All photographs on the NASAcommons Flickr account originate from the compilation of photos from NASA's image and video library and other archived sources.Viewabout NASA HistoryNASA Image ResourcesImage Usage GuidelinesImage and Video LibraryNASA HQ Photo OfficeEarth ObservatoryFaces of NASAHubble Space TelescopeJames Webb Space TelescopeNASA Spacewalkers24 ImagesGo To GalleryGo To GalleryDiscover MoreGames and InteractivesBlogsNewslettersWays to Watch

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TITLE: NASA’s Chandra Releases Deep Cut From Catalog of Cosmic Recordings

Lee MohonJan 23, 2026ArticleContentsVisual DescriptionNews Media ContactLike a recording artist who has had a long career,NASA’s Chandra X-ray Observatoryhas a “back catalog” of cosmic recordings that is impossible to replicate. To access these X-ray tracks, or observations, the ultimate compendium has been developed: theChandra Source Catalog(CSC).The CSC contains theX-raydata detected up to the end of 2020 by Chandra, the world’s premier X-ray telescope and one of NASA’s “Great Observatories.” The latest version of the CSC, known as CSC 2.1, contains over 400,000 unique compact and extended sources and over 1.3 million individual detections in X-ray light.beforeafterThis image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. WolkThis image is the sum of 86 observations added together, representing over three million seconds of Chandra observing time. It spans just about 60 light-years across, which is a veritable pinprick on the entire sky. The underlying image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively. The annotations on the image show where Chandra has detected over 3,300 individual sources in this field of view over a 22-year timeframe.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. WolkbeforeafterThis image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. WolkThis image is the sum of 86 observations added together, representing over three million seconds of Chandra observing time. It spans just about 60 light-years across, which is a veritable pinprick on the entire sky. The underlying image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively. 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This allows scientists using other telescopes — both on the ground and in space including NASA's James Webb and Hubble Space Telescopes — to combine this unique X-ray data with information from other types of light.The richness of the Chandra Source Catalog is illustrated in a new image of the Galactic Center, the region around the supermassive black hole at the center of the Milky Way galaxy called Sagittarius A*. In this image that spans just about 60light-yearsacross, a veritable pinprick on the entire sky, Chandra has detected over 3,300 individual sources that emit X-rays. This image is the sum of 86 observations added together, representing over three million seconds of Chandra observing time.Another new representation of the vast scope of the Chanda Source Catalog is found in a just-released sonification, the translation of astronomical data into sound. This sonification encompasses the new map that includes 22 years of Chandra observations across the sky, beginning from its launch through its observations in 2021. Because many X-ray sources have been observed multiple times over the life of the Chandra mission, this sonification represents those repeat X-ray sightings over time through different notes.To view this video please enable JavaScript, and consider upgrading to a web browser thatsupports HTML5 videoChandra Source Catalog Sonification.NASA/CXC/SAO/K.Arcand, SYSTEM Sounds (M. Russo, A. SantaguidaIn the view of the sky, projected in a similar way to how the Earth is often depicted in world maps, the core of theMilky Wayis in the center and the Galactic plane is horizontal across the middle of the image. A circle appears at the position of each detection and the size of the circle is determined by the number of detections in that location over time. A year counter appears at the top of the frame. Since Chandra continues to be fully operational, the text changes to "… and beyond" after 2021 as the telescope continues to collect observations. During the video, a collage of images produced by Chandra fades in as a background. In the final frames of the video, thumbnail images representing the thousands of Chandra observations taken over the lifetime of the mission appear behind the sky map.The most recent version of the Chandra Source Catalog can be accessed athttps://cxc.cfa.harvard.edu/csc/NASA's Marshall Space Flight Center in Huntsville, Alabama, manages the Chandra program. The Smithsonian Astrophysical Observatory's Chandra X-ray Center controls science operations from Cambridge, Massachusetts, and flight operations from Burlington, Massachusetts.To learn more about Chandra, visit:https://science.nasa.gov/chandraRead more from NASA's Chandra X-ray ObservatoryLearn more about the Chandra X-ray Observatory and its mission here:https://www.nasa.gov/chandrahttps://chandra.si.eduVisual DescriptionA very deep Chandra X-ray Observatory image around the Sagittarius A* supermassive black hole, located in the center of the Milky Way galaxy, is shown. The image is dominated by burnt orange, deep gold and blue hues, with a sprinkling of rich green. The area looks both intricate and full, with a dense population of tiny dots, along with larger clumps and diffuse areas and nebulous areas peeking through.At the center of the image, there is a bright, lumpy area in pale gold showing the intense X-ray radiation emanating from the Sagittarius A* black hole. In the surrounding area, there are more smaller lumps layered throughout, feathering out to a large almost butterfly shape filling much of the screen. The image appears textured, like dozens of blue and orange glow worms are paused in their wriggling.The image offers an unprecedented view of lobes of hot gas extending for a dozen light years on either side of the black hole. These lobes provide evidence for powerful eruptions occurring several times over the last ten thousand years. The image also contains several mysterious X-ray filaments, some of which may be huge magnetic structures interacting with streams of energetic electrons produced by rapidly spinning neutron stars. Such features are known as pulsar wind nebulas. Chandra has detected over 3,300 individual sources that emit X-rays in this field of view. This image is the sum of 86 observations added together, representing over three million seconds of Chandra observing time.News Media ContactMegan WatzkeChandra X-ray CenterCambridge, Mass.617-496-7998mwatzke@cfa.harvard.eduJoel WallaceMarshall Space Flight Center, Huntsville, Alabama256-544-0034joel.w.wallace@nasa.gov

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The underlying image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively. The annotations on the image show where Chandra has detected over 3,300 individual sources in this field of view over a 22-year timeframe.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. WolkbeforeafterThis image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. WolkThis image is the sum of 86 observations added together, representing over three million seconds of Chandra observing time. It spans just about 60 light-years across, which is a veritable pinprick on the entire sky. The underlying image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively. The annotations on the image show where Chandra has detected over 3,300 individual sources in this field of view over a 22-year timeframe.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. 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This allows scientists using other telescopes — both on the ground and in space including NASA's James Webb and Hubble Space Telescopes — to combine this unique X-ray data with information from other types of light.The richness of the Chandra Source Catalog is illustrated in a new image of the Galactic Center, the region around the supermassive black hole at the center of the Milky Way galaxy called Sagittarius A*. In this image that spans just about 60light-yearsacross, a veritable pinprick on the entire sky, Chandra has detected over 3,300 individual sources that emit X-rays. This image is the sum of 86 observations added together, representing over three million seconds of Chandra observing time.Another new representation of the vast scope of the Chanda Source Catalog is found in a just-released sonification, the translation of astronomical data into sound. This sonification encompasses the new map that includes 22 years of Chandra observations across the sky, beginning from its launch through its observations in 2021. Because many X-ray sources have been observed multiple times over the life of the Chandra mission, this sonification represents those repeat X-ray sightings over time through different notes.To view this video please enable JavaScript, and consider upgrading to a web browser thatsupports HTML5 videoChandra Source Catalog Sonification.NASA/CXC/SAO/K.Arcand, SYSTEM Sounds (M. Russo, A. SantaguidaIn the view of the sky, projected in a similar way to how the Earth is often depicted in world maps, the core of theMilky Wayis in the center and the Galactic plane is horizontal across the middle of the image. A circle appears at the position of each detection and the size of the circle is determined by the number of detections in that location over time. A year counter appears at the top of the frame. 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The Smithsonian Astrophysical Observatory's Chandra X-ray Center controls science operations from Cambridge, Massachusetts, and flight operations from Burlington, Massachusetts.To learn more about Chandra, visit:https://science.nasa.gov/chandraRead more from NASA's Chandra X-ray ObservatoryLearn more about the Chandra X-ray Observatory and its mission here:https://www.nasa.gov/chandrahttps://chandra.si.eduVisual DescriptionA very deep Chandra X-ray Observatory image around the Sagittarius A* supermassive black hole, located in the center of the Milky Way galaxy, is shown. The image is dominated by burnt orange, deep gold and blue hues, with a sprinkling of rich green. The area looks both intricate and full, with a dense population of tiny dots, along with larger clumps and diffuse areas and nebulous areas peeking through.At the center of the image, there is a bright, lumpy area in pale gold showing the intense X-ray radiation emanating from the Sagittarius A* black hole. In the surrounding area, there are more smaller lumps layered throughout, feathering out to a large almost butterfly shape filling much of the screen. The image appears textured, like dozens of blue and orange glow worms are paused in their wriggling.The image offers an unprecedented view of lobes of hot gas extending for a dozen light years on either side of the black hole. These lobes provide evidence for powerful eruptions occurring several times over the last ten thousand years. The image also contains several mysterious X-ray filaments, some of which may be huge magnetic structures interacting with streams of energetic electrons produced by rapidly spinning neutron stars. Such features are known as pulsar wind nebulas. Chandra has detected over 3,300 individual sources that emit X-rays in this field of view. 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Before and After

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To learn more about Chandra, visit:

https://science.nasa.gov/chandra

Read more from NASA's Chandra X-ray Observatory

Learn more about the Chandra X-ray Observatory and its mission here:

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https://chandra.si.edu

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AeronauticsDrones & YouLook here for updates about NASA research into making it possible for drones of all types and sizes to safely move through the sky. This includes air taxis that promise to provide new personal flight options and robot drones designed to fly in atmospheres elsewhere in the solar system — among many other related areas of research.Latest News Related to Drones & YouMore Drones & You NewsArticle3 Min ReadNASA, Partners Push Forward with Remotely Piloted Airspace IntegrationArticle2 Min ReadNASA Tests Tools to Assess Drone Safety Over Cities4 Min ReadNASA Tests Research Aircraft to Improve Air Taxi Flight ControlsArticle3 Min ReadNASA Uses Wind Tunnel to Test Advanced Air Mobility Aircraft WingArticle3 Min ReadNASA Drop Test Supports Safer Air Taxi Design and CertificationArticleFeatured StoryNASA Uses Two Worlds to Test Future Mars Helicopter DesignsEngineers will go beyond the ends of the Earth to find more performance for future Mars helicopters. For the first...Read the StoryDrones and You Gallery19 ImagesGo To GalleryGo To GalleryAdvanced Air Mobility MissionThe vast majority of the research detailed on this Drones & You page resides in NASA's Air Mobility (AAM) mission. The AAM vision is to transform our communities by bringing the movement of people and goods off the ground, on demand, and into the sky. This air transportation system of the future will include low-altitude passenger transport, cargo delivery, and public service capabilities. This includes delivering data to guide industry's development of electric air taxis and drones and to assist the Federal Aviation Administration in safely integrating these vehicles into the national airspace. This will set the stage for a flourishing industry by 2030.Read More About the AAM MissionSeveral projects supporting NASA's Advanced Air Mobility (AAM) mission are working on different elements to help make AAM a reality. For example, AAM could be used in future healthcare operations in the form of air taxi ambulances or delivery of medical supplies. This concept graphic shows how a future AAM vehicle could aid in healthcare by carrying passengers to a hospital.NASAHere is Some Other Cool NASA Aeronautics Stuff to Check OutAeronautics @ HomeA collection of activities and lessons to learn more about NASA Aeronautics and the science of flight that you can do at home.STEM Learning ResourcesExplore hands-on activities, interactive lesson plans, educator guides, and other downloadable content related to aeronautics.Aeronautics BackgroundsTake video calls from your home or office in the cockpit of the X-59 or from inside a wind tunnel! And other aviation themes.Free E-Books to DownloadAdd these books covering NASA Aeronautics and other aviation and space topics to your electronic library.NASA X Video FilesThis binge worthy page collects aeronautics-themed topics from the popular NASA X video series.Other Aeronautics topicsEven More AeronauticsSupersonic FlightResearch about flying faster than the speed of sound.Flight InnovationStories about pioneering the frontiers of 21st century flight.Drones & YouAll about new ways to get from here to there in the air.Ultra-Efficient AviationLearn about research to make aviation more sustainable.Air Traffic SolutionsRead about how NASA is opening up the sky for all.Facebook logo@NASA@NASAaero@NASAes@NASA@NASAaero@NASA_esInstagram logo@NASA@NASAaero@NASA_esLinkedin logo@NASAKeep ExploringDiscover More Topics From NASAMissionsArtemisScience MissionsNACA to NASA to Now

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Advanced Air Mobility MissionThe vast majority of the research detailed on this Drones & You page resides in NASA's Air Mobility (AAM) mission. The AAM vision is to transform our communities by bringing the movement of people and goods off the ground, on demand, and into the sky. This air transportation system of the future will include low-altitude passenger transport, cargo delivery, and public service capabilities. This includes delivering data to guide industry's development of electric air taxis and drones and to assist the Federal Aviation Administration in safely integrating these vehicles into the national airspace. This will set the stage for a flourishing industry by 2030.Read More About the AAM MissionSeveral projects supporting NASA’s Advanced Air Mobility (AAM) mission are working on different elements to help make AAM a reality. For example, AAM could be used in future healthcare operations in the form of air taxi ambulances or delivery of medical supplies. This concept graphic shows how a future AAM vehicle could aid in healthcare by carrying passengers to a hospital.NASA

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Chelsea GohdJan 05, 2026ArticleContentsJupiter beams bright, Saturn and the Moon cozy up, and the Beehive Cluster appearsSkywatching HighlightsTranscriptJupiter beams bright, Saturn and the Moon cozy up, and the Beehive Cluster appearsJupiter is at its biggest and brightest all year, the Moon and Saturn pair up, and the Beehive Cluster buzzes into view.Skywatching HighlightsJan. 10:Jupiter at oppositionJan. 23:Saturn and Moon conjunctionJan. (throughout):Beehive ClusterTranscriptJupiter is at its biggest and brightestThe Moon and Saturn share the skyAnd the beehive cluster makes an appearanceThat's what's up, this JanuaryJanuary 10, Jupiter will be at its most brilliant of the entire year!This night, Jupiter will be at what's called "opposition," meaning that Earth will be directly between Jupiter and the Sun.NASA/JPL-CaltechIn this alignment, Jupiter will appear bigger and brighter in the night sky than it will all year - talk about starting off the new year bright!To see Jupiter at its best this year, look to the east and all evening long, you'll be able to see the planet in the constellation Gemini. It will be one of the brightest objects in the night sky (only the moon and Venus will be brighter)Saturn and the Moon will share the sky on January 23rd as part of a conjunction!NASA/JPL-CaltechA conjunction is when objects in the sky look close together even though they're actually far apart.To spot the pair, look to the west and you'll see Saturn just below the moon, sparkling in the night sky.The beehive cluster will be visible in the night sky throughout January!The beehive cluster, more formally known as Messier 44, or M44, is made of at least 1,000 starsIt's an open star cluster, meaning it's a loosely-bound group of stars. There are thousands of open star clusters like the beehive in the Milky Way Galaxy!NASA/JPL-CaltechTo see the beehive cluster, look to the eastern night sky after sunset and before midnight throughout the month - especially great nights to spot the cluster are around the middle of January when the cluster isn't too high or low in the sky to see.With dark skies you might be able to spot the beehive with just your eyes, but binoculars or a small telescope will help.Here are the phases of the Moon for January.NASA/JPL-CaltechYou can stay up to date on all of NASA's missions exploring the solar system and beyond at science.nasa.gov.I'm Chelsea Gohd from NASA's Jet Propulsion Laboratory, and that's What's Up for this month.

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Skywatching Highlights

Transcript

Jupiter beams bright, Saturn and the Moon cozy up, and the Beehive Cluster appearsJupiter is at its biggest and brightest all year, the Moon and Saturn pair up, and the Beehive Cluster buzzes into view.Skywatching HighlightsJan. 10:Jupiter at oppositionJan. 23:Saturn and Moon conjunctionJan. (throughout):Beehive ClusterTranscriptJupiter is at its biggest and brightestThe Moon and Saturn share the skyAnd the beehive cluster makes an appearanceThat's what's up, this JanuaryJanuary 10, Jupiter will be at its most brilliant of the entire year!This night, Jupiter will be at what's called "opposition," meaning that Earth will be directly between Jupiter and the Sun.NASA/JPL-CaltechIn this alignment, Jupiter will appear bigger and brighter in the night sky than it will all year - talk about starting off the new year bright!To see Jupiter at its best this year, look to the east and all evening long, you'll be able to see the planet in the constellation Gemini. It will be one of the brightest objects in the night sky (only the moon and Venus will be brighter)Saturn and the Moon will share the sky on January 23rd as part of a conjunction!NASA/JPL-CaltechA conjunction is when objects in the sky look close together even though they're actually far apart.To spot the pair, look to the west and you'll see Saturn just below the moon, sparkling in the night sky.The beehive cluster will be visible in the night sky throughout January!The beehive cluster, more formally known as Messier 44, or M44, is made of at least 1,000 starsIt's an open star cluster, meaning it's a loosely-bound group of stars. There are thousands of open star clusters like the beehive in the Milky Way Galaxy!NASA/JPL-CaltechTo see the beehive cluster, look to the eastern night sky after sunset and before midnight throughout the month - especially great nights to spot the cluster are around the middle of January when the cluster isn't too high or low in the sky to see.With dark skies you might be able to spot the beehive with just your eyes, but binoculars or a small telescope will help.Here are the phases of the Moon for January.NASA/JPL-CaltechYou can stay up to date on all of NASA's missions exploring the solar system and beyond at science.nasa.gov.I'm Chelsea Gohd from NASA's Jet Propulsion Laboratory, and that's What's Up for this month.

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Jan. 10:Jupiter at opposition

Jan. 23:Saturn and Moon conjunction

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What’s Up

URL: https://www.nasa.gov/directorates/stmd/nasa-back-for-seconds-with-new-food-system-design-challenge/

TITLE: NASA Back for Seconds with New Food System Design Challenge

4 min readNASA Back for Seconds with New Food System Design Challenge

Loura HallJan 13, 2026ArticleThis illustration of Moon to Mars infrastructure shows astronauts living and working on the surface of Mars. NASA’s Moon to Mars Objectives establish an objectives-based approach to the agency’s human deep space exploration efforts; NASA’s Moon to Mars Architecture approach distills the objectives into operational capabilities and elements.NASA is getting ready to send four astronauts around the Moon with Artemis II, laying the foundation for sustainable missions to the lunar surface and paving the way for human exploration on Mars. As the agency considers deep space endeavors that could last months or years, it must develop ways to feed astronauts beyond sending supplies from Earth.That is why NASA is launching theDeep Space Food Challenge: Mars to Table, a new global competition inviting chefs, innovators, culinary experts, higher-education students, and citizen scientists to design a complete, Earth-independent food system for long-duration space missions.“In the future, exploration missions will grow in both duration and distance from Earth. This will make the critical question of feeding our astronauts more complex, requiring innovative solutions to allow for long-term human exploration of space,” said Greg Stover, acting associate administrator of NASA’s Space Technology Missions Directorate at NASA Headquarters in Washington. “Opening the door to ideas from beyond the agency strengthens NASA’s ability to operate farther from Earth with greater independence.”Mars to Table builds on NASA’sfirst Deep Space Food Challengeby seeking to integrate multiple food production and preparation methods into a holistic, self-sustaining system designed for use on Mars. This new challenge is open now until July 31 to the global public and carries a prize purse of up to $750,000.“Future crews on the Moon and Mars will need food systems that are nutritious, sustainable, and fully independent from Earth,” said Jarah Meador, program executive for NASA’s Prizes, Challenges, and Crowdsourcing Program at NASA Headquarters. “Food will play a pivotal role in the overall health and happiness of future deep space explorers. The Mars to Table Challenge is about bringing all those pieces together into one comprehensive design.”Solvers are tasked with creating a complete meal plan suitable for astronauts living on Mars, using a NASA-created mission scenario as their guide. Each team will design a full food system concept, including a detailed operations plan and system design layout that supports a surface mission. Teams must consider every detail – from nutritional balance and taste to safety, usability, and integration with NASA’sEnvironmental Control and Life Support Systems.Participants in the Mars to Table Challenge are also encouraged to address food security on Earth. Innovative growth systems designed for space could make fresh food production possible in harsh, remote, or resource-limited areas, such as research stations located at Earth’s poles or in rural areas with limited access to traditional supply chains.“This challenge isn’t just about feeding astronauts; it’s about feeding people anywhere,” said Jennifer Edmunson, acting program manager for NASA’s Centennial Challenges at NASA’s Marshall Spaceflight Center in Huntsville, Alabama. “Novel meals that are compact, shelf-stable, and nutrient-rich could expand culinary options for groups like military personnel or disaster relief responders. By solving for Mars and future planetary expeditions, we can also find solutions for Earth.”NASA’s Centennial Challenges have a 20-year legacy of engaging the public to solve complex problems that benefit

NASA’s broader initiatives. Past challenges have spurred advances in robotics, additive manufacturing, power and energy, textiles, chemistry, and biology.Mars to Table is a collaborative, cross-program Centennial Challenge with support from NASA’s Division of Biological and Physical Sciences, Heliophysics Division, Planetary Science Program, Human Research Program, Mars Campaign Office, and Earth Science Division. Subject matter experts at the agency’s Johnson Space Center in Houston and Kennedy Space Center in Florida support the challenge. This challenge is part of thePrizes, Challenges and Crowdsourcingprogram within NASA’s Space Technology Mission Directorate. NASA has partnered with the Methuselah Foundation and contracted Floor23 Digital to support the administration and management of this challenge.To learn more about the challenge, including timelines, submission requirements, and future webinar dates, visit:https://www.deepspacefood.org/marstotableBy Savannah BullardFacebook logo@NASATechnology@NASA_Technology

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By Savannah Bullard

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URL: https://www.nasa.gov/learning-resources/for-colleges-universities/nasa-launches-2026-lunabotics-challenge/

TITLE: NASA Launches 2026 Lunabotics Challenge

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Jason CostaSep 08, 2025ArticleStudents prepare their robots to enter Artemis Arena during NASA's Lunabotics competition on May 20, 2025, at the Center for Space Education near the Kennedy Space Center Visitor Complex in Florida.NASA/Isaac WatsonAs college students across the country embark upon the academic year, NASA is giving them something else to look forward to – the agency's 2026 Lunabotics Challenge. Teams interested in participating can submit their applications and supporting materials through NASA' sStem Gateway portalbeginning Monday, Sept. 8.Key dates and challenge details are available in the2026 Lunabotics Challenge Guidebook. Once all applications and supporting materials are received and evaluated, NASA will notify the selected teams to begin the challenge.Student teams participating in this year's challenge will create robots capable of building berms out of lunar regolith – the loose, fragmental material on the Moon's surface. Structures like these will be important during lunar missions as blast protection during lunar landings and launches, shading for cryogenic propellant tank farms, radiation shielding around nuclear power plants, and other uses critical to future Moon missions."We are excited to continue the Lunabotics competition for universities as NASA develops new Moon to Mars technologies for the Artemis program," said Robert Mueller, senior technologist at NASA, as well as co-founder and chief judge of the Lunabotics competition. "Excavating and moving regolith is a fundamental need to build infrastructure on the Moon and Mars and this competition creates 21st century skills in the future workforce."An in-person qualifying event will be held May 12-17, 2026, at the University of Central Florida's Space Institute's Exolith Lab in Orlando. From this round, the top 10 teams will be invited to bring their robots to the final competition on May 19-21, at the Kennedy Space Center Visitor Complex's Artemis Arena in Florida, which has an area filled with a lunar regolith simulant. The team scoring the most points will receive the Lunabotics Grand Prize and participate in an exhibition-style event at NASA Kennedy.By encouraging innovative construction techniques and assessing student designs and data the same way it does its own prototypes, NASA casts a wider net to find innovative solutions to challenges inherent in future Artemis missions, like developing future lunar excavators,in-situ resource utilizationcapabilities, and living on the Moon or Mars. With its multidisciplinary approach, Lunabotics also serves as a workforce pipeline, with teams gaining valuable hands-on experience in computer coding, engineering, manufacturing, fabricating, and other crucial skills, while also receiving technical expertise in space technology development.NASA's Lunabotics Challenge, held annually since 2010, is one of severalArtemis Student Challenges. The two-semester competition provides U.S. college and technical school teams an opportunity to design, build, and operate a prototype lunar robot usingNASA systems engineeringprocesses. Competitions help NASA get innovative design and operational data, reduce risks, and cultivate new ideas needed to return to the Moon under theArtemiscampaign to prepare for human exploration of Mars.To learn more about Lunabotics, visit:https://www.nasa.gov/learning-resources/lunabotics-challenge/

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URL: https://www.nasa.gov/centers-and-facilities/marshall/nasa-seeks-proposals-for-2026-human-exploration-rover-challenge/

TITLE: NASA Seeks Proposals for 2026 Human Exploration Rover Challenge

3 min readNASA Seeks Proposals for 2026 Human Exploration Rover Challenge

Beth RidgewayAug 15, 2025ArticleNASA now is accepting proposals from student teams for a contest to design, build, and test rovers for Moon and Mars exploration through Sept. 15.Known as the Human Exploration Rover Challenge, student rovers should be capable of traversing a course while completing mission tasks. Thechallenge handbookhas guidelines for remote-controlled and human-powered divisions.The cover of the HERC 2026 handbook, which is now available online."Last year, we saw a lot of success with the debut of our remote-controlled division and the addition of middle school teams," said Vemitra Alexander, the activity lead for the challenge at NASA's Marshall Space Flight Center in Huntsville, Alabama. "We're looking forward to building on both our remote-controlled and human-powered divisions with new challenges for the students, including rover automation."This year's mission mimics future Artemis missions to the lunar surface. Teams are challenged to test samples of soil, water, and air from sites along a half-mile course that includes a simulated field of asteroid debris, boulders, erosion ruts, crevasses, and an ancient streambed. Human-powered rover teams will play the role of two astronauts in a lunar terrain vehicle and must use a custom-built task tool to manually collect samples needed for testing. Remote-controlled rover teams will act as a pressurized rover, and the rover itself will contain the tools necessary to collect and test samples onboard."NASA's Human Exploration Rover Challenge creates opportunities for students to develop the skills they need to be successful STEM professionals," said Alexander. "This challenge will help students see themselves in the mission and give them the hands-on experience needed to advance technology and become the workforce of tomorrow."Seventy-five teams comprised of more than 500 students participated in the agency's 31st rover challenge in 2025. Participants represented 35 colleges and universities, 38 high schools, and two middle schools, across 20 states, Puerto Rico, and 16 nations around the world.The 32nd annual competition will culminate with an in-person event April 9-11, 2026, at the U.S. Space & Rocket Center near NASA Marshall.The rover challenge is one of NASA'sArtemis Student Challenges, reflecting the goals of theArtemiscampaign, which seeks to explore the Moon for scientific discovery, technology advancement, and to learn how to live and work on another world as we prepare for human missions to Mars. NASA uses such challenges to encourage students to pursue degrees and careers in the fields of science, technology, engineering, and mathematics.Since its inception in 1994, more than 15,000 students have participated in the rover challenge – with manyformer studentsnow working at NASA or within the aerospace industry.To learn more about HERC, visit:https://www.nasa.gov/roverchallenge/

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The 32nd annual competition will culminate with an in-person event April 9-11, 2026, at the U.S. Space & Rocket Center near NASA Marshall.

The rover challenge is one of NASA’sArtemis Student Challenges, reflecting the goals of theArtemiscampaign, which seeks to explore the Moon for scientific discovery, technology advancement, and to learn how to live and work on another world as we prepare for human missions to Mars. NASA uses such challenges to encourage students to pursue degrees and careers in the fields of science, technology, engineering, and mathematics.

Since its inception in 1994, more than 15,000 students have participated in the rover challenge – with manyformer studentsnow working at NASA or within the aerospace industry.

To learn more about HERC, visit:

https://www.nasa.gov/roverchallenge/

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TITLE: What is AI? (Grades 5-8)

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NASA STEM TeamDec 01, 2025ArticleContentsWhat is AI?How is NASA using AI?Advice From NASA AI ExpertsCareer CornerExplore MoreThis article is for students grades 5-8.What is AI?Artificial intelligence, or AI, is a type of technology that helps machines and computers have “thinking” abilities similar to humans. Devices using AI can learn words and concepts, recognize objects, see patterns, or make predictions. They can also be taught how to workautonomously. AI is often used to help people understand and solve problems more quickly than they could on their own.AI includes:Machine learning:This type of AI looks at large amounts of data and learns how to make fast and accurate predictions based on that data.Deep learning:This type helps computers operate much like the human brain. It uses several layers of “thought” to recognize patterns and learn new information. Deep learning is a type of machine learning.Generative AI:A human can use generative AI to create text, videos, images, and more. It is based on deep learning.NASA’s Perseverance Mars rover alongside the rock nicknamed “Cheyava Falls,” in this July 23, 2024, selfie made up of 62 individual images. “Cheyava Falls,” which has features that may bear on the question of whether the Red Planet was long ago home to microscopic life, is to the left of the rover near the center of the image. The small hole visible in the rock is where Perseverance collected the “Sapphire Canyon” core sample.NASA/JPL-Caltech/MSSSHow is NASA using AI?NASA has found uses for AI in many of its missions and programs.For missions to the Moon, AI can use satellite imagery to create detailed 3D maps of dark craters. This data could help scientists plan missions, spot hazards, and even identify where future crews might find water ice. On Mars, the Perseverance rover uses AI to drive itselfautonomously. 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For example, AI can count tarps on roofs in satellite images to measure damage after a storm. NASA is also supporting flight controllers and pilots by using AI to plan better flight routes, making air travel safer and more efficient.AI is helping NASA explore space, protect people, and make amazing discoveries!The blue tentacle-like arms containing gecko-like adhesive pads, attaBlue tentacle-like arms with gecko-like adhesive pads reach out and grapple a “capture cube” inside the International Space Station. The arms are attached to the cube-shaped Astrobee robotic free-flyer, right. The experimental grippers demonstrated techniques to autonomously perform tasks in low Earth orbit.NASAAdvice From NASA AI Experts“AI is a great field for people who like solving problems, building things, or asking questions about how the world works. People use AI to help doctors understand diseases, to teach robots how to explore space, and to help communities prepare for things like floods or wildfires. If you like using technology to help people and discover new things, AI could be a great career for you!” – Krista Kinnard, NASA’s Deputy Chief AI OfficerIn this illustration, astronauts work on the lunar surface as part of NASA’s Artemis program.NASAStart exploring coding and STEM activities like robotics clubs. Just remember to always stay curious, keep practicing, and don’t be afraid of making mistakes. This really helps you learn.Martin GarciaAI Adoption and Innovation Lead, NASA’s Johnson Space CenterCareer CornerNASA roles that may involve AI include:Astronauts: Astronauts on the International Space Station can use an AI “digital assistant” to get medical recommendations. This is helpful when communication with Earth is interrupted. 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TITLE: Helio Highlight: Comets, the Sun, and You

6 Min ReadHelio Highlight: Comets, the Sun, and YouInterstellar comet 3I/ATLAS passed through the inner solar system in the last months of 2025. By studying comets like 3I/ATLAS, scientists can learn more about the Sun and other stars, and about how planetary systems form in the first place.Credits:NASA/JPL-Caltech

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Nicholas OakesChantil Hunt EstevezDec 02, 2025ArticleContentsThe Sun’s Smallest CompanionsMessengers from AfarThe New Kid on the BlockHow Heliophysics Can HelpWhat We Can Learn TogetherResources for Educators ↓The Sun’s Smallest CompanionsThere's a lot orbiting the Sun. Aside from the eight planets, there are billions of smaller objects, too. One type is theasteroids, hunks of rock left over from the formation of the solar system. They mostly orbit in two belts: one between the orbits of Mars and Jupiter, and one out beyond Neptune.Cometsare icy bodies which swing through the solar system on long, highly elliptical orbits. Most of them orbit far beyond Pluto, in theOort Cloud.When comets swing in close to the Sun, its heat causes their outer shells to melt. Then the solar wind pushes against this loosened material and causes it to trail away from the Sun in a great, glowing tail. These tails are one of the most dynamic images that we associate with comets, as they can bloom up to many times larger than the size of planets as observed from Earth. When the comet departs the inner solar system, the tail fades as the comet freezes up again.Comet C/2013 A1 as seen by the Hubble Space Telescope, showcasing the dusty tail flowing behind the comet.NASAMessengers from AfarMost comets, including the famousHalley's Comet, come from within our solar system. That is to say that they were born in the same protoplanetary disk that birthed the Sun and the other bodies around it. We know this mostly by studying their orbits, since the paths they carve through space could only occur if they began on a closed orbit around the Sun.But these visitors can come from farther away, too. Much farther. On October 19, 2017, the University of Hawaii's Pan-STARRS1 telescope, which is funded by NASA'sNear-Earth Object Observations(NEOO) Program, tracked a possible asteroid in Earth's neighborhood. Continued observations showed that it was travelling at over 196,000 miles per hour (87.3 kilometers per second) and was accelerating into the inner solar system. That's comet behavior.Initially dubbed 1I/2017 U1, the object was given the name'Oumuamuaby the International Astronomical Union (IAU). This means "a messenger arriving from afar and arriving first" in Hawaiian, and recognizes that it is the first interstellar object known to have visited our solar system. It went beyond Saturn's orbit in early 2019 and is now heading back into deep space.An artist's concept of the first detected interstellar object, 1I/2017, later named 'Oumuamua by the International Astronomical Union.NASA/ESA/STScIThe New Kid on the Block'Oumuamua was first, but not last. Scientists now estimate that about one such object flies through our solar system every year – and now we know how to look for them. On July 1, 2025, the NASA-fundedATLAS(Asteroid Terrestrial-impact Last Alert System) survey telescope in Rio Hurtado, Chile, reported a new interstellar visitor. It was given the name3I/ATLAS.3I/ATLAS has been extensively observed since then. TheHubble Space Telescopeobserved it on July 21, 2025, when it was about 277 million miles (446 million kilometers) away from Earth. These observations revealed a teardrop-shaped tail flying off of the nucleus, which scientists estimate to be anywhere up to 3.5 miles (5.6 kilometers) in diameter. Other observations have come in from assets as diverse as theJames Webb Space Telescope, thePerseverance roveron Mars, and ESA/NASA'sSolar and Heliospheric Observatory(SOHO).How Heliophysics Can HelpThe Sun is so bright that its light would burn out many of the telescopes which would otherwise observe 3I/ATLAS as it approaches perihelion, or closest approach to the Sun. But NASA's heliophysics mission fleet is designed to observe the Sun closely, meaning that these spacecraft can keep an eye on 3I/ATLAS as it nears the Sun. This will allow scientists to gather vital data about how comets interact with the Sun, the solar wind, and more, as well as how interstellar objects like 3I/ATLAS differ from objects native to our own solar system.A faint image of comet 3I/ATLAS as observed by ESA/NASA's SOHO mission between Oct. 15-26, 2025. The comet appears as a slight brightening in the center of the image. This mission, launched in December 2025, has discovered many comets in its years of observing the Sun.Lowell Observatory/Qicheng ZhangOne instrument observing 3I/ATLAS is SOHO'sLarge Angle and Spectrometric Coronagraph(LASCO). A coronagraph is a telescope which blocks light from the solar disk in order to see the Sun's faint atmosphere, or corona. LASCO has three coronagraphs that image the corona from 1.1 to 32 solar radii (one solar radius is about 420,000 miles, or 700,000 kilometers).LASCO was able to see 3I/ATLAS when it made its closest approach to the Sun on October 30, 2025. At this closest approach, the object was about 1.36astronomical units (AU)from the Sun, or about 126 millions miles (203 million kilometers). It won't make its closest approach to Earth until sometime around December 19, when it will be about 1.8 AU from our planet. From there, it will proceed back out of the solar system on a long, dark journey through interstellar space.What We Can Learn TogetherWhile 3I/ATLAS makes its close approach to the Sun, LASCO and other such instruments will help gather as much data as possible. After the images they return are processed, this data should help scientists to better understand where this strange visitor comes from and what it is made of. Using data gathered on objects like 3I/ATLAS, scientists can work backwards to make more educated inferences about the origins of planetary systems, including our own.Heliophysics can also use data from comets to make discoveries about how the Sun and other stars interact with objects in their orbit. For example, scientists used observations of the tails of comets to help prove the existence of the solar wind. 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There's a lot orbiting the Sun. Aside from the eight planets, there are billions of smaller objects, too. One type is theasteroids, hunks of rock left over from the formation of the solar system. They mostly orbit in two belts: one between the orbits of Mars and Jupiter, and one out beyond Neptune.Cometsare icy bodies which swing through the solar system on long, highly elliptical orbits. Most of them orbit far beyond Pluto, in theOort Cloud.

When comets swing in close to the Sun, its heat causes their outer shells to melt. Then the solar wind pushes against this loosened material and causes it to trail away from the Sun in a great, glowing tail. These tails are one of the most dynamic images that we associate with comets, as they can bloom up to many times larger than the size of planets as observed from Earth. When the comet departs the inner solar system, the tail fades as the comet freezes up again.

Comet C/2013 A1 as seen by the Hubble Space Telescope, showcasing the dusty tail flowing behind the comet.NASA

Comet C/2013 A1 as seen by the Hubble Space Telescope, showcasing the dusty tail flowing behind the comet.

Most comets, including the famousHalley's Comet, come from within our solar system. That is to say that they were born in the same protoplanetary disk that birthed the Sun and the other bodies around it. We know this mostly by studying their orbits, since the paths they carve through space could only occur if they began on a closed orbit around the Sun.

But these visitors can come from farther away, too. Much farther. On October 19, 2017, the University of Hawaii's Pan-STARRS1 telescope, which is funded by NASA'sNear-Earth Object Observations(NEOO) Program, tracked a possible asteroid in Earth's neighborhood. Continued observations showed that it was travelling at over 196,000 miles per hour (87.3 kilometers per second) and was accelerating into the inner solar system. That's comet behavior.

Initially dubbed 1I/2017 U1, the object was given the name'Oumuamuaby the International Astronomical Union (IAU). This means "a messenger arriving from afar and arriving first" in Hawaiian, and recognizes that it is the first interstellar object known to have visited our solar system. It went beyond Saturn's orbit in early 2019 and is now heading back into deep space.

An artist's concept of the first detected interstellar object, 1I/2017, later named 'Oumuamua by the International Astronomical Union.NASA/ESA/STScI

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'Oumuamua was first, but not last. Scientists now estimate that about one such object flies through our solar system every year – and now we know how to look for them. On July 1, 2025, the NASA-fundedATLAS(Asteroid Terrestrial-impact Last Alert System) survey telescope in Rio Hurtado, Chile, reported a new interstellar visitor. It was given the name3I/ATLAS.

3I/ATLAS has been extensively observed since then. TheHubble Space Telescopeobserved it on July 21, 2025, when it was about 277 million miles (446 million kilometers) away from Earth. These observations revealed a teardrop-shaped tail flying off of the nucleus, which scientists estimate to be anywhere up to 3.5 miles (5.6 kilometers) in diameter. Other observations have come in from assets as diverse as theJames Webb Space Telescope, thePerseverance roveron Mars, and ESA/NASA'sSolar and Heliospheric Observatory(SOHO).

The Sun is so bright that its light would burn out many of the telescopes which would otherwise observe 3I/ATLAS as it approaches perihelion, or closest approach to the Sun. But NASA's heliophysics mission fleet is designed to observe the Sun closely, meaning that these spacecraft can keep an eye on 3I/ATLAS as it nears the Sun. This will allow scientists to gather vital data about how comets interact with the Sun, the solar wind, and more, as well as how interstellar objects like 3I/ATLAS differ from objects native to our own solar system.

A faint image of comet 3I/ATLAS as observed by ESA/NASA's SOHO mission between Oct. 15-26, 2025. The comet appears as a slight brightening in the center of the image. This mission, launched in December 2025, has discovered many comets in its years of observing the Sun.Lowell Observatory/Qicheng Zhang

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Lowell Observatory/Qicheng Zhang

One instrument observing 3I/ATLAS is SOHO'sLarge Angle and Spectrometric Coronagraph(LASCO). A coronagraph is a telescope which blocks light from the solar disk in order to see the Sun's faint atmosphere, or corona. LASCO has three coronagraphs that image the corona from 1.1 to 32 solar radii (one solar radius is about 420,000 miles, or 700,000 kilometers).

LASCO was able to see 3I/ATLAS when it made its closest approach to the Sun on October 30, 2025. At this closest approach, the object was about 1.36astronomical units (AU)from the Sun, or about 126 millions miles (203 million kilometers). It won't make its closest approach to Earth until sometime around December 19, when it will be about 1.8 AU from our planet. From there, it will proceed back out of the solar system on a long, dark journey through interstellar space.

While 3I/ATLAS makes its close approach to the Sun, LASCO and other such instruments will help gather as much data as possible. After the images they return are processed, this data should help scientists to better understand where this strange visitor comes from and what it is made of. Using data gathered on objects like 3I/ATLAS, scientists can work backwards to make more educated inferences about the origins of planetary systems, including our own.

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You can watch the Global Winners Announcementhereto meet these winning teams and learn about the inspiration behind their projects.

2024 NASA Space Apps Challenge Global Winners

Best Use of Science Award:WMPGang

Team Members:Dakota C., Ian C., Maximilian V., Simon S.

Challenge:Create an Orrery Web App that Displays Near-Earth Objects

Country/Territory:Waterloo,Canada

Using their skills in programming, data analysis, and visualization, WMPGang created a web app that identifies satellite risk zones using real-time data on Near-Earth Objects and meteor streams.

Learn more about WMPGang’sSkyShield: Protecting Earth and Satellites from Space Hazardsproject

Best Use of Data Award:GaamaRamma

Team Members:Aakash H., Arun G., Arthur A., Gabriel A., May K.

Challenge:Leveraging Earth Observation Data for Informed Agricultural Decision-Making

Country/Territory:Universal Event, United States

GaamaRamma’s team of tech enthusiasts aimed to create a sustainable way to help farmers efficiently manage water availability in the face of drought, pests, and disease.

Learn more about GaamaRamma’sWaterwiseproject

Best Use of Technology Award:42 QuakeHeroes

Team Members:Alailton A., Ana B., Gabriel C., Gustavo M., Gustavo T., Larissa M.

Challenge:Seismic Detection Across the Solar System

Country/Territory:Maceió, Brazil

Team 42 QuakeHeroes employed a deep neural network model to identify the precise locations of seismic events within time-series data. They used advanced signal processing techniques to isolate and analyze unique components of non-stationary signals.

Learn more about 42 QuakeHeroes' project

Galactic Impact Award:NVS-knot

Team Members:Oksana M., Oleksandra M., Prokipchyn Y., Val K.

Country/Territory:Kyiv, Ukraine

The NVS-knot team assessed planting conditions using surface soil moisture and evapotranspiration data, then created an app that empowers farmers to manage planting risks.

Learn more about NVS-knot's2plant | ! 2plantproject

Best Mission Concept Award:AsturExplorers

Team Members:Coral M., Daniel C., Daniel V., Juan B., Samuel G., Vladimir C.

Challenge:Landsat Reflectance Data: On the Fly and at Your Fingertips

Country/Territory:Gijón, Spain

AsturExplorers created Landsat Connect, a web app that provides a simple, intuitive way to track Landast satellites and access Landsat surface reflectance data. The app also allows users to set a target location and receive notifications when Landsat satellites pass over their area.

Learn more about AsturExplorers'Landsat Connectproject

Most Inspirational Award:Innovisionaries

Team Members:Rikzah K., Samira K., Shafeeqa J., Umamah A.

Challenge:SDGs in the Classroom

Country/Territory:Sharjah, United Arab Emirates

Innovisionaries developed Eco-Metropolis to inspire sustainability through gameplay. This city-building game engages players in making critical urban planning and resource management decisions based on real-world environmental data.

Learn more about Innovisionaries'Eco-Metropolis: Sustainable City Simulationproject

Best Storytelling Award:TerraTales

Team Members:Ahmed R., Fatma E., Habiba A., Judy A., Maya M.

Challenge:Tell Us a Climate Story!

Country/Territory:Cairo, Egypt

TerraTales shared stories of how Earth's changing climate affects three unique regions: Egypt, Brazil, and Germany. The web app also features an artificial intelligence (AI) model for climate forecasting and an interactive game to encourage users to make eco-friendly choices.

Learn more about TerraTale's project

Global Connection Award:Asteroid Destroyer

Team Members:Kapeesh K., Khoi N., Sathyajit L., Satyam S.

Challenge:Navigator for the Habitable Worlds Observatory (HWO): Mapping the Characterizable Exoplanets in our Galaxy

Country/Territory:Saskatoon, Canada

Team Asteroid Destroyer honed in on exoplanets, utilizing data processing and machine learning techniques to map exoplanets based on size, temperature, and distance.

Learn more about Asteroid Destroyer's project

Art & Technology Award:Connected Earth Museum

Team Members:Gabriel M., Luc R., Lucas R., Mattheus L., Pedro C., Riccardo S.

Challenge:Imagine our Connected Earth

Country/Territory:Campinas, Brazil

Team Connected Earth Museum created an immersive virtual museum experience to raise awareness of Earth's changing climate. An AI host guides users through an interactive gallery featuring 3D and 2D visualizations, including a time series on Earth and ocean temperatures, population density, wildfires, and more.

Learn more about Connected Earth Museums' project

Local Impact Award:Team I.O.

Team Members:Frank R., Jan K., Raphael R., Ryan Z., Victoria M.

Challenge:Community Mapping

Country/Territory:Florianópolis, Brazil

Team I.O. bridges the gap between complex Geographic Information Systems data and user-friendly communication, making critical environmental information accessible to everyone, regardless of technical expertise.

Learn more about Team I.O.'sG.R.O.W. (Global Recovery and Observation of Wildfires)project

Want to take part in the 2025 NASA Space Apps Challenge?Mark your calendars for October 4 and 5! Registration will open in July. At that time, participants will be able to register for a local event hosted by NASA Space Apps leads from around the world. You can stay connected with NASA Space Apps onFacebook,Instagram, andX.

Space Apps is funded by NASA'sEarth Science Divisionthrough a contract withBooz Allen Hamilton,Mindgrub, andSecondMuse.

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